Topics Covered

Topic
1. Introduction/ Measure of Information Entropy Joint Entropy Conditional Entropy KL - Divergence Mutual Information Conditioal Divergence Conditional Mutual Information Convexity and Concavity Markov Chains Data Processing Inequality Sufficient Statistics Fano’s Inequality
2. Fixed to Fixed Length Data Compression Fixed to Fixed Length Source Coding Theorem Coding Theorem Asysmptotic Equipartition Property (AEP) Typical Sets
3. Fixed to Variable Length Data Compression Non-Singular Codes Uniequely Decodable Codes Prefix/ Instantaneous Codes Kraft Inequality Prefix Codes with Minimum Expected Codeward Length Shanon Fano Code n - length Blocks of Source Symbols Mismatched Lossless Variable Length Codes UD Codes with Kraft’s Inequality Non Singular Codes and their Versions of Kraft’s Inequality Huffman Codes

4. Entropy Rate
Stationarity
Entropy Rate of a Stationary Process
Entropy Rate of a Markov Chain

5. Universal Source Coding
Lempel Ziv Algorithm

6. Channel Coding and Channel Capacity (&Discrete Memoryless Channels)
Definitions
Preview of CCT
Examples of Channel Capacity
Converse of CCT
Jointly Typical Sequences
Achievability of CCT
Channel Capacity with Feedback
Jonit Source Channel Coding Theorem/ Seperation Theorem

7. Continous RVs and Differential Entropy - Lightly Covered
Differential Entropy
Conditional Differential Entropy
Differential Entropy of a Gaussian RV
Divergence and Mutual Information
Other Results

8. Gaussian Channel - Lightly Covered

9. Rate Distortion Theory
Definitons
Preview of RDT
Calculation of Rate Distortion Function
Converse of RDT
Distortion Typical Sequences
Achievability of RDT
Distortion Rate Function (DRT)
Jonit Source Channel Coding Theorem/ Seperation Theorem with Distortion

Introduction

Assignment - Read Chapters I, II, III of Cover, Thomas

A Brief Historical Perspective

- Information theory was started by "Claude E. Shannon" in the paper "A Mathematical Theory of Communication" 1948.
- Shannon's model is on point-to-point communication.

Two-Basic Sets of Problems

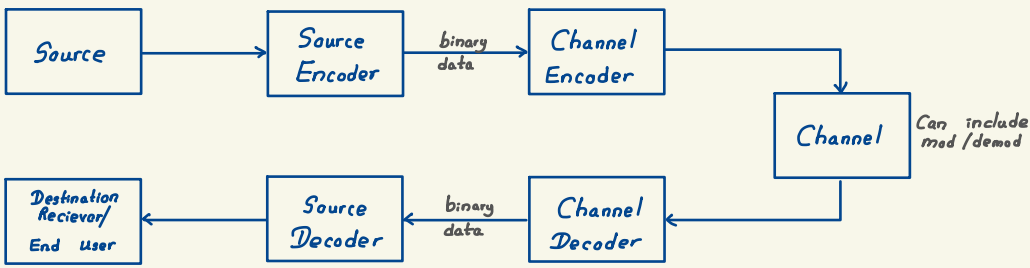
1) Lossless data compression / Lossy data compression.
2) Transmission over a noisy channel.

A Code

- A Code = Encoder + Decoder
- A code accomplishes "reliable transmission", if the joint operation of encoder, channel and decoder results in source messages being delivered to the destination, while satisfying a "QOS" requirement or "fidelity criterion."

Source - Channel Seperation

- When different types of source messages (ex:- images, audio) is sent through different or same channels (ex:- co-axial cables, micro-waves), source - channel seperation is needed



Source Encoder / Decoder

- How many bits are needed to adequately seperate a source message?
- Based solely on source characteristics.
- Data compression / Source coding is done.

Channel Encoder / Decoder

- What is the largest number of information bits, that can be reliably transmitted over the channel?
- Based solely on channel characteristics.
- Channel coding / Forward Errorr Correction (FEC) is done.

Mathematical Models

- Mathematical models are created with under several ideal assumptions to analyse the systems.
- These are "Probabilistic Models for Sources and Channels".
- Have specific applications.

Measure of Information

Fundemental Information Theoretic Measures (Ω, a, P)

H(X)

1) Entropy

- Entropy of a random variable is measure of it's uncertainity.

Definition :- Let X is a finite valued r.v with values in a finite set $\mathcal{X}$. Let

pmf P_X : $P_X(x) = P(X = x) \quad x \in \mathcal{X}$

$\mathcal{X}$ is called alphabet of X

Entropy of r.v. X :

$$H(X) \triangleq \sum_{x \in \mathcal{X}} P_X(x) \cdot \log \frac{1}{P_X(x)}$$

$$\triangleq - \sum_{x \in \mathcal{X}} P_X(x) \cdot \log P_X(x)$$

$$H(X) \triangleq \mathbb{E}_{P_X}\left[- \log P_X(X) \right]$$

Replaced with r.v for notational ease

- $H(X)$ depends only on the p.m.f of X (i.e P_X), but not on the specific realizations of X.
- Convention is, all logs are taken to base 2. Hence, entropy is in bits.
- Looking at the continuity of $x \cdot \log x$ and taking limit to zero,

$$0 \cdot \log 0 = \lim_{x \downarrow 0} x \cdot \log x = 0$$

- Hereafter, P will be used instead of P_X

$$\text{Entropy of r.v. } X = H(X) \triangleq \mathbb{E}\left[- \log P(X) \right]$$

Lemma 1 $H(X) \geq 0$

Proof: $H(X) = \mathbb{E}[-\log P(X)]$

$= \mathbb{E}[\log 1/P(X)]$

≥ 0 ..... since $-\log P(X) \geq 0$ almost surely (a.s.) ■

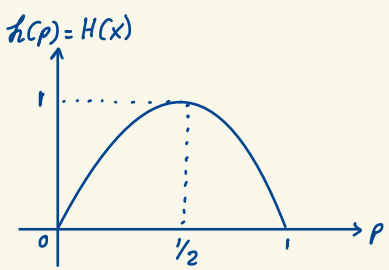
Binary Entropy Function

Example: Let $X \in \mathcal{X} = \{0,1\}$ with $P(X=1) = p = 1 - P(X=0)$, $0 \leq p \leq 1$. (i.e. $X \sim$ Bernoulli(p)). The entropy of this specific r.v. is called "binary entropy function" $h(p)$

Binary Entropy Function $= h(p) = H(X)$

$= -p \log p - (1-p) \cdot \log(1-p)$

- If $p=0$ or $p=1$, we have no uncertainity about the realization of X. $(h(0) = h(1) = 0)$
- When $p = 1/2$, we have the maximum uncertainity of 1 bits. $(h(1/2) = 1 \text{ bit})$
- h is a concave function in p.

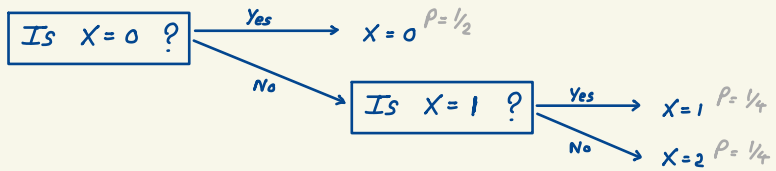


Binary Questions and Entropy

- $H(X)$ has a key role in determining the minimum expected number of 'binary' questions that must be asked to resolve the value of X.

Example:- $X \in \mathcal{X} = \{0, 1, 2\}$. $P_X(0) = 1/2$, $P_X(1) = P_X(2) = 1/4$

$$\Rightarrow H(X) = -\tfrac{1}{2}\cdot\log\tfrac{1}{2} - \tfrac{1}{4}\log\tfrac{1}{4} - \tfrac{1}{4}\cdot\log\tfrac{1}{4} = 1.5 \text{ bits}$$



$$\Rightarrow \mathbb{E}[\text{No of questions needed}] = 1(\tfrac{1}{2}) + 2(\tfrac{1}{4}) + 2(\tfrac{1}{4}) = 1.5$$

$$\Rightarrow H(X) = \mathbb{E}[\text{No of questions needed}]$$

- In general,

$$H(X) \leqslant \text{Minimum}\left(\mathbb{E}[\text{No. of binary questions needed}]\right) \leqslant H(X) + 1$$

$H(X,Y)$

2) Joint Entropy

Definition : X and Y and finite valued r.vs with values in finite set $\mathcal{X}, \mathcal{Y}$. Let

$$\text{Joint pmf of } X, Y = P_{XY} = P_{XY}(x,y) = P(X=x, Y=y) \qquad \begin{matrix} x \in \mathcal{X} \\ y \in \mathcal{Y} \end{matrix}$$

Then,

$$\text{Joint Entropy of } (X,Y) = H(X,Y) \triangleq -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x,y) \cdot \log P_{XY}(x,y)$$

$$\boxed{H(X,Y) \triangleq -\mathbb{E}\left[\log P_{XY}(X,Y)\right]}$$

- $H(X,Y) \geq \begin{cases} H(X) \\ H(Y) \end{cases}$

Proof : From definition,

$$\Rightarrow H(X,Y) = -\mathbb{E}\left[\log P_{XY}(X,Y)\right]$$

(from conditional pmf's in EREE620)

$$= -\mathbb{E}\left[\log P_X(x) \cdot P_{Y|X}(Y|x)\right]$$

$$= -\mathbb{E}\left[\log P_X(x)\right] - \mathbb{E}\left[\log P_{Y|X}(Y|x)\right]$$

$$= H(X) + \underbrace{\mathbb{E}\left[-\log P_{Y|X}(Y|x)\right]}_{\text{This quantity is } \geq 0 \text{ a.s. since log of a probability is negative}}$$

(This quantity inspires conditional entropy)

$$\Rightarrow H(X,Y) \geq H(X)$$

■

3) Conditional Entropy

$H(Y|X)$

Definition: X and Y and finite valued r.vs with values in finite set $\mathcal{X}, \mathcal{Y}$. Let

Conditional pmf of Y given X $= P_{Y|X} = P_{Y|X}(y|x) = P(Y=y \mid X=x)$ $\quad x \in \mathcal{X}, y \in \mathcal{Y}$

Then,

Conditional Entropy of Y given X $= H(Y|X) \triangleq -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x,y) \cdot \log P_{Y|X}(y|x)$

$$H(Y|X) \triangleq -\mathbb{E}\left[\log P_{Y|X}(Y|X)\right]$$

- $H(Y|X) \geq 0$

  Since $-\log P_{Y|X}(Y|X) \geq 0$ and $H(Y|X)$ is the expectation of that

- Conditional Entropy of Y given $X=x$ $\triangleq H(Y|X=x) \triangleq -\sum_{y \in \mathcal{Y}} P_{Y|X}(y|x) \cdot \log P_{Y|X}(y|x)$

  Just a notation, i.e. definition.

- $H(Y|X) = \mathbb{E}\left[H(Y|X=x)\right] = \sum_{x \in \mathcal{X}} P_X(x) \cdot H(Y|X=x)$

  Proof: From the definition of conditional entropy.

$$\begin{aligned} H(Y|X) &\triangleq -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x,y) \cdot \log P_{Y|X}(y|x) \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_X(x) \cdot P_{Y|X}(y|x) \cdot \log P_{Y|X}(y|x) \\ &= \sum_{x \in \mathcal{X}} \left(P_X(x) \left(\sum_{y \in \mathcal{Y}} P_{Y|X}(y|x) \cdot \log P_{Y|X}(y|x) \right) \right) \\ &= \sum_{x \in \mathcal{X}} P_X(x) \cdot H(Y|X=x) \end{aligned}$$

- $H(Y|X=x) \gtreqless H(Y|X)$

  When X is realized, the conditional entropy of Y can increase, decrease or not change at all.

Lemma 2 - Chain Rule for Entropy

$$H(X,Y) = H(X) + H(Y|X)$$
$$= H(Y) + H(X|Y)$$

Proof: Done already from two definitions.

Example: $\mathcal{X} = \mathcal{Y} = \{0,1\}$ are the two alphabets. Assume $P_X(0) = P_X(1) = 1/2$. Assume $P_{Y|X}$ is given in the following stochastic matrix.

$P_{Y|X} \rightarrow$

X \ Y	0	1
0	0.25	0.75
1	0.5	0.5

Find $H(X)$, $H(Y)$, $H(Y|X=x)$ and $H(Y|X)$

$$H(X) = -\sum_{x\in\mathcal{X}} P_X(x)\cdot \log P_X(x) = -0.5 \log 0.25 - 0.5 \log 0.5 = 1 \text{ bit}$$

$$P(Y=0) = \sum P(Y=0|X=x)\cdot P(X=x) = 0.25 \times 0.5 + 0.5 \times 0.5 = 5/8$$
$$P(Y=1) = 3/8$$

$$H(Y) = -\sum_{y\in\mathcal{Y}} P_Y(y)\cdot \log P_Y(y) = -5/8 \log 5/8 - 3/8 \cdot \log 3/8 = 0.95 \text{ bit}$$

$$H(Y|X=1) = -\sum_y P_{Y|X}(y|1)\cdot \log P_{Y|X}(y|1) = -0.5 \log 0.5 - 0.5 \log 0.5 = 1 \text{ bit}$$

$$H(Y|X=0) = -\sum_y P_{Y|X}(y|0)\cdot \log P_{Y|X}(y|0) = -0.25 \log 0.25 - 0.75 \log 0.75 = 0.815 \text{ bit}$$

$$H(Y|X) = \sum_x P_X(x)\cdot H(Y|X=x) = 0.5 \times 1 \text{ bit} + 0.5 \times 0.815 \text{ bit} = 0.910 \text{ bit}$$

$H(Y|X) < H(Y)$ - is this always true?

$H(Y|X)$ = 0.91, $H(Y)$ = 0.95

Given some additional information (i.e X), the uncertainity of Y should reduce

To prove this we need a technical result (log sum inequality) and a notion of divergence.

Lemma 3 - Logsum Inequality

If a_i, b_i, $i = 1, \ldots, r$ are arbitary non-negative numbers, then

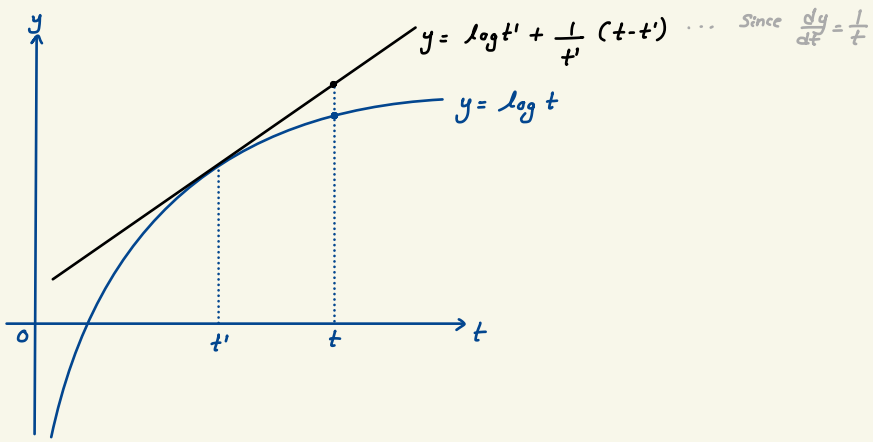
$$\sum_{i=1}^{r} a_i \log\left(\frac{a_i}{b_i}\right) \geq \left(\sum_{i=1}^{r} a_i\right) \log\left(\frac{\sum_{i=1}^{r} a_i}{\sum_{i=1}^{r} b_i}\right)$$

with equality iff (if and only if)

$$\frac{a_i}{b_i} = \frac{\left(\sum_{j=1}^{r} a_j\right)}{\left(\sum_{j=1}^{r} b_j\right)} \qquad i = 1, \ldots, r$$

- Conventions : $0 \cdot \log 0 = 0$ $\quad 0 \cdot \log \frac{0}{0} = 0$ $\quad 0 \cdot \log \frac{a}{0} = +\infty$ if $a > 0$
- All logs are to the base 'e'. But everything can be converted to base 2 with no effect.

Proof: Assume $t' > 0$ is an arbitary non-negative value. Observe that the black tangent at t' always lies above the log function in blue, due to concaveness.



Hence

$$\Rightarrow \log t \leq \log t' + \frac{1}{t'}(t - t') \quad \cdots\cdots \circledast$$ with equality iff $t = t'$

Substitute $t = \frac{b_i}{a_i}$, $t' = \sum_{j=1}^{r} b_j \Big/ \sum_{j=1}^{r} a_j$ for $i = 1, \ldots, r$

$$\Rightarrow \log \frac{b_i}{a_i} \leqslant \log\left(\frac{\sum_{j=1}^{r} b_j}{\sum_{j=1}^{r} a_j}\right) + \left(\frac{\sum_{j=1}^{r} a_j}{\sum_{j=1}^{r} b_j}\right)\cdot\left(\frac{b_i}{a_i} - \frac{\sum_{j=1}^{r} b_j}{\sum_{j=1}^{r} a_j}\right)$$

Multiply both sides by a_i

$$\Rightarrow a_i \log \frac{b_i}{a_i} \leqslant a_i \log\left(\frac{\sum_{j=1}^{r} b_j}{\sum_{j=1}^{r} a_j}\right) + \left(\frac{\sum_{j=1}^{r} a_j}{\sum_{j=1}^{r} b_j}\right)\cdot\left(b_i - a_i \frac{\sum_{j=1}^{r} b_j}{\sum_{j=1}^{r} a_j}\right)$$

Sum above over the $i = 1, \ldots, r$

$$\Rightarrow \sum_i \left(a_i \log \frac{b_i}{a_i}\right) \leqslant \sum_i \left(a_i \cdot \log \frac{\sum b_j}{\sum a_j}\right) + \left(\frac{\sum a_j}{\sum b_j}\right)\cdot\left(\sum_i b_i - \sum_i a_i \frac{\sum b_j}{\sum a_j}\right)$$

$$\Rightarrow \sum_i \left(a_i \log \frac{b_i}{a_i}\right) \leqslant \sum_i \left(a_i \cdot \log \frac{\sum b_j}{\sum a_j}\right) + \left(\frac{\sum a_j}{\sum b_j}\right)\cdot\left(\sum_i b_i\right) - \left(\frac{\sum a_j}{\sum b_j}\right)\left(\sum_i a_i \frac{\sum b_j}{\sum a_j}\right)$$

$$\Rightarrow \sum_i \left(a_i \log \frac{b_i}{a_i}\right) \leqslant \sum_i \left(a_i \cdot \log \frac{\sum b_j}{\sum a_j}\right)$$

$$\Rightarrow \sum_i \left(a_i \log \frac{b_i}{a_i}\right) \leqslant \left(\sum_i a_i\right)\left(\log \frac{\sum b_j}{\sum a_j}\right)$$

multiply by (-1) which flips the numerator inside the logs,

$$\Rightarrow \sum_i \left(a_i \log \frac{a_i}{b_i}\right) \geqslant \left(\sum_i a_i\right)\left(\log \frac{\sum a_i}{\sum b_i}\right)$$

■

Solomon Kullback
Richard Leibler
1951
$D(P\|Q)$

4) Kullback - Leibler Divergence (KL Divergence / Relative Entropy)

- This is a notion to measure the seperation between two probability distributions. Used in diffusion, as a loss function to train the models to learn similar distributions to given images.
- This came after Shanon, in 1951.
- Although it can be viewed as a seperation between distributions, it's not a metric.

Definition: Given two pmfs P, Q on the same finite alphabet $\mathcal{X}$, the KL-divergence $D(P\|Q)$ between P and Q is defined by;

$$D(P\|Q) = \sum_{x \in \mathcal{X}} P(x) \cdot \log \frac{P(x)}{Q(x)}$$

- Convention: $0 \cdot \log \frac{0}{0} \triangleq 0$
  $a \cdot \log \frac{a}{0} \triangleq +\infty$ if $a > 0$
  i.e. $D(P\|Q) \triangleq +\infty$ if $P(x) > 0$ for some $x \in \mathcal{X}$ for which $Q(x) = 0$
- In other words,

$$D(P\|Q) = \mathbb{E}_P \left[\log \frac{P(x)}{Q(x)} \right] \quad \dots P, Q \text{ pmfs on } \mathcal{X}$$

- $D(P\|Q) \neq D(Q\|P)$

Lemma 4 - $D(P\|Q) \geq 0$ with equality iff $P = Q$ (i.e. $P(x) = Q(x) \quad x \in \mathcal{X}$)

Proof:- By using the definition on KL Divergence,

$$D(P\|Q) = \sum_{x\in\mathcal{X}} P(x) \cdot \log \frac{P(x)}{Q(x)}$$

$$\geq \left(\sum_{x'\in\mathcal{X}} P(x')\right) \cdot \log \left(\frac{\sum_{x\in\mathcal{X}'} P(x')}{\sum_{x\in\mathcal{X}'} Q(x')}\right)$$

$$= 1 \cdot \log \frac{1}{1}$$

$$= 0$$

Equality holds according to logsum inequality, iff

$$\Leftrightarrow \frac{P(x)}{Q(x)} = \frac{\sum_{x'\in\mathcal{X}} P(x')}{\sum_{x'\in\mathcal{X}} Q(x')} = 1 \qquad x \in \mathcal{X}$$

$$\Leftrightarrow P(x) = Q(x) \quad \text{ie} \quad P = Q$$

■

This happens even if $\sum_{x\in\mathcal{X}'} Q(x') < 1$ with $Q(x') \geq 0 \quad x' \in \mathcal{X}$.

$$D(P_{XY} \| P_X \times P_Y) = H(Y) - H(Y|X) = H(X) - H(X|Y)$$
$$= H(X) + H(Y) - H(X,Y)$$

Proof:- $H(Y) - H(Y|X) = H(Y) - (H(X,Y) - H(X))$ ..... Lemma 2. Chain rule of entropy

$$= H(X) + H(Y) - H(X,Y)$$
$$= \mathbb{E}[-\log P_X(X)] + \mathbb{E}[-\log P_Y(Y)] - \mathbb{E}[-\log P_{XY}(X,Y)]$$
$$= \mathbb{E}_{P_{XY}}\left[\log \frac{P_{XY}(X,Y)}{P_X(X)\cdot P_Y(Y)}\right]$$
$$= D(P_{XY} \| P_X \times P_Y)$$

p.m.f is on $\mathcal{X}\times\mathcal{Y}$

$\perp\!\!\!\perp$ - Independance
$\perp$ - Uncorrelated

Lemma 5 - Conditioning Reduces Entropy

$H(Y|X) \leq H(Y)$
$H(X|Y) \leq H(X)$ with equality iff $X \perp\!\!\!\perp Y$

Proof: From the previous result, we have

$$\Rightarrow H(Y) - H(Y|X) = \mathcal{D}\left(P_{XY} \,\|\, P_X \times P_Y\right) \geq 0$$

(P_{XY}: p.m.f is on $\mathcal{X} \times \mathcal{Y}$)

$$\Rightarrow H(Y|X) \leq H(Y)$$

Equality holds if

$$\mathcal{D}(P_{XY} \,\|\, P_X \times P_Y) = 0 \iff P_{XY} = P_X P_Y$$
$$\iff P_{XY}(x,y) = P_X(x) \cdot P_Y(y) \quad x \in \mathcal{X},\ y \in \mathcal{Y}$$
$$\iff X \perp\!\!\!\perp Y$$

■

Note: From the above lemma, $H(X|Y) \leq H(X)$. This inspires to define quantity based on how much of a entropy reduction happens when conditioning. This is called Mutual Information

$I(X;Y)$
$I(X \wedge Y)$

5) Mutual Information

Definition: X, Y are r.v.s with joint p.m.f P_{XY} on finite sets $\mathcal{X} \times \mathcal{Y}$. The Mutual Information between X and Y is defined as,

$$I(X \wedge Y) \triangleq H(X) - H(X|Y) = H(Y) - H(Y|X)$$
$$= H(X) + H(Y) - H(X,Y)$$
$$I(X \wedge Y) = \mathcal{D}(P_{XY} \,\|\, P_X \times P_Y)$$

- $I(X \wedge Y)$ = divergence between P_{XY} and $P_X \times P_Y$
- $I(X \wedge X) = H(X) - H(X|X) = H(X)$ (No uncertainty in X if you know X. i.e $P_{X|X}$ is a point mass)

Lemma 6 $I(X \wedge Y) \geq 0$ with equality iff $X \perp\!\!\!\perp Y$

Proof: This follows from Lemma 4, and also from the definition of mutual information

$$I(X \wedge Y) = \mathcal{D}(P_{XY} \,\|\, P_X \times P_Y) \geq 0$$

Lemma 4 showed that equality occurs iff $P_{XY} = P_X \times P_Y$, i.e. $X \perp\!\!\!\perp Y$, which applies here as well. ∎

Example: $\mathcal{X} = \mathcal{Y} = \{0,1\}$ are the two alphabets. Assume $P_X(0) = P_X(1) = 1/2$. Assume $P_{Y|X}$ is given in the following stochastic matrix.

$P_{Y|X} \rightarrow$

X \ Y	0	1
0	0.25	0.75
1	0.5	0.5

We calculated $H(X) = 1$, $H(Y) = 0.95$, $H(Y|X) = 0.91$, $P(Y=0) = 5/8$, $P(Y=1) = 3/8$. Find $I(X \wedge Y)$.

$$I(X \wedge Y) = H(Y) - H(Y|X) = 0.95 - 0.91 = 0.04 //$$

We can do the calculation by taking Y as the given probability. From Baye's Theorem,

$$P(X=1|Y=0) = \frac{P(Y=0|X=1)\cdot P(X=1)}{P(Y=0)} = \frac{1/2 \cdot 1/2}{5/8} = \frac{2}{5}$$

$$P(X=0|Y=0) = \frac{3}{5}$$

$$P(X=1|Y=1) = \frac{P(Y=1|X=1)\cdot P(X=1)}{P(Y=1)} = \frac{1/2 \cdot 1/2}{3/8} = \frac{2}{3}$$

$$P(X=0|Y=1) = \frac{1}{3}$$

$$H(X|Y=0) = -\sum_x P_{X|Y}(x|0)\cdot \log P_{X|Y}(x|0)$$
$$= -2/5 \log 2/5 - 3/5 \log 3/5$$

$$H(X|Y=1) = -\sum_x P_{X|Y}(x|1)\cdot \log P_{X|Y}(x|1)$$
$$= -2/3 \log 2/3 - 1/3 \log 1/3$$

$$H(X|Y) = \sum_y P_Y(y)\cdot H(X|Y=y)$$

$$\therefore H(X) - H(X|Y) = 0.04$$

Example : $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ $P_X(0) = 1/4$, $P_X(1) = 3/4$. Find $H(X|Y)$.

$P_{Y|X} \rightarrow$

X \ Y	0	1
0	1/2	1/2
1	1/2	1/2

$$H(X) = -\sum_x P_X(x)\cdot \log P_X(x) = -1/4 \log 1/4 - 3/4 \log 3/4$$
$$= 0.815$$

Since the probabilities of Y, given X does not change, X and Y are independant. $\therefore$ From Lemma 4,

$$H(X|Y) = H(X) = 0.815 //$$

6) Conditional Divergence

Definition: Given two stochastic matrices V, W on the same finite alphabet $\mathcal{X}$, to finite alphabet $\mathcal{Y}$. P is a pmf on $\mathcal{X}$. The conditional divergence between V and W given P is,

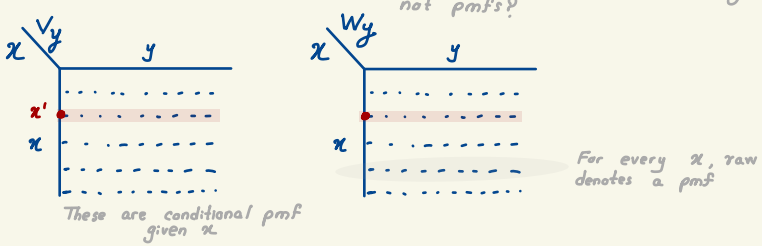
$$D(V\|W|P) = \sum_{x\in\mathcal{X}} P(x)\cdot D\left(V(\cdot|x)\,\|\,W(\cdot|x)\right)$$

$$= \sum_{x\in\mathcal{X}} P(x)\cdot\left[\sum_{y\in\mathcal{Y}} V(y|x)\log\frac{V(y|x)}{W(y|x)}\right]$$

- As done for the KL-Divergence, conditional divergence can also be written in expectation form as

$$D(V\|W|P) = \mathbb{E}_P\left[D\left(V(\cdot|x)\,\|\,W(\cdot|x)\right)\right]$$

- V, W are stochastic matrices. - What is the difference. Why not pmfs?



- Use the usual conventions to handle zeros.

- This is equivalent to finding the difference between 2 conditional distributions $V(y|x)$, $W(y|x)$ given a distribution on $X=x$ called P.

- Pictorially : Take 2 corresponding rows in the stochastic matrices at $X = x'$. These rows denote 2 pmfs. We find the divergence between them, ie $D(V(\cdot|x) \| W(\cdot|x))$ Then get the expectation of this divergences.

Lemma 7 $D(V\|W \mid P) \geq 0$ with equality iff $V(\cdot|x) = W(\cdot|x)$ for every x with $P(x) > 0$

Proof: Easy, by using Lemma 3.

$I(X \wedge Y \mid Z)$

7) Conditional Mutual Information

Definition: Let X, Y, Z be random variables on finite alphabets $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$. Let P_{XYZ} be the joint pmf of (X, Y, Z). Then the Mutual Information of X, Y given Z is

$$I(X \wedge Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z)$$

- $I(X \wedge Y \mid Z) = I(Y \wedge X \mid Z)$
- Conditional Mutual Info. is equivalent to finding KL Divergence (i.e difference) of $P_{XY|Z}$ (joint distribution) and $P_{X|Z} \cdot P_{Y|Z}$ (independant product) and getting the expected KL Divergence

$$I(X \wedge Y \mid Z) = \sum_{z \in \mathcal{Z}} \left[P_Z(z) \left(\sum_{\substack{x \in \mathcal{X} \\ y \in \mathcal{Y}}} P_{XY|Z}(x,y|z) \cdot \log \frac{P_{XY|Z}(x,y|z)}{P_{X|Z}(x|z) \cdot P_{Y|Z}(y|z)} \right) \right]$$

Proof: From the definition,

$$I(X \wedge Y \mid Z) = H(X|Z) - H(X|Y,Z)$$

$$= \mathbb{E}_{P_{XYZ}}\left[-\log P_{X|Z}(X|Z) \right] - \mathbb{E}_{P_{XYZ}}\left[-\log P_{X|Y,Z}(X|Y,Z) \right]$$

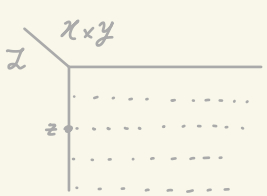
$$= \mathbb{E}_{P_{XYZ}}\left[\log \frac{P_{X|YZ}(X|Y,Z)}{P_{X|Z}(X|Z)} \right]$$

$$= \mathbb{E}_{P_{XYZ}}\left[\log \frac{P_{XY|Z}(X,Y|Z)}{P_{X|Z}(X|Z) \cdot P_{Y|Z}(Y|Z)} \right] \quad \cdots\cdots P_{XY|Z} = P_{Y|Z} P_{X|YZ}$$

$$= \sum_{x,y,z} \underbrace{P_{XYZ}(x,y,z)}_{P_Z(z) \cdot P_{XY|Z}(x,y|z)} \cdot \log \frac{P_{XY|Z}(x,y|z)}{P_{X|Z}(x|z) \cdot P_{Y|Z}(y|z)}$$

$$= \sum_{z} \left[P_Z(z) \cdot \left(\sum_{x,y} P_{XY|Z}(x,y|z) \cdot \log \frac{P_{XY|Z}(x,y|z)}{P_{X|Z}(x|z) \cdot P_{Y|Z}(y|z)} \right) \right]$$

$$P_{XY|Z}(x,y|z) = P(X=x, Y=y \mid Z=z) = \frac{P(X=x, Y=y, Z=z)}{P(Z=z)} = \frac{P(X=x, Y=y, Z=z)}{P(Y=y, Z=z)} \cdot \frac{P(Y=y, Z=z)}{P(Z=z)} = P(X=x \mid Y=y, Z=z) \cdot P(Y=y \mid Z=z) = P_{X|YZ}(x|y,z) \cdot P_{Y|Z}(y|z)$$



$$I(X \wedge Y \mid Z) = \mathcal{D}\left(P_{XY|Z} \,\|\, P_{X|Z}\, P_{Y|Z} \,\middle|\, P_Z\right)$$

Proof: Previous expression can be simplified using Conditional Divergence definition,

$$= \sum_z \left[P_Z(z) \cdot \mathcal{D}\left(P_{XY|Z} \,\|\, P_{X|Z} \cdot P_{Y|Z}\right)\right]$$

$$= \mathcal{D}\left(P_{XY|Z} \,\|\, P_{X|Z} \times P_{Y|Z} \,\middle|\, P_Z\right)$$

$$= \mathcal{D}\left(P_{XY|Z} \,\|\, P_{X|Z} \cdot P_{Y|Z} \,\middle|\, P_Z\right)$$

■

$$I(X \wedge Y \mid Z) = H(X|Z) + H(Y|Z) - H(X,Y|Z)$$

Proof: From the definition

$$I(X \wedge Y \mid Z)$$

$$= H(X|Z) - H(X|Y,Z)$$

$$= H(X|Z) - \left(H(X,Y|Z) - H(Y|Z)\right) \quad \text{.... From Lemma 9 on chain rule of cond. entropy. i.e. } H(X,Y|Z) = H(Y|Z) + H(X|Y,Z)$$

$$= H(X|Z) + H(Y|Z) - H(X,Y|Z)$$

■

Lemma 8: $I(X \wedge Y \mid Z) \geq 0$ with equality iff

X and Y are conditionally Independant given Z i.e

$$P_{XY|Z}(x,y|z) = P_{X|Z}(x|z) \cdot P_{Y|Z}(y|z)$$

$\forall x, y$ and $\forall z$ with $P_Z(z) > 0$.

Proof: $I(X \wedge Y \mid Z) = \sum_z \left[P_Z(z) \cdot \mathcal{D}\left(P_{XY|Z} \,\|\, P_{X|Z} \cdot P_{Y|Z}\right)\right]$

$$\geq \sum_z \left[P_Z(z) \cdot 0\right] \quad \text{.... with equality if all divergences are zeros. i.e } P_{XY|Z} = P_{X|Z} \cdot P_{Y|Z}$$

$$\geq 0$$

Equality occurs iff $P_{X|Z}$ and $P_{Y|Z}$ are independant.

■

Example: Let X, Y are rvs on $\{0,1\}$ and $X \perp\!\!\!\perp Y$. $P_X(0) = P_X(1) = 1/2$ $P_Y(0) = P_Y(1) = 1/2$. Let $Z = X + Y = X+Y \pmod 2$. Compute mutual information, and conditional mutual information given Z, between X and Y

$$X \perp\!\!\!\perp Y \implies I(X \wedge Y) = 0 \quad \cdots \text{ from Lemma 5.}$$

$$\therefore X \not\perp\!\!\!\perp Y \mid Z \implies I(X \wedge Y \mid Z) \neq 0 \quad \cdots \text{ from Lemma 6.}$$

Given Z, (X, Y) are not independant since fixing two variables would determine the third variable as well.

$$I(X \wedge Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z)$$

0 ... fixing 2 will determine the other, ∴ No uncertainity

X	Y	Z = X⊕Y Deterministic	Probability
1/2 0	1/2 0	0	1/4
1/2 0	1/2 1	1	1/4
1/2 1	1/2 0	1	1/4
1/2 1	1/2 1	0	1/4

$\therefore P(Z=0) = 1/2$
$P(Z=1) = 1/2$
$\therefore P(X=x, Z=z) = 1/4 = 1/2 \times 1/2$
$= P(x=x) \cdot P(z=z)$

$$= H(X) \quad \cdots\cdots\cdots \quad X \perp\!\!\!\perp Z \text{ and from Lemma 4}$$

$$= 1 \text{ bit} \quad \cdots \text{ from example in Lecture 1}$$

Here $I(X \wedge Y \mid Z) > I(X \wedge Y)$. So conditioning does not necessarily reduces the mutual independance.

$$I(X \wedge Y \mid Z) \gtrless I(X \wedge Y)$$

Consider a Markov Chain $X \multimap Y \multimap Z$, which is an example where conditioning can reduces mutual information. (In a Markov chain, future and past is independant of present.)

$X \multimap Y \multimap Z$ (Y: Present, X: Past, Z: future)

$$\implies X \perp\!\!\!\perp Z \mid Y \quad \text{i.e. } X|Z \perp\!\!\!\perp Y|Z$$
$$\text{i.e } P_{X|Y|Z} = P_{X|Z}\, P_{Y|Z}$$

$$\implies I(X \wedge Z \mid Y) = 0 \quad \cdots \text{ from Lemm 6.}$$

But we can design a markov chain $X = Y = Z$ since $P(X|Y|Z) = 1$, $P(X|Z) = 1$, $P(Y|Z) = 1$ but $I(X \wedge Y) > 0$ can happen.

Lemma 9 - Additive Chain Rules

X, Y, Z, U are random variables on finite alphabets $\mathcal{X}, \mathcal{Y}, \mathcal{Z}, \mathcal{U}$ with joint pmf P_{XYZU}. Then

(i) $H(X,Y) = H(X) + H(Y|X)$
$= H(Y) + H(X|Y)$

(ii) $H(X,Y|Z) = H(X|Z) + H(Y|X,Z)$
$= H(Y|Z) + H(X|Y,Z)$

(iii) $I((X,Y) \wedge Z) = I(X \wedge Z) + I(Y \wedge Z | X)$
$= I(Y \wedge Z) + I(X \wedge Z | Y)$

(iv) $I((X,Y) \wedge Z | U) = I(X \wedge Z|U) + I(Y \wedge Z | (X,U))$
$= I(Y \wedge Z|U) + I(X \wedge Z | (Y,U))$

Proof: i) Lemma 2

ii) $H(X,Y|Z) = \mathbb{E}_{P_{XYZ}}[-\log P_{XY|Z}(X,Y)|Z)$ .... Definition of conditional entropy

$= \mathbb{E}[-\log P_{X|Z}(X|Z) \cdot P_{Y|XZ}(Y|X,Z)]$

$= \mathbb{E}[-\log P_{X|Z}(X|Z)] + \mathbb{E}[-\log P_{Y|XZ}(Y|X,Z)]$

$= H(X|Z) + H(Y|X,Z)$ ∎

iii) $I((X,Y) \wedge Z) = \mathcal{D}(P_{XYZ} \| P_{XY} \cdot P_Z)$ ... definition of mutual Information

$= \mathbb{E}_{P_{XYZ}}\left[\log \frac{P_{XYZ}(X,Y,Z)}{P_{XY}(X,Y) \cdot P_Z(Z)}\right]$ ... definition of Divergence

Substitute,

$P_{XYZ}(x,y,z) = P_{XZ}(x,z)\, P_{Y|XZ}(y|(x,z)) \Rightarrow P_{XYZ} = P_{XZ} \cdot P_{Y|XZ}$
$P_{YZ|X}((y,z)|x) = P_{Z|X}(z|x) \cdot P_{Y|ZX}(y|(z,x)) \Rightarrow P_{Y|XZ} = \frac{P_{YZ|X}}{P_{Z|X}}$
$\Big\} P_{XYZ} = \frac{P_{XZ} \cdot P_{YZ|X}}{P_{Z|X}}$

$P_{XY}(x,y) = P_{Y|X}(y|x) \cdot P_X(x) \Rightarrow P_{XY} = P_{Y|X} \cdot P_X$

$$I((X,Y)\wedge Z) = \mathbb{E}_{\rho_{XYZ}}\left[\log \frac{\rho_{XZ}(X,Z)\cdot \rho_{YZ|X}((Y,Z)|X)}{\rho_{Y|X}(Y|X)\cdot \rho_X(X)\ \rho_{Z|X}(Z|X)\cdot \rho_Z(Z)}\right]$$

$$= \mathbb{E}_{\rho_{XZ}}\left[\log \frac{\rho_{XZ}(X,Z)}{\rho_X(X)\cdot \rho_Z(Z)}\right] + \mathbb{E}_{\rho_{XYZ}}\left[\log \frac{\rho_{YZ|X}((Y,Z)|X)}{\rho_{Y|X}(Y|X)\cdot \rho_{Z|X}(Z|X)}\right]$$

$$= I(X\wedge Z) + I((Y,Z)|X)$$

■

(iv) $I((X,Y)\wedge Z|U) = H(Z|U) - H(Z|(U,(X,Y)))$ .. definition of conditional mutual inform.

$$= H(Z|U) - H(Z|U,X,Y)$$

$$= H(Z|U) - H(Z|X,U) + H(Z|X,U) - H(Z|U,X,Y)$$

$$= I(X\wedge Z|U) + I((Y\wedge Z)|(X,U))$$

■

Lemma 9 - Chain Rules Continued.

If $X_1, \ldots, X_n$ $n \geq 2$ and Y are random variables on finite alphabets with joint pmf $p_{X_1 \ldots X_n Y}$. Then

(i) $H(X_1, \ldots, X_n) = H(X_1) + \sum_{t=2}^{n} H(X_t \mid X_1, \ldots, X_{t-1})$

(ii) $I((X_1, \ldots, X_n) \wedge Y) = I(X_1 \wedge Y) + \sum_{t=2}^{n} I(X_t \wedge Y \mid X_1, \ldots, X_{t-1})$

Proof: (i) Method I

When $n=2$ (Base Case)

$H(X_1, X_2) = H(X_1) + H(X_2 \mid X_1)$ which is true from Lemma 7, property 1.

Assume $n=k$ is a true statement (Induction hypothesis)

$$\Rightarrow H(X_1, \ldots, X_K) = H(X_1) + \sum_{t=2}^{k} H(X_t \mid X_1, \ldots, X_{t-1})$$

$$\Rightarrow \underbrace{H(X_1, \ldots, X_K) + H(X_{K+1} \mid X_1, \ldots, X_K)}_{\text{From property 1, Lemma 5}} = H(X_1) + \sum_{t=2}^{k+1} H(X_t \mid X_1, \ldots, X_{t-1})$$

$$\Rightarrow H(X_1, \ldots, X_{K+1}) = H(X_1) + \sum_{t=2}^{k+1} H(X_t \mid X_1, \ldots, X_{t-1})$$

$\therefore$ Statement is true for $t = k+1$. $\therefore$ Proved from induction. ∎

Method II

From the conditional chain rule for probabilities, we have

$$\Rightarrow p_{X_1, \ldots, X_n}(x_1, \ldots, x_n) = p_{X_1}(x_1) \cdot \prod_{t=2}^{n} p_{X_t \mid X_1 \ldots X_{t-1}}(x_t \mid x_1, \ldots, x_{t-1})$$

take log of both sides $\Rightarrow \log\left[p_{X_1, \ldots, X_n}(x_1, \ldots, x_n)\right] = \log\left[p_{X_1}(x_1) \cdot \prod_{t=2}^{n} p_{X_t \mid X_1 \ldots X_{t-1}}(x_t \mid x_1, \ldots, x_{t-1})\right]$

$$\Rightarrow \log\left[p_{X_1, \ldots, X_n}(x_1, \ldots, x_n)\right] = \log p_{X_1}(x_1) + \sum_{t=2}^{n} \log\left[p_{X_t \mid X_1 \ldots X_{t-1}}(x_t \mid x_1, \ldots, x_{t-1})\right]$$

take negative expectation $\Rightarrow \mathbb{E}\left[-\log\left[p_{X_1,\ldots,X_n}(x_1,\ldots,x_n)\right]\right] = \mathbb{E}\left[-\log p_{X_1}(x_1)\right] + \sum_{t=2}^{n} \mathbb{E}\left[-\log\left[p_{X_t|X_1\ldots X_{t-1}}(x_t \mid x_1,\ldots,x_{t-1})\right]\right]$

$\Rightarrow H(X_1,\ldots X_n) = H(X_1) + \sum_{t=2}^{n} H(X_t \mid X_1,\ldots X_{t-1})$ ∎

(ii) When $n=2$ (Base Case)

$I(X_1, X_2 \wedge Y) = I(X_1 \wedge Y) + I(X_2 \wedge Y \mid X_1)$ which is true from Lemma 7, property 3.

Assume $n=k$ is a true statement (Induction hypothesis)

$\Rightarrow I(X_1,\ldots,X_K \wedge Y) = I(X_1 \wedge Y) + \sum_{t=2}^{K} I(X_t \wedge Y \mid X_1,\ldots,X_{t-1})$

$\Rightarrow I(X_1,\ldots,X_K \wedge Y) + \underbrace{I(X_{K+1} \wedge Y \mid X_1,\ldots,X_K)}_{\text{From property 3, Lemma 5}} = I(X_1 \wedge Y) + \sum_{t=2}^{K+1} I(X_t \wedge Y \mid X_1,\ldots,X_{t-1})$

$\Rightarrow I((X_1,\ldots,X_{K+1}) \wedge Y) = I(X_1 \wedge Y) + \sum_{t=2}^{K+1} I(X_t \wedge Y \mid X_1,\ldots,X_{t-1})$

$\therefore$ Statement is true for $t = k+1$. $\therefore$ Proved from induction. ∎

Corollary - $H(X_1,..,X_n) \leq \sum_{t=1}^{n} H(X_t)$ with equality $X_1,..,X_n$ are mutually independant

Proof: Method 1

From Lemma 7, property 1 we have

$H(X_1,\ldots,X_K) = H(X_1) + \sum_{t=2}^{K} H(X_t \mid X_1,\ldots,X_{t-1})$

$\leq H(X_1) + \sum_{t=2}^{K} H(X_t)$ ..... Conditioning reduces entropy from Lemma 4.

$= \sum_{t=1}^{K} H(X_t)$

The equality occurs according to Lemma 4, when

$\Rightarrow X_t \perp\!\!\!\perp X_1,\ldots,X_{t-1} \quad \forall\ t \in [2,k]$

$\Rightarrow X_1 \perp\!\!\!\perp X_2 \perp\!\!\!\perp \ldots \perp\!\!\!\perp X_K$ mutually independant ∎

Method 2

From Lemma 3, we have

$$\Rightarrow \quad 0 \leq D\left(P_{X_1 \ldots X_n} \,\|\, P_{X_1} \times \ldots \times P_{X_n}\right)$$

$$= \sum_{x_1,\ldots,x_n} \left(P_{X_1 \ldots X_n}(x_1,\ldots,x_n) \cdot \log \frac{P_{X_1 \ldots X_n}(x_1,\ldots,x_n)}{P_{X_1}(x_1) \cdot \ldots \cdot P_{X_n}(x_n)} \right)$$

$x^n = (x_1,\ldots,x_n)$

$$= \sum_{x^n} P_{X^n}(x^n) \cdot \log P_{X_n}(x^n) - \sum_{x^n} P_{X^n}(x^n) \cdot \left[\sum_{x_i} \log P_{X_i}(x_i) \right]$$

$$= -H(X^n) - \sum_{x_i} \left(\sum_{x^n} P_{X^n}(x^n) \cdot \log P_{X_i}(x_i) \right)$$

$$= -H(X^n) - \sum_{x_i} \left(\sum_{x_i} P_{X_i}(x_i) \cdot \log P_{X_i}(x_i) \right) \quad \ldots \text{ replaced } x^n \text{ by } x_i \text{ since } x_i \in x^n$$

$$= -H(X^n) - \sum_{x_i} \left(-H(X_i) \right)$$

$$= \sum_{i=1}^{n} H(X_i) - H(X_1, \ldots, X_n)$$

$$\Rightarrow \quad H(X_1, \ldots, X_n) \leq \sum_{i=1}^{n} H(X_i)$$

For the equality to occur, divergence should be zero

$$\Leftrightarrow \quad P_{X_1 \ldots X_n} = \prod_{t=1}^{n} P_{X_t}$$

$\Leftrightarrow$ $X_1, \ldots, X_n$ are mutually independant. ■

Remark: $H(X \mid (Y,Z)) \leq \begin{cases} H(X \mid Y) \\ H(X \mid Z) \end{cases}$

Proof: $H(X \mid (Y,Z)) = H((X,Y) \mid Z) - H(Y \mid Z)$ ..... Lemma 7, Property 2

From Conditional mutual information Definition

$$= H(X \mid Z) + H(Y \mid Z) - I(X \wedge Y \mid Z) - H(Y \mid Z)$$

$$\leq H(X \mid Z) \quad \ldots \quad \because \text{Lemma 6.}$$

Equality occurs according to Lemma 6, when X and Y are Conditionally independant given Z. ■

Remark: $I(X \wedge Y \mid Z) \leq \begin{cases} I((X,Z) \wedge Y) \\ I((Y,Z) \wedge X) \end{cases}$

Proof: From property 3 of Lemma 7,

$$I((X,Z) \wedge Y) = I(Z \wedge Y) + I(X \wedge Y \mid Z)$$
$$\geq I(X \wedge Y \mid Z) \quad \cdots\cdots \text{ From Lemma 5 } I(Z \wedge Y) \geq 0$$

Equality occurs when $Z \perp\!\!\!\perp Y$ from Lemma 5. ∎

Lemma 10 - Chain Rule for Divergence

If X, Y are two random variables on finite alphabets $\mathcal{X}, \mathcal{Y}$. And P_{XY}, Q_{XY} be joint pmfs on $\mathcal{X} \times \mathcal{Y}$. Then,

$$D(P_{XY} \| Q_{XY}) = D(P_X \| Q_X) + D(P_{Y|X} \| Q_{Y|X} \mid P_X)$$
$$= D(P_Y \| Q_Y) + D(P_{X|Y} \| Q_{X|Y} \mid P_Y)$$

Proof:- $D(P_{XY} \| Q_{XY})$

$$= \mathbb{E}_{P_{XY}}\left[\log \frac{P_{XY}(X,Y)}{Q_{XY}(X,Y)}\right]$$

$$= \mathbb{E}_{P_{XY}}\left[\log \frac{P_X(X) \cdot P_{Y|X}(Y|X)}{Q_X(X) \cdot Q_{Y|X}(Y|X)}\right] \quad \cdots \text{ joint = marginal × condition}$$

$$= \mathbb{E}_{P_{XY}}\left[\log \frac{P_X(X)}{Q_X(X)}\right] + \mathbb{E}_{P_{XY}}\left[\log \frac{P_{Y|X}(Y|X)}{Q_{Y|X}(Y|X)}\right]$$

Depends only on X

Unbiasness property, $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$ from ENEE 620

$$= \mathbb{E}_{P_X}\left[\log \frac{P_X(X)}{Q_X(X)}\right] + \mathbb{E}_{P_X}\left[\mathbb{E}_{P_{Y|X}(\cdot|x)}\left[\log \frac{P_{Y|X}(Y|x)}{Q_{Y|X}(Y|x)}\right]\right]$$

$$= D(P_X \| Q_X) + \mathbb{E}_{P_X}\left[D\left(P_{Y|X}(Y|X=x) \| Q_{Y|X}(Y|X=x)\right)\right]$$ ← using Divergence definition

$$= D(P_X \| Q_X) + \mathbb{E}_{P_X}\left[D(P_{Y|X}(\cdot|x) \| Q_{Y|X}(\cdot|x)\right]$$

$$= D(P_X \| Q_X) + D(P_{Y|X} \| Q_{Y|X} \mid P_X)$$

∎

Read Section 2.6 in Cover / Thomas (Page 25-28)

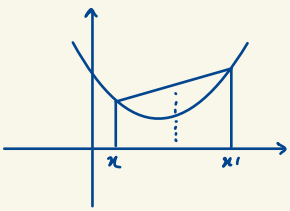
8) Convexity and Concavity

Let $g: \mathbb{R} \longrightarrow \mathbb{R} \cup \{+\infty\}$ be a mapping such that $g(x) < \infty$ for at least one $x \in \mathbb{R}$

Convex Function

g is convex $\iff$ $g(\alpha x + (1-\alpha)x') \leq \alpha g(x) + (1-\alpha) \cdot g(x')$

$\forall\ x < x' \in \mathbb{R}$ and $\forall\ \alpha \in [0,1]$



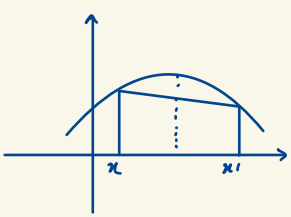
Chord lies above Curve

Concave Function

g is concave $\iff$ $g(\alpha x + (1-\alpha)x') \geq \alpha g(x) + (1-\alpha) \cdot g(x')$

$\forall\ x < x' \in \mathbb{R}$ and $\forall\ \alpha \in [0,1]$

g is concave if $-g$ is convex.



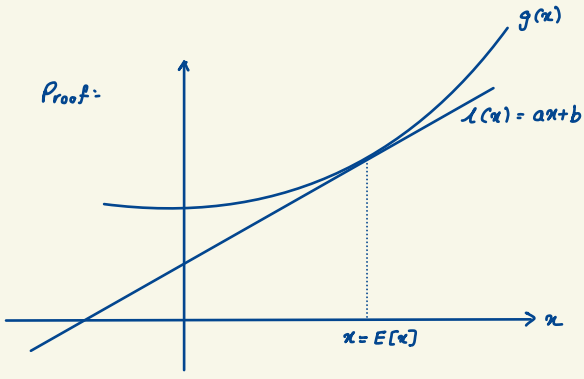
Chord lies below Curve

Jensens' Inequality

Theorem: Let $g: \mathbb{R} \longrightarrow \mathbb{R} \cup \{+\infty\}$ be a convex mapping. Let X be a $\mathbb{R}$-valued rv with $\mathbb{E}[|X|] < \infty$. Then

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$$

If g is concave $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$



Drop a tangent at $E[x]$ point in the x axis. Assume tangent equation is $l(x) = ax+b$.

At the point where we drew the tangent, l and g are same

$\Rightarrow g(E[x]) = l(E[x]) \cdots (1)$

Convex $\Rightarrow$ tangent lies below the curve

$\Rightarrow g(x) \geq l(x) \quad \forall x$ (w.p.1)

$\Rightarrow E[g(x)] \geq E[l(x)] = E[ax+b]$

$= a\,E[x] + b$

$= l(E[x])$

$= g(E[x]) \cdots$ from (1)

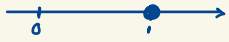
■

Probability Simplex

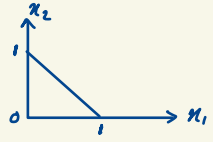
Definition:- The probability simplex in $\mathbb{R}^m$ is the set of points $X = (x_1, \ldots, x_m) \in \mathbb{R}^m$ such that $x_i \geq 0$, $\sum_{i=1}^{m} x_i = 1$.

- Simplex is an $(m-1)$ - dimensional manifold in m-dimensional space.

$m=1$ $\quad X \in \{(x_1) \mid x_1 = 1\}$



$m=2$ $\quad X \in \{(x_1, x_2) \mid x_1 + x_2 = 1\}$



$m=3$ $\quad X \in \{(x_1, x_2, x_3) \mid x_1 + x_2 + x_3 = 1\}$

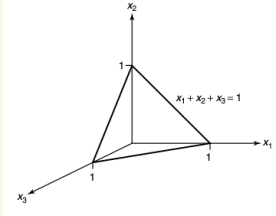


FIGURE 11.1. Probability simplex in $\mathbb{R}^3$.

$H(X) \longrightarrow H(P)$

$H(Y|X) \longrightarrow H(W|P)$

$I(X;Y) \longrightarrow I(P,W)$

IDK? $\xrightarrow{\text{Maybe according to Cover Thomas}}$

$P(x) \sim P$ - Distribution of r.v X

$P(y|x) \sim W$ - Conditional distribution of Y given X.

Lemma II - Convexity / Concavity

1) **Concavity of Entropy** - $H(P)$ is a concave function of P.

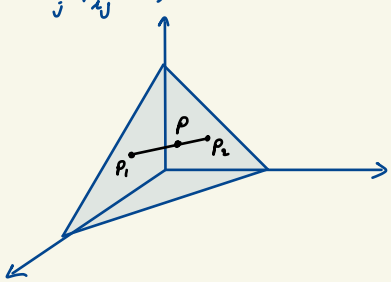
2) $H(Y|X)$ is a concave function of $P(Y|X)$ (W), and linear function of $P(X)$ (P).

3) **Convexity of Divergence** - $D(P\|Q)$ is a convex function of pair (P, Q).

4) Mutual Information $I(X;Y)$ ($I(P, W)$) is a concave function of $P(X)$ (P) for given $P(Y|X)$, and convex function of $P(Y|X)$ (W) for given $P(X)$.

Proof: Consider P_1, P_2, Q_1, Q_2 be pmfs on finite alphabet $\mathcal{X}$ where $\mathcal{X} \in \mathbb{R}^m$. Then the probability simplex is an $(m-1)$ dimensional space, where each point $P_i = (P_{i1}, P_{i2}, \ldots, P_{i(m-1)})$ represents a pmf on $\mathcal{X}$ (Since $\sum_j P_{ij} = 1$)

If $m=2$, simplex is a triangle and P_1, P_2, Q_1, Q_2 lies on it.



All the other pmfs of $\mathcal{X} \in \mathbb{R}^m$ should also lie on this simplex.

$\alpha \in (0,1)$, $x \in \mathcal{X}$

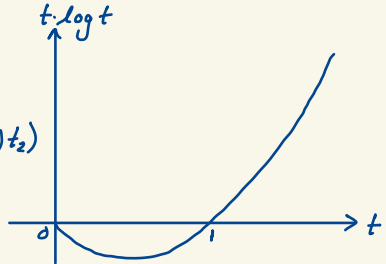
$$P = \alpha P_1 + (1-\alpha) P_2 \implies P(x) = \alpha P_1(x) + (1-\alpha) \cdot P_2(x) \quad \cdots (1)$$
$$Q = \alpha Q_1 + (1-\alpha) Q_2 \implies Q(x) = \alpha Q_1(x) + (1-\alpha) Q_2(x) \quad \cdots (2)$$

1) Consider the convexity of $f(t) = t \log t$ $\quad t \cdot \log t$

$t \geq 0$

$$\implies \alpha t_1 \cdot \log t_1 + (1-\alpha) t_2 \log t_2 \geq (\alpha t_1 + (1-\alpha) t_2) \cdot \log (\alpha t_1 + (1-\alpha) t_2)$$

Substitute $t_1 = P_1(x)$, $t_2 = P_2(x)$



$$\Rightarrow \alpha P_1(x) \log P_1(x) + (1-\alpha)\cdot P_2(x)\cdot \log P_2(x) \geq (\underbrace{\alpha P_1(x) + (1-\alpha) P_2(x)}_{P(x)})\cdot \log (\underbrace{\alpha\cdot P_1(x) + (1-\alpha)\cdot P_2(x)}_{P(x) \text{ from } ①})$$

Summing over x

$$\Rightarrow \alpha \sum_x P_1(x)\cdot \log \alpha P_1(x) + (1-\alpha)\cdot \sum_x P_2(x)\cdot \log P_2(x) \geq \sum_x P(x)\cdot \log P(x)$$

$$\Rightarrow -\alpha H(P_1) - (1-\alpha)\cdot H(P_2) \geq -H(P)$$

$$\Rightarrow \alpha H(P_1) + (1-\alpha) H(P_2) \leq H(\alpha P_1 + (1-\alpha) P_2)$$

■

2) From the definition of conditional entropy, we have

$$\begin{aligned} H(Y|X) &= \mathbb{E}[H(Y|X=x)] \\ &= \sum_x p_x(x)\cdot H(Y|X=x) \quad \cdots\cdots \text{ } p \text{ be the distribution of r.v } X \\ &= \sum_x p_x(x)\cdot H(W(\cdot|x)) \quad \cdots \text{ } W \text{ be the distribution of r.v } Y \text{ given } X. \end{aligned}$$

By observing this expression,

- When W (i.e $Y|X$) is given, conditional distribution $H(Y|X)$ is linear on X (i.e P).
- Since entropy is concave according to part 1), when X is given (i.e. $X=x$), $H(W(\cdot|x))$ is concave in W $\forall x$. Since a linear combination of concave functions with non-negative weights is concave, $H(Y|X)$ is concave on $Y|X$ (i.e. W).

3) $$\alpha\cdot P_1(x) \log \frac{P_1(x)}{Q_1(x)} + (1-\alpha)\cdot P_2(x)\cdot \log \frac{P_2(x)}{Q_2(x)}$$

$$= \underbrace{\alpha\cdot P_1(x)}_{a_1} \log \frac{\alpha\cdot P_1(x)}{\underbrace{\alpha\cdot Q_1(x)}_{b_1}} + \underbrace{(1-\alpha)\cdot P_2(x)}_{a_2}\cdot \log \frac{(1-\alpha)\cdot P_2(x)}{\underbrace{(1-\alpha)\cdot Q_2(x)}_{b_2}}$$

$$\geq \underbrace{[\alpha\cdot P_1(x) + (1-\alpha)\cdot P_2(x)]}_{a_1+a_2}\cdot \log \frac{[\alpha P_1(x) + (1-\alpha)\cdot P_2(x)]}{\underbrace{[\alpha Q_1(x) + (1-\alpha) Q_2(x)]}_{b_1+b_2}} \quad \cdots$$

From log sum inequality for $r=2$ case

$= p(x) \log \frac{p(x)}{Q(x)}$ . . . . . From ① and ②

Take the sum over all x,

$$\Rightarrow \alpha \cdot \sum_x p_1(x) \cdot \log \frac{p_1(x)}{Q_1(x)} + (1-\alpha) \sum_x p_2(x) \cdot \log \frac{p_2(x)}{Q_2(x)} \geq \sum_x p(x) \cdot \log \frac{p(x)}{Q(x)}$$

$$\Rightarrow \alpha \cdot D(P_1 \| Q_1) + (1-\alpha) \cdot D(P_2 \| Q_2) \geq D(P \| Q)$$

$\Rightarrow$ Divergence is convex. ■

Linear can be taken as beath convex and concave

4) From Cover / Thomas

$I(X \wedge Y)$ is Concave in $P(x)$

From part 1) $H(Y)$ is a concave function in $P(Y)$

If $p(Y|x)$ is fixed $P(Y)$ is a linear function of $P(X)$

$\Rightarrow H(Y)$ is a concave function in $P(x)$

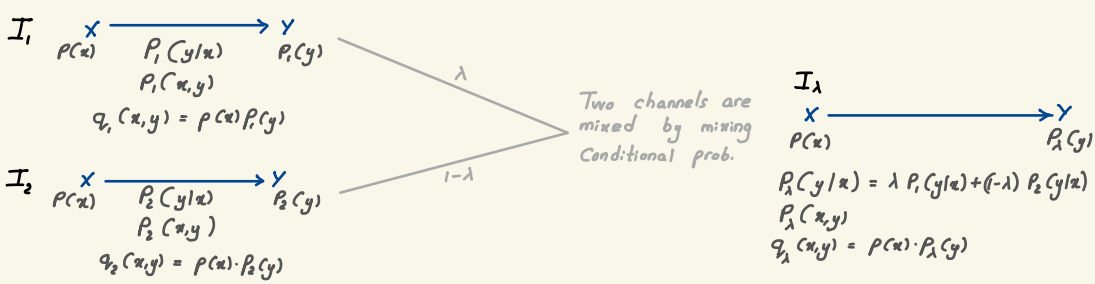
From part 2) $\Rightarrow H(Y|X)$ is a linear function of $P(X)$

⊛ $H(Y) - H(Y|X)$ is a concave function in $P(x)$

From the mutual information definition, $\Longrightarrow I(X \wedge Y) = H(Y) - H(Y|X)$

⊛ $\Longrightarrow I(X \wedge Y)$ is concave in $P(x)$ ■

$I(X \wedge Y)$ is convex in $P(Y|X)$



Setting:- Assume we have 2 channels, both taking X as input, and have conditional distributions $P_1(y|x)$ and $P_2(y|x)$. These 2 channels are mixed with ratio λ to create a new channel. (Mixing two channels with a ratio effectively is equivalent to mixing conditional probabilities.)

To Prove:- Mutual Information is convex on $P(y|x)$

i.e. Mutual Information of mixed channel $\leq$ Mixing the Mutual Information of each channel

i.e. $I_\lambda(X \wedge Y) \leq \lambda \cdot I_1(X \wedge Y) + (1-\lambda) \cdot I_2(X \wedge Y)$

Proof:- From channel mixing with ratio $\lambda \implies P_\lambda(y|x) = \lambda \cdot P_1(y|x) + (1-\lambda) \cdot P_2(y|x)$

$$\begin{aligned} \therefore P_\lambda(x,y) &= P(x) \cdot P_\lambda(y|x) \\ &= P(x) \cdot [\lambda P_1(y|x) + (1-\lambda) \cdot P_2(y|x)] \\ &= \lambda \cdot P(x) \cdot P_1(y|x) + (1-\lambda) P(x) \cdot P_2(y|x) \\ &= \lambda \cdot P_1(x,y) + (1-\lambda) \cdot P_2(x,y) \quad \cdots\cdots\cdots (1) \end{aligned}$$

$$\begin{aligned} \therefore P_\lambda(y) = \sum_x P_\lambda(x,y) &= \lambda \sum_x P_1(x,y) + (1-\lambda) \cdot \sum_x P_2(x,y) \\ &= \lambda \cdot P_1(y) + (1-\lambda) \cdot P_2(y) \end{aligned}$$

Define q_1, q_2, q_λ as the product of the marginals of each channel

$$\begin{aligned} \therefore q_\lambda(y) = P(x) \cdot P_\lambda(y|x) &= P(x) \cdot [\lambda P_1(y|x) + (1-\lambda) \cdot P_2(y|x)] \\ &= \lambda P(x) P_1(y|x) + (1-\lambda) P(x) P_2(y|x) \\ &= \lambda q_1(x,y) + (1-\lambda) \cdot q_2(x,y) \quad \cdots\cdots (2) \text{ definition} \end{aligned}$$

From part 3) we have that relative entropy is convex on (p, q) hence,

$$\implies D(\lambda p_1 + (1-\lambda) p_2 \,\|\, \lambda q_1 + (1-\lambda) q_2) \leq \lambda \cdot D(p_1 \| q_1) + (1-\lambda) \cdot D(p_2 \| q_2)$$

$$\implies D(\underbrace{P_\lambda}_{\text{joint}} \| \underbrace{q_\lambda}_{\text{product of marginals}}) \leq \lambda \cdot D(p_1 \| q_1) + (1-\lambda) \cdot D(p_2 \| q_2) \quad \cdots \text{from } (1), (2)$$

from definition of P_i and q_i

$$\implies D(P_\lambda(x,y) \| P(x) q_\lambda(y)) \leq \lambda \cdot D(P_1(x,y) \| P(x) \cdot q_1(y)) + (1-\lambda) \cdot D(P_2(x,y) \| P(x) q_2(y))$$

(Same input to all channel)

$$\implies I_\lambda(X \wedge Y) \leq \lambda \cdot I_1(X \wedge Y) + (1-\lambda) \cdot I_2(X \wedge Y) \quad \cdots\cdots \text{from definition of MI.}$$

From Narayan

Not clear if X, Y|x to W,P replacement is trivial.

$I(P,W)$ is concave in P

$I(P,W) = H(PW) - H(W|P)$

PW is a pmf on $\mathcal{Y}$.

$PW(y) = \sum_x P(x) \cdot W(y|x) \quad y \in \mathcal{Y}$

$\Rightarrow$ Concave in PW

$\Rightarrow$ PW is linear in P

$\Rightarrow$ Concave in P

$\Rightarrow$ Linear in P

$\Rightarrow$ Concave in P

$I(P,W)$ is concave in P.

$I(P,W)$ is Convex in W

$$
\begin{aligned}
I(P,W) &= I(X \wedge Y) \\
&= D(P_{XY} \,\|\, P_X P_Y) \\
&= D(P_X P_{Y|X} \,\|\, P_X P_Y) \\
&= D(P_X \,\|\, P_X) + D(P_{Y|X} \,\|\, P_Y \,|\, P_X) \quad \cdots \text{Chain rule of Divergence} \\
&\qquad \text{0 Lemma 3} \\
&= D(P_{Y|X} \,\|\, P_Y \,|\, P_X) \\
&= \sum_x P(x) \cdot D(W(\cdot|x) \,\|\, PW)
\end{aligned}
$$

This is also not trivial
*Is $P_{XY} = P_X \cdot P_Y$?

$D(W(\cdot|x) \,\|\, PW)$: Linear in W

$\sum_x P(x) \cdot D(W(\cdot|x) \,\|\, PW)$: Convex in W

Linear combination of convex functions with non negative weights are convex

Lemma 12 X is a random variable with values in finite set $\mathcal{X}$. Then,

$$H(X) \leq \log |\mathcal{X}|$$

with equality iff X is uniformly distributed on $\mathcal{X}$.

Proof: $\log |\mathcal{X}|$ = is the number of bits to represent all the elements in the set $\mathcal{X}$.

Let P = pmf of X on $\mathcal{X}$

U = uniform distributions over $\mathcal{X}$ $\Rightarrow$ $U(x) = 1/|\mathcal{X}|$

From non-negativity of Divergence,

$$\begin{aligned} 0 \leq D(P \| U) &= \sum_{x \in \mathcal{X}} p(x) \cdot \log \frac{p(x)}{u(x)} \\ &= \sum_{x \in \mathcal{X}} p(x) \cdot \log |\mathcal{X}| \cdot p(x) \\ &= \underbrace{\sum_{x \in \mathcal{X}} p(x) \log p(x)}_{-H(X)} + \sum_{x \in \mathcal{X}} p(x) \cdot \log |\mathcal{X}| \\ &= -H(X) + \log |\mathcal{X}| \cdot \underbrace{\sum_{x} p(x)}_{1} \\ &= -H(X) + \log |\mathcal{X}| \end{aligned}$$

$$\Rightarrow H(X) \geq \log |\mathcal{X}|$$

The equality occurs iff $D(P\|U) = 0 \Leftrightarrow P = U$ ∎

9) Markov Chains

A Recap from ENEE 620 and More

- Why use Markov Chain assumptions in Information Theory - When it is required to show $X \perp\!\!\!\perp Z \mid Y$, the general way requires to use joint distributions. But considering the Conditional Mutual Information ($H(X \wedge Y \mid Z)$) and playing with these inequalities we might be able to show conditional independance of X, Y easily, by showing that it is a Markov Chain ($X \multimap Y \multimap Z$)

Definition :- A finite or infinite sequence of rvs $X_1, X_2, \ldots,$ forms a Markov Chain (MK), denoted by

$$X_1 \multimap X_2 \multimap X_3 \multimap \ldots$$

if for every $t = 2, 3, \ldots,$ X_{t+1} is Conditionally Independant of $(X_1, \ldots, X_{t-1})$ given X_t. Thus

$$P_{X_{t+1} \mid X_t, X_{t-1}, \ldots, X_1}(x_{t+1} \mid x_t, x_{t-1}, \ldots, x_1) = P_{X_{t+1} \mid X_t}(x_{t+1} \mid x_t)$$

(future | present, past)

$\forall\ x_1, \ldots, x_{t-1}, x_t, x_{t+1} \in \mathcal{X}$

Lemma 13 - $X_1 \multimap X_2 \multimap X_3 \multimap \ldots \iff I(\underbrace{X_1, \ldots, X_{t-1}}_{\text{past}} \wedge \underbrace{X_{t+1}}_{\text{future}} \mid \underbrace{X_t}_{\text{present}}) = 0$ $\forall\ t = 2, 3, \ldots$

Proof :- Let past $= P = X_{t-1}, \ldots, X_1$

present $= S = X_t$

future $= F = X_{t+1}$

Claim 1 :- $X_1 \multimap X_2 \multimap \ldots \iff P(P, F \mid S) = P(F \mid S) \cdot P(P \mid S)$

From chain rule for conditional probability.

$$\Rightarrow P(x_1, \ldots, x_{t-1}, x_{t+1} \mid x_t) = p(x_1, \ldots, x_{t-1} \mid x_t) \cdot p(x_{t+1} \mid x_1, \ldots, x_{t-1}, x_t)$$

$= p(x_1, \ldots, x_{t-1} \mid x_t) \cdot p(x_{t+1} \mid x_t)$ ..... only the present is needed for future in a Markov chain.

$\Rightarrow \quad X_1, \ldots, X_{t-1} \perp\!\!\!\perp X_{t+1} \mid X_t$

$\Rightarrow \quad P \perp\!\!\!\perp F \mid S \quad$ ie $\quad P(P,F \mid S) = P(P \mid S) \cdot P(F \mid S)$

Claim 2 : $P(P,F \mid S) = P(P \mid S) \cdot P(F \mid S) \iff I(P \wedge F \mid S) = 0$

From the definition of conditional mutual information

$\Leftrightarrow \quad I(P \wedge F \mid S) = \sum_{p,s,f} P_{PSF}(p,s,f) \cdot \log \left(\frac{P_{P,F|S}(p,f \mid s)}{P_{P|S}(p \mid s) \cdot P_{F|S}(f \mid s)} \right)$

$\Leftrightarrow \quad I(X_1, \ldots, X_t \wedge X_{t+1} \mid X_t)$

$= \sum_{x_1, \ldots, x_{t+1}} p(x_1, \ldots, x_{t-1}, x_t, x_{t+1}) \cdot \log \frac{p(x_1, \ldots, x_{t+1} \mid x_t)}{p(x_1, \ldots, x_{t-1} \mid x_t) \cdot P(x_{t+1} \mid x_t)}$

$= \sum_{x_1, \ldots, x_{t+1}} p(x_1, \ldots, x_{t-1}, x_t, x_{t+1}) \cdot \log \frac{p(x_1, \ldots, x_{t-1} \mid x_t) \cdot p(x_{t+1} \mid x_t)}{p(x_1, \ldots, x_{t-1} \mid x_t) \cdot P(x_{t+1} \mid x_t)}$

$\log 1 = 0$

$= 0$

Hence proved from claim 1 and 2 ∎

Remarks -

1) $X \multimap Y \multimap Z \iff P_{XYZ}(x,y,z) = P_X(x) \cdot P_{Y|X}(y|x) \cdot P_{Z|Y}(z|y)$

Proof : When $X \multimap Y \multimap Z$, we have $P(z \mid x,y) = P(z \mid y)$

$\therefore P_{XYZ}(x,y,z) = P_{Z|X,Y}(z \mid x,y) \cdot P_{XY}(x,y)$

$= P_{Z|Y}(z|y) \cdot P_{XY}(x,y)$ .... from Markov property $X \multimap Y \multimap Z$

$= P_{Z|Y}(z|y) \cdot P_{Y|X}(y|x) \cdot P_X(x)$ ∎

2) $X \multimap Y \multimap Z \iff P_{XZ|Y}(x,z|y) = P_{X|Y}(x|y) \cdot P_{Z|Y}(z|y)$

Proof : $P_{XZ|Y}(x,z|y) = \frac{P(x,z,y)}{P(y)} = \frac{P(x,y,z)}{P(x,y)} \cdot \frac{P(x,y)}{P(y)}$

$= P_{Z|X,Y}(z \mid x,y) \cdot P_{X|Y}(x|y)$

$= P_{Z|Y}(z|y) \, P_{X|Y}(x|y)$ ... from Markov property $X \multimap Y \multimap Z$ ∎

3) $X \multimap Y \multimap Z \iff Z \multimap Y \multimap X$

Proof: $P_{X|Z,Y}(x \mid z,y) = P(x \mid z,y) = \frac{P(x,z,y)}{P(z,y)}$

$= \frac{P(x,y,z)}{P(x,y)} \cdot \frac{P(x,y)}{P(z,y)}$

$= P_{Z|X,Y}(z \mid x,y) \cdot \frac{P(x,y)}{P(z,y)}$

$= P_{Z|Y}(z|y) \cdot \frac{P(x,y)}{P(z,y)}$ ... from Markov property of $X \multimap Y \multimap Z$

$= \frac{\cancel{P(z,y)}}{P(y)} \cdot \frac{P(x,y)}{\cancel{P(z,y)}} = \frac{P(x,y)}{P(y)}$

$= P_{X|Y}(x|y)$

Hence $Z \multimap Y \multimap X$ is a markov chain ∎

4) $X \multimap Y \multimap Z = f(Y)$ forms a Markov chain when f is deterministic.

Proof 1: $P_{Z|X,Y}(z \mid x,y) = \begin{cases} 1 & \text{if } z = f(y) \\ 0 & \text{if } z \neq f(y) \end{cases} = \mathbb{1}(z = f(y))$

$P_{Z|Y}(z \mid y) = \begin{cases} 1 & \text{if } z = f(y) \\ 0 & \text{if } z \neq f(y) \end{cases} = \mathbb{1}(z = f(y))$

$\therefore P(z \mid x,y) = P(z|y)$. Hence this forms a Markov chain. ∎

Proof 2: $I(X \wedge Z \mid Y) = I(X \wedge f(Y) \mid Y)$

$= H(f(Y) \mid Y) - H(f(Y) \mid X,Y)$

$= 0 - 0$ ..... when f is deterministic, have no uncertainty

$= 0$

$\therefore$ from lemma 13, $X \multimap Y \multimap Z$. ∎

5) If $X_1 \multimap X_2 \multimap X_3 \multimap X_4 \implies X_1 \multimap X_2 \multimap X_4$

6) If $X_1 \multimap X_2 \multimap X_3 \implies I(X_1 \wedge (X_2, X_3)) = I(X_1 \wedge X_2)$

7) If $X_1 \multimap X_2 \multimap X_3 \implies I(X_1 \wedge X_2 \mid X_3) \leq I(X_1 \wedge X_2)$

Can be derived from Data Processing Inequality proof

Time Invariant Markov Chains

Definition - The Markov Chain is said to be time invariant if the conditional probability $P(X_{n+1} | X_n)$ does not depend on n.

i.e. $P(X_{n+1} = j \mid X_n = i) = P(X_2 = j \mid X_1 = i)$

- If $\{X_i\}_{i=1}^{\infty}$ is a Markov Chain, X_n is called the state at time n.
- If $X_i \in \mathcal{X}$ i.e all possible realizations of X_i $\forall i$ is comming from a finite alphabet then, no. of possible states are finite. Assume that finite cardinality of X_i $\forall i$ is m.

$$|\mathcal{X}| = m < \infty \quad \text{where} \quad X_i \in \mathcal{X} \ \forall i$$

- Assume a MC is going from (state i) $\longrightarrow$ (state j) $i, j \ll m$ If this MC is <u>time invarient</u> then it does not matter on which time step that this is happening.

$$\therefore P_{ij} = P[X_{n+1} = j \mid X_n = i] \quad \text{invariant on } n$$

Definition - For a time invariant Markov Chain,

Probability Transition Matrix $= P =$

$$\begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1m} \\ P_{21} & P_{22} & \cdots & P_{2m} \\ \vdots & \vdots & & \vdots \\ P_{m1} & P_{m2} & \cdots & P_{mm} \end{bmatrix} = [P_{ij}]$$

where, $P_{ij} = P[X_{n+1} = j \mid X_n = i]$

- For a Time Invariant Markov chain, probability transition matrix P does not depend on t.

Invariant / Stationary Distribution on a Markov Chain

- At a given time n, X_n can take m states with certain probabilities. Let this probability distribution of X_n over m states be $P_{X_n} = \mu_n$

Definition - If the probability distribution of states at time $n+1$ is same as the distribution at time n, that MC has a stationary distribution,

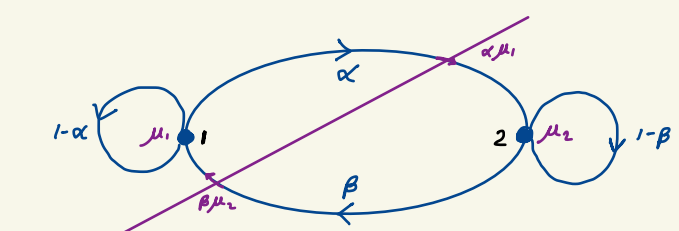
$$\mu = P_{X_n} \quad \forall n$$

- If initial state of a MC is drawn from a Stationary Distribution $\Longrightarrow$ MC forms a Stationary Process

Fact - For a Time Invariant MC with Stationary distribution μ and the Probability Transition Matrix P $\Longrightarrow$ $\mu P = \mu$

just the vector notation of, $p(x_{n+1}) = \sum_{x_n} p(x_n) \cdot p_{x_n x_{n+1}}$ ($p(x_{n+1})$: μ, $p(x_n)$: μ, $p_{x_n x_{n+1}}$: P)

Example :- Consider a $m = 2$ state Time Invariant Markov Chain below. The probability transition matrix is



$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$$

Probability flow accross any cross section of a stationary MC is 0. $\Longleftrightarrow \mu P = \mu$

If this has a stationary distribution $\mu = [\mu_1, \mu_2] = [P_{X_1=1}, P_{X_1=2}] = P_{X_1}$

$$\Rightarrow \mu P = \mu$$

$$\Rightarrow [\mu_1, \mu_2] \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} = [\mu_1, \mu_2]$$

$$\Rightarrow \mu_1 = \beta/(\alpha+\beta), \quad \mu_2 = \alpha/(\alpha+\beta)$$

Irreducible Markov Chains - It is possible to go from any state of the MC to any other state in finite steps.

Aperiodic Markov Chains - If the largest common factor of different paths from a state to itself is 1.

Fact - If the finite-state Markov chain is irreducible and aperiodic, the stationary distribution is unique and from any starting distribution X_1, the distribution of X_n tends to stationary distribution as $n \to \infty$.

This page is from ENEE 620 "Stochastic Processes"

Definition: Markov Processes

A $\mathbb{R}^n$ valued process $\{\underline{X}_t, t \in T\}$ is a Markov Process (or has the Markov Property) if for every $t_1 < t_2 < \dots < t_K < t$ and $\forall\, k \geq 2$

$$P\left(\underline{X}_t \in B \,\middle|\, \underline{X}_{t_1}, \dots, \underline{X}_{t_K}\right) = P\left(\underline{X}_t \in B \,\middle|\, \underline{X}_{t_K}\right)$$

Precent — Past — Most recent past

Almost Surely (P-a.s.)

$\forall\, B \in \mathcal{B}^n$

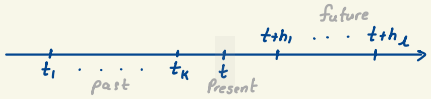
i.e.

$\forall\, \underline{x}_1, \dots, \underline{x}_K \in \mathbb{R}^n$

$$P\left(\underline{X}_t \in B \,\middle|\, \underline{X}_{t_1} = \underline{x}_1, \dots, \underline{X}_{t_K} = \underline{x}_K\right) = P\left(\underline{X}_t \in B \,\middle|\, \underline{X}_{t_K} = \underline{x}_K\right)$$

Conditional Independence

Theorem: Past and Future are conditionally independant given the present in a markov process.



$\forall\, t \in T \quad \forall\, t_1 < \dots < t_K \quad \forall\, k \geq 1$

$\forall\, 0 < h_1 < \dots < h_L \quad \forall\, L \geq 1$

$$P\left(\underline{X}_{t_1} \in B_1, \dots, \underline{X}_{t_K} \in B_K, \underline{X}_{t+h_1} \in A_1, \dots, \underline{X}_{t+h_L} \in A_L \,\middle|\, \underline{X}_t\right)$$

$$= P\left(\underline{X}_{t_1} \in B_1, \dots, \underline{X}_{t_K} \in B_K \,\middle|\, \underline{X}_t\right) + P\left(\underline{X}_{t+h_1} \in A_1, \dots, \underline{X}_{t+h_L} \in A_L \,\middle|\, \underline{X}_t\right)$$

almost surely

$\forall\, B_1, \dots, B_K, A_1, \dots, A_K \in \mathcal{B}^n$

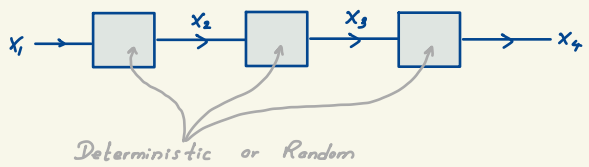
Proof: Homework.

10) Data Processing Inequality
(Markov Processing Inequality)

- At it's heart this means that amount of information in a system can't be improved by manipulating it. But we might always be able to make better inference.

What is Data Processing?

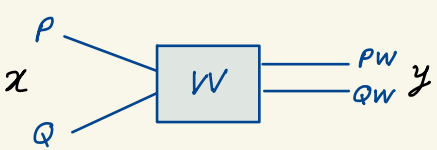
Data Processing $\tilde{\Longleftrightarrow}$ Markov Processing



If this processes are Deterministic, we trivially have $\Longrightarrow$ Markov - chain
$X_1 \multimap X_2 \multimap X_3 \multimap X_4$

If Random, $\Longrightarrow$ We assume Markov property.
$X_1 \multimap X_2 \multimap X_3 \multimap X_4$

Insight from ENEE 621



Assume 'P' and 'Q' are two signals respective to detecting and non-detecting cases. Then 'W' is the stochastic matrix denoting the noise induced when observing P, Q. ∴ The signals analysed at the end are 'PW' and 'QW'.

∴ Data processing lemma, intuitively suggest the difference (divergence) of (P, Q) only decreases under noisy observations, (PW, QW)

$$PW(y) = \sum_x P(x) \cdot W(y|x) \quad \cdots\cdots \circledast$$

$$QW(y) = \sum_x Q(x) \cdot W(y|x)$$

Lemma 14 - Data Processing Inequalities

1) If $X_1 - o - X_2 - o - X_3 - o - X_4$, then

(i) $I(X_1 \wedge X_2) \geq I(X_1 \wedge X_3)$ $\quad (\Leftrightarrow H(X_1|X_2) \leq H(X_1|X_3))$

(ii) $I(X_2 \wedge X_3) \geq I(X_1 \wedge X_4)$

2) For any pmfs P and Q on $\mathcal{X}$, and stochastic matrix $W = \{W(y|x),\ x \in \mathcal{X}\ y \in \mathcal{Y}\}$. we have,

$$D(PW \| QW) \leq D(P \| Q)$$

Proof :- 1) (i) From Lemma 7, additive chain rules,

$$I(X_1 \wedge (X_2, X_3)) = I(X_1 \wedge X_2) + \underbrace{I(X_1 \wedge X_3 | X_2)}_{=0 \text{ since } X_1 - o - X_2 - o - X_3 \text{ and Lemma 13}}$$

$$= I(X_1 \wedge X_2) \quad \cdots \textcircled{1}$$

Similarly,

$$I(X_1 \wedge (X_2, X_3)) = I(X_1 \wedge X_3) + \underbrace{I(X_1 \wedge X_2 | X_3)}_{\geq 0 \text{ from Lemma 6}}$$

$$\leq I(X_1 \wedge X_3) \quad \cdots \textcircled{2}$$

$\textcircled{1}, \textcircled{2} \Rightarrow I(X_1 \wedge X_2) \leq I(X_1 \wedge X_3)$ ∎

(ii) $I(X_1 \wedge X_4) \leq I((X_1, X_2) \wedge X_4)$ ..... Since having more info (i.e X_2) should increase the knowledge we have on X_4

Not proved before? Just intuition

$$= I(X_2 \wedge X_4) + \underbrace{I(X_1 \wedge X_4 | X_2)}_{=0 \text{ from lemma 13 and } X_1 - o - X_2 - o - X_3 - o - X_4 \Rightarrow X_1 - o - X_2 - o - X_4} \quad \cdots \text{ Lemma 7}$$

$$= I(X_2 \wedge X_4)$$

$$\leq I(X_2 \wedge (X_3, X_4)) \quad \cdots \text{ more info doesn't hurt}$$

$$= I(X_2 \wedge X_3) + \underbrace{I(X_2 \wedge X_4 | X_3)}_{=0 \text{ from lemma 13}}$$

$$= I(X_2 \wedge X_3)$$

∎

2) $D(P\|Q) = \sum_x P(x) \cdot \log \frac{P(x)}{Q(x)}$

$= \sum_x P(x) \cdot \log \frac{P(x) \cdot W(y|x)}{Q(x) \cdot W(y|x)}$

$= \sum_x P(x) \cdot \underbrace{\left(\sum_y W(y|x)\right)}_{=1} \cdot \log \frac{P(x) \cdot W(y|x)}{Q(x) \cdot W(y|x)}$ .... When x is fix $\sum_y W(y|x) = 1$

$= \sum_y \left[\sum_x \left(P(x) \cdot W(y|x) \cdot \log \frac{P(x) \cdot W(y|x)}{Q(x) \cdot W(y|x)} \right) \right]$

$\geqslant \sum_y \left[\left(\sum_x P(x) \cdot W(y|x)\right) \cdot \log \frac{\left(\sum_x P(x) \cdot W(y|x)\right)}{\left(\sum_x Q(x) \cdot W(y|x)\right)} \right]$ ... from Lemma 3 Logsum inequality

$= \sum_y PW(y) \cdot \log \frac{PW(y)}{QW(y)}$ .... from ⊛

$= D(PW \,\|\, QW)$ .... definition of KL divergence

■

The function of data Y, cannot increase the information abot X

Corollary :- If $Z = f(Y) \implies I(X \wedge Y) \geqslant I(X \wedge Z)$

Proof :- When $Z = f(Y)$ we know $X \multimap Y \multimap f(Y)$ forms a Markov chain. ∴ Result follows from part 1(i) of Data Processing Inequality.

11) Sufficient Statistics

- Assume X is a distribution. We observe Y and try to get conclusions on X. (ie $X|Y$). In order to do that we manipulate our observation Y. (ie take $\mathbb{E}[Y]$, $\text{Var}[Y]$ ~ $f(Y)$ etc.). Then we know that $X \multimap Y \multimap f(Y)$.

  Or trained model $Y \xrightarrow{f} f(Y)$

- But Data Processing Inequality says we loose info. when we go down the Markov Chain. (i.e $I(X \wedge f(Y)) \leq I(X \wedge Y)$.)

- What is we find some functions where we don't loose info that was in Y $\Longrightarrow$ Those are called "Sufficient Statistics"

Definition - $f(Y)$ is a sufficient statistic for X on the basis of Y, if

$$X \multimap f(Y) \multimap Y$$

Remark: $X \multimap f(Y) \multimap Y \iff I(X \wedge f(Y)) = I(X \wedge Y)$

Proof: From Chain Rule

$$\Rightarrow I(X \wedge Y, f(Y)) = I(X \wedge Y) + \underbrace{I(X \wedge f(Y) \mid Y)}_{= 0 \text{ from lemma 13 and } X \multimap Y \multimap f(Y)}$$
$$= I(X \wedge Y)$$

$$\Rightarrow I(X \wedge Y, f(Y)) = I(X \wedge f(Y)) + \underbrace{I(X \wedge Y \mid f(Y))}_{= 0 \text{ from } X \multimap f(Y) \multimap Y \text{ in Sufficient Statistics}}$$
$$= I(X \wedge f(Y))$$

$$\therefore I(X \wedge Y) = I(X \wedge f(Y))$$

■

- When tossing a coint, we don't need to observe all the outcomes of all tosses. Just calculating the no. of heads is enough. ∴ It's a sufficient statistics.

Probability of Estimation / Detection Error

12) Fano's Inequality

$g(Y) = Y$ case of real Fano's Inequality

Lemma 15: Fano's Inequality

Let rv X takes values in the finite set $\mathcal{X}$. Let Y be another rv on $\mathcal{Y}$, where $\mathcal{Y} \subseteq \mathcal{X}$. Then,

$$H(X|Y) \leq P(X \neq Y) \cdot \log(|\mathcal{X}|-1) + h(P(X \neq Y))$$

Note :- If $\mathcal{Y} \not\subseteq \mathcal{X}$, then replace $\log(|\mathcal{X}|-1)$ by $\log(|\mathcal{X}|)$

Proof :- Let E be the rv denoting an 'error' event. i.e. $E = \mathbb{1}(X \neq Y)$. From Lemma 7- additive chain rules,

$$\Rightarrow H(X,E|Y) = H(X|Y) + H(E|(X,Y)) \quad (= 0)$$

$$\text{also} \rightarrow = H(E|X) + H(X|(E,Y))$$

When X, Y is known, we don't have any uncertainity on E

Hence $H(E|X,Y) = 0$

$$\Rightarrow H(X|Y) = H(E|X) + H(X|(E,Y)) \quad \ldots\ldots (*)$$

From Lemma 4, about conditioning reducing entropy,

$$\Rightarrow H(E|X) \leq H(E) \quad \text{with equality iff } E \perp X$$
$$= h(P(X \neq Y)) \quad \text{from binary entropy definition} \quad \ldots\ldots (1)$$

From the definition of conditional entropy we have, $H(Y|X) = \sum_x P_X(x) \cdot H(Y|X=x)$. We can modify this to have $X \rightarrow Y, E$ and $Y \rightarrow X$

$$\Rightarrow P(X|(E,Y)) = \sum_{e,y} P(E=e, Y=y) \cdot H(X|E=e, Y=y)$$

$$= \sum_y P(E=0, Y=y) \cdot \cancel{H(X \mid E=0, Y=y)}^{\,0} \quad \text{If } E=0 \Rightarrow X=Y \Rightarrow X=Y=y \Rightarrow \text{No uncertainty}$$
$$+ \sum_y P(E=1, Y=y) \cdot H(X \mid E=1, Y=y)$$

$$= \sum_y P(E=1, Y=y) \cdot H(X \mid E=1, Y=y) \quad \cdots\cdots (2)$$

Consider the event $(X \mid E=1, Y=y)$. So X can be anything but the specific value X took.

If $y \subseteq \mathcal{X} \Rightarrow |(X \mid E=1, Y=y)| = |\mathcal{X}| - 1$

If $y \nsubseteq \mathcal{X} \Rightarrow |(X \mid E=1, Y=y)| = |\mathcal{X}|$

$\therefore$ From Lemma 11,

If $y \subseteq \mathcal{X} \Rightarrow H(X \mid E=1, Y=y) \leq \log(|\mathcal{X}| - 1)$

If $y \nsubseteq \mathcal{X} \Rightarrow H(X \mid E=1, Y=y) \leq \log(|\mathcal{X}|)$

(3) equality iff $X \mid E=1, Y=y$ is uniformly distributed

From (2), (3)

$$\Rightarrow P(X \mid (E,Y)) \leq \sum_y P(E=1, Y=y) \cdot \log(|\mathcal{X}| - 1)$$
$$= \log(|\mathcal{X}| - 1) \cdot \sum_y P(E=1, Y=y)$$
$$= \log(|\mathcal{X}| - 1) \cdot P(E=1) \quad \cdots\cdots (4)$$

Substitute (4), (1) to (*)

$$\Rightarrow H(X \mid Y) \leq h(P(X \neq Y)) + \log(|\mathcal{X}| - 1) \cdot P(X \neq Y)$$

■

Equality Condition

Equality holds iff $\forall y \in \mathcal{Y}$ with $P_Y(y) > 0$, it should hold

$$P(X = x \mid Y = y) = \begin{cases} P(X = Y) & \text{if } x = y \\ \dfrac{P(X \neq Y)}{|\mathcal{X}| - 1} & \text{if } x \neq y \end{cases}$$

Proof: For the equality, boath equality conditions of ① and ③ should occur.

$$① \longrightarrow X \perp E \iff X \perp \mathbb{1}(X \neq Y) \iff X \perp Y$$
$$\iff P(x|y) = P(x)$$

② $\longrightarrow (X|E=1, Y=y)$ is uniformly distributed $\iff (X \mid X \neq Y, Y=y)$ is unif.

$$\iff P(x \mid Y=y, X \neq Y) = \frac{1}{|(X \mid X \neq Y, Y=y)|}$$

$$\iff P(x \mid Y=y, x \neq Y) = \frac{1}{|\mathcal{X}|-1}$$

$$\iff P(x|y=y) = \frac{P(x \neq Y)}{|\mathcal{X}|-1} \text{ if } x \neq y$$

Hence proved from combining boath cases. ∎

Remarks :-

1) $$P(X \neq Y) \geq \frac{H(X|Y) - h(P(X \neq Y))}{\log(|\mathcal{X}|-1)}$$

2) Assume X, Y has same alphabet i.e. $\mathcal{X} = \mathcal{Y}$ and suppose we don't want any errors, ie. $P(X \neq Y) = 0$,

∴ From Fano's Inequality

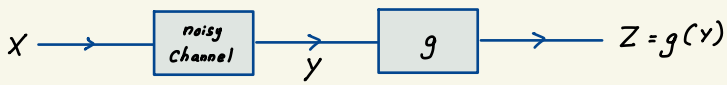
$$\Rightarrow H(X|Y) \leq h(0) + 0 \cdot \log(|\mathcal{X}|-1)$$
$$\leq 0$$
$$\Rightarrow H(X|Y) = 0 \text{ and likewise } H(Y|X) = 0$$
$$\Rightarrow X, Y \text{ should be functions of each other.}$$

Setting

- Let X, Y with joint pmf P_{XY}. We wish to estimate X on the basis of observation Y. For that we will use a deterministic mapping g (called 'estimator')

$$g: \mathcal{Y} \longrightarrow \mathcal{X}$$

Then $g(Y)$ is an estimator of X on the basis Y.



- Then from previous results, $X \multimap Y \multimap g(Y)$.

Fano's Inequality (Powered with Data Processing Inequality)

Let rv X takes values in the finite set $\mathcal{X}$. Let Y be another rv on $\mathcal{Y}$. Let $g: Y \longrightarrow X$ be any deterministic estimator. Then,

$$g(\mathcal{Y}) \subseteq \mathcal{X} \Rightarrow H(X|Y) \leq P(X \neq g(Y)) \cdot \log(|\mathcal{X}|-1) + h(P(X \neq g(Y))$$

$$g(\mathcal{Y}) \nsubseteq \mathcal{X} \Rightarrow H(X|Y) \leq P(X \neq g(Y)) \cdot \log(|\mathcal{X}|) + h(P(X \neq g(Y))$$

Which can be weakened to

$$H(X|Y) \leq P(g(Y) \neq X) \cdot \log|\mathcal{X}| + 1$$

or,

$$P(g(Y) \neq X) \geq \frac{H(X|Y) - 1}{\log|\mathcal{X}|}$$

Proof: ▸ Since g is deterministic $\Rightarrow$ $X \multimap Y \multimap g(Y)$

$\Rightarrow$ $I(X \wedge Y) \geq I(X \wedge g(Y))$

$\Rightarrow$ $H(X|Y) \leq H(X|g(Y))$ ..... ① from Data Processing Ineq.

In Lemma 15, substitute $(X, Y) \longrightarrow (X, g(Y))$

$$P(X \neq g(Y)) \log(|\mathcal{X}|-1) + h(P(X \neq g(Y)) \geq H(X|g(Y)) \geq H(X|Y) \quad \text{.... from ①}$$

▸ If $g(y) \notin \mathcal{X}$ then $|\mathcal{X}|-1$ will become $|\mathcal{X}|$

$\Rightarrow$ $P(X \neq g(Y)) \cdot \log|\mathcal{X}| + \underbrace{h(P(X \neq g(Y))}_{\leq 1 \text{ from Binary Entropy Function.}} \geq H(X|Y)$

$\Rightarrow$ $P(X \neq g(Y)) \cdot \log|\mathcal{X}| + 1 \geq H(X|Y)$

$\Rightarrow$ $P(X \neq g(Y)) \geq \dfrac{H(X|Y) - 1}{\log|\mathcal{X}|}$ ■

Note: If $X \sim \text{Unif}(\mathcal{X})$, then $\forall$ estimators $g: \mathcal{Y} \longrightarrow \mathcal{X}$

$$P(X \neq g(Y)) \geq 1 - \frac{I(X \wedge Y)+1}{\log|\mathcal{X}|}$$

Proof: From Fano's Inequality,

$\Rightarrow$ $P(X \neq g(Y)) \geq \dfrac{H(X|Y) - 1}{\log|\mathcal{X}|}$

$\Rightarrow$ $P(X \neq g(Y)) \geq \dfrac{H(X)}{\log|\mathcal{X}|} - \dfrac{I(X \wedge Y)+1}{\log|\mathcal{X}|}$ .... $I(X \wedge Y) = H(X) - H(X|Y)$

$\Rightarrow$ $= 1 - \dfrac{I(X \wedge Y)+1}{\log|\mathcal{X}|}$ ..... from lemma 11 equality holds when $X \sim \text{unif}(\mathcal{X})$

■

Data Compression or "Lossles" Source Coding by

Fixed - to - fixed length "block" Source Codes

1) Introduction

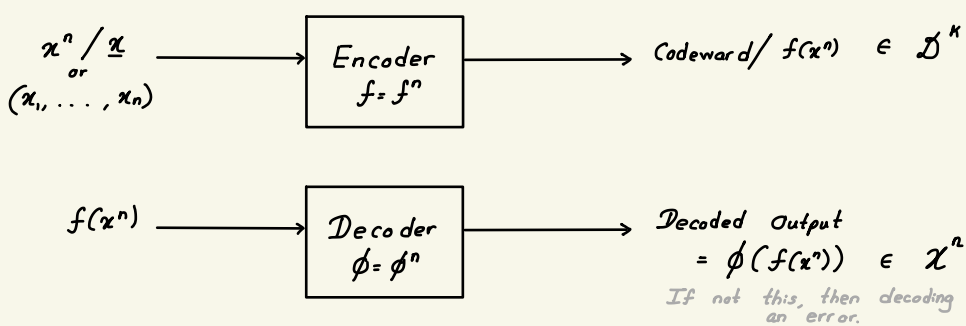
- **Discrete Memoryless Sources (DMS)**
  Here we assume the source symbols $\{x_1, \dots\}$ (ie $\{X_i\}_{i=1}^{\infty}$) are i.i.d. The p.m.f. X on finite alphabet is denoted as P_X.

- **n-to-k block source code**
  fixed-to-fixed length source code. This is a pair of mappings. One from source to codebook. Other from codebook, back to source.

$$\text{Encoder}: \quad f = f^{(n)} : \mathcal{X}^n \longrightarrow \mathcal{D}^k$$
$$\text{Decoder}: \quad \phi : \mathcal{D}^k \longrightarrow \mathcal{X}^n$$

- **Codebook**
  $\{f(x^n), x^n \in \mathcal{X}^n\}$. The codebook should have a lesser cardinality than all possible source combinations. i.e. $< |\mathcal{X}|^n$

- **Code alphabet - $\mathcal{D}$**
  A finite set of given symbols used in transmition. (ex: $\mathcal{D} = \{0,1\}$)



Compression / Rate

$$\text{Compression / rate of code } f\text{-}\phi \triangleq \frac{k}{n} \quad (\mathcal{D}\text{-ary symbols / source symbols})$$

If $\mathcal{D} = \{0,1\}$ then bits / source symbol

Probability of Error

$$\text{Probability of Error} = e(f, \phi) = e(f^{(n)}, \phi^{(n)}) = P(\phi(f(x^n)) \neq x^n)$$

Randomness come from X.

Objective

Finding codes $(f, \phi) = (f^{(n)}, \phi^{(n)})$ of smallest rate (i.e compression) and of smallest error probability $P(\phi(f(x^n)) \neq x)$

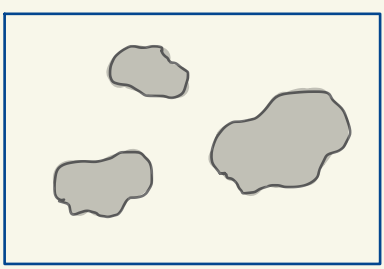
- This is studied primarily with the assumption of large 'n' i.e. large block lengths of source outputs.
- Single shot analysis or $n=1$ base case also gives good insight.

$k_{min}(n, \varepsilon)$ = Smallest k for which $\exists$ n-to-k block code (f, ϕ) with $e(f, \phi) \leq \varepsilon$

$0 \leq \varepsilon < 1$, $\forall\, n \geq 1$, = Minimum with respect to k over all n-to-k block codes with error probability ε.

Motivation

$P(B) \geq 1-\varepsilon$



$\mathcal{X}^n$

$B = B^{(n)} \subseteq \mathcal{X}^{(n)}$

Cardinality of B = $|B| = |\mathcal{D}|^k$

- To optimally compress $\mathcal{X}^n$ to $\mathcal{D}^n$ without wasting $\mathcal{D}$-ary symbols, the most intutive way is to figure out most important (i.e. highest probable) source sequences and prioritize them. These source symbol set is called B.
- Then each n-sequences in B is assigned a unique $\mathcal{D}$-ary codeward of length k.
- Each n-sequences not in B (ie $B^c = \mathcal{X}^n \setminus B$) is assigned a dummy codeward, neglecting them to purposefully result in errors.

$$
\begin{aligned}
& P(B) && \geq 1-\varepsilon \\
\Rightarrow\ & P(B^c) && < \varepsilon \\
\Rightarrow\ & e(f,\phi) && < \varepsilon
\end{aligned}
$$

New Objective

What is the smallest cardinality of the highest probable set B ($\Longleftrightarrow$ smallest k)
What are algorithms for the compression.

- Important question is what happens to $\lim_{n\to\infty} \frac{k(n,\varepsilon)}{n}$ which brings us to next Theorem.

2) Fixed to Fixed Length Source Coding Theorem

Theorem :- FFLSCT

For a Discrete Memoryless System (DMS) $\{X_t\}_{t=1}^{\infty}$ with pmf P_X and $k_{min}(n,\varepsilon)$ - smallest k with n-to-k code with error ε, $(0 < \varepsilon < 1)$

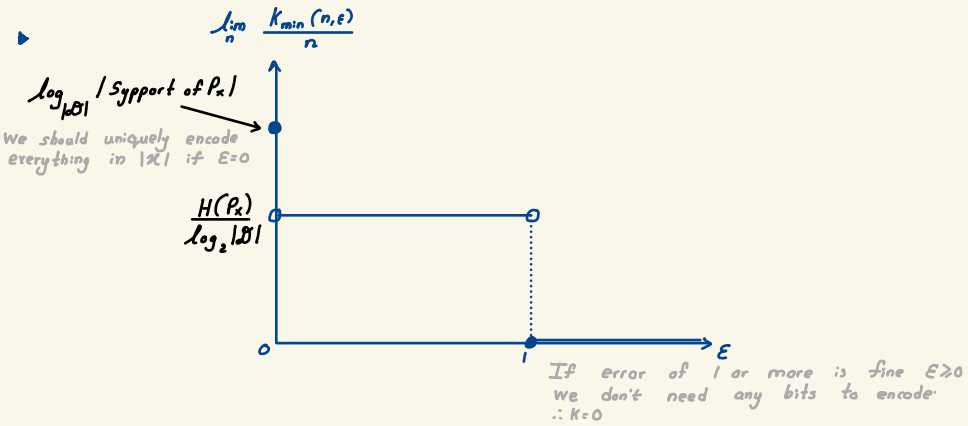
$$\lim_{n\to\infty} \frac{k_{min}(n,\varepsilon)}{n} = \frac{H(P_X)}{\log_2 |\mathcal{X}|}$$

where $H(P_X)$ = Shanon entropy of X (P_X).

Proof:- We will build up to this proof by first going through a concept by proving it for $n=1$ single shot analysis. Proof involves

- → "achievability"
- → "converse" / "optimality"

Remark:-

- This Theorem establishes the operational significance of entropy in figuring out the best source coding with a small error rate (ε) is possible.
- 

Corollary : $0 \leq H(P_X) \leq \log_2 |\mathcal{X}|$

Proof : **Claim :** $\frac{k_{min}(n,\varepsilon)}{n} \leq \log_{|\mathcal{B}|} |\mathcal{X}|$

For fixed n, $k_{min}(n,\varepsilon)$ is non-decreasing in $\varepsilon \in [0,1)$

$$\Rightarrow \frac{k_{min}(n,\varepsilon)}{n} \leq \frac{k_{min}(n,0)}{n} \quad \cdots\cdots (1)$$

When error probability is $\varepsilon = 0$, we need set $\mathcal{B}$ to be entire $\mathcal{X}^n$.
But we have $|\mathcal{B}| = |\mathcal{B}|^k$

$$\Rightarrow |\mathcal{B}| = |\mathcal{B}^k| = |\mathcal{X}|^n$$

$$\Rightarrow \log_{|\mathcal{B}|} |\mathcal{X}| = k/n$$

$$\Rightarrow \frac{k_{min}(n,\varepsilon)}{n} \leq \log_{|\mathcal{B}|} |\mathcal{X}| \quad \cdots\cdots \text{from } (1)$$

Claim Proved.

From the above claim and F-F Length Source Coding

$$\Rightarrow \lim_{n\to\infty} \frac{k_{min}(n,\varepsilon)}{n} = \frac{H(P_X)}{\log_2 |\mathcal{B}|} \leq \lim_{n\to\infty} \log_{|\mathcal{B}|} |\mathcal{X}| = \log_{|\mathcal{B}|} |\mathcal{X}|$$

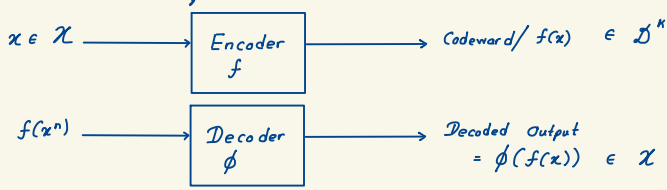
$$\Rightarrow H(P_X) \leq \log_2 |\mathcal{B}| \cdot \log_{|\mathcal{B}|} |\mathcal{X}| = \log_2 |\mathcal{X}|$$

■

2) F·FLSC Theorem for $n=1$

Proof By Himanshu Tyagi (IISc)

- Also called Non asymptotic "Single-shot" approach.
- We study $\frac{k_{min}(n,\varepsilon)}{n}$, $0 \leq \varepsilon < 1$ for joint $n=1$
- Consider $k_{min}(1,\varepsilon)$, $0 \leq \varepsilon < 1$



- Lesson from following examples. If $\varepsilon = 0$ or $X \sim$ unifor $(\mathcal{X})$ with very small error probability Then compression is infeasible.

Example 1 :- $X \sim$ unif $\{1, \ldots, 8\}$ with $P_X(x) = 0.125 \ \forall x \in \mathcal{X}$ $\varepsilon = 0.01$

$\Rightarrow k_{min}(1, 0.01) = \log_2 8 = 3$ bits //

$\Rightarrow$ No compression is possible for this uniform source.

Example 2 :- Is it possible to compress the following non-unifor source if a small error probability of $\varepsilon = 0.01$ is allowed.

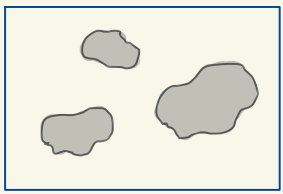
x	1	2	3	4	5	6	7	8
$P_X(x)$	0.33	0.33	0.33	$\frac{0.01}{5}$	$\frac{0.01}{5}$	$\frac{0.01}{5}$	$\frac{0.01}{5}$	$\frac{0.01}{5}$

- $k_{min}(1, 0) = \log_2 8 = 3$ bit
- If $\varepsilon = 0.01$ we can just encode high probable 1,2,3 and ignore others.

Encoder (f)	Decoder (ϕ)
$1 \longrightarrow 00$	$\phi(00) = 1$
$2 \longrightarrow 01$	$\phi(01) = 2$
$3 \longrightarrow 10$	$\phi(10) = 3$
$4, \ldots, 8 \longrightarrow 11$	$\phi(11) =$ Some error

- $e(f, \phi) = P(X \in \{4,5,6,7,8\})$
  $= 0.01$
  $\Rightarrow k_{min}(1, 0.01) = 2$ bit //
- Compression is possible.

Idea



- Want smallest B with $P(B) \geq 1-\varepsilon$. Assign codewords only to sets in B. $\Rightarrow$ $k_{min}(1,\varepsilon) = \lceil \log_2 |B| \rceil$ bits.
- Arrange symbols in $\mathcal{X}$, according to the descending order of probability.

$$\mathcal{X} = \{x_1, \ldots, x_n\} \qquad P_X(x_1) \geq P_X(x_2) \geq \ldots \geq P_X(x_{|\mathcal{X}|})$$

- Assign symbols starting from x_1, to B until $P(B) \geq 1-\varepsilon$. Then,

$$B = \left\{ x_k \;\middle|\; k < \min\left\{ i \;\middle|\; \sum_{j<i} P_X(x_j) \geq 1-\varepsilon \right\} \right\}$$

- (Encoder f) - Encode symbols in B uniquely using $\lceil \log |B| \rceil$ no. of bits. For all other symbols in $B^c = \mathcal{X} \setminus B$, assign a dummy codeword.
- (Decoder ϕ) - Decode the codewords into symbols if they are in B. Decode everything else to an error.

$$e(f,\phi) = P_X(\phi(f(x)) \neq x) = P(x \notin B) < \varepsilon$$

- This idea boils down to a minimum k_{min} of,

$$k_{min}(1,\varepsilon) = \left\lceil \log_2 \left(\min\left\{ i \;\middle|\; \sum_{j<i} P_X(x_j) \geq 1-\varepsilon \right\} + 1 \right) \right\rceil \quad (*)$$

($\min\{\ldots\}$: for source symbols in B; $+1$: for B^c)

Ex: In previous example $k_{min}(1, 0.01) \Rightarrow 1-\varepsilon = 1-0.01 = 0.99$. We can put 3 symbols (1,2,3) into set B, before $P(B)$ exceeds 0.99. $\therefore$ $(*)$ becomes, $k_{min} = \log_2(3+1) = 2$ bits //

Large Probability Upper Bound - λ_u
Large Probability Lower Bound - λ_l

Large Probability Bounds

- To give a mathematical meaning to above ⊛, we need a formal word to the first ρ_i that makes the set $\mathcal{B}$ exceed in $(1-\varepsilon)$ probability. That is l.p.l.b / l.p.u.b.

- Given $0 < \varepsilon < 1$, let $\wedge = \wedge(\varepsilon) > 1$ be such that

$$P_x\left(\{x \in \mathcal{X} \mid P_x(x) > \wedge\}\right) \geq 1-\varepsilon$$

The largest value possible for $\wedge$ is called the "Large Probability Lower Bound" for r.v $P_x(x)$

$\wedge \leftrightarrow P_x(x)$
$\lambda_l/\lambda_u \leftrightarrow -\log P_x$

Definition - Given $0 < \varepsilon < 1$, let $\lambda_u = \lambda_u(\varepsilon)\left(= \log \wedge^{-1}\right) > 0$ be such that

$$P_x\left(\{x \in \mathcal{X} \mid -\log(P_x(x)) \leq \lambda_u\}\right) \geq 1-\varepsilon$$

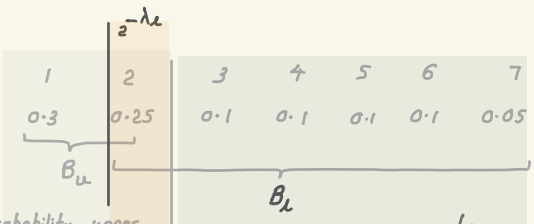
The smallest value possible for λ_u is called the "Large Probability Upper Bound" for $\log\left(P_x(x)^{-1}\right)$

Definition - Given $0 < \varepsilon < 1$, let $\lambda_l = \lambda_l(\varepsilon) > 0$ be such that

$$P_x\left(\{x \in \mathcal{X} \mid -\log(P_x(x)) \geq \lambda_l\}\right) \geq 1-\varepsilon$$

The largest value possible for λ_l is called the "Large Probability Lower Bound" for $\log\left(P_x(x)^{-1}\right)$

Example :-



Assume $\varepsilon < 0.5$ is sufficient.

Large probability upper bound speaks about how much lower prob. Symbols should be included if we are staying at high values.

Large probability lowe bound is about how much symbols should we need if we limit ourselves to lower probable symbols

Closer the better

Remarks

1) Such a Λ exists $(\Leftrightarrow \lambda_u)$ ..... $E_x := \Lambda = P_x(x_i^*)$

2) By using l.p.l.b and l.p.u.b definitions on (*), we need at most
   - $\simeq$ Large Probability Upper Bound on $(-\log P_x(x) + 1)$ bits
   - $\simeq$ Large Probability Upper Bound on $\lceil (-\log P_x(x)) \rceil$ bits

   to compress, store or represent X, almost losslessly.

3) Upper bound on codeward length $\Longleftrightarrow$ Large probability upper bound on $-\log P_x(x)$.

4) Lower bound on codeward length $\Longleftrightarrow$ Large probability Lower bound on $-\log P_x(x)$.

5) $k_{min}(1, \varepsilon) \leq \left\lceil \frac{H(P_x(x))}{\varepsilon} \right\rceil + 1$

k=1 case from ENEE 620 "Inequalities"

Proof Sketch: By Markov's Inequality for a r.v. $P\left(Z \leq \frac{\mathbb{E}[Z]}{\varepsilon}\right) \geq 1-\varepsilon$
Then choose $Z = \log \frac{1}{P_x(x)}$. Take expectation of boath sides

$$\Rightarrow \mathbb{E}[Z] = \mathbb{E}\left[\log 1/P_x(x)\right] = H(P_x) \quad \text{... Definition of Entropy}$$

Substitute $\mathbb{E}[Z]$ from above and Z to Markov Inequality

$$\Rightarrow P_x\left(-\log P_x(x) \leq \frac{H(P_x)}{\varepsilon}\right) \geq 1-\varepsilon$$

λ_u here is comparable to $H(P_x)/\varepsilon$ here. Hence we need

$$\Rightarrow \lambda_u \leq H(P_x)/\varepsilon$$

From (*) and previous remark we need $k_{min}(1, \varepsilon) \leq \lceil \lambda_u \rceil + 1$.

$$\Rightarrow k_{min}(1, \varepsilon) \leq \left\lceil \frac{H(P_x(x))}{\varepsilon} \right\rceil + 1$$

■

Lemma 1: (Upper Bound on $k_{min}(1,\varepsilon)$ / Achievability)

$$k_{min}(1,\varepsilon) \leqslant \lceil \lambda_u \rceil + 1$$

Operational definition — From achievability scheme

Proof:- **Method 1**

Suppose i^* attains the minimum in ⊛ after sorting the probabilities. ∴ Then from Largest probability lower bound definition

$$\wedge = P_X(x_{i^*}) \quad \cdots\cdots ①$$

Since only the elements having higher than $\wedge$ probability is selected to B. Since P_{X_i} is already sorted $B = \{x_1, x_2 \ldots, x_{i^*}\}$

$$\Rightarrow P_X(x) \geqslant P_X(x_{i^*}) = \wedge \quad \ldots \quad \forall x \in B$$

$$\Rightarrow \sum_{x} P_X(x) \geqslant \sum_{x \in B} P_X(x) \geqslant |B| \cdot \wedge \quad \ldots \quad B \subseteq X$$

$$\Rightarrow 1 \geqslant |B| \cdot \wedge$$

$$= i^* \cdot \wedge \quad \ldots\ldots\ldots\ldots |B| = i^*$$

$$\Rightarrow \frac{1}{\wedge} \geqslant i^*$$

$$\Rightarrow = \min\left\{ i \,\middle|\, \sum_{j<i} P_X(x_i) \geqslant 1-\varepsilon \right\} \quad \cdots\cdots \text{This is what we assumed } i^* \text{ to be.}$$

$$\Rightarrow \left\lceil \log\left(\frac{1}{\wedge} + 1\right) \right\rceil \geqslant \underbrace{\left\lceil \log\left(\min\left\{ i \,\middle|\, \sum_{j<i} P_X(x_i) \geqslant 1-\varepsilon \right\} + 1\right) \right\rceil}_{\text{from ⊛}}$$

$$\Rightarrow \lceil \log \wedge^{-1} + 1 \rceil \geqslant k_{min}(1,\varepsilon)$$

$$\Rightarrow \lceil \log \wedge^{-1} \rceil + 1 \geqslant k_{min}(1,\varepsilon)$$

$$\Rightarrow \lceil \lambda_u \rceil + 1 \geqslant k_{min}(1,\varepsilon) \quad \cdots\cdots \lambda_u = \log \lambda_L^{-1}$$

■

Method 2

- Set $B_u = \{ x \in \mathcal{X} \mid P_X(x) \geqslant \wedge \}$
  $= \{ x \in \mathcal{X} \mid -\log_K P(x) \leqslant \lambda_u \}$ $\cdots\cdots$ $\lambda_u = \log \wedge^{-1} \Rightarrow \wedge^{-1} = 2^{\lambda_u}$

- Encoder : Enumerates all the symbols in B_u. And
  $x \in B_u \longrightarrow f(x) = (0, b(x))$ - binary representation of enumeration of x in B_u.
  $x \notin B_u \longrightarrow f(x) = (1, 0 \ldots 0)$

- **Decoder**: If codeword start with 0, $\phi(0,b) = x_b$
  , , , , 1, $\phi(1,0..0) =$ some $x \in \mathcal{X} \setminus B_u$

- Since only the elements in B_u are coded and all the other codes in $\mathcal{X} \setminus B_u$ need one code
  length of codeward $= \lceil \log |B_u| \rceil + 1$

- Error probability for this B_u $= P(\phi(f(x)) \neq x) \leq P(x \notin B_u) \leq \varepsilon$
  $P(B_u) \geq 1-\varepsilon$

$$\Rightarrow 1 = \sum_{x \in \mathcal{X}} P_X(x) \geq \sum_{x \in B_u} P_X(x) \geq |B_u| \cdot \wedge$$

$$\Rightarrow \wedge^{-1} \geq |B_u|$$

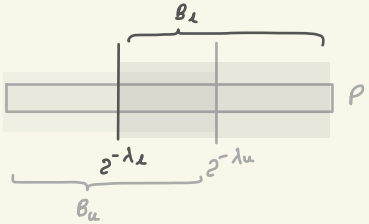
$$\Rightarrow 2^{\lambda_u} \geq |B_u|$$

$$\Rightarrow \lambda_u \geq \log |B_u|$$

$$\Rightarrow \lceil \lambda_u \rceil + 1 \geq \lceil \log B_u \rceil + 1$$

$$\Rightarrow \text{length of codeward} = k(1,\varepsilon) = \lceil \log B_u \rceil + 1 \leq \lceil \lambda_u \rceil + 1$$

■



Lemma 2:- (Lower Bound on $K_{min}(1,\varepsilon)$ / Converse)

Given $0 < \varepsilon, \delta$ with $\varepsilon + \delta < 1$

$$K_{min}(1,\varepsilon) \geq \lambda_L - \log \frac{1}{1-\varepsilon-\delta}$$

Proof:

- This states that any 1-to-k code (f,ϕ) with $e(f,\phi) \leq \varepsilon$ has $k \geq \lambda_L - \log(1/1-\varepsilon-\delta)$.
- Let (f,ϕ) be any given source code of length $k(1,\varepsilon)$ with $P(\phi(f(x)) \neq x) \leq \varepsilon$.
- Suppose $\tilde{\mathcal{X}} = \{x \in \mathcal{X} \mid \phi(f(x)) = x\}$ i.e all the elements in $\tilde{\mathcal{X}}$ is decoded correctly. Since this is supposed to be a ε error code, $P(\tilde{\mathcal{X}}) \geq 1-\varepsilon$. ····①
- We assume $k(1,\varepsilon)$ is codeward length. But $\tilde{\mathcal{X}}$ is having all one to one encoded correct source symbols

$$\Rightarrow 2^{k(1,\varepsilon)} \geq |\tilde{\mathcal{X}}|$$

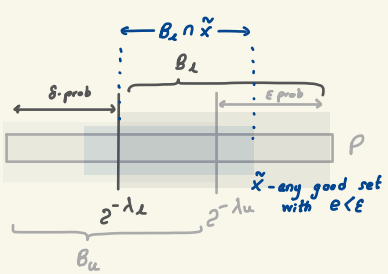
$$\Rightarrow k(1,\varepsilon) \geq \log_2 |\tilde{\mathcal{X}}| \cdots\cdots ②$$

- Denote $B_L = \{x \in \mathcal{X} \mid -\log P_X(x) \geq \lambda_L\}$

$$= \{x \in \mathcal{X} \mid P_X(x) \leq 2^{-\lambda_L}\} \cdots\cdots ③$$

The the leftout probability from set B_L (called δ) should be smaller than that from the optimal set B_u (called ε) and $\delta + \varepsilon < 1$

$$P_X(B_L) \geq 1-\delta \cdots\cdots ④$$



Then,

$$\Rightarrow P_x(B_1 \cup \tilde{x}) = P(B_1) + P(\tilde{x}) - P(B_1 \cap \tilde{x})$$

$$\Rightarrow P_x(B_1 \cap \tilde{x}) = P(B_1) + P(\tilde{x}) - P(B_1 \cup \tilde{x})$$

$$\geq (1-\delta) + (1-\varepsilon) - 1 \quad \cdots\cdots \text{①, ④ and axiom of prob.}$$

$$= 1 - \varepsilon - \delta \quad \cdots\cdots (>0 \text{ since } \varepsilon+\delta<1)$$

$$\Rightarrow |B_1 \cap \tilde{x}| \cdot 2^{-\lambda_1} \geq 1 - \varepsilon - \delta \quad \cdots\cdots \text{from ③ all elements in } B_1 \text{ has less than } 2^{-\lambda_1} \text{ prob.}$$

$$\Rightarrow |\tilde{x}| \geq |B_1 \cap \tilde{x}| \geq 2^{\lambda_1}(1-\varepsilon-\delta)$$

$$\Rightarrow \log|\tilde{x}| \geq \log\left(2^{\lambda_1}(1-\varepsilon-\delta)\right)$$

$$= \lambda_1 + \log(1-\varepsilon-\delta)$$

$$= \lambda_1 - \log\frac{1}{1-\varepsilon-\delta}$$

$$\text{②} \Rightarrow k(1,\varepsilon) \geq \log|\tilde{x}| \geq \lambda_1 - \log\frac{1}{1-\varepsilon-\delta}$$

$$\Rightarrow k_{min}(1,\varepsilon) \geq \lambda_1 - \log\frac{1}{1-\varepsilon-\delta}$$

■

Remarks

(i) λ_u and λ_l are the quantities of interest. Closer these are, the better the estimates of $k_{min}(1,\varepsilon)$.

(ii) Example 1:

$X \sim$ unif $\{1,\ldots,8\}$ with $P_x(x) = 0.125 \ \forall x \in \mathcal{X}$ $\quad \varepsilon = 0.01$

$\Rightarrow k_{min}(1, 0.01) = \log_2 8 = 3$ bits //

$\Rightarrow$ No compression is possible for this uniform source.

$\mathcal{B}_u = \mathcal{X}$ with $\lambda_u = \log|\mathcal{X}| = 3$ and $\varepsilon =$ any value in $(0,1)$

$(\Lambda = 1/|\mathcal{X}| = 1/8)$

$\mathcal{B}_l = \mathcal{X}$ with $\lambda_l = \log|\mathcal{X}| = 3$ and $\delta =$ any value in $(0,1)$.

Hence,

By Lemma 1 $\Rightarrow$ $k_{min}(1,\varepsilon) \leq \lceil \log|\mathcal{X}| \rceil + 1 = 3+1$

By Lemma 2 $\Rightarrow$ $k_{min}(1,\varepsilon) \geq \lceil \log|\mathcal{X}| \rceil - \log\frac{1}{1-\varepsilon-\delta} > \lceil \log|\mathcal{X}| \rceil - 3$

Hence $k_{min}(1,\varepsilon) \cong \lceil \log|\mathcal{X}| \rceil$ as expected.

(iii) Example 2

Is it possible to compress the following non-unifor source if a small error probability of $\varepsilon = 0.01$ is allowed.

x	1	2	3	4	5	6	7	8
$P_x(x)$	0.33	0.33	0.33	$\frac{0.01}{5}$	$\frac{0.01}{5}$	$\frac{0.01}{5}$	$\frac{0.01}{5}$	$\frac{0.01}{5}$

$\Rightarrow k_{min}(1, 0.01) = 2$ bit //

$\Rightarrow$ Compression is possible.

$\mathcal{B}_u = \{1,2,3\}$ with $\lambda_u = \log 0.33 = 1.599$ and $\varepsilon =$ any value in $(0,1)$

$(\Lambda = 1/0.33)$

(least probability needed when picking starting from top. ~ 0.33)

$\mathcal{B}_l = \{1,2,..,8\}$ with $\lambda_l = \log 0.33 = 1.599$ and $\delta =$ any value in $(0, 1-\varepsilon) = [0, 0.9)$

(highest probability needed when picking starting from bottom. ~ 0.33)

$2^{-\lambda_l} \cong 0.33$

$\lambda_l \cong \log 0.33$

Hence,

By Lemma 1 $\Rightarrow$ $k_{min}(1,\varepsilon) \leq \lceil \log|\mathcal{B}_u| \rceil + 1 = 2.599$

By Lemma 2 $\Rightarrow$ $k_{min}(1,\varepsilon) \geq \lceil \log|\mathcal{X}| \rceil - \log\frac{1}{1-\varepsilon-\delta} = 1.599 - \log\frac{1}{1-0.01-\delta}$

Hence $k_{min}(1,\varepsilon) \cong \lceil \lambda_u \rceil \simeq \lceil 1.599 \rceil = 2$ bits as expected.

3) Extending $n=1$ Results to General n

- To go from $n=1$ to general n, replace X by X^n.
- Take Large Probability Upper Bound ($\lambda_u^{(n)}$) and Large Probability Lower Bound ($\lambda_l^{(n)}$) for $-\log P_{X^n}(x^n)$
- In some clases of sources, the upper bound and lower bound match asymptotically. (as $n \to \infty$). For that the structure of $\{X_t\}_{t=1}^{\infty}$ plays a role. (i.e we need WLLN).

$n=1$	General n
λ_u - Large Probability Upper Bound for $-\log P_X(x)$	$\lambda_u^{(n)}$ - Large Probability Upper Bound for $-\log P_{X^n}(x^n)$
λ_l - Large Probability Lower Bound for $-\log P_X(x)$.	$\lambda_l^{(n)}$ - Large Probability Lower Bound for $-\log P_{X^n}(x^n)$
Lemma 1 - Achievability $k_{min}(n,\varepsilon) \leq \lceil \lambda_u \rceil + 1$	(1) $\frac{k_{min}(n,\varepsilon)}{n} \leq \frac{\lceil \lambda_u^{(n)} \rceil}{n} + \frac{1}{n} \leq \frac{\lambda_u^n}{n} + \frac{2}{n}$
Lemma 2 - Converse $k_{min}(n,\varepsilon) \geq \lambda_l - \frac{1}{1-\varepsilon-\delta}$	(2) $\frac{k_{min}(n,\varepsilon)}{n} \geq \frac{\lambda_l^{(n)}}{n} - \frac{1}{n}\log\frac{1}{1-\varepsilon-\delta}$

Weak Law of Large Numbers (WLLN)

This page is from ENEE 620

Theorem: $\{X_k\}_{k=1}^{n}$ is a sequence of $\mathbb{R}$ valued uncorrelated rvs on the same $(\Omega, \mathcal{A}, P)$ probability measure with

$$\left.\begin{array}{ll} \text{mean} & \mathbb{E}[X_k] = \mu_k \\ \text{Variance} & \mathbb{E}[(X_k - \mu_k)^2] = \sigma_k^2 \end{array}\right\} \forall k$$

Assume $\mathbb{E}[X_k^2] < \infty \ \forall k$. Define a sequence of sample / sampled means as,

$$\mu_n(\omega) = \frac{1}{n} \sum_{k=1}^{n} X_k(\omega)$$

Then

$$\frac{1}{n^2} \sum_{k=1}^{n} \sigma_k^2 \xrightarrow[n\to\infty]{} 0 \quad \Longrightarrow \quad \mu_n - \mathbb{E}[\mu_n] \xrightarrow[n\to\infty]{\text{q.m.}} 0$$

(Bounded Variance)

Proof:

$$\mathbb{E}[\mu_n(\omega)] = \mathbb{E}\left[\frac{1}{n}\sum_{k=1}^{n} X_k(\omega)\right] = \frac{1}{n}\sum_{k=1}^{n} \mathbb{E}[X_k(\omega)] = \frac{1}{n}\sum_{k=1}^{n} \mu_k \quad \cdots (1)$$

When $\{X_n\}_{n=1}^{\infty}$ is uncorrelated $\Longrightarrow$ $Cov[X_l, X_m] = \sigma_l^2 \delta_{lm}$

where $\delta_{lm} = \begin{cases} 1 & l = m \\ 0 & \text{o.w} \end{cases}$

$$Var[\mu_n] = \mathbb{E}\left[(\mu_n - \mathbb{E}[\mu_n])^2\right]$$

$$= \mathbb{E}\left[\left(\frac{1}{n}\sum_{k=1}^{n} X_k(\omega) - \frac{1}{n}\sum_{k=1}^{n} \mu_k\right)^2\right] \quad \cdots \text{ from } (1)$$

$$= \frac{1}{n^2}\, \mathbb{E}\left[\left(\sum_{k=1}^{n} (X_k(\omega) - \mu_k)\right)^2\right]$$

$$= \frac{1}{n^2} \cdot \sum_{k=1}^{n} \mathbb{E}\left[(X_k - \mu_k)^2\right] + \frac{1}{n^2}\left[\sum_{1 \le l \ne m \le n} \mathbb{E}[(X_l - \mu_l)(X_m - \mu_m)]\right]$$

$$= \frac{1}{n^2}\sum_{k=1}^{n} Var[X_k] + \frac{1}{n^2}\left[\sum_{1 \le l \ne m \le n} Cov[X_l, X_m]\right]$$

$$= \frac{1}{n^2}\sum_{k=1}^{n} \sigma_k^2 + 0 \quad \cdots \quad X_i \text{ is uncorrelated}$$

$$\Rightarrow \therefore \lim_n Var[\mu_n] = \lim_n \mathbb{E}\left[(\mu_n - \mathbb{E}[\mu_n])^2\right] = \lim_n \frac{1}{n^2}\sigma_k^2 = 0 \quad \cdots \text{ from bounded Variance hypothesis}$$

$$\Rightarrow \lim_n \mathbb{E}\left[(\mu_n - \mathbb{E}[\mu_n])^2\right] = 0$$

$$\Rightarrow \mu_n - \mathbb{E}[\mu_n] \xrightarrow[n\to\infty]{\text{q.m}} 0$$

■

Corollary 1 - $\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2 \xrightarrow[n\to\infty]{} 0 \implies \mu_n - \mathbb{E}[\mu_n] \xrightarrow[n\to\infty]{\rho} 0$

(Bounded Variance)

Proof: Since Converge in r^{th} mean for any $r>0$ $\implies$ Converge in Probability proof follow

Corollary 2 - If $\mathbb{E}[X_k] = \mu \ \forall k$

$\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2 \xrightarrow[n\to\infty]{} 0 \implies \mu_n \xrightarrow[n\to\infty]{q.m.} \mu$

(Bounded Variance)

Corollary 3 - If $\mathbb{E}[X_k] = \mu \ \forall k$

$\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2 \xrightarrow[n\to\infty]{} 0 \implies \mu_n \xrightarrow[n\to\infty]{\rho} \mu$

(Bounded Variance)

Rephrase

$\{Y_t\}_{t=1}^{\infty}$ is $\mathcal{Y}$-valued i.i.d process with $\mathcal{Y}$ being finite set. Then, $\forall n$ sufficiently large i.e. $n \geq N(\tau, \varepsilon)$

$$\frac{1}{n}\sum_{t=1}^{n} Y_t \xrightarrow[\text{in probability}]{n\to\infty} \mathbb{E}[Y]$$

i.e. $\lim_{n\to\infty} \rho\left(\left|\frac{1}{n}\sum_{t=1}^{\infty} Y_t - \mathbb{E}[Y]\right| < \tau\right) = 1 \quad \forall \tau > 0$

i.e. $\rho\left(\left|\frac{1}{n}\sum_{t=1}^{\infty} Y_t - \mathbb{E}[Y]\right| < \tau\right) \geq 1-\varepsilon \quad \forall 0<\varepsilon<1, \tau>0$

:... (#)

Proof of FFLSCT

Theorem :- FFLSCT For a Discrete Memoryless System (DMS) $\{X_t\}_{t=1}^{\infty}$ with pmf P_X and $k_{min}(n,\varepsilon)$ - smallest k with n-to-k code with error ε, $(0 < \varepsilon < 1)$

$$\lim_{n\to\infty} \frac{k_{min}(n,\varepsilon)}{n} = \frac{H(P_X)}{\log_2 |\mathcal{X}|}$$

where $H(P_X)$ = Shanon entropy of X (P_X).

Proof :- ▸ Set $Y_t \triangleq -\log P_X(X_t)$. Then

$$\Rightarrow \mathbb{E}[Y] = \mathbb{E}[-\log P_X(X)] = H(P_X) \quad \cdots\cdots (1)$$

▸ Apply WLLN to sequence $\{Y_t\}_{t=1}^{\infty}$ assuming they are i.i.d and bounded variance,

$$\Rightarrow P\left(\left| \frac{1}{n}\sum_{t=1}^{n} Y_t - \mathbb{E}[Y] \right| < \tau \right) \geq 1-\varepsilon \quad \forall\ 0<\varepsilon<1,\ \tau>0 \text{ from } (*)$$

$$\Rightarrow P\left(\left| \frac{1}{n}\sum_{t=1}^{n} \underbrace{(-\log P_X(X_t))}_{\text{Why ?}} - H(P_X) \right| < \tau \right) \geq 1-\varepsilon \quad \cdots \text{ from } (1)$$

Answer.
Since the system is DMS, X_t's are i.i.d.
$\Rightarrow P_X(x^n) = P_X(x_1,\ldots,x_n) = \prod_{t=1}^{n} P_X(x_t)$
$\Rightarrow -\log P_X(x^n) = -\log\left(\prod_{t=1}^{n} P_X(x_t)\right) = \sum_{t=1}^{n} -\log P_X(x_t)$

$$\Rightarrow P\left(\left| \frac{1}{n}\left(-\log P_X(x^n)\right) - H(P_X) \right| < \tau \right) \geq 1-\varepsilon$$

$$\Rightarrow P\left(n\left(H(P_X)-\tau\right) < -\log P_X(x^n) < n\left(H(P_X)+\tau\right) \right) \geq 1-\varepsilon \quad \cdots\cdots (2)$$

Typical Set (shall see later)

▸ Applying Lemma (1). From (2)

$$\Rightarrow P\left(-\log P_X(x^n) < n\left(H(P_X)+\tau\right) \right) \geq 1-\varepsilon$$

This qualifies as the set B_u that we were interested in the proof of Lemma 1. Set

$$B_u^{(n)} = \left\{ x^n \in \mathcal{X}^n \;\middle|\; -\log P_X(x^n) < \underbrace{n\left(H(P_X)+\tau\right)}_{\lambda_u} \right\}$$

Then for every $0<\varepsilon<1$, $\tau>0$ with $\lambda_u^{(n)} = n(H(P_X)+\tau)$. ∴ From (*) for n large enough, $n \geq N_1(\tau,\varepsilon)$

$$P(B_u^{(n)}) \geq 1-\varepsilon \qquad n \geq N_1$$

∴ Applying Lemma (1) with this B_u and λ_u

$$\Rightarrow k_{min}(n,\varepsilon) \leq \lceil \lambda_u^n \rceil + 1 = \lceil n(H(P_X)+\tau) \rceil + 1 \quad \cdots\cdots (3)$$

- Applying Lemma (2). From (2)

$$\Rightarrow P\left(-\log P_X(x^n) \geq n(H(P_X)-\tau) \right) \geq \frac{1+\varepsilon}{2}$$? why

This qualifies as the set B_l that we were interested in the proof of Lemma 2. Set

$$B_u^{(n)} = \{ x^n \in \mathcal{X}^n \mid -\log P_X(x^n) \geq n(H(P_X)-\tau) \}$$
$$\lambda_l$$

Then for every $0<\varepsilon<1$, $\tau>0$ with $\lambda_l^{(n)} = n(H(P_X)-\tau)$. ∴ From (*) for n large enough, $n \geq N_2(\tau,\varepsilon)$

$$P(B_l^{(n)}) \geq \frac{1+\varepsilon}{2} = 1-\frac{1-\varepsilon}{2} = 1-\delta \qquad n \geq N_2(\tau,\varepsilon)$$

∴ Applying Lemma (2) with this B_l and λ_l

$$\Rightarrow k_{min}(n,\varepsilon) \leq \lambda_l - \frac{1}{1-\delta-\varepsilon} = n(H(P_X)-\tau) - \frac{1}{1-\left(\frac{1-\varepsilon}{2}\right)-\varepsilon}$$

$$= n(H(P_X)-\tau) - \log\frac{2}{1-\varepsilon} \quad \cdots (4)$$

- From (3), (4) ∀ n large enough i.e $n \geq \max(N_1, N_2)$

$$\Rightarrow n(H(P_X)-\tau) - \log\frac{2}{1-\varepsilon} \leq k_{min}(n,\varepsilon) \leq \lceil n(H(P_X)+\tau) \rceil + 1$$

$$\Rightarrow H(P_X)-\tau - \frac{\log\frac{2}{1-\varepsilon}}{n} \leq \frac{k_{min}(n,\varepsilon)}{n} \leq \frac{\lceil n(H(P_X)+\tau) \rceil}{n} + \frac{1}{n}$$

$\forall\ 0<\varepsilon<1 \quad \tau>0$

$$\Rightarrow \overline{\lim_n} = \lim_n \sup \frac{k_{min}(n,\varepsilon)}{n} \leq H(P_x)+\tau$$

$$\Rightarrow \underline{\lim_n} = \lim_n \inf \frac{k_{min}(n,\varepsilon)}{n} \geq H(P_x)-\tau$$

Then τ can be chosen arbitarily small,

$$\Rightarrow \lim_n \frac{k_{min}(n,\varepsilon)}{n} = H(P_x) \qquad \forall\ 0<\varepsilon<1$$

■

4) Coding Theorem → Not exactly

(Equivalent Formulation of F-FLSC Theorem)

- This is a version of Fixed to Fixed Length Source Coding Theorem, by replacing the k-codeward length from R - rate / compression of code.

Recall

- Source code is a pair of mappings (f_n, ϕ_n) such that, $f_n : \mathcal{X}^n \longrightarrow \mathcal{D}^{k_n}$ and $\phi_n : \mathcal{D}^{k_n} \longrightarrow \mathcal{X}^n$. Then,

$$\text{Code Rate / Compression} = R = \frac{k_n}{n} \longrightarrow$$

R always get better (i.e lower) when n increase, since more info can help modelling f to reduce k

$$\text{Error Probability} = e(f_n, \phi_n) = P\left(\phi_n(f_n(x^n)) \neq x^n\right)$$

$$\text{Number of Codewards} = \exp(k_n) = \exp(nR)$$

ε-Achievable Code Rate - R

Definition - A code rate $R > 0$ is called ε-achievable if $\exists$ a sequence of codes $\{(f_n, \phi_n)\}_{n=1}^{\infty}$ with $\lim_n \frac{k_n}{n} = R$ and $e(f_n, \phi_n) \leq \varepsilon$ $\forall n$ large enough.

R_ε^*

$$R_\varepsilon^* = \inf\{R \mid e \leq \varepsilon, \lim k_n/n = R\} = \text{Infimum of all } \varepsilon\text{-achievable code rates } R.$$

When room for error (i.e. ε) decreases, we should be forced to use more $k(n, \varepsilon)$ to denote the source. $\therefore$ R should increase.

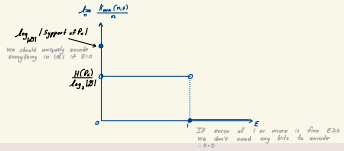
$$\therefore \quad \varepsilon \geq \varepsilon' \implies R_\varepsilon^* \leq R_{\varepsilon'}^*$$

R^*

Suprimum is achieved when ε reduces.

$$R^* \triangleq \sup_{\varepsilon \in (0,1)} \{R_\varepsilon^*\} = \text{rate of an almost noiseless source code}$$

Due to definition of suprimum $R^* \geq R_\varepsilon^*$



Theorem - Coding Theorem (Terminologically)

For a Discrete Memoryless System (DMS) $\{X_t\}_{t=1}^{\infty}$ with pmf P_X and error ε, $(0 < \varepsilon < 1)$

$$R^* = R_\varepsilon^* = H(P_X) \qquad \forall\, 0 < \varepsilon < 1$$

Proof: Like FFLSC Theorem, the proof is of two parts

$\longrightarrow$ $R^* \lesssim H(P_X) + (\cdot)$ .... achievability

$\longrightarrow$ $R^* \gtrsim H(P_X) - (\cdot)$ .... converse / optimality

We will build up to this proof by disscussing about Asymptotic Equipartition Property (AEP) which characterize ε-Typical sets.

Converse - of Coding Theorem

Where did the Infimum go?

If (f_n, ϕ_n) is a sequence of codes with $\lim_n \frac{k_n}{n} = R < H(P_X)$, then

$\lim_n e(f_n, \phi_n) \longrightarrow 1 \iff$ "Strong Converse"

$\lim_n e(f_n, \phi_n) \neq 0 \iff$ "Weak Converse"

and so > 0

Note:

- Minimum rate of fixed-to-fixed length source code for DMS $\{X_t\}_{t=1}^{\infty} \sim P_X$, for every fixed error probability $0 < \varepsilon < 1$ is $H(P_X)$.
- $X^n = (X_1, \ldots, X_n)$ can be compressed by such a code to $\cong n \cdot H(P_X)$ bits $\forall n$ sufficiently large. $R = k/n \cong H(P_X) \Rightarrow k \simeq nH(P_X)$

5) Asymptotic Equipartition Property (AEP)

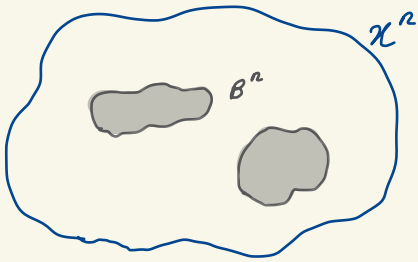
- AEP is the Information Theoretic version of Weak Law of Large Numbers. (WLLN).

Recall

For $n=1$

$P(B) \geq 1-\varepsilon$

For General n



$P(B^n) \geq 1-\varepsilon$

Theorem:- Asymptotic Equipartition Property (AEP) / Shannon - McMillan Brieman Theorem

For a Discrete Memoryless System (DMS) with $\{X_t\}_{t=1}^{\infty}$ $\mathcal{X}$-valued i.i.d process with pmf P_X

$$-\frac{1}{n} \log P_X(X_1, \ldots, X_n) \xrightarrow[n \to \infty]{\text{in Probability}} H(P_X)$$

Proof: Since X_i is an i.i.d (DMS) process

$\Rightarrow$ $X_1, \ldots, X_n$ are i.i.d

$\Rightarrow$ $\log P(X_1), \ldots, \log P(X_n)$ are i.i.d ..... (functions of i.i.d processes are i.i.d)

$\Rightarrow$ $\log P(X_1, \ldots, X_n) = \log \left(\prod_i P(X_i)\right)$ ..... (joint pmf is product of marginals)

$\Rightarrow$ $\log P_X(x^n) = \sum_i \log P(X_i)$ ... (1)

Apply WLLN to $-\log P(X_1)$

$$\Rightarrow \frac{1}{n} \sum_i -\log P(X_i) \xrightarrow[n \to \infty]{\text{in } P} \mathbb{E}[-\log P(X)] = H(P_X)$$

$$\Rightarrow -\frac{1}{n} \log P_X(x^n) \xrightarrow[n \to \infty]{\text{in } P} H(P_X) \quad \text{.... from (1)}$$

■

- This can be stated in another form as,

$$\lim_{n \to \infty} P\left(\left|-\frac{1}{n} \log P_X(X_t) - H(P_X)\right| \leq \varepsilon\right) = 1$$

$A_\varepsilon^{(n)}$ $(= B_L \cup B_U)$

6) Typical or ε-typical Set

Weakly Typical Set / Entropy Typical Set

AEP $\xrightarrow{\text{to}}$ Typical Sets

- From AEP, for every fixed $\varepsilon > 0$,

$$\Rightarrow \lim_n P_X\left(\left\{x^n \in \mathcal{X}^n \;\middle|\; \left|-\frac{1}{n}\log P_X(x_t) - H(P_X)\right| \leqslant \varepsilon\right\}\right) = 1$$

expand the modulus

$$\Rightarrow \lim_n P_X\left(\left\{x^n \in \mathcal{X}^n \;\middle|\; \exp\left[-n(H(P_X)+\varepsilon)\right] \leqslant P_X(x^n) \leqslant \exp\left[-n(H(P_X)-\varepsilon)\right]\right\}\right) = 1 \quad (*)$$

upon ignoring small ε's

$$\Rightarrow \lim_n P_X\left(\left\{x^n \in \mathcal{X}^n \;\middle|\; P_X(x^n) \cong \exp\left(-n \cdot H(P_X)\right)\right\}\right) = 1$$

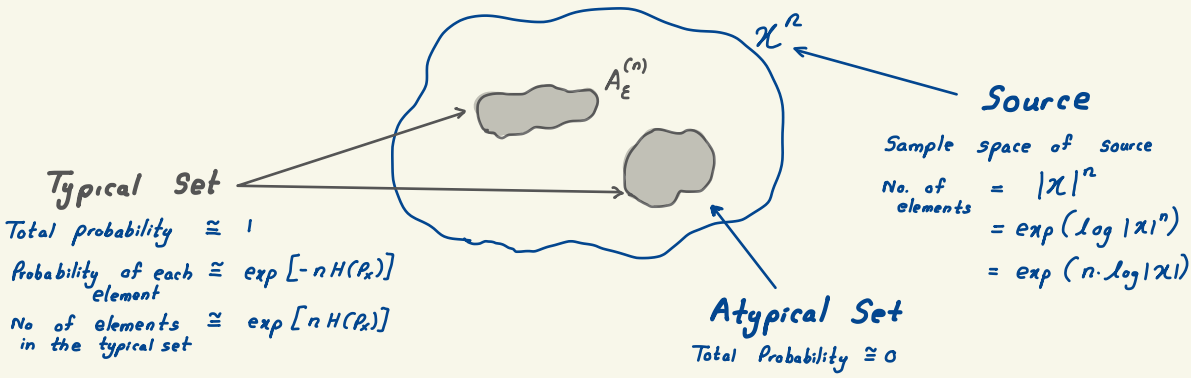
$\Rightarrow$ For large enough n,

$$P_X\left(\left\{x^n \in \mathcal{X}^n \;\middle|\; P_X(x^n) \cong \exp\left(-n \cdot H(P_X)\right)\right\}\right) = 1$$

Typical Set

$\Rightarrow$ For sufficiently large n, all the likely or 'typical' observations or realizations $x^n \in \mathcal{X}^n$ of X^n cumulatively carry all the probabilities and are roughly equiprobable to $\simeq \exp[-nH(P_X)]$

- For large enough n,



Definition:- ε- Typical Set $(A_\varepsilon^{(n)})$

For a Discrete Memoryless System (DMS) with $\{X_t\}_{t=1}^{\infty}$ $\mathcal{X}$-valued i.i.d process with pmf P_X. For every $\varepsilon > 0$ ε-typical set $\mathcal{A}_\varepsilon^{(n)} \subseteq \mathcal{X}^{(n)}$ is defined as,

$$\mathcal{A}_\varepsilon^{(n)} = \left\{x^n \in \mathcal{X}^n \;\middle|\; \exp\left[-n\cdot(H(P_X)+\varepsilon)\right] \leqslant P_X(x^n) \leqslant \exp\left[-n(H(P_X)-\varepsilon)\right]\right\}$$

Properties of Typical Sets

Theorem: 1) For $x^n \in A_\varepsilon^{(n)}$

$$H(P_x) - \varepsilon \leq -\tfrac{1}{n} \cdot \log P_x(x^n) \leq H(P_x) + \varepsilon \qquad \forall n$$

2) $P_x(A_\varepsilon^{(n)}) \geq 1-\varepsilon \qquad \forall n$ sufficiently large.

3) $|A_\varepsilon^{(n)}| \leq \exp[\, n(H(P_x)+\varepsilon)\,] \qquad \forall n$

4) $|A_\varepsilon^{(n)}| \geq (1-\varepsilon)\cdot \exp[\, n(H(P_x)-\varepsilon)\,] \qquad \forall n$ sufficiently large.

Proof:- 1) Follows from the definition of $A_\varepsilon^{(n)}$

2) From AEP we have,

$\Rightarrow -\tfrac{1}{n} \log P_x(X_1, \ldots, X_n) \xrightarrow[n \to \infty]{\text{in Probability}} H(P_x)$

$\Rightarrow \lim_n P_x\left(\left\{x^n \in \mathcal{X}^n \;\middle|\; \left|-\tfrac{1}{n}\log P_x(X_i) - H(P_x)\right| \leq \varepsilon\right\}\right) = 1$

$\Rightarrow \lim_n P_x\left(\left\{x^n \in \mathcal{X}^n \;\middle|\; \exp[-n(H(P_x)+\varepsilon)] \leq P_x(x^n) \leq \exp[-n(H(P_x)-\varepsilon)]\right\}\right) = 1$

$\Rightarrow \lim_n P_x(A_\varepsilon^{(n)}) = 1$ ... from definition of Typical set

$\Rightarrow \forall \delta > 0 \quad \exists N = N(\delta)$ such that $\forall n > N \quad P_x(A_\varepsilon^{(n)}) \geq 1-\delta$

$\Rightarrow$ Setting $\delta = \varepsilon \quad \forall \delta \; \exists N \; \forall n > N \quad P_x(A_\varepsilon^{(n)}) \geq 1-\varepsilon$

$\Rightarrow P_x(A_\varepsilon^{(n)}) \geq 1-\varepsilon$ for sufficiently large n.

∎

3) $$\begin{aligned} 1 &= P(x^n \in \mathcal{X}^n) \\ &\geq P(x^n \in A_\varepsilon^{(n)}) \\ &= \sum_{x^n \in A_\varepsilon^{(n)}} P_X(x^n) \\ &\geq \sum_{x^n \in A_\varepsilon^{(n)}} \exp\left[-n\cdot(H(P_X)+\varepsilon)\right] \quad \text{---- from definition of Typical set} \\ &= |A_\varepsilon^{(n)}| \cdot \exp\left[-n\cdot(H(P_X)+\varepsilon)\right] \end{aligned}$$

$$\Rightarrow |A_\varepsilon^{(n)}| \leq \exp\left[n\cdot(H(P_X)+\varepsilon)\right]$$ ∎

4) From 2) $\forall\, n$ sufficiently large

$$\begin{aligned} \Rightarrow 1-\varepsilon &\leq P(A_\varepsilon^{(n)}) \\ &\leq \sum_{x^n \in A_\varepsilon^{(n)}} \exp\left[-n(H(P_X)-\varepsilon)\right] \quad \text{---- from definition of Typical set} \\ &= |A_\varepsilon^{(n)}| \cdot \exp\left[-n(H(P_X)-\varepsilon)\right] \end{aligned}$$

$$\Rightarrow |A_\varepsilon^{(n)}| \geq (1-\varepsilon)\cdot \exp\left[n(H(P_X)-\varepsilon)\right]$$ ∎

Using AEP and Typical Sets to Prove Coding Theorem

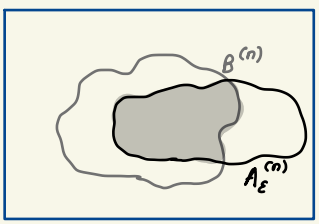
1) Achievability $\quad R^* \lesssim H(P_X) + C.)$

$$|A_\varepsilon^{(n)}| \cong \exp\left[n\cdot H(P_X)\right]$$
$$|\mathcal{X}|^n \cong \exp\left[n\cdot \log|\mathcal{X}|\right]$$

$H(P_X) < \log|\mathcal{X}|$. Realistically $H(P_X)$ can be much lower in cardinality. Hence we can get away with only compressing small set $A_\varepsilon^{(n)}$, which can be proved that to result in desired error probability.

2) Convers / Optimality $\quad R^* \gtrsim H(P_X) - C.)$

Converse is to show that nothing for smaller that $A_\varepsilon^{(n)}$ will exist with similar code rate. For that we take any set $B^{(n)}$ with ε error. Then show a significant potion of $B^{(n)}$ intersects with $A_\varepsilon^{(n)}$



For all sets
$P(B^{(n)}) > \delta > 0$
$\forall\, n$

Data Compression

or

"Noiseless" / "Lossles" Source Coding

by

Fixed - to - Variable Length Codes

1) Introduction

- Pros of having a variable length encoding.
  - Large probability codewords can be assigned shorter codes, resulting in higher rate. Source coding theorem avoided this by sending $n \to \infty$ and making all the probabilities in a set $A_\epsilon^{(n)}$ equal. (i.e making all have high prob. at once).
  - If n is not fixed we can do processing on the fly.

- A con of variable length is we should be able to detect the start and the end of a code. This gives rise to the notion of "prefix freeness." ∴ Cons are
  - Delays
  - Complexity in algorithms.

- Start with a single rv $\mathcal{X}$, with a D-ary code alphabet. $\mathcal{D} = \{0, \ldots, D-1\}$

Definition of Fixed to Variable Length Code

- A source code $\mathcal{C}$ for the source rv X with values in finite set $\mathcal{X}$ (source alphabet) is a pair of mappings (f, ϕ) where,

$$\text{Encoder} - \quad f : \mathcal{X} \longrightarrow \{0, 1, \ldots, D-1\}^* = \mathcal{D}^*$$

Here $\mathcal{D}^*$ is the set of all finite sequences formed from the elements of $\{0, 1, \ldots, D-1\}$. $\mathcal{D}^*$ does not have fixed length elements.

$$\text{Decoder} - \quad \phi : \mathcal{D}^* = \{0, 1, \ldots, D-1\}^* \longrightarrow \mathcal{X}$$

These (f, ϕ) are called fixed to variable length code.

- $\{ f(x) \mid x \in \mathcal{X} \}$ are called codewords.
- $l(f(x))$ = length in D-ary symbols of the codeword $f(x)$, $x \in \mathcal{X}$
- We want $P(\phi(f(x)) = x) \cong 1$
  $\mathbb{E}_P[l(f(x))]$ to be small.

Definition: Expected length - $L(f)$

Let P_x be the pmf of the source rv X. Then the expected length of the source code (f, ϕ) for the rv X is,

$$\bar{l}(f) = \mathbb{E}_P[l(f(x))] = \sum_{x \in \mathcal{X}} p(x) \cdot l(f(x))$$

in D-ary digits per source symbol.

Example 1: Source alphabet $\mathcal{X} = \{a, b, c, d\}$, codebook $\mathcal{D} = \{0, 1\}$. Assume

x	a	b	c	d
$f(x)$	0	10	110	111
$p(x)$	$1/2$	$1/4$	$1/8$	$1/8$

- variable length

- $\bar{l}(f) = \sum p(x) \cdot l(f(x)) = 1/2 \cdot 1 + 1/4 \cdot 2 + 1/8 \cdot 3 + 1/8 \cdot 3 = 1.75$ bits per source symbol
- $H(x) = \sum p(x) \cdot \log \frac{1}{p(x)} = 1/2 \cdot 1 + 1/4 \cdot 2 + 1/8 \cdot 3 + 1/8 \cdot 3 = 1.75$
- In this example $\bar{l}(f) = H(x)$. We call these sequence "uniquely decodable". Observe any sequence of bits can be uniquely decoded into a sequence of source symbols from $\mathcal{X}$.

Ex: 0 0 | 11 | 0 | 0 0 | 10 | 11 | 0 | 01 | 0 | 10 | 0 | 10
a a | c | a | a a | b | c | a | b | b | a | b

Example 2:- Source alphabet $\mathcal{X} = \{a, b, c, d\}$, code book $\mathcal{D} = \{0, 1\}$. Assume

x	a	b	c	d
$f(x)$	00	01	10	11
$p(x)$	$1/2$	$1/4$	$1/8$	$1/8$

- fixed length

- $\bar{l}(f) = \sum p(x) \cdot l(f(x)) = 1/2 \cdot 2 + 1/4 \cdot 2 + 1/8 \cdot 2 + 1/8 \cdot 2 = 2$ bits per source symbol

- $\log |\mathcal{X}| = \log |4| = 2$

- From previous section of course we know $H(X) \leq \log |\mathcal{X}|$. So in this instance $\bar{l}(f)$ has become the highest possible value for entropy. (Here $H(X) = 1.75 < \log |\mathcal{X}| = 2$)

2) Non - Singular Codes

Definition - A code (f, ϕ) is non-singular if every element of $\mathcal{X}$ is mapped into a distinct codeword. i.e.

$$x \neq x' \implies f(x) \neq f(x') \qquad \forall\, x, x' \in \mathcal{X}$$

Then decoder $\phi = f^{-1}$. Else those codes are called singular.

- Non sigularity suffices for an unambiguos description of a single value X, but not a sequence of values $X_1, X_2, X_3, \ldots$ all of which are $\mathcal{X}$-valued.

$$f_2(x_1, x_2, x_3) \triangleq f_2(x_1) \cdot f_2(x_2) \cdot f_2(x_3)$$

- Sometimes, the singularity is deliberately used, as seen previously in Atypical sets. All low probable source symbols are bunched together and enroded to a super symbol, only sacrificing some error

Example 3 -

$\mathcal{X}$	a	b	c	d	
f_1	0	0	0	1	→ Singular
f_2	0	010	01	10	→ non-singular

- The decoders of $f_1^{-1} = \phi_1$ and $f_2^{-1} = \phi_2$ be

$\mathcal{B}^*$	ϕ_1
0	a
1	d

$$P(\phi_1(f_1(x)) = x) = P(a) + P(d)$$ and is small

$\mathcal{B}^*$	ϕ_2
0	a
010	b
01	c
10	d

$$P(\phi_2(f_2(x)) = x) = 1$$

- f_2 is non singular, but we might recieve a concatanated sequence which will cause ambiguity. Consider the sequence 00101 0, this can be,

00 1 0 10 → a a d d
00 1 0 10 → a c c a
00 1 0 10 → a b d

- This shows the need of 'uniquely decodable' codes.

3) Uniquely Decodable Codes

Seperable Codes

- **Notation :** Let $x^n \triangleq (x_1, x_2, \ldots, x_n)$ be a sequence of n symbols from $\mathcal{X}$.

Definition - Extension

An extension $(f, \phi)^*$ of a code (f, ϕ) is a mapping from all finite length concatanated sequences of $\mathcal{X}$ to finite length sequences from $\mathcal{D}$.

$$f(x^n) \triangleq f(x_1) f(x_2) \ldots f(x_n) \qquad x_i \in \mathcal{X}$$

Definition - Uniquely Decodable Code

A code is uniquely decodable (u.d) if its extension is non-singular.

- A uniquely decodable code has the property that any encoded source sequence can be decoded uniquely to get the original source signal.
- But decoder may have to wait long to clarify the ambiguity.

Example 4 :-

x	a	b	c	d
$f(x)$	10	00	11	110

- If the encoded sequence begins with,
  - 10 ⟶ decode to a
  - 00 ⟶ decode to b
- If the encoded sequence begins with 11, it is ambigus to say c or d. In that case, the decoder waits and count the number of trailing zeros.
  - d: 11 0 0 ⋯ 0 (odd) ⟶ No. of trailing 0's are odd, decode to d
  - c: 11 0 ⋯ 0 (even) ⟶ No. of trailing 0's are even, decode to c
- This waiting is a waste. ∴ This shows the need of self punctuation.

Self Punctuating Codes

4) Prefix / Instantaneous Codes

- To avoid ambiguity in decoding a sequence of concatanated codewards, we need a system.
- Need to be able to identify the start and the end of a codeward. ⟶ i.e. self punctuating / instantaneous
- Inserting a seperate symbol to mark start of a codeward is too expensive.

Definition:- **Prefix Code**
A code is a prefix (free) code if no codeward is a prefix of any other codeward.

- A prefix code ⟹ Uniquely decodable code
- The end of a prefix codeword is instantly recognizable
- By example 4, prefix property is sufficient, but not necessary for unique decodability.
- <u>Prefix free code always exists</u>

Proof:- Consider the trivial fixed to fixed length code where $|\mathcal{X}|$ source symbols are encoded to $\lceil \log_D |\mathcal{X}| \rceil$. Since when all symbols are equal in length. Nothing can be a prefix of other.

0	000
1	001
2	010
3	011
⋮	⋮
7	111

$x \in \mathcal{X} \xrightarrow{f}$ D-ary codeward of length $\lceil \log_D |\mathcal{X}| \rceil$

$\bar{l}(f) = \sum_{x \in \mathcal{X}} p(x) \lceil \log_D |\mathcal{X}| \rceil = \lceil \log_D |\mathcal{X}| \rceil$ D-ary symbols.

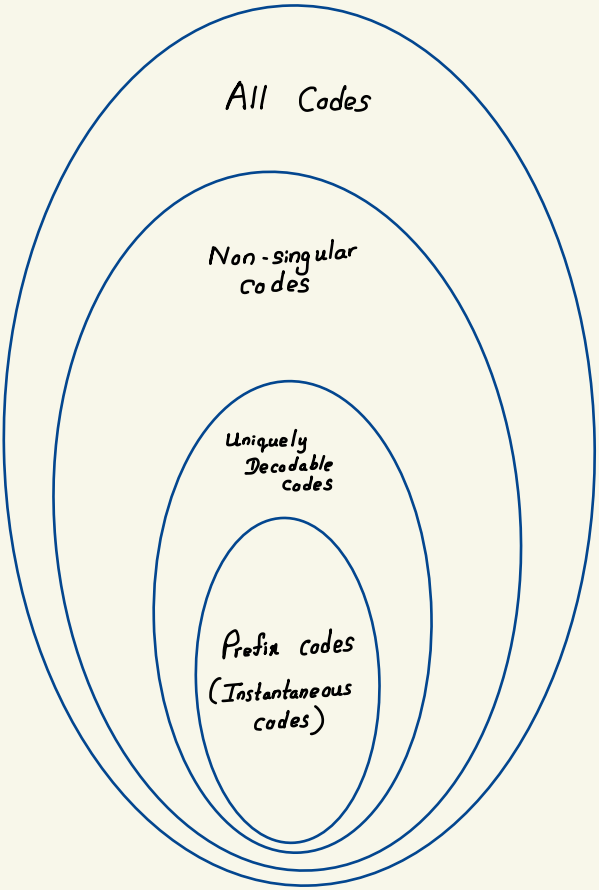
Not efficient.

- Shall see that expected length of a prefix code $\bar{l} \geq H(x)$.

Example 5 -

x	a	b	c	d
$f(x)$	0	10	110	111

$f(x^n) = \underbrace{0}_{a}\underbrace{0}_{a}\underbrace{1}_{b}\underbrace{0}_{b}\underbrace{1}_{b}\underbrace{0}_{a}\underbrace{111}_{d}\underbrace{0}_{a}\underbrace{0}_{a}$ ⟹ self punctuating.


All Codes
Non-singular codes
Uniquely Decodable codes
Prefix codes
(Instantaneous codes)

5) Kraft Inequality

Kraft Inequality places a restriction on the codeward lengths of a prefix code.

Theorem 1 - Kraft Inequality

$\Rightarrow$ Consider a prefix code (f, ϕ) with code alphabet of size D, and codeward lengths $\{l(f(x)), x \in \mathcal{X}\}$. Then the codeward lengths must satisfy,

$$\sum_{x \in \mathcal{X}} D^{-l(f(x))} \leq 1$$

$\Leftarrow$ Conversely, if there is a given set of codeward lengths which satisfy the Kraft Inequality, $\exists$ a prefix code with these codeward lengths.

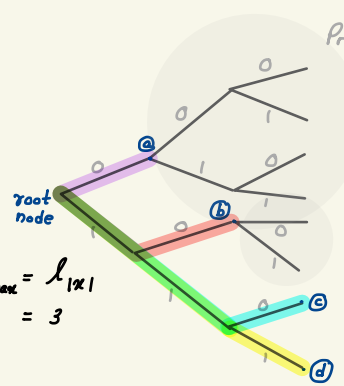
Note:

- The converse part of the theorem does not assest that any code with codeward lengths that satisfy KI constitutes a prefix code. Rather there will exist at least one prefix code with these exact lengths.
- KI is a necessary condition of a prefix code and sufficient condition for the existance of a prefix code with given length.

$D = 2$

x	$f(x)$	$l(f(x))$
a	0	1
b	10	2
c	110	3
d	111	3

$l_{max} = l_{|x|} = 3$



KI: $\sum_{i=1}^{4} D^{-l_i} = 1$

Proof :- Prefix Code $\Longrightarrow$ $\sum D^{-l(f(x))} \leq 1$

- Consider a D-ary tree in which each node has D branches eminating from it.

  Each branch $\longrightarrow$ corresponds to a symbol from the code alphabet $\mathcal{D} = \{0, 1, \ldots, D-1\}$.

  A node of the tree $\longrightarrow$ represents each codeward.

  Path from root to node $\longrightarrow$ traces out a codeward.

- In this setting,

  Prefix Code $\Longleftrightarrow$ No codeward can be the ancestor of any other codeward

  $\Longleftrightarrow$ Decendants of any codeward cannot be codewards themselves. i.e can be pruned.

- Let l_{max} = length of the longest codeward. Then all the given prefix codewards can be embeded in a D-ary tree of level l_{max}.

- A codeward at level l_i, $1 \leq l_i \leq l_{max}$ will have $D^{l_{max} - l_i}$ desendants. By the above prefix code requirement, each desendent set of nodes at level l_{max} must be disjoint from all the other.

$$\Rightarrow \sum_i D^{l_{max} - l_i} \leq D^{l_{max}}$$

$$\Rightarrow D^{l_{max}} \cdot \sum_i D^{-l_i} \leq D^{l_{max}}$$

$$\Rightarrow \sum_i D^{-l_i} \leq 1$$

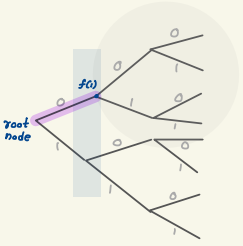
■

$\sum D^{-l(f(x))} \leq 1 \implies \exists$ of a Prefix Code

- Arrange the set of codeword lengths satisfying KI in lexicographical order. (i.e. nondecreasing).

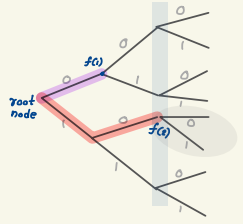
$$l_1 \leq l_2 \leq \dots \leq l_{|\mathcal{X}|} \ (= l_{max})$$

- Consider a D-ary of level (deapth) $= l_{|\mathcal{X}|}$



Label the uppermost node at level l_1 as the codeword $f(1)$. Prune it's descendants from the tree.

$\therefore$ No of terminal nodes removed $= D^{l_{|\mathcal{X}|} - l_1}$



Next, name the uppermost remaining node at level l_2 to be codeword $f(2)$, and prune its descendants.

$\therefore$ No of terminal nodes removed $= D^{l_{|\mathcal{X}|} - l_2}$

This is doable due to KI For an example we won't be forced to choose 5 codewords $f(1), f(2), f(3), f(4), f(5)$ at level 2 in this specific case. If so,
$2^{-1} + 2^{-1} + 2^{-2} + 2^{-2} + 2^{-2} > 1$
$\therefore$ We can always find a remaining node at level l_i.

- Continue this process with codeword lengths $l_3, \dots, l_{|\mathcal{X}|}$

Total no. of terminal nodes removed $= \sum_{i=1}^{|\mathcal{X}|} D^{l_{|\mathcal{X}|} - l_i}$ .... pruned sets are disjoint

$$= D^{l_{|\mathcal{X}|}} \underbrace{\sum_{i=1}^{|\mathcal{X}|} D^{-l_i}}$$

$$\leq D^{l_{|\mathcal{X}|}} \cdot 1 \quad \dots\dots \ l_i\text{'s are assumed to satisfy KI}$$

$$= \text{Total no. of terminal nodes available.}$$

$\implies$ This process makes sure we are prefix free and also will not run out of nodes to remove. ∎

Remarks

1) Shall see that Uniquely Decodable codes must satify Kraft's Inequality. i.e. we are not loosing much by going from uniquely decodable codes $\longrightarrow$ prefix codes

2) The codeword lengths of a countably ∞ prefix code must also satisfy the KI.

6) Prefix Codes with Minimum Expected Codeword Length

- **Problem :-** Let $\mathcal{X} = \{1, \dots, |\mathcal{X}|\}$. Find a prefix code with minimum expected codeword length. ($\min \mathbb{E}[\ell(f(x))]$)
- Powered by Kraft's Inequality, above statement turns into,
- **New Problem :-** Find positive integers $\{l_1, \dots, l_{|\mathcal{X}|}\}$ that minimize

$$\bar{l} = \mathbb{E}[\ell(f(x))] = \sum_{i=1}^{|\mathcal{X}|} p_i \cdot l_i = \sum_{i=1}^{|\mathcal{X}|} P(X=i) \cdot \ell(f(i))$$

subject to the constraints of KI

$$\sum_{i=1}^{|\mathcal{X}|} D^{-l_i} \leq 1$$

- No general formula in terms of $\{p_i, i = 1, \dots, |\mathcal{X}|\}$ is known for the minimum value of $\bar{l}$.
- These integer programming problems are notoriously hard to optimize. One way is to relax the integer requirement and round them at the end.

Theorem 2: Consider a prefix code (any code that satisfy the KI) with a D-ary code alphabet, and codeword lengths $l_1, \dots, l_{|\mathcal{X}|}$. Let

$$\bar{l} = \text{average codeword length} = \sum_{i=1}^{|\mathcal{X}|} p_i l_i \qquad i = (1, \dots, |\mathcal{X}|)$$

Then,

$$\boxed{\bar{l} \geq H_D(X) = H_D(P_X)}$$

with equality iff $D^{-l_i} = p_i$ $\left(\iff l_i = \log_D \frac{1}{p_i} = -\log_D p_i\right)$

Can we attain this minimum expected length ⟶ Yes the answer is Huffman codes.

Proof:

$$\Rightarrow \bar{l} - H_D(X) = \sum_i p_i l_i - \sum_i p_i \log \frac{1}{p_i}$$

$$= \sum_i p_i \log_D D^{l_i} - \sum_i p_i \log \frac{1}{p_i}$$

$$= \sum_i p_i \log_D \left(\frac{p_i}{D^{-l_i}} \right)$$

By KI $\sum_{i=1}^{|\mathcal{X}|} D^{-l_i} \leq 1$, so that $(D^{-l_1}, \ldots, D^{-l_{|\mathcal{X}|}})$ need not be a pmf on $\mathcal{X}$. Define Q (pmf on $\mathcal{X}$) by normalizing D^{-l_i} to make a pmf.

$$Q = \left(\frac{D^{-l_1}}{\sum_{i=1}^{|\mathcal{X}|} D^{-l_i}}, \ldots, \frac{D^{-l_{|\mathcal{X}|}}}{\sum_{i=1}^{|\mathcal{X}|} D^{-l_i}} \right)$$

$$\Rightarrow \bar{l} - H_D(X) = \sum_{i=1}^{|\mathcal{X}|} p_i \log_D \left(\frac{p_i}{D^{-l_i} / \sum_j D^{-l_j}} \cdot \frac{1}{\sum_j D^{-l_j}} \right)$$

$$= \sum_{i=1}^{|\mathcal{X}|} p_i \log_D \left(\frac{p_i}{D^{-l_i} / \sum_j D^{-l_j}} \right) + \sum_{i=1}^{|\mathcal{X}|} p_i \log_D \left(\frac{1}{\sum_j D^{-l_j}} \right)$$

$$= D(P \| Q) + \sum_i p_i \left(\log \frac{1}{\sum D^{-l_i}} \right) \quad \cdots (*)$$

From KI we have $\sum D^{-l_i} \leq 1 \Rightarrow \log 1/\sum D^{-l_i} \geq 0$

$$\Rightarrow \sum_i p_i \log 1/\sum D^{-l_i} \geq 0 \quad \cdots (1)$$

$(*), (1) \Rightarrow \bar{l} - H_D(X) \geq D(P\|Q)$

$\geq 0 \quad \cdots (2) \quad \cdots$ KL Divergence is positive.

Equality holds iff the two non-negative terms in $(*)$ equal 0. From (1) and (2), for equality,

$$\left.\begin{array}{l} (1) \cdots \sum D^{-l_i} = 1 \\ (2) \cdots P = Q \Leftrightarrow p_i = \dfrac{D^{-l_i}}{\sum D^{-l_i}} \end{array}\right\} \Rightarrow p_i = D^{-l_i} \Leftrightarrow l_i = -\log_D p_i$$

■

Definition:- D-adic Distributions

A pmf on $\mathcal{X}$ is D-adic if each of the probability values is of form D^{-n} for some positive integers $n = 1, 2, \ldots$

- By the Theorem 2, $\bar{l}_{min} = H_D(X) \iff$ pmf of X is D-adic
- $\mathcal{X} = \{a, b, c, d\}$, $D = (2^{-1}, 2^{-2}, 2^{-3}, 2^{-3})$ is 2-adic.
- It is not straight forward to model any given distibation using a D-adic function. So the other alternative is to find a code with expected length closer to $H(X)$.
- Two fixed to variable length coding procedures we will discuss are,
  1) Shanon - Fano Code (Suboptimal)
  2) Huffman code (Optimal)

7) Shanon Fano Code

Shanon Fano Code : pick $l_i = \lceil \log_D \frac{1}{p_i} \rceil$

- Shanon Fano Code satisfies the Krafft's Inequality.

Proof :- Let $l_i = \lceil \log_D 1/p_i \rceil$ be codeward lengths of SF code

$$\Rightarrow \sum_{i=1}^{|X|} D^{-l_i} = \sum_{i=1}^{|X|} D^{-\lceil \log_D 1/p_i \rceil}$$

$$\leq \sum_{i=1}^{|X|} D^{-\log_D 1/p_i} \quad \ldots \quad a \leq \lceil a \rceil$$

$$= \sum_{i=1}^{|X|} p_i$$

$$= 1$$

■

- A prefix code for Shanon Fano code can be created following the tree prooning method in inverse proof of Krafft's Inequality.

- $H_D(x) \leq \bar{l}_{SF} < H_D(x) + 1$

Proof :- Since Shanon Fano Code satisfies Krafft's Inequality it is a Prefix code and it should be bounded below by entropy from Theorem 2,

$$\therefore \bar{l}_{SF} \geq H_D(x)$$

$$\bar{l}_{SF} = \sum_{i=1}^{|X|} p_i l_i = \sum_{i=1}^{|X|} p_i \lceil \log_D 1/p_i \rceil < \sum_{i=1}^{|X|} p_i (\log_D 1/p_i + 1)$$

$$= \sum_{i=1}^{|X|} p_i \cdot \log_D \frac{1}{p_i} + \sum_{i=1}^{|X|} p_i$$

$$= H_D(x) + 1$$

Thus $H_D(x) \leq \bar{l}_{SF} < H_D(x) + 1$

■

Theorem 3: Given a pmf $P = (p_1, \ldots, p_{|\mathcal{X}|})$. Let $l_1^*, \ldots, l_{|\mathcal{X}|}^*$ be the length of an optimal prefix code from a D-ary alphabet. Let

$$\bar{l}^* = \sum_{i=1}^{|\mathcal{X}|} p_i l_i^*$$

be the corresponding (minimal) expected codeward length. Then,

$$H_D(X) \leq \bar{l}^* < H_D(X) + 1$$

Proof :- $H_D(X) \leq \bar{l}^*$ follows from Theorem 2 and
$\bar{l}^* \leq \bar{l}_{SF} < H_D(X) + 1$ follows from facts about SF codes

8) Encoding n-length Blocks of Source Symbols

- According to Theorem 3, since P_X may not be D-adic, an overhead of at most one D-ary digit can exist. This is very ineffecient when $|\mathcal{X}|$ is small.

- This overhead can be reduced by encoding n-length block of source symbols.

P_{X^n} = Source pmf on $\mathcal{X}^n$

$l(f(x^n))$ = length in D-ary digits of the (prefix) codeward for x^n, $(x^n \in \mathcal{X}^n)$

$\bar{l}_n \triangleq$ Expected codeward length per a souce symbol, when n source symbols are encoded at once. (i.e Expected or average rate of code.)

Theorem 4: Given p.m.f P_{X^n} on $\mathcal{X}^n$, Let $l_1^*, \dots, l_{|\mathcal{X}^n|}^*$ be the length of an optimal prefix code from a D-ary alphabet. Then for $n = 1, 2, \dots$ the minimum expected code length per source symbol satisfies

$$\frac{1}{n} H_D(X_1, \dots, X_n) \leq \bar{l}_n^* \leq \frac{1}{n} H_D(X_1, \dots, X_n) + \frac{1}{n}$$

Proof: $\bar{l}_n = \frac{1}{n} \mathbb{E}_{P_{X^n}}[l(f(x^n))] = \frac{1}{n} \sum_{x^n \in \mathcal{X}^n} P_{X^n}(x^n) \cdot l(f(x^n))$

From previous results of Theorem 2 and Shanon Fano Code, we can find a prefix code $(f, \phi) = (f^{(n)}, \phi^{(n)})$ with

$f: \mathcal{X}^n \longrightarrow \mathcal{D}^{(n)}$ s.t.

$$\Rightarrow H_D(X_1, \ldots, X_n) \leq E_{P_{X^n}}[l(f(X^n))] < H_D(X_1, \ldots, X_n) + 1$$

$$\Rightarrow \frac{1}{n} H_D(X_1, \ldots, X_n) \leq \frac{1}{n} E_{P_{X^n}}[l(f(X^n))] < \frac{1}{n} H_D(X_1, \ldots, X_n) + \frac{1}{n}$$

$$\Rightarrow \frac{1}{n} H_D(X_1, \ldots, X_n) \leq \bar{l}_n < \frac{1}{n} H_D(X_1, \ldots, X_n) + \frac{1}{n}$$

Since SF code is not better than optimal code, we have $\bar{l}_n^* \leq \bar{l}_n$. Hence

$$\Rightarrow \frac{1}{n} H_D(X_1, \ldots, X_n) \leq \bar{l}_n^* < \frac{1}{n} H_D(X_1, \ldots, X_n) + \frac{1}{n}$$

Remarks

1) In theorem 4, $X_1, \ldots, X_n$ need not be i.i.d.

2) If $X_1, \ldots, X_n$ are i.i.d, then theorem 4 becomes

$$H_D(X) \leq \bar{l}_n^* \leq H_D(X) + 1/n$$

hence for larger n, $\bar{l}_n^* \cong H_D(X)$

3) If $\{X_n\}_{n=1}^{\infty}$ is a stationary process, then shall see,

$$\underset{\text{(For large } n\text{)}}{\bar{l}_n^*} \cong \frac{1}{n} H_D(X_1, \ldots X_n) \xrightarrow{n \to \infty} \bar{H}_D(X)$$

where $\bar{H}_D(X)$ = entropy rate.

9) 'Mismatched' Lossless Variable Length Codes

P and Q are D-adic

- Assume that the true pmf of a D-addic distribution is $P = (p_1, \ldots, p_{|\mathcal{X}|})$
- Assume that the compressor is tuned to another D-adic distribution $Q = (q_1, \ldots, q_{|\mathcal{X}|})$

Redundancy in optimal coding resulting from imperfect knowledge Q of true pmf P is

$$= \mathbb{E}_P\left[\log_D \frac{1}{Q(X)}\right] - \mathbb{E}_P\left[\log_D \frac{1}{P(X)}\right]$$

$$= \mathbb{E}_P\left[\log_D \frac{P(X)}{Q(X)}\right]$$

$$= D(P\|Q)$$

$$\geq 0$$

= Increase in expected codeward length when a source $\sim P$ is encoded optimally $\sim Q$.

P and Q are not D-adict

Shanon-Fano Code

Theorem 5: The codeward lengths assigned with respect to $Q = (q_1, \ldots, q_{|\mathcal{X}|})$. according to length $l_i = \left\lceil \log_D \frac{1}{q_i} \right\rceil$ $i = 1, \ldots, |\mathcal{X}|$, the expected codeward length under $P = (p_1, \ldots, p_{|\mathcal{X}|})$ satisfies

$$H_D(P) + D(P\|Q) \leq \mathbb{E}_P[l(X)] < H_D(P) + D(P\|Q) + 1$$

Proof :- From properties of ceiling.

$$\Rightarrow \sum_i p_i \log \frac{1}{q_i} < \sum_{i \in \mathcal{X}} p_i \left\lceil \log \frac{1}{q_i} \right\rceil < \sum_i p_i \left(\log \frac{1}{q_i} + 1\right)$$

$$\Rightarrow \sum_i p_i \left(\log \frac{p_i}{q_i} + \log \frac{1}{p_i}\right) < \mathbb{E}_P[l(X)] < \sum_i p_i \left(\log \frac{p_i}{q_i} + \log \frac{1}{p_i}\right) + \sum_i p_i$$

$$\Rightarrow D(P\|Q) + H_D(P) < \mathbb{E}_P[l(X)] < D(P\|Q) + H_D(P) + 1$$

■

10) Uniquely Decodable Codes and Kraft's Inequality

- When searching for a optimal prefix code, we were limited by entropy (i.e. $H_D(P)$) and the KI. A natural question is can we do better by relaxing from prefix to larger class of uniquely decodable codes. ∴ Answer - No

Theorem 6: KI for UD Codes / McMillan

Let $(l_1, \ldots, l_{|\mathcal{X}|})$ represent the codeword length of a uniquely decodable code. Then, these lengths satisfy Krafts Inequality. i.e.

$$\sum_{i=1}^{|\mathcal{X}|} D^{-l_i} \leq 1$$

Conversely, given any set of codeword lengths that satisfy KI, ∃ a uniquely decodable code with these codeword length.

Proof: $\Longleftarrow$ Converse

Already done since ∀ lengths that satisfy KI, we can find a prefix code. All prefix codes are U.D codes.

$\Longrightarrow$ Forward - Cover Thomas - Due to J. Kaush 1961

Remark ∴ The codeword length of a countably ifinite uniquely decodable code also satisfy KI

Cover Thomas
Page 105

Theorem 5.5.1 (*McMillan*) *The codeword lengths of any uniquely decodable D-ary code must satisfy the Kraft inequality*

$$\sum D^{-l_i} \leq 1. \tag{5.48}$$

Conversely, given a set of codeword lengths that satisfy this inequality, it is possible to construct a uniquely decodable code with these codeword lengths.

Proof: Consider C^k, the kth extension of the code (i.e., the code formed by the concatenation of k repetitions of the given uniquely decodable code C). By the definition of unique decodability, the kth extension of the code is nonsingular. Since there are only D^n different D-ary strings of length n, unique decodability implies that the number of code sequences of length n in the kth extension of the code must be no greater than D^n. We now use this observation to prove the Kraft inequality.

Let the codeword lengths of the symbols $x \in \mathcal{X}$ be denoted by $l(x)$. For the extension code, the length of the code sequence is

$$l(x_1, x_2, \ldots, x_k) = \sum_{i=1}^{k} l(x_i). \tag{5.49}$$

The inequality that we wish to prove is

$$\sum_{x \in \mathcal{X}} D^{-l(x)} \leq 1. \tag{5.50}$$

The trick is to consider the kth power of this quantity. Thus,

$$\left(\sum_{x \in \mathcal{X}} D^{-l(x)} \right)^k = \sum_{x_1 \in \mathcal{X}} \sum_{x_2 \in \mathcal{X}} \cdots \sum_{x_k \in \mathcal{X}} D^{-l(x_1)} D^{-l(x_2)} \cdots D^{-l(x_k)} \tag{5.51}$$

$$= \sum_{x_1, x_2, \ldots, x_k \in \mathcal{X}^k} D^{-l(x_1)} D^{-l(x_2)} \cdots D^{-l(x_k)} \tag{5.52}$$

$$= \sum_{x^k \in \mathcal{X}^k} D^{-l(x^k)}, \tag{5.53}$$

by (5.49). We now gather the terms by word lengths to obtain

$$\sum_{x^k \in \mathcal{X}^k} D^{-l(x^k)} = \sum_{m=1}^{kl_{\max}} a(m) D^{-m}, \tag{5.54}$$

where $l_{\max}$ is the maximum codeword length and $a(m)$ is the number of source sequences x^k mapping into codewords of length m. But the code is uniquely decodable, so there is at most one sequence mapping into each code m-sequence and there are at most D^m code m-sequences. Thus, $a(m) \leq D^m$, and we have

$$\left(\sum_{x \in \mathcal{X}} D^{-l(x)} \right)^k = \sum_{m=1}^{kl_{\max}} a(m) D^{-m} \tag{5.55}$$

$$\leq \sum_{m=1}^{kl_{\max}} D^m D^{-m} \tag{5.56}$$

$$= kl_{\max} \tag{5.57}$$

and hence

$$\sum_j D^{-l_j} \leq (kl_{\max})^{1/k}. \tag{5.58}$$

Since this inequality is true for all k, it is true in the limit as $k \to \infty$. Since $(kl_{\max})^{1/k} \to 1$, we have

$$\sum_j D^{-l_j} \leq 1, \tag{5.59}$$

which is the Kraft inequality.

Conversely, given any set of $l_1, l_2, \ldots, l_m$ satisfying the Kraft inequality, we can construct an instantaneous code as proved in Section 5.2. Since every instantaneous code is uniquely decodable, we have also constructed a uniquely decodable code. □

11) Non-singular and Singular Codes and Their Versions of Kraft's Inequality and Expected Codeword Length

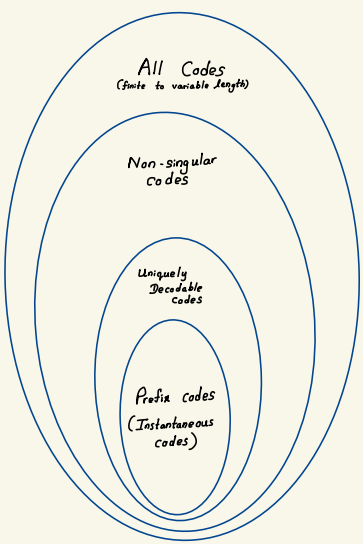
- Uniquely decodable codes are of benifit when compressing a source sequence symbol-by-symbol.

- When Coding just one rv symbol or a rv sequence considered as a large super symbol, assume we need no error.

  Probability of Error = 0

  Then if we are searching over the class of Prefix codes only, Theorem 4 showed that our expected codeward length is bounded by,

$$\frac{1}{n} H_D(X_1, \ldots, X_n) \leq \frac{1}{n} \mathbb{E}[\ell(f(x^n))] = \bar{\ell}_n^* \leq \frac{1}{n} H_D(X_1, \ldots, X_n) + \frac{1}{n}$$



A natural question to ask is,

- i.e. What is the lowest expected codeward length when we relax from prefix → Singular → non-singular large class.?

- Can we achieve a far better code rate if we go to a less ideal larger class.?

Example:- Let $X \sim \{a, b, c\}$. $X \sim$ uniform. $\mathcal{D} = \{0, 1\}$ Consider the non-singular, non UD (Uniquely decodable) code,

$a \leftrightarrow 0$

$b \leftrightarrow 1$

$c \leftrightarrow 01$

$P(a) = P(b) = P(c) = 1/3$

Entropy $= H(X) = \sum 1/3 \log_2 3 = 1.58$ bits //

Expected codeword length $= \mathbb{E}[l(f(x))] = 4/3 = 1.33$ bits //

$\mathbb{E}[l(f(x))] < H(x)$

$\therefore$ Intution suggest entropy is limiting even the larger class.

2024 PS2 - Problem 24

Lemma 1:- $\mathcal{X}$ - finite source alphabet. Let (f, ϕ) is a nonsingular code i.e $f: \mathcal{X} \longrightarrow \{0,1\}^*$ is one-to-one. Then, the codeword length satisfy.

$$\sum_{x \in \mathcal{X}} 2^{-l(f(x))} \leq \log |\mathcal{X}|$$

+1 is not necessary

Proof: Let $M = |\mathcal{X}|$, and codeword lengths of binary non-singular code f as

$$l_1 \leq l_2 \leq \dots \leq l_M$$ after-sorting

Since f is non-singular / one-to-one first i codewords must fit among all binary strings of length at most l_i

$\Rightarrow$ first i codewords (No. of) $\leq$ all binary strings of length at most l_i $\quad \forall i \in (1, \dots, M)$

$\Rightarrow \quad i \leq \sum_{k=1}^{l_i} 2^k$

$\Rightarrow \quad = 2^{l_i+1} - 2 < 2^{l_i+1}$

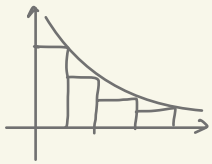
$\Rightarrow \quad \log_2 i < l_i + 1$

$\Rightarrow \quad \log_2 i \leq l_i$

$\Rightarrow \quad 1/i \geq 2^{-l_i}$

$\Rightarrow \quad \sum_{i=1}^{M} 1/i \geq \sum_{i=1}^{M} 2^{-l_i}$

$\Rightarrow \quad \sum_{i=1}^{M} 1/i \geq \sum_{x \in \mathcal{X}} 2^{-l(f(x))} \quad \cdots (1)$



from discrete sum vs integration $\Rightarrow \int_1^M 1/x \, dx \geq \sum_{i=1}^{M} 1/i$

$\Rightarrow \quad \ln M \geq \sum_{i=1}^{M} 1/i \quad \cdots (2)$

from (1), (2) $\Rightarrow \quad \ln M \geq \sum_{x} 2^{-l(f(x))}$

$\Rightarrow \quad \log_2 |\mathcal{X}| \geq \sum_{x} 2^{-l(f(x))}$

■

Lemma 2 - $\mathbb{E}_p[l(f(x))] \geq H(P) - \log \log |\mathcal{X}|$

Proof: Let $C = \sum 2^{-l(f(x))}$ and $Q(x) = 2^{-l(f(x))}/C$. Since KL Divergence is positive,

$\Rightarrow \quad D(P \| Q) \geq 0$

$\Rightarrow \quad \sum_{x} P(x) \cdot \log \frac{P(x)}{Q(x)} \geq 0$

$\Rightarrow \quad \sum_{x} P(x) \cdot \log \frac{P(x)}{2^{-l(f(x))}/C} \geq 0$

$\Rightarrow \quad \sum_{x} P(x) \cdot \log P(x) - \sum_{x} P(x) \cdot \log\left(\frac{2^{-l(f(x))}}{C}\right) \geq 0$

$\Rightarrow \quad \sum_{x} P(x) \cdot \log P(x) - \sum_{x} P(x) \cdot (-l(f(x)) - \log C) \geq 0$

$\Rightarrow \quad \sum_{x} P(x) \cdot \log P(x) + \sum_{x} P(x) \cdot l(f(x)) + \underbrace{\sum_{x} P(x)}_{1} \cdot \log C \geq 0$

$\Rightarrow \quad \sum_{x} P(x) \cdot l(f(x)) \geq - \sum_{x} P(x) \cdot \log P(x) - \log C$

$\Rightarrow \quad \mathbb{E}[l(f(x))] \geq H(P) - \log C$

From previous part $C = \sum_{x} 2^{-l(f(x))} < \log |\mathcal{X}| + 1$

Not sure about this and everything regarding $+1$ in upper bound

$\Rightarrow \quad \mathbb{E}[l(f(x))] \geq H(P) - \log \log |\mathcal{X}| - \underbrace{\log 1}_{0}$

■

Lemma 3- $\{X_t\}_{t=1}^{\infty}$ is an arbitary $\mathcal{X}$-valued source. Let (f_n, ℓ_n) is a non-singular code for $\mathcal{X}^n$ $n \geq 1$. Then,

$$\frac{1}{n} \mathbb{E}[\ell(f_n(X^n))] \geq \frac{1}{n} H(X_1, \ldots, X_n) - \frac{\log n}{n} - \frac{\log \log |\mathcal{X}|}{n}$$

Proof: $\log \log |\mathcal{X}^n| = \log (\log |\mathcal{X}|^n) = \log (n \cdot \log |\mathcal{X}|)$

$$= \log n + \log \log |\mathcal{X}| \quad \cdots (1)$$

Apply Lemma 2 to the super symbol $X^n = (X_1, \ldots, X_n)$ instead on X. Thus,

$$\Rightarrow \mathbb{E}[\ell_n(f(X^n))] \geq H(X^n) - \log \log |\mathcal{X}^n|$$

$$= H(X^n) - \log n - \log \log |\mathcal{X}| \quad \cdots (2)$$

$$\Rightarrow \frac{1}{n} \mathbb{E}[\ell_n(f(X^n))] \geq \frac{1}{n} H(X^n) - \frac{\log n}{n} - \frac{\log \log |\mathcal{X}|}{n}$$

Remarks

- $f(x^n)$, $x^n \in \mathcal{X}^n$ are distinct for fixed n.
- When $n \to \infty$, Lemma 3 is bounded below by entropy (loosly speaking).

$$\frac{1}{n} \mathbb{E}[\ell(f_n(X^n))] \geq \frac{1}{n} H(X_1, \ldots, X_n) - \frac{\log n}{n} - \frac{\log \log |\mathcal{X}|}{n}$$

For $\{X_t\}_{t=1}^{\infty}$ iid this term is $H(X)$

When $n \to \infty$ this term $\to 0$

- If $\{X_t\}_{t=1}^{\infty}$ is stationary and ergodic, above result turns into

(To have $\lim \frac{1}{n} H(X^n) = \bar{H}(X)$; actually not needed)

$$\liminf_n \frac{1}{n} \mathbb{E}[\ell(f(X^n))] \geq \bar{H}(X)$$

- Hence, entropy reigns supreme. :) . Even when we search over the larger class of non-singular codes.

- Finally even for a singular code with $\mathcal{D} = \{0,1\}$ with arbitarily small error probability ε,

$$\lim_n \inf \frac{1}{n} \mathbb{E}[\ell(f_n(X^n))] \geq H(P) - \underbrace{(\varepsilon \cdot \log(|\mathcal{X}|-1) + h(\varepsilon))}_{\underset{\text{Fano}}{\approx} H(X^n \mid \phi(f(X^n)))/n}$$

Proof: $\Rightarrow I(X^n, \phi_n(f_n(X^n))) \leq I(X^n, f_n(X^n))$ .... from Data Processing Inequality

$\Rightarrow I(X^n, \phi_n(f_n(X^n))) \leq H(f_n(X^n))$

$\Rightarrow H(X^n) - H(X^n \mid \phi_n(f_n(X^n))) \leq H(f_n(X^n))$ .... $I(X,Y) = H(X) - H(X|Y)$

$\Rightarrow H(X^n) \leq H(f_n(X^n)) + H(X^n \mid \phi_n(f_n(X^n)))$

If X_i is i.i.d and is P $\Rightarrow n \cdot H(P) \leq H(f_n(X^n)) + H(X^n \mid \phi_n(f_n(X^n)))$

$\Rightarrow n H(P) \leq H(f_n(X^n)) + \sum_{t=1}^{n} H(X_t \mid \phi_n(f_n(X^n)))$

$\Rightarrow n H(P) \leq H(f_n(X^n)) + \sum_{t=1}^{n} H(X_t \mid \phi(f(X_t)))$ Conditioning reduce entropy ①

From Fano's Inequality for state t, $Pr(X_t \neq \phi(f(X_t))) = \varepsilon_t$ $\quad \varepsilon_t \leq \varepsilon$

$\Rightarrow H(X_t \mid \phi(f(X_t))) \leq h(\varepsilon_t) + \varepsilon_t \log(|\mathcal{X}|-1)$

$\Rightarrow \sum_{t=1}^{n} H(X_t \mid \phi(f(X_t))) \leq \sum_{t=1}^{n} h(\varepsilon_t) + \sum_{t=1}^{n} \varepsilon_t \log(|\mathcal{X}|-1)$

$\Rightarrow \leq \underbrace{n \cdot h\left(\frac{1}{n} \cdot \sum_{t=1}^{n} \varepsilon_t\right)}_{\text{from concavity of } h} + n \cdot \log(|\mathcal{X}|-1) \underbrace{\left(\frac{1}{n} \sum_{t=1}^{n} \varepsilon_t\right)}_{\leq 1/n \cdot n \cdot \varepsilon = \varepsilon}$

$\Rightarrow \leq n \cdot h(\varepsilon) + n \cdot \log(|\mathcal{X}|-1) \cdot \varepsilon$ .. ②

From ① and ②

$\Rightarrow n H(P) \leq H(f_n(X^n)) + n \cdot h(\varepsilon) + n\varepsilon \log(|\mathcal{X}|-1)$

$\Rightarrow H(P) - h(\varepsilon) - \varepsilon \cdot \log(|\mathcal{X}|-1) \leq \frac{1}{n} H(f_n(X^n))$ ... ✱

From Lemma ③,

$\Rightarrow \frac{1}{n} \mathbb{E}[\ell(f_n(X^n))] \geq \frac{1}{n} H(f_n(X^n)) - \frac{\log n}{n} - \frac{\log\log|\mathcal{X}|}{n}$

$\Rightarrow \geq H(P) - h(\varepsilon) - \varepsilon \log(|\mathcal{X}|-1) - \underbrace{\frac{\log n}{n} \cdot \frac{\log\log|\mathcal{X}|}{n}}_{0}$

Taking lim inf

$\Rightarrow \lim_n \inf \mathbb{E}[\ell(f_n(X^n))] \geq H(P) - (\varepsilon \cdot \log(|\mathcal{X}|-1) + h(\varepsilon))$

■

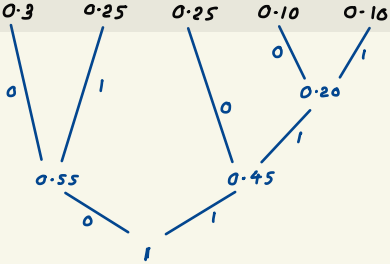
12) Huffman Code

- A simple algorithm for constructing a prefix code with the shortest expected code length.
- First arrange the symbols in $\mathcal{X}$ in non-decreasing order of probability.

Example 1 : $\mathcal{D} = \{0, 1\}$

$\mathcal{X}$	x_1	x_2	x_3	x_4	x_5
$p_X(x)$	0.3	0.25	0.25	0.10	0.16

First arrange the in ascending order.



Huffman code is not unique, which is evident from our freedom in combining equaly probable symbols.

$x_1 \leftrightarrow 00$

$x_2 \leftrightarrow 01$

$x_3 \leftrightarrow 10$

$x_4 \leftrightarrow 110$

$x_5 \leftrightarrow 111$

Expected codeword length $= \bar{l}$

$= 2(0.3 + 0.25 + 0.25) + 3(0.1 + 0.1)$

$= 2.2$ bits / source symbol

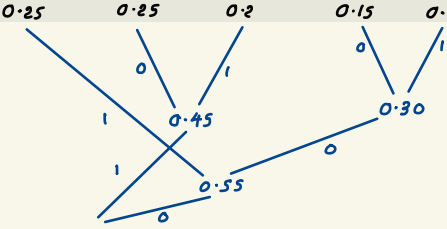
Entropy of source symbols $= H(x)$

$= \sum p_x \log(1/p_x) = 2.185$ bits

$\therefore$ We have $H(x) < \bar{l} < H(x) + 1$ in this example.

Example 2 : $\mathcal{D} = \{0, 1\}$

$\mathcal{X}$	x_1	x_2	x_3	x_4	x_5
$p_X(x)$	0.25	0.25	0.2	0.15	0.15



$x_1 \leftrightarrow 01$

$x_2 \leftrightarrow 10$

$x_3 \leftrightarrow 11$

$x_4 \leftrightarrow 000$

$x_5 \leftrightarrow 001$

Expected codeword length $= \bar{l} = 2.3$ bits / source symbol

Adding Dummy Symbols

In the example 3 below,

- Problem - Now we have only two branches from the root node
- Solution - Add dummy symbols (with zero probability) to $\mathcal{X}$. But how many dummy symbols should we add?
- Suppose that the procedure has k mergins. In each merging we get a new symbol and loose D previous symbols,

Net loss $D-1$ symbols

Total number of source symbols including the dummy symbols $= D + (k-1)(D-1) = \boxed{k(D-1)+1}$

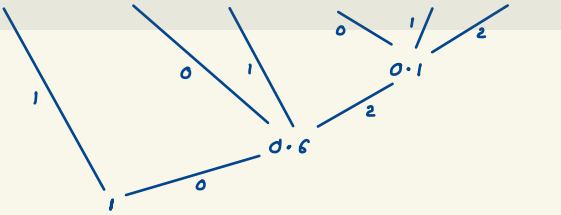
D → Should remain D symbols for final merge

$(k-1)(D-1)$ → loss in $k-1$ steps

- We should find the lowest k, such that $k(D-1)+1$ exceeds $|\mathcal{X}|$. Then we should add $k(D-1)+1 - |\mathcal{X}|$ symbols

Example 3 : $\mathcal{D} = \{0, 1, 2\}$

$\mathcal{X}$	x_1	x_2	x_3	x_4	x_5	x_6
$P_X(x)$	0.4	0.3	0.2	0.05	0.03	0.02



Expected codeword length $= \bar{L} = 1.7$ ternary digits per source symbol.

Adding Dummy Symbols

$|\mathcal{X}| = 6$

$k=1 \rightarrow k(D-1)+1 = 1(2)+1 = 3 < 6$

$k=2 \rightarrow k(D-1)+1 = 2(2)+1 = 5 < 6$

$k=3 \rightarrow k(D-1)+1 = 2(3)+1 = 7 > 6$ ✓

$\therefore$ Symbols to add $= 7-6 = 1$

$\mathcal{X}$	x_1	x_2	x_3	x_4	x_5	x_6	dummy
$P_X(x)$	0.4	0.3	0.2	0.05	0.03	0.02	0

$x_1 \leftrightarrow 0$

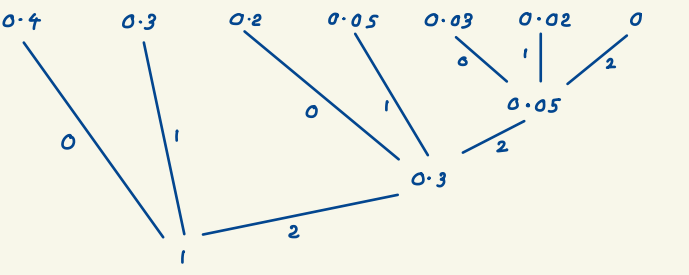
$x_2 \leftrightarrow 1$

$x_3 \leftrightarrow 20$

$x_4 \leftrightarrow 21$

$x_5 \leftrightarrow 220$

$x_6 \leftrightarrow 221$



Expected codeword length $= \bar{L} = 1.35$ ternary digits per source symbol.

Remarks:

I) Hufman code is not unique. But expected codeword length of all hufman codes are optimal. $\bar{l}_{opt} = \bar{l}_{Huffman}$

II) Comparing Huffman and Shanon Fano Code for $\mathcal{D} = \{0,1\}$. $\bar{l}_{SF}$ and $\bar{l}_{Huffman}$ i.e between $H(P_x)$ and $H(P_x)+1$. i.e. 1 bit of each other.

		Shanon Fano $\lceil \log 1/p(x) \rceil$	Huffman
x_1	$P_x(x_1) = 0.9999$	1 bit	0
x_2	$P_x(x_2) = 0.0001$	14 bit	1
Expected codeword length		$\bar{l}_{SF}$ $= 1 \times 0.9999 + 14 \times 0.0001$ $= 1.0013$ bits	$\bar{l}_{Huffman} = 1$ bit

III) For a paticular source symbol, codeword length of an optimal Hufman code for it can be longer than it's Shanon Fano code. (ie $\lceil \log 1/p_j \rceil$). However in expected value,

$$\bar{l}_{Hffman} \leq \bar{l}_{Shanon\text{-}Fano}$$

IV) Huffman codes need a source code pmf to operate. Adaptive Huffman code is a method where this distribution is estimated on the fly

Optimality of the Hufman Codes

Assume w.l.o.g : $p_1 \geq p_2 \geq \dots \geq p_{|\mathcal{X}|}$. Optimal prefix code is the one where $\sum_{i=1}^{|\mathcal{X}|} p_i \cdot l_i$ is minimum and the code is prefix free.

Cover Thomas
Section 5.8

5.8 OPTIMALITY OF HUFFMAN CODES

We prove by induction that the binary Huffman code is optimal. It is important to remember that there are many optimal codes: inverting all the bits or exchanging two codewords of the same length will give another optimal code. The Huffman procedure constructs one such optimal code. To prove the optimality of Huffman codes, we first prove some properties of a particular optimal code.

Without loss of generality, we will assume that the probability masses are ordered, so that $p_1 \geq p_2 \geq \cdots \geq p_m$. Recall that a code is optimal if $\sum p_i l_i$ is minimal.

Lemma 5.8.1 *For any distribution, there exists an optimal instantaneous code (with minimum expected length) that satisfies the following properties:*

1. *The lengths are ordered inversely with the probabilities (i.e., if $p_j > p_k$, then $l_j \leq l_k$).*
2. *The two longest codewords have the same length.*
3. *Two of the longest codewords differ only in the last bit and correspond to the two least likely symbols.*

Proof: The proof amounts to swapping, trimming, and rearranging, as shown in Figure 5.3. Consider an optimal code C_m:

- *If $p_j > p_k$, then $l_j \leq l_k$.* Here we swap codewords. Consider C'_m, with the codewords j and k of C_m interchanged. Then

$$L(C'_m) - L(C_m) = \sum p_i l_i' - \sum p_i l_i \tag{5.62}$$

$$= p_j l_k + p_k l_j - p_j l_j - p_k l_k \tag{5.63}$$

$$= (p_j - p_k)(l_k - l_j). \tag{5.64}$$

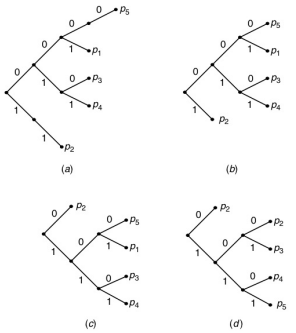


FIGURE 5.3. Properties of optimal codes. We assume that $p_1 \geq p_2 \geq \cdots \geq p_m$. A possible instantaneous code is given in (a). By trimming branches without siblings, we improve the code to (b). We now rearrange the tree as shown in (c), so that the word lengths are ordered by increasing length from top to bottom. Finally, we swap probability assignments to improve the expected depth of the tree, as shown in (d). Every optimal code can be rearranged and swapped into canonical form as in (d), where $l_1 \leq l_2 \leq \cdots \leq l_m$ and $l_{m-1} = l_m$, and the last two codewords differ only in the last bit.

But $p_j - p_k > 0$, and since C_m is optimal, $L(C'_m) - L(C_m) \geq 0$. Hence, we must have $l_k \geq l_j$. Thus, C_m itself satisfies property 1.

- *The two longest codewords are of the same length.* Here we trim the codewords. If the two longest codewords are not of the same length, one can delete the last bit of the longer one, preserving the prefix property and achieving lower expected codeword length. Hence, the two longest codewords must have the same length. By property 1, the longest codewords must belong to the least probable source symbols.
- *The two longest codewords differ only in the last bit and correspond to the two least likely symbols.* Not all optimal codes satisfy this property, but by rearranging, we can find an optimal code that does. If there is a maximal-length codeword without a sibling, we can delete the last bit of the codeword and still satisfy the prefix property. This reduces the average codeword length and contradicts the optimality of the code. Hence, every maximal-length codeword in any optimal code has a sibling. Now we can exchange the longest codewords so that the two lowest-probability source symbols are associated with two siblings on the tree. This does not change the expected length, $\sum p_i l_i$. Thus, the codewords for the two lowest-probability source symbols have maximal length and agree in all but the last bit.

Summarizing, we have shown that if $p_1 \geq p_2 \geq \cdots \geq p_m$, there exists an optimal code with $l_1 \leq l_2 \leq \cdots \leq l_{m-1} = l_m$, and codewords $C(x_{m-1})$ and $C(x_m)$ that differ only in the last bit. □

Thus, we have shown that there exists an optimal code satisfying the properties of the lemma. We call such codes *canonical* codes. For any probability mass function for an alphabet of size m, $\mathbf{p} = (p_1, p_2, \ldots, p_m)$ with $p_1 \geq p_2 \geq \cdots \geq p_m$, we define the Huffman reduction $\mathbf{p}' = (p_1, p_2, \ldots, p_{m-2}, p_{m-1} + p_m)$ over an alphabet of size $m - 1$ (Figure 5.4). Let $C^*_{m-1}(\mathbf{p}')$ be an optimal code for $\mathbf{p}'$, and let $C^*_m(\mathbf{p})$ be the canonical optimal code for $\mathbf{p}$.

The proof of optimality will follow from two constructions: First, we expand an optimal code for $\mathbf{p}'$ to construct a code for $\mathbf{p}$, and then we condense an optimal canonical code for $\mathbf{p}$ to construct a code for the Huffman reduction $\mathbf{p}'$. Comparing the average codeword lengths for the two codes establishes that the optimal code for $\mathbf{p}$ can be obtained by extending the optimal code for $\mathbf{p}'$.

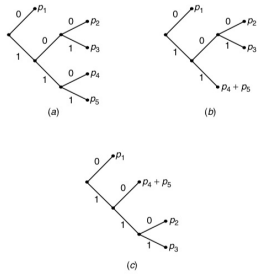


FIGURE 5.4. Induction step for Huffman coding. Let $p_1 \geq p_2 \geq \cdots \geq p_5$. A canonical optimal code is illustrated in (a). Combining the two lowest probabilities, we obtain the code in (b). Rearranging the probabilities in decreasing order, we obtain the canonical code in (c) for $m - 1$ symbols.

From the optimal code for $\mathbf{p}'$, we construct an extension code for m elements as follows: Take the codeword in C^*_{m-1} corresponding to weight $p_{m-1} + p_m$ and extend it by adding a 0 to form a codeword for symbol $m - 1$ and by adding 1 to form a codeword for symbol m. The code construction is illustrated as follows:

	$C^*_{m-1}(\mathbf{p}')$		$C_m(\mathbf{p})$	
p_1	w'_1	l'_1	$w_1 = w'_1$	$l_1 = l'_1$
p_2	w'_2	l'_2	$w_2 = w'_2$	$l_2 = l'_2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
p_{m-2}	w'_{m-2}	l'_{m-2}	$w_{m-2} = w'_{m-2}$	$l_{m-2} = l'_{m-2}$
$p_{m-1} + p_m$	w'_{m-1}	l'_{m-1}	$w_{m-1} = w'_{m-1}0$	$l_{m-1} = l'_{m-1} + 1$
			$w_m = w'_{m-1}1$	$l_m = l'_{m-1} + 1$

(5.65)

Calculation of the average length $\sum_i p'_i l'_i$ shows that

$$L(\mathbf{p}) = L^*(\mathbf{p}') + p_{m-1} + p_m. \tag{5.66}$$

Similarly, from the canonical code for $\mathbf{p}$, we construct a code for $\mathbf{p}'$ by merging the codewords for the two lowest-probability symbols $m - 1$ and m with probabilities p_{m-1} and p_m, which are siblings by the properties of the canonical code. The new code for $\mathbf{p}'$ has average length

$$L(\mathbf{p}') = \sum_{i=1}^{m-2} p_i l_i + p_{m-1}(l_{m-1} - 1) + p_m(l_m - 1) \tag{5.67}$$

$$= \sum_{i=1}^{m} p_i l_i - p_{m-1} - p_m \tag{5.68}$$

$$= L^*(\mathbf{p}) - p_{m-1} - p_m. \tag{5.69}$$

Adding (5.66) and (5.69) together, we obtain

$$L(\mathbf{p}') + L(\mathbf{p}) = L^*(\mathbf{p}') + L^*(\mathbf{p}) \tag{5.70}$$

or

$$(L(\mathbf{p}') - L^*(\mathbf{p}')) + (L(\mathbf{p}) - L^*(\mathbf{p})) = 0. \tag{5.71}$$

Now examine the two terms in (5.71). By assumption, since $L^*(\mathbf{p}')$ is the optimal length for $\mathbf{p}'$, we have $L(\mathbf{p}') - L^*(\mathbf{p}') \geq 0$. Similarly, the length of the extension of the optimal code for $\mathbf{p}'$ has to have an average length at least as large as the optimal code for $\mathbf{p}$ [i.e., $L(\mathbf{p}) - L^*(\mathbf{p}) \geq 0$]. But the sum of two nonnegative terms can only be 0 if both of them are 0, which implies that $L(\mathbf{p}) = L^*(\mathbf{p})$ (i.e., the extension of the optimal code for $\mathbf{p}'$ is optimal for $\mathbf{p}$).

Consequently, if we start with an optimal code for $\mathbf{p}'$ with $m - 1$ symbols and construct a code for m symbols by extending the codeword corresponding to $p_{m-1} + p_m$, the new code is also optimal. Starting with a code for two elements, in which case the optimal code is obvious, we can by induction extend this result to prove the following theorem.

Theorem 5.8.1 *Huffman coding is optimal; that is, if C^* is a Huffman code and C' is any other uniquely decodable code, $L(C^*) \leq L(C')$.*

Although we have proved the theorem for a binary alphabet, the proof can be extended to establishing optimality of the Huffman coding algorithm for a D-ary alphabet as well. Incidentally, we should remark that Huffman coding is a "greedy" algorithm in that it coalesces the two least likely symbols at each stage. The proof above shows that this local optimality ensures global optimality of the final code.

Reading Assignment
Cover, Thomas chapter 7 - (on DMCs and Channel Coding)

Why is "Channel Capacity" relevant here

13) Other Fixed to Variable length Codes

1) Shanon Fano Elias Code

In SF code the source code pmf is required. These methods bypass it by assuming a cdf. But the compromise is

$$H_D(x) \leq \bar{L} \leq H_D(x) + 2$$ (D ary bits)

2) Arithmatic Code

The +2 is SF-Elios code can be reduced by taking n bits at once. But estimating cdf for n-length source sequences become exponentially computationally complex with n.

These arithmatic codes computation of probabilities and cdf grows linearly with n, and not exponentially. Also uses fixed precision arithmatic rather than precision that grows with n.

3) Gilbert-Moore Code

Summary

I) Fixed - to - Fixed Length Codes,

if $e(f_n, \phi_n) = 0$, need the minimum code rate = $\log |\mathcal{X}|$
No compression and f_n is one-to-one nonsingular.

if $e(f_n, \phi_n) = \varepsilon$, minimum code rate = $H(P_X)$.
Instantaneous codes

II) Fixed - to - Variable Length Codes,

if $e(f_n, \phi_n) = 0$ Expected code rate $\cong H(P_X)$ and codes are uniquely decodable and istantaneous.

III) Also we can talk about

Variable - to - Fixed Length codes ⟶ Tunstall Algorithm

Variable - to - Variable Length Codes ⟶ Run length Codes, Universal Source coding.

Cover Thomas
Chapter 4

Entropy Rate of a Stochastic Process

1) Introduction

Motivation

- Given source sequence $X^n \triangleq (X_1, \ldots, X_n)$, $n \geq 1$ with pmf p_{X^n}. From Theorem 4 in "Fixed-to-Variable Length Codes" we have seen, $\forall n$

$$\frac{1}{n} H_D(X_1, \ldots, X_n) \leq \bar{l}_n^* \triangleq \frac{1}{n} E_{p_{X^n}}[l(f(X^n))] < \frac{1}{n} H_D(X_1, \ldots, X_n) + \frac{1}{n}$$

- In theorem 4, $X_1, \ldots, X_n$ need not be i.i.d.

- If $X_1, \ldots, X_n$ are i.i.d, then theorem 4 becomes

$$H_D(X) \leq \bar{l}_n^* \leq H_D(X) + 1/n$$

hence for larger n, $\bar{l}_n^* \cong H_D(X)$. Hence we can associate $H_D(X_1)$ with the entropy of the i.i.d process $\{X_t\}_{t=1}^{\infty}$ since,

$$H_D(X_1) = \frac{1}{n} H_D(X^n)$$

- If $\{X_n\}_{n=1}^{\infty}$ is a stationary process, then shall see,

$$\underset{\text{(For large } n\text{)}}{\bar{l}_n^* \cong} \quad \frac{1}{n} H_D(X_1, \ldots X_n) \xrightarrow{n \to \infty} \bar{H}_D(X)$$

where $\bar{H}_D(X)$ = entropy rate.

- If $X_1, \ldots, X_n$ are not i.i.d what happens to $\frac{1}{n} H_D(X^n)$ for large n. Especially in limit as $n \to \infty$. Can we define a notion of the entropy of a process $\{X_t\}_{t=1}^{\infty}$.

This page is from ENEE 620 "Stationarity Concepts".

Stationarity

$\mathbb{R}^n$- valued stochastic process $\{\underline{X}_t,\ t \in T\}$ is

Stationary of Order K

Definition:- The $\mathbb{R}^n$- valued stochastic process $\{\underline{X}_t,\ t \in T\}$ is stationary of order k $(k = 1, 2, \ldots)$ if $\forall\ t_1 < \ldots < t_k$ and $\forall h \in T$ the rvs,

$$\underline{X}_{t_1}, \ldots, \underline{X}_{t_k} \quad \text{and} \quad \underline{X}_{t_1+h}, \ldots, \underline{X}_{t_k+h}$$

have the same cdf.

Theorem:- If $\{\underline{X}_t,\ t \in T\}$ is stationary of order (at least) 2 and is a second order process, then

$$\mu_{\underline{X}}(t) = \text{Constant in } t$$
$$C_{\underline{X}}(s,t) = C_{\underline{X}}(s+h, t+h) = C_{\underline{X}}(|t-s|)$$

$h \in T$ $\forall s,t \in T$

This motivates a weaker notion of Stationarity. (Covariance Stationarity)

Strict Stationarity

Definition:- The $\mathbb{R}^n$- valued stochastic process $\{\underline{X}_t,\ t \in T\}$ has strict stationarity if it is stationary of order k $\forall$ $k = 1, 2, \ldots$

Ex:- • $\{X_t\}_{t=1}^{\infty}$ i.i.d sequence of rvs.

• $\{X_t\}_{t=1}^{\infty}$ $X_t = X$ $\forall t$

Definition: $\{X_t\}_{t=1}^{\infty}$ is stationary if the joint pmf of the rvs $X_t, X_{t+1}, \ldots, X_{t+k}$ does not depend on t $\forall$ $k = 1, 2, \ldots$

we will use this version

2) Entropy Rate

Definition:- $\{X_t\}_{t=1}^{\infty}$ is a stochastic process with values from finite alphabet $\mathcal{X}$.

Entropy rate of $\{X_t\}_{t=1}^{\infty} \triangleq \bar{H} \triangleq \lim_{t\to\infty} \frac{1}{t} H(X_1, \ldots, X_t)$

$\sim \lim \frac{1}{t} H(X^t)$

whenever the limit exists.

$\bar{H}(X) = H(X)$ - per symbol entropy of the n random variables.

Examples

1) If $X_1, X_2, \ldots,$ are i.i.d rvs. Then,

$$\bar{H}(X) = \lim \frac{H(X_1, \ldots, X_t)}{t} = \lim \frac{t \cdot H(X_1)}{t} = H(X_1)$$

2) Consider $X_1, X_2, \ldots$ independant, but not identical stochastic process with,

$$p_i = P(X_i = 1) = \begin{cases} 1/2 & \text{if } 2k < \log\log i \leq 2k+1 \\ 0 & \text{if } 2k+1 < \log\log i \leq 2k+2 \end{cases}$$

In this example we have arbitarily long stretches where $H(X_i) = 1$, followed by exponentially longer segments where $H(X_i) = 0$. $\therefore$ running average of $H(X_i)$ will oscilate between 0 and 1.

$$\therefore \quad \frac{1}{t} H(X_1, \ldots, X_t) = \frac{1}{t} \sum_{i=1}^{t} H(X_i) \quad \text{oscilates.}$$

$\therefore \lim \frac{1}{n} H(X^t)$ does not exist.

3) Shall see that if $\{X_t\}_{t=1}^{\infty}$ is stationary, then $\bar{H}$ exists and admits an alternate expression. But $\bar{H}$ may exist even if $\{X_t\}_{t=1}^{\infty}$ is not stationary.

$\bar{H}'$ = Conditional entropy of the last r.v given past.

$$\bar{H}' = H'(\mathcal{X}) = \lim_{t\to\infty} H(X_t \mid X_{t-1}, X_{t-2}, \ldots, X_1)$$

Entropy Rate of a Stationary Process

Theorem - Cesaro Mean Convergence

If $\lim_n a_n = a$

$b_n = \frac{1}{n}\sum_{i=1}^{n} a_i$

$\Longrightarrow \lim_n b_n = a$

Proof:- $\lim_n a_n = a$

$\Longrightarrow \forall \varepsilon/2 > 0 \quad \exists N(\varepsilon) \quad \forall n \geq N(\varepsilon) \quad |a_n - a| \leq \varepsilon/2$

$\Longrightarrow \forall \varepsilon/2 > 0 \quad \exists N(\varepsilon) \quad \forall n \geq N(\varepsilon) \quad \frac{1}{n}\sum_{i=N(\varepsilon)+1}^{n} |a_i - a| \leq \frac{n-N(\varepsilon)}{n}\varepsilon/2 < \varepsilon/2$

$\Longrightarrow \forall \varepsilon > 0 \quad \exists N(\varepsilon) \quad \forall n \geq N(\varepsilon) \quad \frac{1}{n}\sum_{i=1}^{n} |a_i - a| \leq \underbrace{\frac{1}{n}\sum_{i=1}^{N(\varepsilon)} |a_i - a|}_{\text{make } n\to\infty\text{. This term } \to 0 \text{ i.e. } < \varepsilon/2} + \varepsilon/2$

$\Longrightarrow \forall \varepsilon > 0 \quad \exists N(\varepsilon) \quad \forall n \geq N(\varepsilon) \quad \frac{1}{n}\sum_{i=1}^{n} |a_i - a| \leq \varepsilon/2 + \varepsilon/2 = \varepsilon$

$\Longrightarrow \forall \varepsilon > 0 \quad \exists N(\varepsilon) \quad \forall n \geq N(\varepsilon) \quad \left| \underbrace{\frac{1}{n}\sum_{i=1}^{n} (a_i - a)}_{b_n - a} \right| \leq \frac{1}{n}\sum_{i=1}^{n} |a_i - a| \leq \varepsilon$

$\Longrightarrow \forall \varepsilon > 0 \quad \exists N(\varepsilon) \quad \forall n \geq N(\varepsilon) \quad |b_n - a| \leq \varepsilon$

$\Longrightarrow \lim_n b_n = a$ ∎

Lemma 1:- For a stationary stochastic process $\{X_t\}_{t=1}^{\infty}$ $H(X_t \mid X^{t-1})$ is non-increasing in n and has a limit $H'(X)$.

Proof:-

$H(X_t \mid X^{t-1}) = H(X_t \mid X_1, X_2, \ldots, X_{t-1})$

$\leq H(X_t \mid X_2, \ldots, X_{t-1})$ ..... conditioning reduces entropy

$= H(X_{t-1} \mid X_1, \ldots, X_{t-1})$ ... reduce all subscripts by 1, since stationary

$= H(X_{t-1} \mid X^{t-2})$

Thus $\{H(X_t \mid X^{t-1})\}_{t=1}^{\infty}$ is non decreasing and bounded below by 0. Hence $\lim_{t\to\infty} H(X_t \mid X^{t-1})$ exists. ∎

Lemma 2 - For a stationary stochastic process $\{X_t\}_{t=1}^{\infty}$ $\frac{1}{t} \cdot H(X^t)$ is non-increasing in n and has a limit.

Proof: By chain rule of conditional entropy,

$$\frac{1}{t+1} H(X_{t+1}, X_t, \ldots, X_1) = \frac{1}{t+1}\left[H(X_1) + H(X_2|X_1) + \ldots + H(X_{t+1}|X_1,\ldots,X_t)\right]$$

$$= \frac{1}{t+1} \sum_{i=1}^{t+1} H(X_i \mid X^{i-1})$$

$$\geq \frac{1}{t+1} \sum_{i=1}^{t+1} H(X_{t+1} \mid X^t) \quad \ldots\ldots \text{from non-increasing property in Lemma 1 which use stationarity}$$

$$= \frac{1}{t+1} (t+1) \cdot H(X_{t+1} \mid X^t)$$

$$= H(X_{t+1} \mid X^t) \quad \ldots\ldots (1)$$

Consider,

$$\frac{1}{t+1} H(X_{t+1}, X_t, \ldots, X_1) = \frac{1}{t+1} H(X_1,\ldots,X_t) + \frac{1}{t+1} H(X_{t+1} \mid X_1,\ldots,X_t)$$

$$= \frac{t}{t+1} \cdot \frac{H(X^t)}{t} + \frac{1}{t+1} H(X_{t+1} \mid X^t)$$

$$\leq \frac{t}{t+1} \cdot \frac{H(X^t)}{t} + \frac{1}{t+1}\left[\frac{1}{t+1} H(X^{t+1})\right]$$

$$\Rightarrow \frac{1}{t+1} H(X^{t+1}) \left[1 - \frac{1}{t+1}\right] \leq \frac{t}{t+1} \cdot \frac{H(X^t)}{t}$$

$$\Rightarrow \frac{1}{t+1} H(X^{t+1}) \leq \frac{1}{t} H(X^t)$$

$\Rightarrow$ Hence $\left\{\frac{1}{t} H(X^t)\right\}_{t=1}^{\infty}$ is non-increasing sequence, bounded below by 0. Hence converge and limit exists

■

Theorem - For a stationary stochastic process $\{X_t\}_{t=1}^{\infty}$, $\bar{H}(X) = \lim_t \frac{1}{t} \cdot H(X^t)$ exists.

Proof:- From chain rule,

$$\frac{1}{t} H(X^t) = \frac{1}{t} \sum_{i=1}^{t} H(X_i | X^{i-1})$$

From Lemma 1, $H(X_i | X^{i-1})$ is non-increasing and has a limit. From Lemma 2, $\frac{1}{t} H(X^t)$ also has a limit. From Cesaro Mean Convergence on RHS of above equation,

$$\lim_t \frac{1}{t} H(X^t) = \lim_t \left[H(X_i | X^{i-1}) \right] = \bar{H}$$

■

Remarks

- If $\lim_{t\to\infty} H(X_t | X^{t-1})$ exists due to stationarity or any other reason, then $\lim_{t\to\infty} \frac{1}{t} H(X^t)$ also exists. Hence

  $\lim_{t\to\infty} H(X_t | X^{t-1})$ exists $\Longrightarrow$ $\lim_{t\to\infty} \frac{1}{t} H(X^t)$ exists and limits are same.

- It is possible that $\lim_{n\to\infty} \frac{1}{t} H(X^t)$ exists, but $\lim_{t\to\infty} H(X_t | X^{t-1})$ does not exist

- $\bar{H}$ is usually defined as $\lim_{t\to\infty} \frac{1}{t} \cdot H(X^t)$.

- Shall see later that entropy rate can be used as average description length of a stationary ergodic process.

Recap ENEE 627
2) Measure of Information
Topic 9 on Markov Chains.

Entropy Rate of a Markov Chain

For a stationary Markov chain

$$\text{Entropy rate} = \bar{H}(\mathcal{X}) = H(X_2 | X_1)$$

($\bar{H}(\mathcal{X})$ denotes a MC)

Proof: $\bar{H}(\mathcal{X}) = \lim_n H(X_n | X_{n-1}, \ldots, X_1)$

$= \lim_n H(X_n | X_{n-1})$ ..... Since a MC

$= H(X_2 | X_1)$ ..... Since stationary

$P_{ij} = P_{j|i}$ = Probability of going from state i to state j.

Theorem: $\{X_t\}_{t=1}^{\infty}$ is a time invariant, stationary Markov Chain with transition probability matrix P, and stationary (or invariant) distribution μ. Then,

$$\bar{H}(\mathcal{X}) = \sum_{i \in \mathcal{X}} \mu_i \left[\sum_{j \in \mathcal{X}} (-P_{j|i} \cdot \log P_{j|i}) \right]$$

$$= \sum_{i,j \in \mathcal{X}} \mu_i [-P_{ij} \log P_{ij}]$$

Proof: From the previous result, entropy rate of a stationary MC is,

$$\bar{H}(\mathcal{X}) = H(X_2 | X_1)$$

$$= -\sum_{i \in \mathcal{X}_1} \sum_{j \in \mathcal{X}_2} P_{X_1 X_2}(i,j) \cdot \log P_{X_2|X_1}(j|i)$$ .... from definition of conditional entropy

$i, j \in \mathcal{X} = X_1 = X_2$ (states of MC)

$$= -\sum_{i,j \in \mathcal{X}} P(X_1 = i, X_2 = j) \cdot \log P(X_2 = j | X_1 = i)$$

$$= -\sum_{i,j \in \mathcal{X}} \underbrace{P(X_2 = j | X_1 = i)}_{P_{ij}} \cdot \underbrace{P(X_1 = i)}_{\mu_i} \cdot \log \underbrace{P(X_2 = j | X_1 = i)}_{P_{ij}}$$ from State Transition Matrix

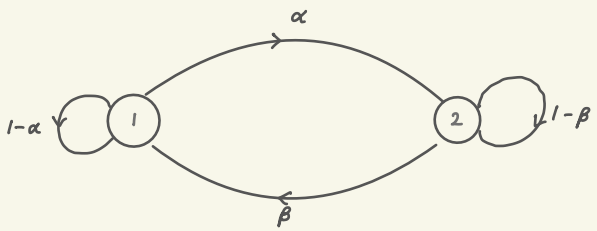
$$= -\sum_{i,j} P_{ij} \cdot \mu_i \cdot \log P_{ij}$$

$$= \sum_i \mu_i \left(\sum_j (-P_{ij} \cdot \log P_{ij}) \right)$$

■

Example: Consider a Time Invariant, $m=2$ state Markov Chain with Probability Transition Matrix $P = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$. Assume it has a stationary distribution μ.

Find the entropy of X_n and Entropy rate of MC.



$0 < \alpha, \beta < 1$

If this has a stationary distribution $\mu = [\mu_1, \mu_2] = [P_{X_1=1}, P_{X_1=2}] = P_{X_1}$

$$\Rightarrow \mu P = \mu$$

$$\Rightarrow [\mu_1, \mu_2] \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} = [\mu_1 \ \mu_2]$$

$$\Rightarrow \mu_1 = \beta/(\alpha+\beta), \quad \mu_2 = \alpha/(\alpha+\beta)$$

Entropy rate at time n $= H(\underline{X}_n) = H(X_n=1, X_n=2) = H(\mu_1, \mu_2)$

$$= \underline{\underline{H\left(\beta/(\alpha+\beta), \alpha/(\alpha+\beta)\right)}}$$

Entropy rate of MC from Theorem,

$$\bar{H}(\mathcal{X}) = \sum_{i \in \mathcal{X}} \mu_i \left[\sum_{j \in \mathcal{X}} \left(-P_{j|i} \cdot \log P_{j|i} \right) \right]$$

($\bar{H}(\mathcal{X})$: To denote a MC)

$$= \mu_1 \left[-P_{1|1} \log P_{1|1} - P_{2|1} \log P_{2|1} \right] + \mu_2 \left[-P_{1|2} \log P_{1|2} - P_{2|2} \cdot \log P_{2|2} \right]$$

$$= \mu_1 \left[-(1-\alpha) \cdot \log(1-\alpha) - \alpha \log \alpha \right] + \mu_2 \left[-\beta \cdot \log \beta - (1-\beta) \cdot \log(1-\beta) \right]$$

$$= \mu_1 \cdot H(X_2 | X_1 = 1) + \mu_2 \cdot H(X_2 | X_1 = 2)$$

$$= \mu_1 \cdot h(\alpha) + \mu_2 \, h(\beta)$$

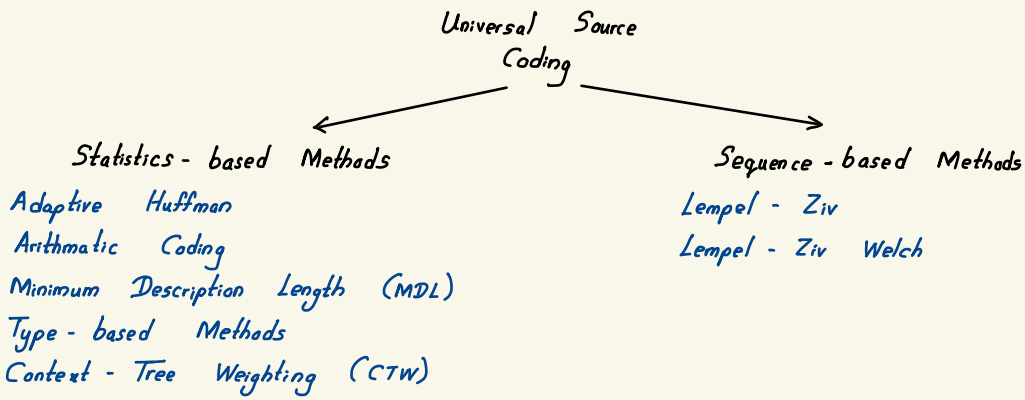
$$= \underline{\underline{\frac{\beta}{\alpha+\beta} h(\alpha) + \frac{\alpha}{\alpha+\beta} \cdot h(\beta)}}$$

Cover Thomas
Chapter 13

Universal Source Coding or Universal Data Compression

- Sometimes source statistics are not known a priori. Source statistis do not exist, or are known to belong to a class of distributions. (but not known exactly)

- There are 2 types of universal source coding methods.



Lempel - Ziv (LZ) Coding or Compression

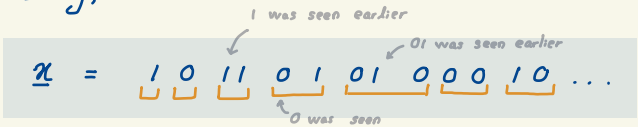
- A universal compression algorithm.

- Simple to implement.

- Ex :-
  Compressing txt, PDF files ⟶ gzip, Winzip
  Compress (Command in UNIX)
  Compressing image files ⟶ GIF : Graphical Interchange Format
  TIFF · Tag Image File Format.

Algorithm via an example

- Consider a string,



Step 1 - Source string $\underline{x}$ is passed sequentially into phrases that have not appeared here-to-fore. The phrases are the shortest, and are seperated by commas.

- Such a parsing has the feature that in the current phrase, Concatanation of all but the last symbol have appeared before as a phrase.

Step 2 - Current phrase is coded by a pair consisting of a pointer (in bits) indicating the location of the earlier occurance of the prefix of the phrase and the (binary) value of the last symbol.

- Given the source sequence $\underline{x} = (x_1, \ldots, x_n) \in \mathcal{X}^n$. let

$c_n(\underline{x})$ – No of distinct phrases obtained via the LZ-parsing.

$\lceil \log_2 c_n(x) \rceil$ – bits suffice to describe the pointer to the location of the prefix of each pointer.

$|\mathcal{X}| = r$ – is cardinality of the alphabet

$\lceil \log_2 r \rceil$ – bits suffice to represent the last symbol of a phrase.

- Back to our example,

$\underline{x}$ = 1 0 11 0 1 01 0 0 0 1 0

Length of sequence = n = 13

$c_n(\underline{x})$ = 7

r = 2 bits

Bits to represent the pointer to the prefix location of each phrase = $\lceil \log_2 c_n(\underline{x}) \rceil = \lceil \log_2 7 \rceil$ = 3 bits

Bits to represent the last symbol = 1 bits

Length of a codeward = 3+1 = 4 bits

- Hence the codebook looks as below

1	⟶	(000, 1)
0	⟶	(000, 0)
11	⟶	(001, 1)
01	⟶	(010, 1)
010	⟶	(100, 0)
00	⟶	(010, 0)
10	⟶	(001, 0)

Prefix '1' appeared in position 1.

prefix '0' appeared in position 0.

} 3 bit representation of the occurrence of prefix.

Decoding - Straightforward and the source sequence can be recovered without error.

Remarks

1) LZ algorithm above requires two passes over the string

2) Original source sequence of length n = 13 bits is expanded into encoded sequence of $7 \times 4 = 28$ bits. But as coding proceeds further into the sequence, phrases become longer and a description of the prefixes by means of location pointers become increasingly efficient.

Theorem - For an unknown stationary, ergodic source, the LZ algorithm asymptotically attains or achieves entropy.

Proof: 13.5.2 Section / 448 - 455 Pages. This is a rephrasing of Theorem 13.5.3 in Cover Thomas

Lemma 13.5.3 (*Lempel and Ziv [604]*) *The number of phrases $c(n)$ in a distinct parsing of a binary sequence $X_1, X_2, \ldots, X_n$ satisfies*

$$c(n) \le \frac{n}{(1-\epsilon_n)\log n}, \tag{13.84}$$

where $\epsilon_n = \min\{1, \frac{\log(\log n)+4}{\log n}\} \to 0$ as $n \to \infty$.

Lemma 13.5.4 *Let Z be a nonnegative integer-valued random variable with mean μ. Then the entropy $H(Z)$ is bounded by*

$$H(Z) \le (\mu+1)\log(\mu+1) - \mu \log \mu. \tag{13.97}$$

Proof: The lemma follows directly from the results of Theorem 12.1.1, which show that the geometric distribution maximizes the entropy of a nonnegative integer-valued random variable subject to a mean constraint. □

Lemma 13.5.5 (*Ziv's inequality*) *For any distinct parsing (in particular, the LZ78 parsing) of the string $x_1 x_2 \cdots x_n$, we have*

$$\log Q_k(x_1, x_2, \ldots, x_n|s_1) \le -\sum_{l,s} c_{ls} \log c_{ls}. \tag{13.104}$$

Note that the right-hand side does not depend on Q_k.

Proof: We write

$$Q_k(x_1, x_2, \ldots, x_n|s_1) = Q_k(y_1, y_2, \ldots, y_c|s_1) \tag{13.105}$$

$$= \prod_{i=1}^{c} P(y_i|s_i) \tag{13.106}$$

or

$$\log Q_k(x_1, x_2, \ldots, x_n|s_1) = \sum_{i=1}^{c} \log P(y_i|s_i) \tag{13.107}$$

$$= \sum_{l,s} \sum_{i:|y_i|=l, s_i=s} \log P(y_i|s_i) \tag{13.108}$$

$$= \sum_{l,s} c_{ls} \sum_{i:|y_i|=l, s_i=s} \frac{1}{c_{ls}} \log P(y_i|s_i) \tag{13.109}$$

$$\le \sum_{l,s} c_{ls} \log \left(\sum_{i:|y_i|=l, s_i=s} \frac{1}{c_{ls}} P(y_i|s_i) \right), \tag{13.110}$$

where the inequality follows from Jensen's inequality and the concavity of the logarithm.

Now since the y_i are distinct, we have $\sum_{i:|y_i|=l,s_i=s} P(y_i|s_i) \le 1$. Thus,

$$\log Q_k(x_1, x_2, \ldots, x_n|s_1) \le \sum_{l,s} c_{ls} \log \frac{1}{c_{ls}}, \tag{13.111}$$

proving the lemma. □

Theorem 13.5.2 *Let $\{X_n\}$ be a binary stationary ergodic process with entropy rate $H(\mathcal{X})$, and let $c(n)$ be the number of phrases in a distinct parsing of a sample of length n from this process. Then*

$$\limsup_{n\to\infty} \frac{c(n)\log c(n)}{n} \le H(\mathcal{X}) \tag{13.112}$$

with probability 1.

Proof: We begin with Ziv's inequality, which we rewrite as

$$\log Q_k(x_1, x_2, \ldots, x_n|s_1) \le -\sum_{l,s} c_{ls} \log \frac{c_{ls} c}{c} \tag{13.113}$$

Writing $\pi_{ls} = \frac{c_{ls}}{c}$, we have

$$\sum_{l,s} \pi_{ls} = 1, \qquad \sum_{l,s} l\pi_{ls} = \frac{n}{c}, \tag{13.115}$$

from (13.102) and (13.103). We now define random variables U, V such that

$$\Pr(U = l, V = s) = \pi_{ls}. \tag{13.116}$$

Thus, $EU = \frac{n}{c}$ and

$$\log Q_k(x_1, x_2, \ldots, x_n|s_1) \le cH(U,V) - c\log c \tag{13.117}$$

or

$$-\frac{1}{n}\log Q_k(x_1, x_2, \ldots, x_n|s_1) \ge \frac{c}{n}\log c - \frac{c}{n}H(U,V). \tag{13.118}$$

Now

$$H(U,V) \le H(U) + H(V) \tag{13.119}$$

and $H(V) \le \log|\mathcal{X}|^k = k$. By Lemma 13.5.4, we have

$$H(U) \le (EU+1)\log(EU+1) - (EU)\log(EU) \tag{13.120}$$

$$= \left(\frac{n}{c}+1\right)\log\left(\frac{n}{c}+1\right) - \frac{n}{c}\log\frac{n}{c} \tag{13.121}$$

$$= \log\frac{n}{c} + \left(\frac{n}{c}+1\right)\log\left(\frac{c}{n}+1\right). \tag{13.122}$$

Thus,

$$\frac{c}{n}H(U,V) \le \frac{c}{n}k + \frac{c}{n}\log\frac{n}{c} + o(1). \tag{13.123}$$

For a given n, the maximum of $\frac{c}{n}\log\frac{n}{c}$ is attained for the maximum value of c (for $\frac{c}{n} \le \frac{1}{e}$). But from Lemma 13.5.3, $c \le \frac{n}{\log n}(1+o(1))$. Thus,

$$\frac{c}{n}\log\frac{n}{c} \le O\left(\frac{\log\log n}{\log n}\right), \tag{13.124}$$

and therefore $\frac{c}{n}H(U,V) \to 0$ as $n \to \infty$. Therefore,

$$\frac{c(n)\log c(n)}{n} \le -\frac{1}{n}\log Q_k(x_1, x_2, \ldots, x_n|s_1) + \epsilon_k(n), \tag{13.125}$$

where $\epsilon_k(n) \to 0$ as $n \to \infty$. Hence, with probability 1,

$$\limsup_{n\to\infty} \frac{c(n)\log c(n)}{n} \le \lim_{n\to\infty} -\frac{1}{n}\log Q_k(X_1, X_2, \ldots, X_n|X_{-(k-1)}^0) \tag{13.126}$$

$$= H(X_0|X_{-1}, \ldots, X_{-k}) \tag{13.127}$$

$$\to H(\mathcal{X}) \qquad \text{as } k \to \infty. \quad \square \tag{13.128}$$

We now prove that LZ78 coding is asymptotically optimal.

Theorem 13.5.3 *Let $\{X_i\}_{-\infty}^{\infty}$ be a binary stationary ergodic stochastic process. Let $l(X_1, X_2, \ldots, X_n)$ be the LZ78 codeword length associated with $X_1, X_2, \ldots, X_n$. Then*

$$\limsup_{n\to\infty} \frac{1}{n} l(X_1, X_2, \ldots, X_n) \le H(\mathcal{X}) \quad \textit{with probability 1,} \tag{13.129}$$

where $H(\mathcal{X})$ is the entropy rate of the process.

Proof: We have shown that $l(X_1, X_2, \ldots, X_n) = c(n)(\log c(n) + 1)$, where $c(n)$ is the number of phrases in the LZ78 parsing of the string $X_1, X_2, \ldots, X_n$. By Lemma 13.5.3, $\limsup c(n)/n = 0$, and thus Theorem 13.5.2 establishes that

$$\limsup \frac{l(X_1, X_2, \ldots, X_n)}{n} = \limsup \left(\frac{c(n)\log c(n)}{n} + \frac{c(n)}{n}\right)$$

$$\le H(\mathcal{X}) \quad \text{with probability 1.} \quad \square \tag{13.130}$$

Assume w.l.o.g that
$\mathcal{X}$ = source alphabet = $\{0,1\}$

Definition - A parsing of a binary string $\underline{x} = (x_1, \ldots, x_n) \in \{0,1\}^n$ is a division of $\underline{x}$ into phrases seperated by commas.

Definition - A parsing is distinct if no two phrases are identical.

Lemma :- LZ parsing of a sequence is distinct.

Proof:- Obvious from the construction of codebook by taking shortest sequence not in here to fore.

Theorem :- $\{X_n\}_{n=1}^{\infty}$ in $\{0,1\}$ valued stationary, ergodic process generated by an unknown stationary ergodic probability measure P on $\{0,1\}^{\infty}$. The mean entropy rate $\bar{H}$ is

$$\bar{H} = \lim_n \frac{1}{n} H(X_1, \ldots, X_n) = \lim_n H(X_n \mid X^{n-1})$$

Then,

$$\lim_n \sup \frac{c_n(x^n)\left(\lceil \log_2 c_n(x^n) \rceil + 1\right)}{n} \cong \bar{H} \quad \text{in } P\text{-a.s.}$$

$\forall$ stationary ergodic P on $\{0,1\}^{\infty}$.

Proof : Cover · Thomas, theorem 13.5.2.

Reading Assignment

What's done here is Not a one-to-one mapping from references

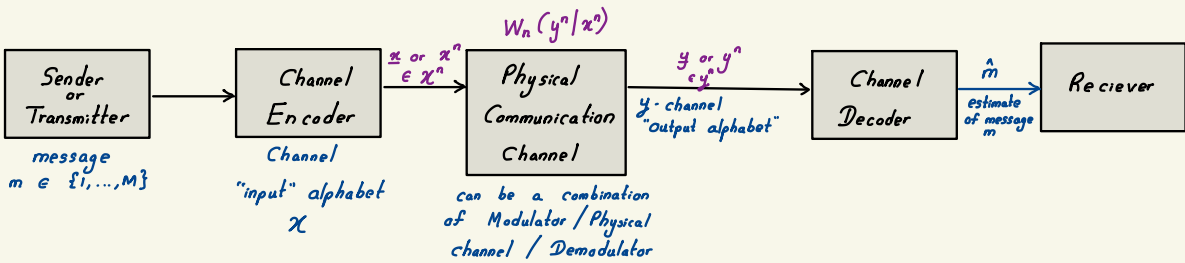
- Cover Thomas Chapter 7
- Csiszar-Korner chapter 2

Channel Coding and Channel Capacity

and DMCs

1) Introduction

A Schematic for a Point-to-Point Communication



- $\text{Rate} = \frac{\log_D M}{n}$ = D ary digits / Channel use
  = No. of information digits per channel use.
  = Encoding rate / Code rate

  Typically $D = 2$, hence R is in bits per channel use.

Objective- Make rate large, i.e make $(\log M/n)$ large while keeping probability of error $P(\hat{m} \neq m)$ to be small.

- This problem is solved in asymptotic sense. i.e for n large.
- Maximum rate at which this objective can be met is called "Capacity of Channe" or "Channel Capacity".
- Shall see that for a class of channels called "Discrete Memoryless Channels" (DMCs), this maximum rate, i.e. Channel Capacity is,

  $$\max \; I(X;Y)$$

  where $P_X = P$, $P_{Y|X}$, W
- Stochastic matrix $W = P_{Y|X}$ is used since to know $P_{Y|X}$ we need joint probabilities. But W is lesser and easier thing to know.

2) Definitions

$(\mathcal{X}, W, \mathcal{Y})$

Definition - Descrete Channel

A descrete channel, denoted by $(\mathcal{X}, W, \mathcal{Y})$ consists of a finite input alphabet $\mathcal{X}$, a finite output alphabet $\mathcal{Y}$, and a stochastic matrix $W: \mathcal{X} \longrightarrow \mathcal{Y}$ where $W(y|x)$, $x \in \mathcal{X}, y \in \mathcal{Y}$ is the probability that channel output is y if the input is x.

$(\mathcal{X}^n, W_n, \mathcal{Y}^n)$, $(\mathcal{X}, W_n, \mathcal{Y})$

Definition - n^{th} Extension of a Discrete Channel

A n^{th} extension of a descrete channel, denoted by $(\mathcal{X}^n, W_n, \mathcal{Y}^n)$

$n = 1, 2, \ldots$

$W_n : \mathcal{X}^n \longrightarrow \mathcal{Y}^n$ Stochastic Matrix

$W_n(y^n|x^n)$ = probability that channel output is y^n, given that input is x^n $\forall$ $x^n \in \mathcal{X}^n$, $y^n \in \mathcal{Y}^n$

- We abrivate the n^{th} extension as, $(\mathcal{X}^n, W_n, \mathcal{Y}^n) \longrightarrow (\mathcal{X}, W_n, \mathcal{Y})$
- Channel : Given $\mathcal{X}, \mathcal{Y}, \{W_n : \mathcal{X}^n \longrightarrow \mathcal{Y}^n\}_{n=1}^{\infty}$
- n^{th} extension for $n = 1, 2, \ldots$ of a discrete channel is also denoted by $\{(\mathcal{X}, W_n, \mathcal{Y})\}_{n=1}^{\infty} \Longleftrightarrow (\mathcal{X}, \{W_n\}_{n=1}^{\infty}, \mathcal{Y})$

$(\mathcal{X}, W, \mathcal{Y})$

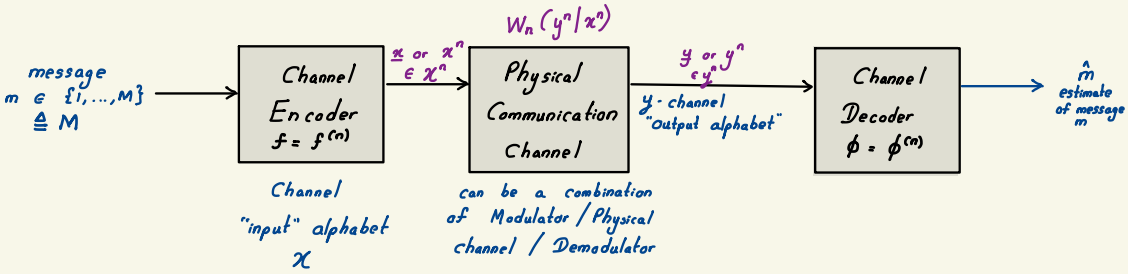
Definition - Discrete Memoryless Channel (DMC)

A Discrete channel $(\mathcal{X}, W, \mathcal{Y})$, $W: \mathcal{X} \longrightarrow \mathcal{Y}$ stochastic matrix, is a DMC if $\forall n^{th}$ extension, i.e. $\{(\mathcal{X}, W_n, \mathcal{Y})\}_{n=1}^{\infty}$ / $(\mathcal{X}, \{W_n\}_{n=1}^{\infty}, \mathcal{Y})$

$$W_n(y^n|x^n) = \prod_{t=1}^{n} W(y_t|x_t) \quad (*) \qquad \forall x^n \in \mathcal{X}^n,\ y^n \in \mathcal{Y}^n \quad \forall n \geq 1$$

- $W: \mathcal{X} \longrightarrow \mathcal{Y}$ in $(*)$ does not change with t.
- A single stochastic matrix $W: \mathcal{X} \longrightarrow \mathcal{Y}$ suffices to specify $W_n : \mathcal{X}^n \longrightarrow \mathcal{Y}^n$, $n \geq 1$ in $(*)$.

Schematic (for General W_n)



(M, n)

Definition - Block Code

An n-length block code for channel $(\mathcal{X}, W_n, \mathcal{Y})$ is (f, ϕ) where

$$f: f^{(n)}: M = M_f = (1, \dots, M = M_f) \longrightarrow \mathcal{X}^n$$
$$\phi: \phi^{(n)}: \mathcal{Y}^n \longrightarrow M' \quad (M' \geq M)$$
finite set

M = messages
$f = f^{(n)}$ = encoder
$\phi = \phi^{(n)}$ = decoder
$\{f(m), m \in M\} \subseteq \mathcal{X}^n$: Codebook
Elements of codebook : Codewards
Block code : (M, n)-code

e_m

Definition - Probability of Error

Given code (f, ϕ) and channel $(\mathcal{X}, W_n, \mathcal{Y})$, probability of error, given that message $m \in M$ was sent:

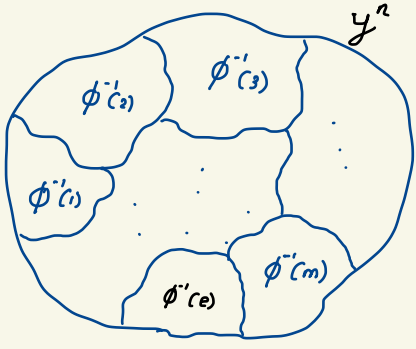
All the regions except $\phi^{-1}(m)$ region

e_m = probability of error given that message $m \in M$ was sent.

$$\triangleq W_n\left(\bigcup_{m' \neq m} \phi^{-1}(m') \,\middle|\, f(m)\right)$$
($\phi^{-1}(m')$: It's decoded to different region; $f(m)$: message m was sent)

$$= \sum_{m' \neq m} \underbrace{W_n\left(\phi^{-1}(m') \,\middle|\, f(m)\right)}_{\sum_{y^n \in \phi^{-1}(m')} W_n(y^n | f(m))}$$

$$= 1 - \underbrace{W_n\left(\phi^{-1}(m) \,\middle|\, f(m)\right)}_{\sum_{y^n \in \phi^{-1}(m)} W_n(y^n | f(m))}$$



$$\phi^{-1}(m) = \{y^n \in \mathcal{Y}^n \mid \phi(y^n) = m\} = \text{Decoding region for message } m.$$

Definition - Maximum Probability of Decoding Error

$$e = e(W_n, f, \phi) \triangleq \max_{m \in M} e_m$$

Definition - Average Probability of Error

$$\bar{e} = \bar{e}(W_n, f, \phi) \triangleq \frac{1}{M} \sum_{m=1}^{M} e_m$$

$$= P(\phi(Y^n) \neq \tilde{M}) \quad \text{? Not sure}$$

Remarks

1) Obviously, for a given code (f, ϕ)

$$\bar{e}(W_n, f, \phi) \leq e(W_n, f, \phi)$$

2) Shall see for a DMC, a code (f, ϕ) with small $\bar{e}$, leads to a code $(\tilde{f}, \tilde{\phi})$ of "effectively the same rate" with small e.

($\bar{e}$: average error; e: maximum error)

Definition - Rate of Code - R

Given a n length block code (f, ϕ) i.e. a (M, n) code for $(\mathcal{X}, W_n, \mathcal{Y})$

$$\text{The rate of the code} = \frac{\log_2 M}{n}$$

bits / Channel transmission or channel use

Definition - Achievable Rate

A non-negative number R is an achievable rate for channel $\{\mathcal{X}, \{W_n\}_{n=1}^{\infty}, \mathcal{Y}\}$ if $\exists$ a sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes, such that

$$\lim_n e = \lim_n e_n = 0$$

What is the difference of e vs e_n? Different lengths of the block?

Remark: If $R > 0$ is achievable, then $R' < R$ is also achievable.

Definition: Capacity of a Channel

Capacity of a channel $(\mathcal{X}, \{W_n\}_{n=1}^{\infty}, \mathcal{Y})$ is the supremum of all achievable rates, and denoted by C.

Operational Meaning / Significance of Capacity

- The Block Codes (M, n) of rate less than capacity exist, where maximum error probability is arbitarily small for large block lengths n.
- Hereafter shall focus on DMCs.

3) Preview of Channel Coding Theorem

Theorem - Shannon's Channel Coding Theorem

Let $(\mathcal{X}, W, \mathcal{Y})$ be a DMC,

i) Achievability : For any rate $R < \max_{P: P_X = P} I(X \wedge Y)$ there exist a sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes with

$$\lim_n e = \lim_n e_n = 0$$

(e = maximum probability of error)

i.e. Every rate below $\max_{P: P_X = P} I(X \wedge Y)$ is achievable.

ii) Converse : Any sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes with

$$\lim_n e = \lim_n e_n = 0$$

must have a rate $R \leq \max_{P: P_X = P} I(X \wedge Y)$

Remark :- The capacity of the DMC channel with W $= C = C(W) = \max_{P: P = P_X} I(X \wedge Y) = \max_P I(P, W)$

(We have no control over W - specified by channel)

Properties of $\max_P I(X \wedge Y) = \max_P I(P, W)$

1) $I(X \wedge Y) = I(P, W)$ is a concave function of P and the maximum is over all pmfs P on finite set $\mathcal{X}$, which is a closed, convex set (a simplex as well). Hence maximum is achieved,

i.e. pmf P^* on $\mathcal{X}$ with $I(P^*, W) \geq I(P, W) \quad \forall P$ on $\mathcal{X}$

2) $\max_{P: P_X = P} I(X \wedge Y) \geq 0 \quad \ldots \because I(X \wedge Y) \geq 0 \quad \forall P$ on $\mathcal{X}$

3) $\max_{P: P_X = P} I(X \wedge Y) \leq \min \{ \log |\mathcal{X}|, \log |\mathcal{Y}| \} < \infty$

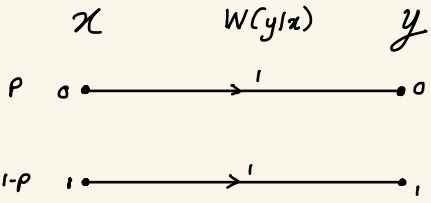
Proof :- Since $\max_{P: P_X = P} I(X \wedge Y) \leq \max_{P: P_X = P} H(X) = \log |\mathcal{X}|$

$\max_{P: P_X = P} I(X \wedge Y) \leq \max_{P: P_X = P} H(Y) \leq \log |\mathcal{X}|$

On $\mathcal{X}$ we know X can have uniform distribution to achieve $\log |\mathcal{X}|$ maximum. But we do not know Y has a Uniform distribution on $\mathcal{X}$.

4) Examples of Channel Capacity

1) Noiseless Binary Channel



$P_X(0) = 1-p = 1 - P_X(1)$

$P_Y(0) = 1-p = 1 - P_Y(1)$

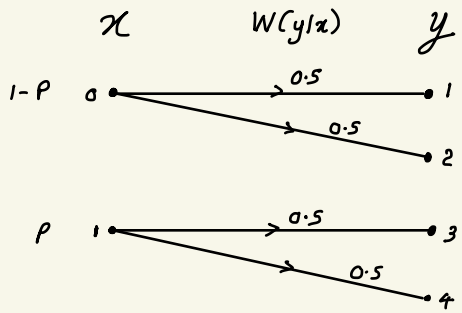
$I(X \wedge Y) = H(Y) - H(Y|X) = H(Y) = h(p)$

(0 — When X is known, Y is fixed)

$C = \max_p I(X \wedge Y) = \max_{0 \le p \le 1} h(p) = h(0.5) = 1$

- Hence, best code rate = 1 bit / channel use while having 0 error probability.
- No coding is needed. i.e. no redundancy is needed.
- C = 1 bit / channel use for p = (0.5, 0.5)

2) Noisy Channel With Non-overlapping Outputs



2 inputs, 4 outputs

$$I(X \wedge Y) = H(X) - H(X|Y) = H(X) = h(\rho)$$

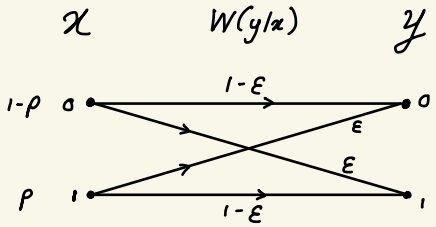
When Y is given, X is known

$$C = \max_{\rho} I(X \wedge Y) = \max_{0 \le \rho \le 1} h(\rho) = h(0.5) = 1$$

$$= \underline{1 \text{ bit / channel use}}$$

- No coding is needed. i.e. no redundancy is needed. with bit error probability 0.

3) Binary Symmetric Channel (BSC)



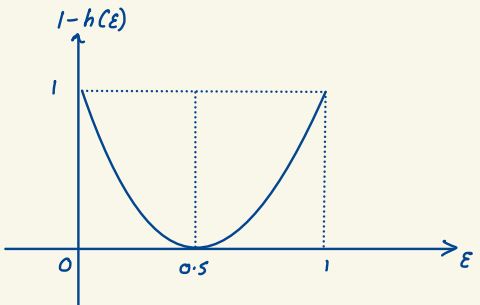
- ε - "Crossover probability". Since it's a probability $0 \le \varepsilon \le 1$. Assume w.l.o.g $0 \le \varepsilon \le 1/2$.

- Shall see that, can still transmit information at positive rate < capacity for arbitarily small error probability with suitable coding.

$$I(X \wedge Y) = H(Y) - H(Y|X)$$
$$= H(Y) - \sum_{x=0,1} p(x) \cdot \underbrace{H(Y|X=x)}_{h(\varepsilon)}$$
$$= H(Y) - h(\varepsilon)$$

$$C = \max_{\rho} I(X \wedge Y) = \max_{\rho} H(Y) - h(\varepsilon) \le 1 - h(\varepsilon)$$

- The upperbound is achieved by $P = (1-\rho, \rho) = (0.5, 0.5)$



Observations

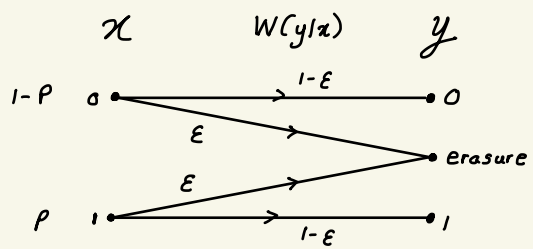
1) A good "capacity achieving code" ($R \lesssim C$, $e_n = 0$ for large n) will have roughly equal number of 0's and 1's.

2) $\varepsilon = 0$ or $1 \implies \max_p I(X \wedge Y) = 1$ bit $\implies$ noiseless channel.

3) $\varepsilon = 0.5 \implies h(\varepsilon) = 1 \implies C = \max_p I(X \wedge Y) = 0$
$\implies X \perp\!\!\!\perp Y$ for every P on $\mathcal{X}$.

4) For $\varepsilon \neq 0$ or 0.5 or 1 (ex:- $0 < \varepsilon < 0.5$), coding is necessary to attain capacity, as otherwise sending unencoded bits will lead to error probability $= \varepsilon$ that is too high.

4) Binary Erasure Channel (BEC)



2 inputs, 3 outputs

- Reciever is assumed to know the erased bits, including their locations.

$$I(X \wedge Y) = H(Y) - H(Y|X)$$
$$= H(Y) - \sum_{x=0,1} p(x) \cdot \underbrace{H(Y|X=x)}_{h(\varepsilon)}$$
$$= H(Y) - h(\varepsilon) \quad \ldots\ldots (1)$$

$$C = \max_p I(X \wedge Y) = \max_p \left(H(Y) - h(\varepsilon)\right) = \max_p H(Y) - h(\varepsilon)$$

Exercise:- $H(Y)$ attains it's maximum when $p = (0.5, 0.5)$

Solution 1

$$P(Y=0) = P(Y=0|X=0)\cdot P(X=0) + \cancel{P(Y=0|X=1)}^{0}\cdot P(X=1)$$
$$= (1-\varepsilon)\cdot(1-p)$$

$$P(Y=1) = \cancel{P(Y=1|X=0)}^{0}\cdot P(X=0) + P(Y=1|X=1)\cdot P(X=1)$$
$$= (1-\varepsilon)\cdot p$$

$$P(Y=e) = P(Y=e|X=0)\cdot P(X=0) + P(Y=e|X=1)\cdot P(X=1)$$
$$= \varepsilon\cdot p + \varepsilon\cdot(1-p)$$
$$= \varepsilon \quad \cdots \quad \cdots \quad (2)$$

$$\Rightarrow H(Y) = \sum P_Y(y)\cdot \log\frac{1}{P_Y(y)}$$
$$= (1-\varepsilon)(1-p)\cdot\log\frac{1}{(1-\varepsilon)(1-p)} + (1-\varepsilon)p\log\frac{1}{(1-\varepsilon)p} + \varepsilon\log\frac{1}{\varepsilon}$$

$$\Rightarrow \frac{d}{dp}H(Y) = (1-\varepsilon)\cdot\log\frac{1}{(1-\varepsilon)(1-p)}(-1) + (1-\varepsilon)\cdot(1-p)\cdot(1-\varepsilon)(1-p)(1-\varepsilon)(-1)$$
$$+ (1-\varepsilon)\cdot p\cdot(1-\varepsilon)\cdot p\,(1-\varepsilon)\,1 + (1-\varepsilon)\cdot\log\frac{1}{(1-\varepsilon)p}\cdot(1)$$
$$= (1-\varepsilon)\cdot\left[\log(1-\varepsilon)(1-p)\cdot + \log\frac{1}{(1-\varepsilon)p}\right] + (1-\varepsilon)^3\left[p^2-(1-p)^2\right]$$
$$= (1-\varepsilon)\cdot\log\left(\frac{1-p}{p}\right) + (1-\varepsilon)^3(2p-1) = 0$$

At the global maximum (which exists) $\Rightarrow p = 1/2$ ∎

Solution 2 Consider the indicator function $E \triangleq \mathbb{1}(Y=e) = \begin{cases} 1 & y=e \\ 0 & y\in\{0,1\} \end{cases}$

$$H(E) = h\left(P(Y=e)\right) = h(\varepsilon) \quad \cdots \text{ from } (2) \cdots (3)$$

$$H(Y|E) = \cancel{H(Y|E=1)}^{0}\cdot P(E=1) + H(Y|E=0)\cdot \underbrace{P(E=0)}_{1-\varepsilon}$$

(E=1 ⇒ error occured, so Y=e is fixed)

$$= H(Y|E=0)(1-\varepsilon) \quad \cdots (4)$$

$$H(Y,E) = H(Y) + \cancel{H(E|Y)}^{0} \quad \text{(when Y is given we know if error or not)}$$
$$= H(E) + H(Y|E)$$

from (3) and (4)

$$\Rightarrow H(Y) = H(E) + H(Y|E) = h(\varepsilon) + H(Y|E=0)(1-\varepsilon)$$

$$\Rightarrow H(Y) - h(\varepsilon) = (1-\varepsilon)H(Y|E=0)$$

Using above result on (1)

max of this can also be found by differentiat.. and setting to '0'.

$$\Rightarrow I(X \wedge Y) = H(Y) - h(\varepsilon) = (1-\varepsilon) \cdot H(Y/E=0) \quad \cdots (5)$$

To compute $H(Y \mid E=0)$

when Y=1 no error has occured so prob. 1

$$P(Y=1 \mid E=0) = \frac{P(Y=1) \cdot P(E=0 \mid Y=1)}{P(E=0)} = \frac{p(1-\varepsilon) \cdot 1}{1-\varepsilon} = p$$

$P(Y=1)$: $P(Y=0) + P(Y=1) = (1-\varepsilon)p + (1-\varepsilon)(1-p)$

$$P(Y=0 \mid E=0) = 1 - P(Y=1 \mid E=0) = 1-p$$

$$\Rightarrow H(Y \mid E=0) = h(p)$$

Hence from (5), $I(X \wedge Y) = (1-\varepsilon) \cdot h(p)$

$$\Rightarrow C = \max_{0 \le p \le 1} I(X \wedge Y)$$

$$= (1-\varepsilon) \max_{0 \le p \le 1} h(p)$$

$$= \underline{\underline{(1-\varepsilon)}} \quad \text{when } p = (0.5,\ 0.5)$$

- Heuristic meaning : Since ε fraction of the input bits are erased at the output, at most $(1-\varepsilon)$ - fraction of the bits are recieved without error

5) Symmetric Channels

Definition - Symmetric Channels

$\{W(y|x),\ x \in \mathcal{X},\ y \in \mathcal{Y}\}$ is a symmetric channel if it has the following properties,

1) $\{W(\cdot|x),\ x \in \mathcal{X}\}$ are permutations of each other, i.e. the raws are permutations of each other.

2) $\{W(y|\cdot),\ x \in \mathcal{X}\}$ are permutations of each other, i.e. the columns are permutations of each other.

Examples

1)

x \ y	0	1	2
0	0.3	0.2	0.5
1	0.2	0.5	0.3
2	0.5	0.3	0.2

2)

x \ y	0	1	2
0	$1/3$	$1/3$	$1/3$
1	$1/3$	$1/3$	$1/3$

We can interchange $x = 0, 1$ without effect, hence in this channel $X \perp\!\!\!\perp Y$.
$\therefore I(X,Y) = 0 \Rightarrow C = 0$

- If W is symmetric. The rows sum to 1, and the column sums are the same.

6) Weakly Symmetric Channels

Definition - Weakly Symmetric Channel

$\{W(y|x), x \in \mathcal{X}, y \in \mathcal{Y}\}$ is a weakly symmetric channel if it has the following properties,

1) $\{W(\cdot|x), x \in \mathcal{X}\}$ are permutations of each other, i.e. the raws are permutations of each other.

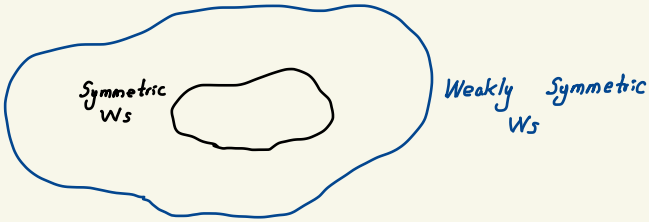
2) Column sums, i.e. $\sum_x W(y|x)$ are the same $\forall y \in \mathcal{Y}$

- W Symmetric $\Rightarrow$ W weakly symmetric, but $\not\Leftarrow$

-

x \ y	0	1	2
0	$1/3$	$1/6$	$1/2$
1	$1/3$	$1/2$	$1/6$

Weakly symmetric since every column sums to $1/4$. But not symmetric since columns are not permutations of fixed set of probabilities.



Theorem - If $\{W(y|x),\ x \in \mathcal{X},\ y \in \mathcal{Y}\}$ is a weakly symmetric channel, then

$$\max_{P} I(X \wedge Y) = \log |\mathcal{Y}| - H(\text{row of } W)$$

and is achieved when input pmf P is uniform on $\mathcal{X}$.

Proof:- $I(X \wedge Y) = H(Y) - H(Y|X)$

Since rows are permutations of each other, entropy per raw is constant

$= H(Y) - H(\text{raw of } W)$

$\leq \log |\mathcal{Y}| - H(\text{raw of } W)$ ..... upperbound for entropy

for equality Y should be uniform,

P_X is uniform on P $\Longrightarrow$ $P_X(x) = 1/|\mathcal{X}|$

$\Longrightarrow$ $P_Y(y) = \sum_x P_X(x) \cdot W(y|x) = \sum_x \frac{1}{|\mathcal{X}|} W(y|x)$

$\Longrightarrow$ $= \frac{1}{|\mathcal{X}|} \underbrace{\sum_x W(y|x)}$

Due to week symmetry every column add up to a constant

$\Longrightarrow$ $= \frac{1}{|\mathcal{X}|} \cdot c$

Constant w.r.t y

$\Rightarrow$ $= \frac{1}{|\mathcal{Y}|}$

$\Rightarrow P_Y$ is uniformly distributed

$\therefore$ for equality P_X being uniform is sufficient.

■

5) Back to Channel Coding Theorem

Converse / Weak Converse

Theorem - Shannon's Channel Coding Theorem

Let $(\mathcal{X}, W, \mathcal{Y})$ be a DMC,

i) Achievability : For any rate $R < \max_{P: P_X = P} I(X \wedge Y)$ there exist a sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes with

$$\lim_n e = \lim_n e_n = 0$$

i.e. Every rate below $\max_{P: P_X = P} I(X \wedge Y)$ is achievable.

ii) Converse : Any sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes with

$$\lim_n e = \lim_n e_n = 0$$

must have a rate $R \leq \max_{P: P_X = P} I(X \wedge Y)$

- The above converse has two versions of it, depending on the $\lim_n e_n = \lim_n$ (max error probability).
- Weak Converse - Any sequence of $(\lfloor 2^{nR} \rfloor, n)$ - codes with rate $R > \max I(X \wedge Y)$

$$\liminf_{n \to \infty} e > 0$$

- Strong Converse - Any sequence of $(\lfloor 2^{nR} \rfloor, n)$ - codes with rate $R > \max I(X \wedge Y)$

$$\liminf_{n \to \infty} e = 1$$

- This course covers only weak converse, using Fano. Strong converse needs sharper tools like divergence bound, information spectrum.

Setting

- Consider any n-length block code (f, ϕ) where, '0' denotes the error case.

$$f = f^{(n)} : \{1, \ldots, M\} \longrightarrow \mathcal{X}^n$$
$$\phi = \phi^{(n)} : \mathcal{Y}^n \longrightarrow \{0, 1, \ldots, M\}$$

- Consider the DMC has the stochastic matrix W,

$$W : \mathcal{X} \longrightarrow \mathcal{Y}$$

- Assume the average error probability of this DMC is,

$$\bar{e}(f, \phi) = \varepsilon \quad \cdots (1)$$

Definition - Empirical p.m.f of Codebook - μ

$$\mu(x) = \frac{1}{M} \sum_{m=1}^{M} \left[\frac{1}{n} \sum_{t=1}^{n} \mathbb{1}\left((f(m))_t = x \right) \right] \quad x \in \mathcal{X} \quad \cdots (2)$$

- μ = is a pmf on $\mathcal{X}$
  = is 'empirical' pmf of the codebook
  = is average 'type' of the codewards.

- The inner summation in the above definition is the average no. of occurances of a specific symbol 'x' in a specific codeword '$f(m)$'. We call this quantity $Q_{f(m)}(x)$

$$Q_{f(m)}(x) = \frac{1}{n} \sum_{t=1}^{n} \mathbb{1}\left((f(m))_t = x \right)$$

Definition - Message Random Variable - J

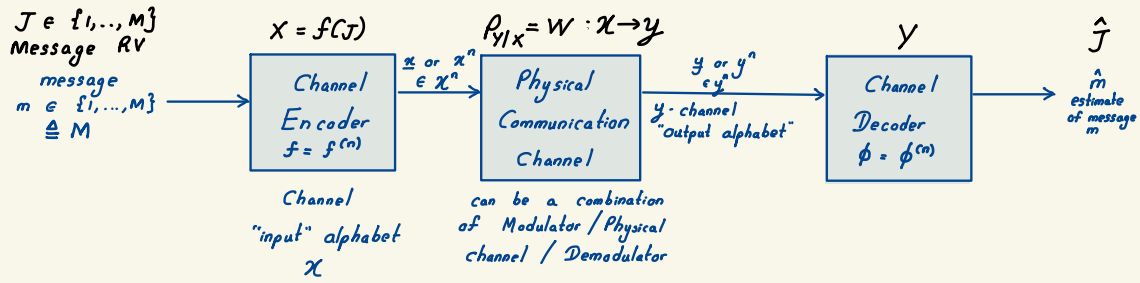
J is a Random Variable taking values in the message set $\{1, 2, \ldots, M\}$

- Coding is about transmitting one of M messages. Turning it to rv allows us to use the Entropy, Mutual information and Fano's inequality.

- From the upper bound on entropy $\Longrightarrow \log M \geq H(J)$ ..... ①
  From the definition of rate on n bolocks $\Rightarrow R = \frac{\text{bits}}{\text{channel use}} = \frac{\log M}{n}$
  $\Rightarrow nR = \log M \geq H(J)$
  $\Rightarrow R \geq H(J)/n$ ..... ②

- In the converse of Coding Theorem we are looking for a lower bound of R. If so it make sense to consider the case where R is suppose to be lowest (i.e. the worst case.) and find a lower bound for it.
  $\Rightarrow$ Worst case R (i.e. lowes R)
  $\Rightarrow$ equality condition in ②
  $\Rightarrow$ equality condition in ①
  $\Rightarrow$ J is uniform.

Assumption : $J \sim \text{Uniform}\{1, 2, \ldots, M\} \Longrightarrow P(J=m) = \frac{1}{M} \Longrightarrow H(J) = \log M$ ..... ③



- $X \triangleq f(J)$ - f-encoder
- Y = $\mathcal{Y}$-valued rv connected to X by stochastic matrix $W: \mathcal{X} \longrightarrow \mathcal{Y}$.
- $P_{Y|X} = W$
- Since channel output Y depends only on X, but not the message distribution J,

$$J \multimap X \multimap Y \quad \ldots\ldots \quad ④$$

Lemma 1- For a DMC

$$I(\underline{X} \wedge \underline{Y}) \leq \sum_{t=1}^{n} I(X_t \wedge Y_t)$$

Proof: By the chain rule of mutual information,

$$I(X^n \wedge Y^n) = I(X^n \wedge Y_1) + \sum_{t=2}^{n} I(X^n \wedge Y_t \mid Y^{t-1})$$

$$= I(X^n \wedge Y_1) + \sum_{t=2}^{n} \left[H(Y_t \mid Y^{t-1}) - H(Y_t \mid X^n, Y^{t-1}) \right]$$

$I(X,Y) \wedge Z) \geq I(X \wedge Z)$

Conditioning reduces entropy

DMC. Memoriless. Current output depend only on current input $P_{Y_t \mid X^n, Y^{t-1}} = P_{Y_t \mid X_t}$

$$\leq I(X_1 \wedge Y_1) + \sum_{t=2}^{n} \left[H(Y_t) - H(Y_t \mid X_t) \right]$$

$$= I(X_1 \wedge Y_1) + \sum_{t=2}^{n} I(X_t \wedge Y_t)$$

$$= \sum_{t=1}^{n} I(X_t \wedge Y_t)$$

■

Lemma 2- $\frac{1}{n} \sum_{t=1}^{n} P_{(f(J))_t}(x) = \mu(x) \quad \forall\, x \in \mathcal{X}$

$$\frac{1}{n} \sum_{t=1}^{n} P_{(f(J))_t}(x)$$

$$= \frac{1}{n} \sum_{t=1}^{n} P\left((f(J))_t = x\right)$$

$$= \frac{1}{n} \sum_{t=1}^{n} \left[\sum_{m=1}^{M} P\left((f(J))_t = x \mid J = m\right) \cdot \underbrace{P(J=m)}_{1/M \text{ from } ③} \right] \quad \cdots \text{ law of total probability}$$

$$= \frac{1}{n} \sum_{t=1}^{n} \left[\frac{1}{M} \sum_{m=1}^{M} P\left((f(J))_t = x \mid J = m\right) \right]$$

$$= \frac{1}{n} \sum_{t=1}^{n} \left[\frac{1}{M} \sum_{m=1}^{M} P\left((f(m))_t = x\right) \right]$$

$$= \frac{1}{n} \sum_{t=1}^{n} \frac{1}{M} \sum_{m=1}^{M} \mathbb{1}\left((f(m))_t = x\right)$$

$$= \frac{1}{M} \sum_{m=1}^{M} \frac{1}{n} \sum_{t=1}^{n} \mathbb{1}\left((f(m))_t = x\right)$$

$$= \mu(x) \quad \cdots \text{ from definition of } \mu(x) \text{ in } ②$$

■

Lemma 3 - For any sequence of $(\lfloor 2^{nR} \rfloor, n)$ - codes

$$\frac{1}{n} \log M \leq \frac{1}{(1-\varepsilon)} \left[I(\mu, W) + \frac{1}{n} \right]$$

Proof:-

Single - Shot ($n=1$) Case

- For $n=1$ case, ① becomes,

$$\Rightarrow P(\phi(Y) \neq J) = \varepsilon \quad \cdots \cdots ⑤$$

- Since $J \leftrightarrow Y \leftrightarrow \phi(Y)$ forms a Markov chain from Data Processing Inequality,

$$\Rightarrow P(J|Y) \leq P(J|\phi(Y)) \quad \cdots \cdots ⑥$$

- $\Rightarrow I(J, X \wedge Y) = I(J, f(J) \wedge Y) \quad \cdots\cdots\cdots\cdots x = f(J)$

$\Rightarrow I(X \wedge Y) + \underbrace{I(J \wedge Y | X)}_{=0 \because \text{from ④ and MC property}} = I(J \wedge Y) + \underbrace{I(f(J) \wedge Y | J)}_{=0 \text{ since } J \text{ is known}}$

$$\Rightarrow I(X \wedge Y) = I(J \wedge Y) \quad \cdots\cdots ⑦$$

- From Fano's Inequality taking $X \to J$ and $Y \to \phi(Y)$

$$\Rightarrow H(J|\phi(Y)) \leq \underbrace{P(J \neq \phi(Y))}_{=\varepsilon \text{ from } ⑤} \cdot \log \underbrace{|J|}_{=M \text{ from definition}} + h\left(\underbrace{P(J \neq \phi(Y))}_{=\varepsilon \text{ from } ⑤}\right)$$

$$\Rightarrow H(J|\phi(Y)) \leq \varepsilon \cdot \log M + \underbrace{h(\varepsilon)}_{h(\varepsilon) \leq 1}$$

$$\Rightarrow H(J|Y) \leq \varepsilon \log M + 1 \quad \cdots\cdots\cdots \because \text{from ⑥ and } h(\varepsilon) \leq 1$$

$$\Rightarrow \underbrace{H(J) - H(J|Y)}_{} \geq H(J) - \varepsilon \cdot \log M - 1$$

$$\Rightarrow I(J \wedge Y) \geq \log M - \varepsilon \cdot \log M - 1 \quad \cdots\cdots\cdots \because \text{from assumption ③}$$

$$\Rightarrow I(J \wedge Y) \geq (1-\varepsilon) \cdot \log M - 1$$

$$\Rightarrow \log M \leq \frac{1}{1-\varepsilon} [I(J \wedge Y) + 1]$$

$$\Rightarrow \log M \leq \frac{1}{1-\varepsilon} \underbrace{[I(X \wedge Y) + 1]}_{= I(\mu, W) \text{ when } n=1} \quad \cdots\cdots \because ⑦ \cdots ⑧$$

A usecase of this inequality } - ENEE 621, in hypothesis testing problem with M, hypothesies. X - observation Y - hypothesis

General $n \geq 1$

- We replace X and Y, by $\underline{X}$ and $\underline{Y}$ which are n letter versions,

$$X \longrightarrow \underline{X} = f(J) = f^{(n)}(J) \quad \in \mathcal{X}^n$$
$$Y \longrightarrow \underline{Y} \in \mathcal{Y}^n$$

- These $\underline{X}$ and $\underline{Y}$ are connected by DMC 'W', i.e.

$$P_{\underline{Y}|\underline{X}} = W^n$$

- The Markov Chain in ④ becomes,

$$J \multimap \underline{X} \multimap \underline{Y} \quad \cdots\cdots \text{④@}$$

- The property ①/⑤ becomes,

$$P(\phi(\underline{Y}) \neq J) = \varepsilon \quad \cdots\cdots \text{⑤@}$$

- Then property ⑧ becomes,

$$\frac{\log M}{n} \leq \frac{1}{1-\varepsilon}\left[\frac{1}{n} I(\underline{X} \wedge \underline{Y}) + \frac{1}{n}\right] \quad \cdots\cdots \text{⑧@}$$

- Next by the DMC property in Lemma 1,

We didn't need the DMC property so far.

$$\Rightarrow \frac{1}{n} I(\underline{X} \wedge \underline{Y}) \leq \frac{1}{n} \sum_{t=1}^{n} I(X_t \wedge Y_t)$$
$$= \frac{1}{n} \sum_{t=1}^{n} I\left((f(J))_t \wedge Y_t\right)$$
$$= \frac{1}{n} \sum_{t=1}^{n} I\left(P_{(f(J))_t} \wedge W\right) \quad \text{why ?}$$
$$\leq I\left(\frac{1}{n}\sum_{t=1}^{n} P_{(f(J))_t}, W\right) \quad \cdots\cdots \text{⑨}$$

From concavity of mutual info

From ⑨ and ⑧@,

$$\Rightarrow \frac{\log M}{n} \leq \frac{1}{1-\varepsilon}\left[I\Big(\underbrace{\frac{1}{n}\sum_{t=1}^{n} P_{(f(J))_t}}_{\mu(x)}, W\Big) + \frac{1}{n}\right]$$

Substituting from Lemma 2. to above,

$$\Rightarrow \frac{\log M}{n} \leq \frac{1}{1-\varepsilon}\left[I(\mu, W) + \frac{1}{n}\right]$$

Converse : of Shanon's Channel Coding Theorem

Any sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes with $\lim_n e = \lim_n e_n = 0$ must have a rate $R \leq \max_{P: P_x = P} I(X \wedge Y)$

Proof :- Consider we have a $(\lfloor 2^{nR} \rfloor, n) = (M, n)$ - code with $\lim_n e = 0$.

Then,

$$\frac{\log M}{n} = \frac{\log \lfloor 2^{nR} \rfloor}{n}$$

$$\geq \frac{\log (2^{nR} - 1)}{n}$$

$$\geq \frac{\log (2^{nR}) - 1}{n} \quad \ldots\ldots \because \log(x-1) \geq \log x - 1 \;\forall\; x \geq 2$$

$$= R - 1/n \quad \ldots\ldots \textcircled{1}$$

For such a code, from Lemma 3 we have that,

$$\Rightarrow \frac{1}{(1-\varepsilon)} \left[I(\mu, W) + \frac{1}{n} \right] \geq \frac{1}{n} \log M$$

Since ε is average error and e_n is max error when ε is very small $e_n \simeq \varepsilon$?

$$\Rightarrow \frac{1}{(1-e_n)} \left[I(\mu, W) + \frac{1}{n} \right] \geq R - 1/n \quad \ldots \because \textcircled{1}$$

take $\lim_{n \to \infty}$

$$\Rightarrow \frac{1}{1 - \lim_n e_n} \left[I(\mu, W) + \underbrace{\lim_n 1/n}_{0} \right] \geq R - \underbrace{\lim_n 1/n}_{0}$$

$$\Rightarrow \frac{1}{1-0} \left[I(\mu, W) + 0 \right] \geq R - 0$$

$$\Rightarrow \max_{P: P = P_X} I(P, W) \geq I(\mu, W) \geq R$$

Why ?

■

Read Cover Thomas Sections 7.6 / 7.7

6) Jointly Typical Sequences

Definition - Jointly Typical Sequences - $A_\varepsilon^{(n)}$

Given $\varepsilon > 0$, then ε-Jointly Typical Set (Jointly Typical Set, Set of Jointly Typical Sequences) $A_\varepsilon^{(n)} \subseteq \mathcal{X}^n \times \mathcal{Y}^n$ with respect to p_{xy} on $\mathcal{X} \times \mathcal{Y}$ is defined as,

$$A_\varepsilon^{(n)} = \left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n \;\middle|\; \begin{array}{l} \left| -\frac{1}{n} \cdot \log p_{x^n}(x^n) - H(X) \right| < \varepsilon, \\ \left| -\frac{1}{n} \cdot \log p_{y^n}(y^n) - H(Y) \right| < \varepsilon, \\ \left| -\frac{1}{n} \cdot \log p_{x^n y^n}(x^n, y^n) - H(X,Y) \right| < \varepsilon, \end{array} \right\}$$

$H(X)$ → $H(p_X)$; $H(Y)$ → $H(p_Y)$; $H(X,Y)$ → $H(p_{XY})$

where $\forall\ x^n \in \mathcal{X}^n \quad y^n \in \mathcal{Y}^n$

$$p_{x^n y^n}(x^n, y^n) = \prod_{t=1}^{n} p_{xy}(x_t, y_t)$$

Theorem - Joint Assymptotic Equipartition Property (AEP)

$\{X_t, Y_t\}_{t=1}^{\infty}$ is a $\mathcal{X} \times \mathcal{Y}$ valued i.i.d process with p.m.f. p_{xy} on $\mathcal{X} \times \mathcal{Y}$. Then given $\varepsilon > 0$

I) $\lim_{n \to \infty} P\left((X^n, Y^n) \in A_\varepsilon^{(n)}\right) = 1$

II) $\left|A_\varepsilon^{(n)}\right| \leq 2^{n(H(X,Y)+\varepsilon)} \quad \forall\ n$

III) $\left|A_\varepsilon^{(n)}\right| \geq (1-\varepsilon) \cdot 2^{n(H(X,Y)-\varepsilon)} \quad \forall\ n$ large.

IV) If $\{\tilde{X}_t, \tilde{Y}_t\}_{t=1}^{\infty}$ is a $\mathcal{X} \times \mathcal{Y}$ valued i.i.d process with p.m.f $p_X p_Y$ on $\mathcal{X} \times \mathcal{Y}$. (i.e. $\tilde{X}^n$ and $\tilde{Y}^n$ are $\perp\!\!\!\perp$, $n = 1, 2, \ldots$ with marginal p.m.fs p_{X^n} and p_{Y^n} respectively.) then,

$$P_{\tilde{X}\tilde{Y}}\left((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\right) \leq 2^{-n[I(X;Y)-3\varepsilon]} \quad \forall\ n$$

$$P_{\tilde{X}\tilde{Y}}\left((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\right) \geq (1-\varepsilon) \cdot 2^{-n[I(X;Y)+3\varepsilon]} \quad \forall\ n \text{ sufficiently large}$$

$P_{\tilde{X}\tilde{Y}} \triangleq p_X p_Y$ — New process is the independant version of the inputs and outputs of old version

Typical set under p_{XY} old process

$D(p_{XY} \| p_X p_Y)$ = $D(p_{XY} \| P_{\tilde{X}\tilde{Y}})$ = Divergence between old and new process

A simple version of Large Deviation Principle from Statistics.

Proofs

I)

Claim :- If $\{X_t, Y_t\}_{t=1}^{\infty}$ is i.i.d $\Longrightarrow$ $\{X_t\}_{t=1}^{\infty}$ is i.i.d

Need all 3 properties in the definition of AEP

Proof :- $\{X_t, Y_t\}_{t=1}^{\infty}$ is i.i.d

$$\Longrightarrow p_{x^n y^n}(x^n, y^n) = \prod_{t=1}^{n} p_{xy}(x_t, y_t)$$

$$\Longrightarrow p_{x^n}(x^n) = \sum_{y^n} p_{x^n y^n}(x^n, y^n) = \sum_{y^n} \prod_{t=1}^{n} p_{xy}(x_t, y_t) \quad \cdots \text{ marginalize}$$

$$= \prod_{t=1}^{n} \sum_{y_t} p_{xy}(x_t, y_t) \quad \cdots \text{ seperate product across } t$$

$$= \prod_{t=1}^{n} p_x(x_t)$$

$\Rightarrow \{X_t\}_{t=1}^{\infty}$ is i.i.d ∎

Claim :- $-\frac{1}{n} \log p_{x^n}(x^n) \xrightarrow{p} H(X)$

Proof :- Let $z_t = -\log p_x(x_t)$ and $\mu_n = \frac{1}{n} \sum_{t=1}^{n} z_t$

$$\Longrightarrow \mathbb{E}[\mu_n] = \mathbb{E}\left[\frac{1}{n} \sum_{t=1}^{n} z_t\right]$$

$$= \frac{1}{n} \cdot n \cdot \mathbb{E}[z_t] \quad \cdots z_t\text{'s are also i.i.d.}$$

$$= \mathbb{E}[z_t]$$

$$= \mathbb{E}[-\log p_x(x)] \quad \cdots X_t = X \text{ due to i.i.d}$$

$$= H(X) \quad \cdots (1)$$

$\{X_t, Y_t\}_{t=1}^{\infty}$ is i.i.d $\Longrightarrow$ $\{X_t\}_{t=1}^{\infty}$ is i.i.d

$$\Longrightarrow p_{x^n}(x^n) = \prod_{t=1}^{n} p_x(x_t)$$

$$\Longrightarrow -\frac{1}{n} \log p_{x^n}(x^n) = -\frac{1}{n} \sum_{t=1}^{n} \log p_x(x_t)$$

$$= \mu_n \quad \cdots (2)$$

From WLLN $\Rightarrow \mu_n - \mathbb{E}[\mu_n] \xrightarrow{p} 0$

$\Rightarrow -\frac{1}{n} \cdot \log p_{x^n}(x^n) - H(X) \xrightarrow{p} 0$ .. from (1), (2)

$\Rightarrow -\frac{1}{n} \cdot \log p_{x^n}(x^n) \xrightarrow{p} H(X)$

∎

$$\lim_n \left(-\frac{1}{n} \log P_{X^n}(x^n) \right) = H(X) \quad \text{in prob.}$$

$$\lim_n \left(-\frac{1}{n} \log P_{Y^n}(y^n) \right) = H(Y) \quad \text{in prob.}$$

$$\lim_n \left(-\frac{1}{n} \log P_{X^nY^n}(x^n y^n) \right) = H(X,Y) \quad \text{in prob.}$$

- Thus, from definition of limit and convergence in probability, $\forall \varepsilon$ and $\forall \delta (= \varepsilon/3)$ we can find a sufficiently large N s.t.

$$P\left(\left\{ x^n \in \mathcal{X}^n \;\middle|\; \left| -\frac{1}{n} \log P_{X^n}(x^n) - H(X) \right| > \varepsilon \right\} \right) \leq \varepsilon/3 \quad \forall n \geq N_1(\varepsilon)$$

$$P\left(\left\{ y^n \in \mathcal{Y}^n \;\middle|\; \left| -\frac{1}{n} \log P_{Y^n}(y^n) - H(Y) \right| > \varepsilon \right\} \right) \leq \varepsilon/3 \quad \forall n \geq N_2(\varepsilon)$$

$$P\left(\left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n \;\middle|\; \left| -\frac{1}{n} \log P_{X^nY^n}(x^n, y^n) - H(X,Y) \right| > \varepsilon \right\} \right) \leq \varepsilon/3 \quad \forall n \geq N_3(\varepsilon)$$

- Thus $\forall n \geq \max(N_1(\varepsilon), N_2(\varepsilon), N_3(\varepsilon))$, at least 1 of 3 criterias needed to not be in a jointly typical sequence are satisfied if that is an element of one of the above sets,

$$\Rightarrow P\left(A_\varepsilon^{(n)\,c} \right) \triangleq P\left((\mathcal{X}^n \times \mathcal{Y}^n) \setminus A_\varepsilon^{(n)} \right) \quad (\text{Complement})$$

$$= P\left(\left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n \;\middle|\; \begin{array}{c} \left| -\frac{1}{n} \log P_{X^n}(x^n) - H(X) \right| > \varepsilon \\ \text{or} \Leftrightarrow \cup \\ \left| -\frac{1}{n} \log P_{Y^n}(y^n) - H(Y) \right| > \varepsilon \\ \text{or} \Leftrightarrow \cup \\ \left| -\frac{1}{n} \log P_{X^nY^n}(x^n y^n) - H(X,Y) \right| > \varepsilon \end{array} \right\} \right)$$

$$\leq P\left(\left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n \;\middle|\; \left| -\frac{1}{n} \log P_{X^n}(x^n) - H(X) \right| > \varepsilon \right\} \right) + P\left(\left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n \;\middle|\; \left| -\frac{1}{n} \log P_{Y^n}(y^n) - H(Y) \right| > \varepsilon \right\} \right) + P\left(\left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n \;\middle|\; \left| -\frac{1}{n} \log P_{X^nY^n}(x^n y^n) - H(X,Y) \right| > \varepsilon \right\} \right)$$

(union bound property)

$$\leq \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon$$

$$\Rightarrow P\left(A_\varepsilon^{(n)} \right) > 1 - \varepsilon$$

$$\Rightarrow \lim_n P\left((x^n, y^n) \in A_\varepsilon^{(n)} \right) = 1 \qquad \forall n \geq \max(N_1, N_2, N_3)$$

Only need the last property of AEP definition

II)

Claim : $(x^n, y^n) \in A_\varepsilon^{(n)} \implies 2^{-n(H(x,y)+\varepsilon)} \geqslant p_{x^n y^n}(x^n, y^n) \geqslant 2^{-n(H(x,y)+\varepsilon)}$

Proof : $(x^n, y^n) \in A_\varepsilon^{(n)} \implies \left| -\frac{1}{n} \log p_{x^n y^n}(x^n, y^n) - H(X,Y) \right| < \varepsilon$

$\implies H(X,Y) - \varepsilon < -\frac{1}{n} \log p_{x^n y^n}(x^n, y^n) < H(X,Y) + \varepsilon$

$\implies -n(H(X,Y)-\varepsilon) \geqslant \log p_{x^n y^n}(x^n, y^n) \geqslant -n(H(X,Y)+\varepsilon)$

$\implies 2^{-n(H(X,Y)-\varepsilon)} \geqslant p_{x^n y^n}(x^n, y^n) \geqslant 2^{-n(H(X,Y)+\varepsilon)}$ ∎

$\therefore 1 = \sum_{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n} p_{x^n y^n}(x^n, y^n)$

$\geqslant \sum_{(x^n, y^n) \in A_\varepsilon^{(n)}} p_{x^n y^n}(x^n, y^n)$

$\geqslant \sum_{(x^n, y^n) \in A_\varepsilon^{(n)}} 2^{-n(H(X,Y)+\varepsilon)}$

$= |A_\varepsilon^{(n)}| \cdot 2^{-n(H(X,Y)+\varepsilon)} \implies |A_\varepsilon^{(n)}| < 2^{n(H(X,Y)+\varepsilon)}$ ∎

III)

By part I, $\forall\ n \geqslant \max(N_1(\varepsilon), N_2(\varepsilon), N_3(\varepsilon))$

$\implies 1 - \varepsilon < P(A_\varepsilon^{(n)})$

$= \sum_{(x^n, y^n) \in A_\varepsilon^{(n)}} p_{x^n y^n}(x^n, y^n)$

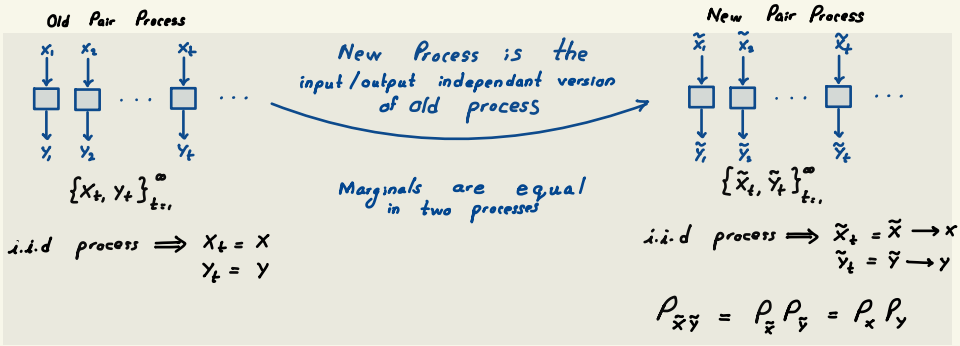
$< \sum_{(x^n, y^n) \in A_\varepsilon^{(n)}} 2^{-n(H(X,Y)-\varepsilon)}$ ... ∵ by the claim before

$= |A_\varepsilon^{(n)}| \cdot 2^{-n(H(X,Y)-\varepsilon)}$

$\implies |A_\varepsilon^{(n)}| > (1-\varepsilon)\, 2^{n(H(X,Y)-\varepsilon)}$ $\forall\ n \geqslant \max(N_1(\varepsilon), N_2(\varepsilon), N_3(\varepsilon))$ ∎

This uses all 3 properties in definition

IV)



i.i.d process $\Rightarrow$ $X_t = X$, $Y_t = Y$

i.i.d process $\Rightarrow$ $\tilde{X}_t = \tilde{X} \to x$, $\tilde{Y}_t = \tilde{Y} \to y$

$$P_{\tilde{X}\tilde{Y}} = P_{\tilde{X}} P_{\tilde{Y}} = P_X P_Y$$

▶ $$P_{\tilde{X}\tilde{Y}}\left((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\right) = \sum_{(x^n,y^n)\in A_\varepsilon^{(n)}} P_{\tilde{X}^n\tilde{Y}^n}(x^n, y^n)$$

$$= \sum_{(x^n,y^n)\in A_\varepsilon^{(n)}} P_{\tilde{X}^n}(x^n)\cdot P_{\tilde{Y}^n}(y^n) \quad \cdots\cdots \text{Input } \perp\!\!\!\perp \text{ output in new processes}$$

$$= \sum_{(x^n,y^n)\in A_\varepsilon^{(n)}} P_{X^n}(x^n)\cdot P_{Y^n}(y^n) \quad \cdots\cdots \text{New proccess has same marginals as old}$$

From the definition of Jointly Typical sets

$$\left|-\tfrac{1}{n}\cdot \log P_{X^n}(x^n) - H(X)\right| \leq \varepsilon$$

$$H(X)-\varepsilon \leq \tfrac{-1}{n}\log P_{X^n}(x^n) \leq H(X)+\varepsilon$$

$$-n(H(X)-\varepsilon) \geq \log P_{X^n}(x^n) \geq -n(H(Y)+\varepsilon) \quad \cdots\cdots$$

$$2^{-n(H(X)-\varepsilon)} \geq P_{X^n}(x^n) \geq 2^{-n(H(Y)+\varepsilon)}$$

$$\leq |A_\varepsilon^{(n)}|\cdot \max\left(P_{X^n}(x^n)\right)\cdot \max\left(P_{Y^n}(y^n)\right)$$

$$= |A_\varepsilon^{(n)}|\cdot 2^{-n(H(X)-\varepsilon)}\cdot 2^{-n(H(Y)-\varepsilon)}$$

From part II of Joint AEP $\cdots\cdots$

$$\leq 2^{n(H(X,Y)+\varepsilon)}\cdot 2^{-n(H(X)-\varepsilon)}\cdot 2^{-n(H(Y)-\varepsilon)}$$

$$= 2^{-n(-H(X,Y)+H(X)+H(Y)-3\varepsilon)}$$

$$= 2^{-n(I(X\wedge Y)-3\varepsilon)} \quad \forall n$$

▶ $$P_{\tilde{X}\tilde{Y}}\left((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}\right) = \sum_{(x^n,y^n)\in A_\varepsilon^{(n)}} P_{\tilde{X}^n\tilde{Y}^n}(x^n, y^n)$$

$$= \sum_{(x^n,y^n)\in A_\varepsilon^{(n)}} P_{\tilde{X}^n}(x^n)\cdot P_{\tilde{Y}^n}(y^n) \quad \cdots\cdots \text{Input } \perp\!\!\!\perp \text{ output in new processes}$$

$$= \sum_{(x^n,y^n)\in A_\varepsilon^{(n)}} P_{X^n}(x^n)\cdot P_{Y^n}(y^n) \quad \cdots\cdots \text{New proccess has same marginals as old}$$

$$\geqslant |A_\varepsilon^{(n)}|\cdot \min\left(P_{X^n}(x^n)\right)\cdot \min\left(P_{Y^n}(y^n)\right)$$

$$= |A_\varepsilon^{(n)}|\cdot 2^{-n(H(X)+\varepsilon)}\cdot 2^{-n(H(Y)+\varepsilon)}$$

From part III of Joint AEP for $\forall n$ sufficiently large $\cdots\cdots$

$$\geqslant (1-\varepsilon)\cdot 2^{n(H(X,Y)-\varepsilon)}\cdot 2^{-n(H(X)+\varepsilon)}\cdot 2^{-n(H(Y)+\varepsilon)}$$

$$= (1-\varepsilon)\cdot 2^{-n(-H(X,Y)+H(X)+H(Y)+3\varepsilon)}$$

$$= (1-\varepsilon)\cdot 2^{-n(I(X\wedge Y)-3\varepsilon)} \quad \forall n \text{ large enough}$$

■

Interpretation

- To apply joint AEP, $\{X_t, Y_t\}_{t=1}^{\infty}$ pair process should be i.i.d accross time. This can be achieved if input is independant accross time and channel is DMC.

- **Lemma**: Input $\{X_t\}_{t=1}^{\infty}$ is i.i.d and Channel is DMC $\implies$ $\{X_t, Y_t\}_{t=1}^{\infty}$ is i.i.d

Proof: The input to the channel is i.i.d $\implies p_{X^n}(x^n) = \prod_{t=1}^{n} p_{X_t}(x_t) \quad \cdots ①$

The Channel is DMC $\implies p_{Y^n|X^n}(y^n|x^n) = W_n(y^n|x^n) = \prod_{t=1}^{n} W(y_t|x_t) \quad \cdots ②$

Coosider,

$$\implies p_{X^nY^n}(x^n, y^n) = p_{X^n}(x^n)\, p_{Y^n|X^n}(y^n|x^n)$$

$$= \left(\prod_{t=1}^{n} p_{X_t}(x_t)\right)\left(\prod_{t=1}^{n} W(y_t|x_t)\right) \quad \because \text{i.i.d input and DMC}$$

$$= \prod_{t=1}^{n} \left(p_{X_t}(x_t) \cdot W(y_t|x_t)\right)$$

$$= \prod_{t=1}^{n} p_{XY}(x_t, y_t)$$

$\implies (X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. ∎

Visualization

- Total number of $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ sequences such that x^n is X-typical and y^n is Y-typical

$$\cong 2^{nH(X)} \cdot 2^{nH(Y)}$$

$$= 2^{n(H(X)+H(Y))}$$

- Total number of $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ sequences such that (x^n, y^n) is jointly (X, Y)-typical

$$\cong 2^{nH(X,Y)}$$

- All Jointly typical sequences should come from set .

- If (x^n, y^n) are randomly picked with uniform pmf from , the probability that $(x^n, y^n) \in$ Jointly Typical is

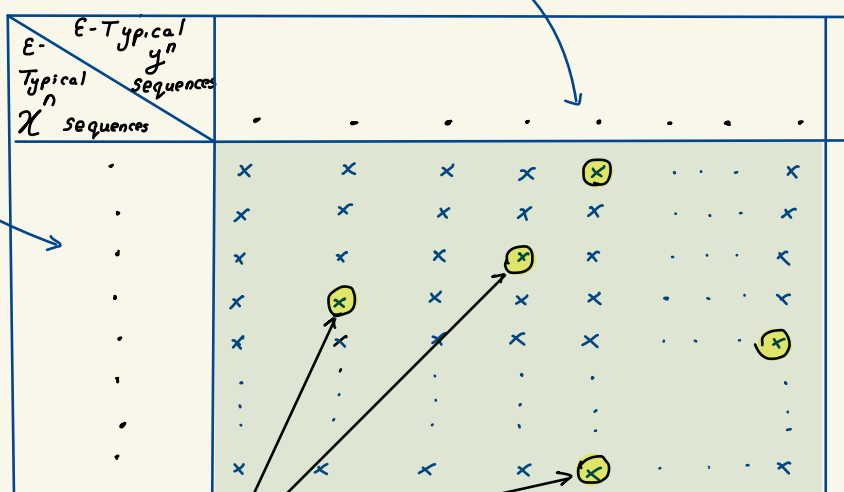
$$\cong \frac{2^{nH(X,Y)}}{2^{n(H(X)+H(Y))}} \cong 2^{-nI(X;Y)}$$

From the property II of typical Sets

$|\varepsilon\text{-Typical } y^n| \text{ sequences} \simeq 2^{nH(Y)}$

From the property II of typical Sets

$|\varepsilon\text{-Typical } x^n| \simeq 2^{nH(X)}$ sequences



Jointly ε-Typical (x^n, y^n) sequences

From property IV of Joint AEP $|\varepsilon\text{-Jointly Typical sequences}| \simeq 2^{nH(X,Y)}$

Three Proof for Achievability of Channel Coding Theorem

Gallager book → Uses the notion of maximum likelihood decoding at the decoder (Random Coding argument) } Will be in HWs.

Cover/Thomas book → Uses joint typicality arguments

Cieser/Corner book → Uses method of types. More mathematically rigurous.

7) Back to Channel Coding Theorem "Achievability" / Forward Part

Theorem - Shanon's Channel Coding Theorem

Let $(\mathcal{X}, W, \mathcal{Y})$ be a DMC,

i) Achievability : For any rate $R < \max_{P: P_x = P} I(X \wedge Y)$ there exist a sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes with

$$\lim_n e = \lim_n e_n = 0$$

i.e. Every rate below $\max_{P: P_x = P} I(X \wedge Y)$ is achievable.

ii) Converse : Any sequence of $(\lfloor 2^{nR} \rfloor, n)$ codes with

$$\lim_n e = \lim_n e_n = 0$$

must have a rate $R \leq \max_{P: P_x = P} I(X \wedge Y)$

Typical Sets and Achievability

- We are going to use the geometry of typical sets under Joint AEP. (i.e. volume packing argument in probability) to prove Channel Coding Achievability.

- Consider a $\{X^n, Y^n\}$ pair process, where X denotes the i.i.d. codewards with p.m.f P, sent through a DMC channel. Y denotes the output. Then corresponding joint p.m.f on $\mathcal{X}^n \times \mathcal{Y}^n$ is, $\prod_{t=1}^{n} [P(\cdot) W(\cdot|\cdot)]$

⇓

From the lemma under joint AEP section above, $\{X^n, Y^n\}$ pair process is i.i.d.

⇓

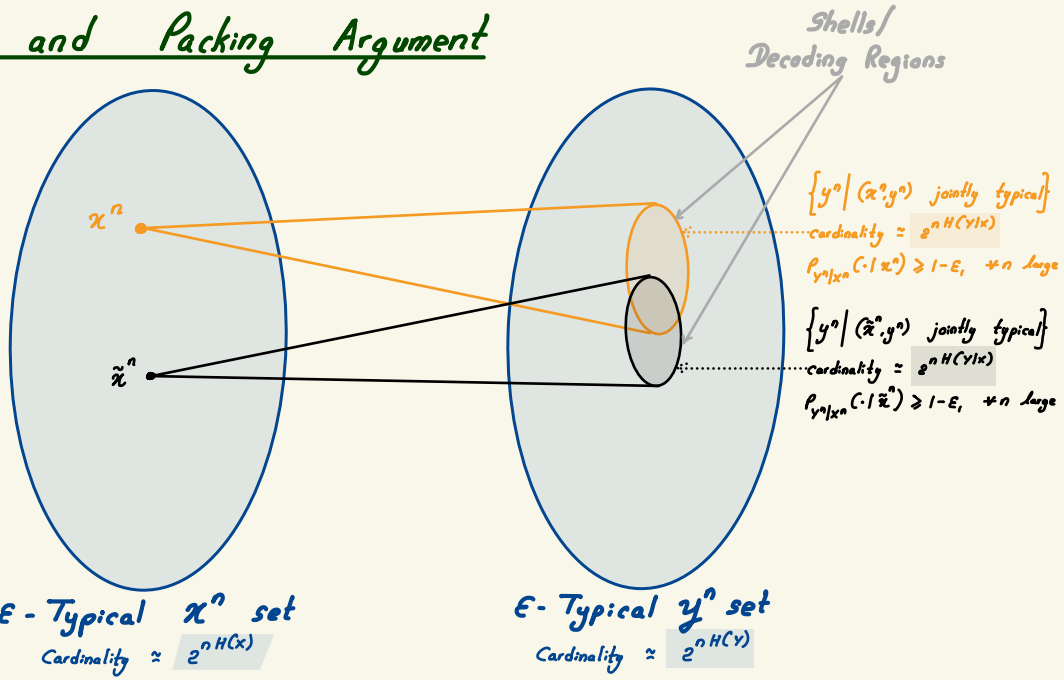
From Joint-AEP part - I, the probability of pair (x^n, y^n) being in ε-Typical set of $\mathcal{X}^n \times \mathcal{Y}^n$ - $A_\varepsilon^{(n)}$ is close to 1.

⇓

$$\lim_{n \to \infty} P\left((X^n, Y^n) \in A_\varepsilon^{(n)}\right) = 1$$

- Loosly speeking, we can limit the search of a codeward for the achievability part of channel coding to $A_\varepsilon^{(n)}$. i.e. in our proof, if an error occurs outside the $A_\varepsilon^{(n)}$ that is effectively having '0' probability.

Shells and Packing Argument



- From condition 1 and 3 in Joint-Typical set definition, if a message x^n is sent, only certain y^n's are likely and they have the approximately equal probability of

$$P(y^n|x^n) = \frac{P(y^n, x^n)}{P(x^n)} \simeq \frac{2^{nH(X,Y)}}{2^{nH(X)}} \simeq 2^{-nH(Y|X)}$$

- According to previous section, we already neglected the errors outside $A_\varepsilon^{(n)}$ - jointly typical set. (i.e. outside the 'shells').

- All errors we care will happen inside the 'shells,' i.e. by 'shells' being intersecting.

- i.e we want shells to have small overlap with small probability of error.

$$\text{Maximum no. of Shells that can be packed} = \text{Maximum no. of Codewords that can be sent (M)} \simeq \frac{2^{nH(Y)}}{2^{nH(Y|X)}} = 2^{nI(X\wedge Y)}$$

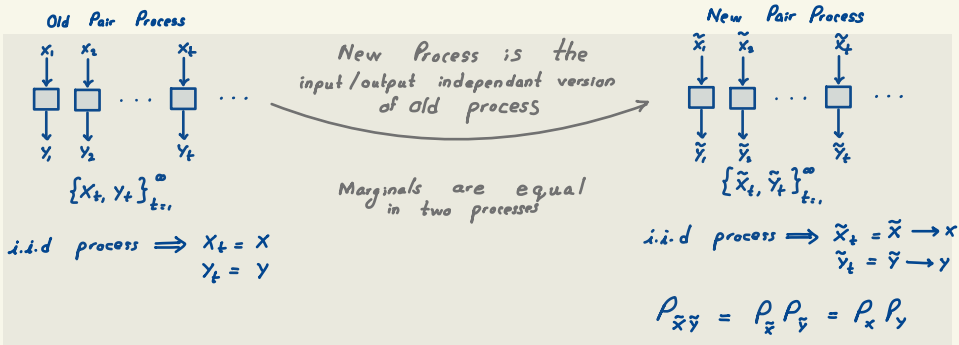
$$\text{Rate of code (R)} \cong \frac{\log M}{n} = \frac{\log 2^{nI(X\wedge Y)}}{n} = I(X\wedge Y)$$

$$C \simeq \max R = \max I(X\wedge Y)$$

Independant Pair Processes (AEP- Part IV) and Wrong Decodings

Correct Decoding	Incorrect Decoding
$X^n(m) \longrightarrow Y^n$ Sent → Observed	$X^n(m') \; m' \neq m \longrightarrow Y^n$ Sent → Observed
$(X^n(m), Y^n)$ is generated through the channel so has a true joint distribution and process is i.i.d over time.	$X^n(m')$ is independant of Y^n. But the process is i.i.d over time.
$\therefore$ from Joint- AEP part I, $(X^n(m), Y^n)$ likely belongs to $A_\varepsilon^{(n)}$.	
Joint distribution, P_{XY}	Joint distribution $P_{\tilde{X}\tilde{Y}} = P_X P_Y$

Incorrect case is the Input - Output Independant Pair- Process of correct case



- From AEP part IV, such an input - output independant new pair process has a probability,
$$P((\tilde{x}^n, \tilde{y}^n) \in A_\varepsilon^{(n)}) = P((X^n(m'), Y^n) \in A^n) \cong 2^{-nI(X\wedge Y)}$$
- Higher the connection $\Rightarrow$ $\uparrow I(X\wedge Y)$ $\Rightarrow$ $\downarrow$ prob. of incorrect pair
- Every pair (X^n, Y^n) has 2^{-nI} probability of confusion. If we should send at least M codewards to send 1 codeward correctly,
$$\Rightarrow M \cdot 2^{-nI} \lesssim 1 \Rightarrow M \lesssim 2^{nI}$$
$$\Rightarrow R = \log M/n \lesssim I$$
$$\Rightarrow C = \max R \simeq I(X\wedge Y)$$

"Random Coding" Argument

- For achievability we should show that $\forall R < \max I(X \wedge Y)$ there exists a $(\lfloor 2^{nR} \rfloor, n)$ code with $\lim_n \epsilon = 0$.
- But designing such a code is hard. So we take all the $(\lfloor 2^{nR} \rfloor, n)$ codes that there is (say that set is S). Then randomly sample a codebook $\mathcal{C} \in S$, with equal probability.
  P_e = Probability of error of codeboow $\mathcal{C}$.
- We prove that the on average (ie expected value), the error of this sampled codebook $\mathcal{C}$ goes to 0.
  $\lim_n \mathbb{E}[P_e] = 0$
- Then if average for the set S, itself is good, we should have at least one code with desired property.

Assumption - Random Coding

Generate the entire codebook ($\mathcal{C}$) by randomly and independantly generating $(2^{nR}, n)$ block codes of block length n according to pmf $P_{X^n}(x^n) = \prod_{t=1}^{n} P_X(x_t) \quad \cdots x^n \in \mathcal{X}^n$

$$\mathcal{C} = \begin{bmatrix} x^n(1) \\ x^n(2) \\ \vdots \\ x^n(\lfloor 2^{nR} \rfloor) \end{bmatrix} = \begin{bmatrix} x_1(1) & x_1(2) & \cdots & x_n(1) \\ x_1(1) & x_2(2) & \cdots & x_n(2) \\ \vdots & \vdots & & \vdots \\ x_1(\lfloor 2^{nR} \rfloor) & x_2(\lfloor 2^{nR} \rfloor) & \cdots & x_n(\lfloor 2^{nR} \rfloor) \end{bmatrix}$$

$$P(\mathcal{C}) = \prod_{m=1}^{\lfloor 2^{nR} \rfloor} \prod_{t=1}^{n} P(x_t(m)) \quad \cdots\cdots (1)$$

Decoding Rules

- Note the decoder is converting a recieved y^n to a codeward M'.
  $\phi: y^n \longrightarrow M' \quad$ where $\quad M \subseteq M' = \{0, \ldots, M\}$

1) Maximum A- Posteriori (MAP) rule $\longrightarrow$ $m' = \arg\max_m P(y^n \mid x^n(m))$

2) Joint - Typicality Decoding rule $\longrightarrow$ $m' = \underset{m \text{ where}}{\text{value of}} \; -\frac{1}{n} \log P(x^n(m), y^n) \approx H(X,Y)$

Proof of Achievability

P_X / P - fixed but arbitary p.m.f on $\mathcal{X}$

R - rate of code. $0 < R < I(X \wedge Y) = I(P_X, W)$ ..... (*)

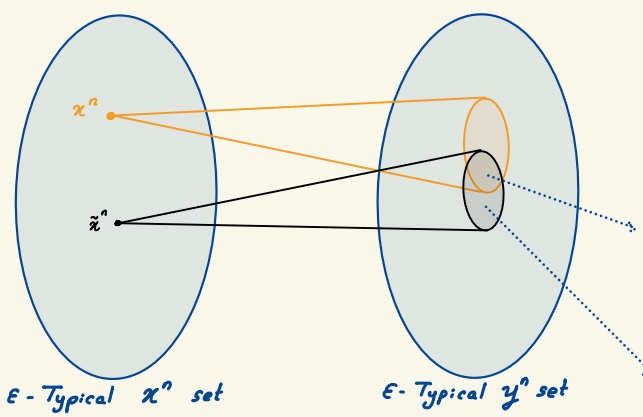
ϵ - fixed error probability. $\epsilon > 0$

- Since each codeward and symbol are independant from **random coding assumption** and from **DMC property**, Reciever gets $y^n \in \mathcal{Y}^n$ ~

$$W(y^n | x^n(m)) = \prod_{t=1}^{n} W(y_t | x_t(m)) \qquad \text{if } m \in \{1, \dots, M\} \text{ is the message sent}$$

- Since we only want to prove the existance of such codeward, we can choose any decoding logic at the decoder. Let reciever uses the **Joint-Typicality Decoding rule**. (rather than optimal MAP rule). Hence, for each recieved $y^n \in \mathcal{Y}^n$

$$\phi(y^n) = \begin{cases} m' & \text{..... if } m' \text{ is the only element } \in \{1, \dots, \lfloor 2^{nR} \rfloor\} \text{ s.t. } (x^n(m'), y^n) \in A_\epsilon^{(n)} \quad \text{(True)} \\ 0 & \text{..... if no such } m' \text{ exists or if } \exists \text{ more than one such } m \in \{1, \dots, \lfloor 2^{nR} \rfloor\} \quad \text{(Error; Error 1; Error 2)} \end{cases}$$



Error 1 $\Rightarrow$ No codeward matches
$\Rightarrow$ True pair $(x^n(m), y^n) \notin A_\epsilon^{(n)}$
$\Rightarrow$ Very rare. We only consider thing inside $A_\epsilon^{(n)}$ in this figure.

Error 2 $\Rightarrow$ $x^n(m)$ and $\tilde{x}^n(m)$ are boath jointly typical with y^n.
$\Rightarrow$ i.e. intersecting regions of shells.

True $\Rightarrow$ Non intersecting regions inside shells.

Probability of Error Calculations

- Average probability of error for a fixed codebook $\mathcal{C}$ is

$$\bar{e}(w, c) = \frac{1}{\lfloor 2^{nR} \rfloor} \sum_{m=1}^{\lfloor 2^{nR} \rfloor} e_m(w, c) \quad \cdots\cdots (2)$$

($e_m(w, c)$: error for message m specifically.)

- Claim: $\sum_{c \in \mathcal{C}} P(c) \cdot e_m(w, c)$ is constant for each message 'm'.

  Proof: Each codeword $x^n(m)$ is generated i.i.d from same distribution and all codebooks are randomly picked. $\because$ (1) ■

- Assume 'c' is the realized codebook out of all random codebooks '$\mathcal{C}$'. Hence,

$$\text{If } c \text{ is realized for } \mathcal{C} \implies \mathcal{C} = c \implies X^n(m) = x_c^n(m) \quad m = 1, \ldots, \lfloor 2^{nR} \rfloor$$

- Define the joint-Typicality event for $m = 1, \ldots, \lfloor 2^{nR} \rfloor$ as follows,

$$\mathcal{E}_m \triangleq \left\{ (X^n(m), Y^n) \in \mathcal{A}_\epsilon^{(n)} \right\} \quad \cdots\cdots (3)$$

- The expected value of average error probability for ensemble of codes (i.e. for all possible codebooks).

$$\tilde{e} = \sum_{c \in \mathcal{C}} P(c)\, \bar{e}(w, c)$$

$$= \sum_{c \in \mathcal{C}} P(c) \frac{1}{\lfloor 2^{nR} \rfloor} \sum_{m=1}^{\lfloor 2^{nR} \rfloor} e_m(w, c) \quad \cdots \because (2)$$

$$= \frac{1}{\lfloor 2^{nR} \rfloor} \sum_{m=1}^{\lfloor 2^{nR} \rfloor} \underbrace{\sum_{c \in \mathcal{C}} P(c)\, e_m(w, c)}_{\text{is same for every message from the claim above}}$$

$$= \sum_{c \in \mathcal{C}} P(c) \cdot e_1(w, c)$$

$$= \sum_{c \in \mathcal{C}} P(c) \cdot P\left(\phi_c(y^n) \neq 1 \;\middle|\; \overbrace{X^n(1) = x_c^n(1)}^{\mathcal{C}=c} \right)$$ where y^n is connected to $x^n(1)$ by w^n

$$= \sum_{c \in \mathcal{C}} P(c)\, P\left((x_c^n(1), y^n) \notin A_\varepsilon^{(n)} \text{ (Error 1) or } (x_c^n(m), y^n) \in A_\varepsilon^{(n)} \text{ (Error 2) for } m \neq 1 \;\middle|\; \overbrace{X^n(1) = x_c^n(1)}^{\mathcal{C}=c} \right)$$

$$= \sum_{c} P(c)\, P\left((x_c^n(1), y^n) \notin A_\varepsilon^{(n)} \text{ or } (x_c^n(m), y^n) \in A_\varepsilon^{(n)} \text{ for } m \neq 1 \;\middle|\; \overbrace{X^n(1) = x_c^n(1),\ X^n(m) = x_c^n(m),\ m = 2, \ldots \lfloor 2^{nR} \rfloor}^{\mathcal{C}=c} \right)$$

(First message $X^n(1)$ being realised to $x_c^n(1)$ means, $c \in \mathcal{C}$. Hence identical to all messages being realised.)

where $P_{Y^n|X^n(1)} = W$

$$= P\left((X^n(1), Y^n) \notin A_\varepsilon^{(n)} \text{ or } (X^n(m), Y^n) \in A_\varepsilon^{(n)} \text{ for } m \neq 1 \right) \quad \because \text{Law of Total Probability}$$

$$= P\left(\mathcal{E}_1^c \cup \left(\bigcup_{m=2}^{\lfloor 2^{nR} \rfloor} \mathcal{E}_m \right) \right) \quad \because \text{From Definition (3)}$$

$$\tilde{e} \leq P(\mathcal{E}_1^c) + \sum_{m=2}^{\lfloor 2^{nR} \rfloor} P(\mathcal{E}_m) \quad \because \text{Union Bound} \quad \ldots (4)$$

- By part I of Joint-AEP
  - $\Rightarrow \lim_n P\left((X^n(1), Y^n) \in A_\varepsilon^{(n)} \right) = 1$
  - $\Rightarrow \lim_n P(\mathcal{E}_1) = 1$
  - $\Rightarrow \lim_n P(\mathcal{E}_1^c) = 0$
  - $\Rightarrow P(\mathcal{E}_1^c) \leq \varepsilon/4 \quad \forall\ n$ sufficiently large $\ldots (5)$

- From DMC property and Random Coding assumption
  - $\Rightarrow$ Given $X^n(1)$, output Y^n depends only on $X^n(1)$, not on any other $X^n(m), m \neq 1$
  - $\Rightarrow Y^n \perp\!\!\!\perp X^n(m),\ m \neq 1 \;|\; X^n(1)$
    - i.e. $X^n(m),\ m \neq 1 \multimap X^n(1) \multimap Y^n$

Since $X^n(1) \perp\!\!\!\perp X^n(m),\ m \neq 1$, $\Rightarrow Y^n \perp\!\!\!\perp X^n(m),\ m \neq 1 \quad \ldots (6)$

- The second term in ④ can be simplified by using independant property in ⑥, and Part IV of joint AEP, where the rareness of input-output independant version of a pair-process is proved.

$$\sum_{m=2}^{\lfloor 2^{nR} \rfloor} P(\mathcal{E}_m) = \sum_{m=2}^{\lfloor 2^{nR} \rfloor} P\left((X^n(m), Y^n) \in A_\varepsilon^{(n)} \right)$$

$$\sum_{m=2}^{\lfloor 2^{nR} \rfloor} P(\mathcal{E}_m) = \sum_{m=2}^{\lfloor 2^{nR} \rfloor} 2^{-n(I(X\wedge Y)-3\varepsilon)} \quad \cdots ⑦$$

from $X^n(m) \perp\!\!\!\perp Y^n$ and Joint-AEP

- From ④, ⑤ and ⑦,

$$\tilde{e} \leqslant \varepsilon/4 + \sum_{m=2}^{\lfloor 2^{nR} \rfloor} 2^{-n(I(X\wedge Y)-3\varepsilon)}$$

$$= \varepsilon/4 + \left(\lfloor 2^{nR} \rfloor - 1 \right) 2^{-n(I(X\wedge Y)-3\varepsilon)}$$

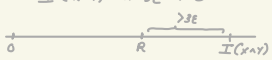
$$\leqslant \varepsilon/4 + 2^{-n(I(X\wedge Y)-3\varepsilon-R)}$$

$$\leqslant \varepsilon/4 + \varepsilon/4 \quad \cdots \exists \text{ a sufficiently large } n, \text{ such that } 2^{-n(I(X\wedge Y)-R-\varepsilon)} \leqslant \varepsilon/4$$

$$= \varepsilon/2 \quad \forall n \text{ sufficiently large with } R < I(X\wedge Y) - 3\varepsilon$$

$$\Rightarrow \tilde{e} \leqslant \varepsilon/2$$

From the initial conditions in ⑧, $R < I(X\wedge Y)$. Hence we can find a small ε where $I(X\wedge Y) - R - 3\varepsilon > 0$



Hence when $n \to \infty$ for that ε, $-n(I(X\wedge Y)-R-\varepsilon) \to -\infty$, $2^{-n(I(X\wedge Y)-R-\varepsilon)} \to 0$

- To complete the proof we can replace the $I(X\wedge Y)$ with $\max I(X\wedge Y)$. Let

$$I(P^*, W) = \max_P I(P, W) = \max_P I(X\wedge Y)$$

if $R < I(P^*, W)$ we can choose $\varepsilon > 0$ to be arbitarily small and, n sufficiently large such that

$$\tilde{e} = \sum_{c \in \mathcal{C}} P(c) \cdot \bar{e}(w, c) \leqslant \varepsilon/2 \quad \forall n \text{ sufficiently large} \quad \cdots ⑧$$

Expugation of Undesired Codewards

By ② $\Rightarrow \bar{e} = \sum_{c \in \mathcal{C}} P(c) \cdot \bar{e}(w, c) \leq \varepsilon/2$ $\forall$ n sufficiently large

$\Rightarrow \exists\ c^* \in \mathcal{C}$ $\bar{e}(w, c^*) \leq \varepsilon/2$ $\forall$ n sufficiently large ····· ③

- The identification c^* involves an exhaustive search over all $(2^{\lfloor nR \rfloor}, n)$ codes.

  No. of codebooks to search $= \left(|\mathcal{X}|^n\right)^{\lfloor 2^{nR} \rfloor} = |\mathcal{X}|^{n\lfloor 2^{nR} \rfloor}$

- Consider c^* and throw away half the codewards (i.e $\lfloor 2^{nR} \rfloor / 2$) that are worst. (i.e. $x^n(m)$ codewards in c^* for which e_m is largest). Then the remaining $\lfloor 2^{nR} \rfloor / 2$ codeward set is called Expurgated Codeward set.

Claim: Over the half expurgated codeward set in c^*, max error probability, $e \leq \varepsilon$

Proof: Let X - r.v denoting the distribution of error for codebook c^* among messages $\{1, 2, \ldots, \lfloor 2^{nR} \rfloor\}$ ∴ From ①, ③,

$\Rightarrow \mathbb{E}[X] = \bar{e}(w, c^*) \leq \varepsilon/2$

from Markov Inequality $\Rightarrow P(|X| > 2\mathbb{E}[X]) = P(|X| > 2\varepsilon/2) = P(|X| > \varepsilon) \leq \frac{\mathbb{E}[X]}{2\,\mathbb{E}[X]} = \frac{1}{2}$

$\Rightarrow P(|X| \leq \varepsilon) \geq 1 - 1/2 = 1/2$

$\Rightarrow$ at least half the codewards satisfy $e_m(w, c^*) \leq \varepsilon$.

$\Rightarrow$ Expurgating bad half yields a subcode of size $\geq \frac{\lfloor 2^{nR} \rfloor}{2}$ with $e_m \leq \varepsilon$.

■

Only half the codewords are used

$$\text{Rate of expergated code} = \frac{1}{n} \log \left(\frac{\lfloor 2^{nR} \rfloor}{2} \right)$$

$$\simeq \frac{1}{n} \log 2^{nR-1}$$

$$= \frac{1}{n} \log 2^{n(R-1/n)}$$

$$= R - 1/n$$

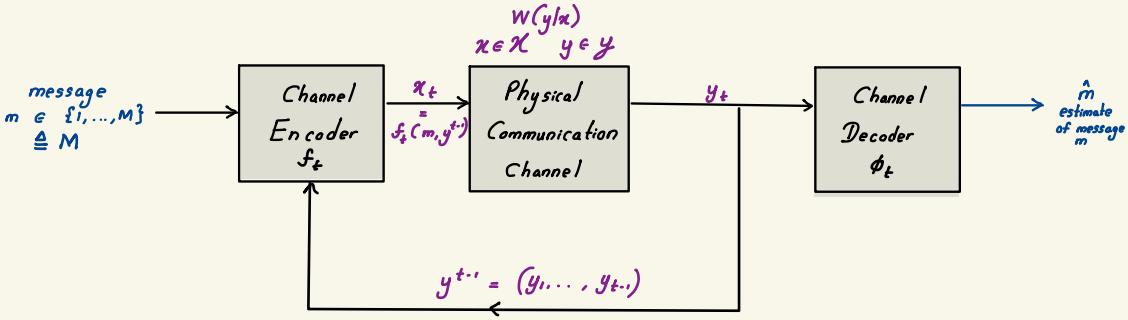
When $n \to \infty$, we have $1/n \longrightarrow 0$. $\therefore$ i.e. negligable for sufficiently large n.

Thus we have shown the existance of a code of rate $R - 1/n \cong R$ for large n with $e = e_n \longrightarrow 0$ for $R < \max_{P : P_X = P} I(X \wedge Y) = \max_{P} I(P, W)$

■

8) Channel Capacity With Feedback

Idealized Model



Theorem - For a DMC with stochastic matrix W, if C_{FB} is the capacity of the channel with feedback,

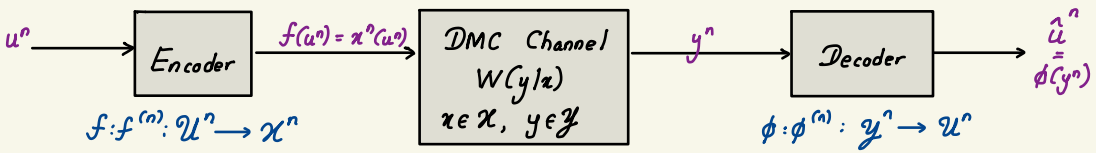
$$C_{FB} = C = \max_{P : P_X = P} I(X \wedge Y)$$

Proof :- Not done in the course.

9) Joint Source Channel Coding Theorem and Seperation Theorem

Setting

- $\{U_t\}_{t=1}^{\infty}$ is a discrete source with finite alphabet $\mathcal{U}$.
- Assume $\{U_t\}_{t=1}^{\infty}$ satisfies AEP. i.e.
  $-\frac{1}{n} \cdot \log P(u^n) \longrightarrow \bar{H}(u)$ in probability.



- Once we know what was transmitted (x^n), the channel output (y^n), does not care from which message it was generated.

$$U^n \longrightarrow X^n = f(U^n) \longrightarrow Y^n \quad \cdots\cdots ①$$

- Error probability for a source sequence u^n is

$$e(u^n) \triangleq P\left(\phi(y^n) \neq u^n \,\middle|\, x^n = f(u^n)\right) \quad \cdots\cdots ②$$

$$= 1 - P\left(\phi(y^n) = u^n \,\middle|\, x^n = f(u^n)\right)$$

$$= 1 - W\left(\phi^{-1}(u^n) \,\middle|\, f(u^n)\right)$$

where $\phi^{-1}(u^n) = \left(y^n \in \mathcal{Y}^n \,\middle|\, \phi(y^n) = u^n\right)$

- Error Probability for all n-block sequences

$$e_n \triangleq P(\phi(y^n) \neq U^n)$$

$$= \sum_{u^n \in \mathcal{U}^n} P(\phi(y^n) \neq u^n, U^n = u^n) \quad \cdots\cdots \text{law of total Probability}$$

$$= \sum_{u^n \in \mathcal{U}^n} \sum_{x^n \in \mathcal{X}} P(\phi(y^n) \neq u^n, U^n = u^n, X^n = x^n) \quad \cdots\cdots \text{law of total Probability}$$

$$= \sum_{u^n \in \mathcal{U}^n} \sum_{\substack{x^n \in \mathcal{X} \\ x^n = f(u^n)}} P\left(\phi(y^n) \neq u^n \mid U^n = u^n, X^n = x^n\right) \cdot \underbrace{P(X^n = x^n \mid U^n = u^n)}_{\mathbb{1}(x^n = f(u^n))} \cdot P(U^n = u^n)$$

The summation terms in $\sum_{x^n}$ disappears when $x^n \neq f(u^n)$.

$$= \sum_{u^n \in \mathcal{U}^n} P\left(\phi(y^n) \neq u^n \mid U^n = u^n, X^n = f(u^n)\right) \cdot P(U^n = u^n)$$

$$= \sum_{u^n \in \mathcal{U}^n} P\left(\phi(y^n) \neq u^n \mid X^n = f(u^n)\right) \cdot P(U^n = u^n) \quad \cdots \because \text{from } ①$$

$$= \sum_{u^n \in \mathcal{U}^n} e(u^n) \cdot P(U^n = u^n) \quad \cdots \text{from } ② \quad \cdots ③$$

Theorem - Joint Source - Channel Coding

Achievability - Let $\{U_t\}_{t=1}^{\infty}$ be a $\mathcal{U}$-valued discrete source with finite alphabet $\mathcal{U}$ and entropy rate $\bar{H}(U)$. Assume $\{U_t\}_{t=1}^{\infty}$ satisfies AEP. Let $(\mathcal{X}, W, \mathcal{Y})$ be a DMC with channel capacity C. Then there exists a source-channel codes $\{(f_n, \phi_n)\}$ with

$$\text{if } \bar{H}(U) < C \implies \lim_n e_n = 0$$

(Same units)

Converse - For any sequence of source channel codes $\{(f_n, \phi_n)\}_{n=1}^{\infty}$

$$\text{if } \lim_n e_n = 0 \implies \bar{H}(U) \leq C$$

- Equivalent statement of converse, Any sequence of $\{f_n, \phi_n\}_{n=1}^{\infty}$ if $\bar{H}(U) > C$ will have $e_n \geq \delta > 0 \quad \forall n$.

Achievability Proof

- This uses a two stage process. Let $\varepsilon > 0$ be fixed but arbitrary.

Step 1 - Source Encoding

- From the properties 2), 3) of the Typical sets, $\exists$ a $A_\varepsilon^{(n)}$ with

$$\Rightarrow (1-\varepsilon)\cdot 2^{n(H(P_u)-\varepsilon)} \leq |A_\varepsilon^{(n)}| \leq 2^{n(H(P_u)+\varepsilon)} \quad \forall\, n \text{ large} \quad \cdots\cdots (1)$$

Since $\{U_t\}_{t=1}^{\infty}$ satisfies the AEP $-\frac{1}{n}\log P(U^n) \longrightarrow \bar{H}(U)$. Hence with high probability $P(U^n)$ is roughly $2^{-n\bar{H}(U)}$. i.e around $2^{n\bar{H}(U)}$ sequences have equal, high probability. Hence $H(P_x) \cong \bar{H}(X)$ in (1),

$$\Rightarrow (1-\varepsilon)\cdot 2^{n(\bar{H}(U)-\varepsilon)} \leq |A_\varepsilon^{(n)}| \leq 2^{n(\bar{H}(U)+\varepsilon)} \quad \forall\, n \text{ large} \quad \cdots\cdots (2)$$

- From property 1) of Typical sets

$$\Rightarrow P(A_\varepsilon^{(n)}) = P(U^n \in A_\varepsilon^{(n)}) \geq 1-\varepsilon \quad \forall\, n \text{ large} \quad \cdots\cdots (3)$$

- Encode sequences in $A_\varepsilon^{(n)}$ by a fixed-to-fixed length code. Represent the sequences in $\mathcal{U}^n \setminus A_\varepsilon^{(n)}$ by a dummy sequence of 0's.

- This source code entails indexing of all elements of $A_\varepsilon^{(n)}$, requiring $\leq n(\bar{H}(U)+\varepsilon)$ bits.

Step 2 - Channel Encoding

- Each of these indices represented by $\leq n(\bar{H}(U)+\varepsilon)$ bits is treated as a message for channel encoding.

- By the Channel Coding Theorem, $\forall \varepsilon > 0$ $\exists$ a sequence of n-length block codes with

$$\text{maximum error probability} \leq \varepsilon \quad \cdots\cdots (4)$$

provided the channel rate of code (R) is $< C$. i.e.

$$\Rightarrow R \cong \frac{\log 2^{n(\bar{H}(U)+\varepsilon)}}{n} < C$$

$$\Rightarrow \bar{H}(U)+\varepsilon < C \quad \cdots\cdots (*)$$

Decoding

- Reciever first reconstruct the transmitted sequence by decoding the message index. (Channel Decoding) and then chooses the sequence in $A_\epsilon^{(n)}$, corresponding to the message index. (Source Decoding).

- Note :-
  - Souce - channel encoder is the composition of source encoder + channel encoder.
  - Souce - channel decoder is the composition of source decoder + channel decoder

If $\bar{H}(U) < C$

$\Rightarrow$ Error probability $e_n = P(\phi(Y^n) \neq U^n)$

$$= P\left(\phi(Y^n) \neq U^n,\ U^n \in A_\epsilon^{(n)}\right) + P\left(\phi(Y^n) \neq U^n,\ U^n \notin A_\epsilon^{(n)}\right)$$

$$\leq P\left(\phi(Y^n) \neq U^n,\ U^n \in A_\epsilon^{(n)}\right) + P\left(U^n \notin A_\epsilon^{(n)}\right)$$

$$\leq P\left(\phi(Y^n) \neq U^n,\ U^n \in A_\epsilon^{(n)}\right) + \epsilon \quad \text{.... from (3)}$$

$$= \underbrace{P\left(\phi(Y^n) \neq U^n \mid U^n \in A_\epsilon^{(n)}\right)}_{\text{error probability of the channel}} P\left(U^n \in A_\epsilon^{(n)}\right) + \epsilon$$

$$\leq \epsilon \cdot P\left(U^n \in A_\epsilon^{(n)}\right) + \epsilon \quad \text{.... from (4)}$$

$$\leq \epsilon \cdot 1 + \epsilon$$

$$= 2\epsilon$$

$\Rightarrow \exists$ a sequence of source channel with $\lim_n e_n = 0$

- Note :- Achievability proof entails source-channel seperation.

Converse Proof

- Let $\{(f^{(n)}, \phi^{(n)})\}_{n=1}^{\infty}$ be any sequence of source channel codes. Assume

$$\lim_n e_n = 0. \quad \cdots\cdots (*)$$

- By Fano's Inequality, $\forall\ n \geq 1$

$$H(U^n \mid \phi(Y^n)) \leq P(U^n \neq \phi(Y^n)) \cdot \log |\mathcal{U}^n| + 1$$

$$= e_n \cdot \log |\mathcal{U}^n| + 1$$

$$= 1 + e_n \cdot n \cdot \log |\mathcal{U}|. \quad \cdots\cdots (1)$$

- By definition of the Entropy Rate,

$$\bar{H}(U) \leq \frac{1}{n} \cdot H(U^n)$$

$$= \frac{1}{n} \cdot \left[H(U^n \mid \phi(Y^n)) + I(U^n \wedge \phi(Y^n)) \right] \quad \cdots I(X \wedge Y) = H(X) - H(X|Y)$$

$$\leq \frac{1}{n} \cdot \left[1 + n \cdot e_n \cdot \log |\mathcal{U}| \right] + \frac{1}{n} \cdot I(U^n \wedge \phi(Y^n)) \quad \cdots \text{from } (1)$$

$$= \frac{1}{n} + e_n \cdot \log |\mathcal{U}| + \frac{1}{n} \cdot I(U^n \wedge \phi(Y^n)) \quad \cdots\cdots (2)$$

- We had the Markov chain, $U^n \multimap X^n(U^n) \,(= f(U^n)) \multimap Y^n \multimap \phi(Y^n)$. From the Markov Processing Inequality,

$$I(U^n \wedge \phi(Y^n)) \leq I(X^n(U^n) \wedge Y^n)$$

$$= n \cdot I(X(U) \wedge Y) \quad \cdots \text{by DMC}$$

$$\leq n \cdot C \quad \cdots\cdots (3) \quad \cdots\cdots \text{by definition of channel capacity } C$$

$(2), (3) \Rightarrow \quad \bar{H}(U) \leq \frac{1}{n} + e_n \cdot \log |\mathcal{U}| + \frac{1}{n} \cdot nC$

when $n \to \infty \Rightarrow \quad \bar{H}(U) \leq \underbrace{\lim_n 1/n}_{0} + |\mathcal{U}| \cdot \underbrace{\lim_n e_n}_{0 \text{ from } (*)} + C = C$

■

Reading Assignment
Cover Thomas
Chapter 8

Continuos RVs and Differential Entropy

1) Differential Entropy

Assume X is a $\mathbb{R}$-valued rv with CDF F_X, i.e. $F_X(x) = P(X \leq x)$, $x \in \mathbb{R}$. Assume F_X admits a density w.r.t Lebesgue measure and let f_X be the corresponding pdf of X.

Definition - Differential Entropy

Differential Entropy of $\mathbb{R}$-valued rv X with pdf f_X is

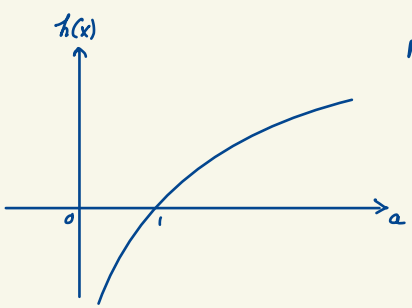
$$h(X) = -\int_{\mathbb{R}} f_X(x) \cdot \log f_X(x)\, dx = \mathbb{E}\left[-\log f_X(X)\right]$$

provided the integral exists.

Example 1- $X \sim \text{Unif}(0, a)$, $a > 0$

$$\therefore f_X(x) = \frac{1}{a} \qquad 0 < x < a.$$

$$\therefore h(X) = -\int_0^a \frac{1}{a} \log \frac{1}{a}\, dx = -\frac{1}{a} \cdot \log \frac{1}{a} \cdot [x]_0^a = \log a \;//$$



Note
- for $0 < a < 1$ we have $h(X) < 0$, negative entropy. $\therefore$ no uncertainity.
- As $a \uparrow$, i.e interval increase, uncertainity increases.

Example 2- $X \sim N(\mu, \sigma^2)$, $\sigma^2 > 0$. Then $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$ Compute differential entropy.

Computing $h(X)$ in nats,

$$h(X) = -\int_R f_X(x) \ln f_X(x)\, dx$$

$$= -\int_R f_X(x) \cdot \left(-\ln \sqrt{2\pi\sigma^2} - \frac{(x-\mu)^2}{2\sigma^2} \right) dx$$

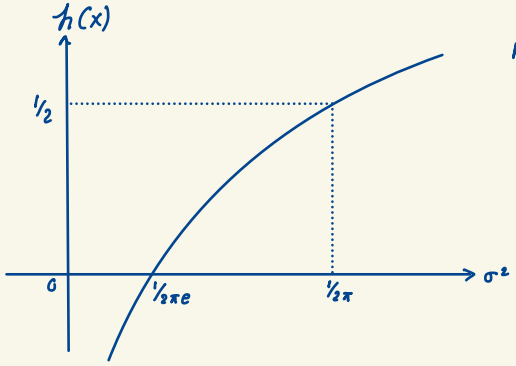
$$= \frac{1}{2} \ln (2\pi\sigma^2) + \frac{1}{2\sigma^2} E[(X-\mu)^2]$$

$$= \frac{1}{2} \ln (2\pi\sigma^2) + \frac{1}{2\sigma^2} \sigma^2 \quad = 1$$

$$= \frac{1}{2} [\ln (2\pi\sigma^2) + \ln(e)]$$

$$= \frac{1}{2} \ln (e 2\pi\sigma^2) \text{ nats} \quad \cdots\cdots \text{ does not depend on } \mu.$$

$$h(X) = \frac{1}{2} \log_2 (e 2\pi\sigma^2) \text{ bits}$$



Note
- As variance $\sigma^2 \uparrow$, uncertainty $h(x)$ goes up.
- $h(x)$ i.e. uncertainty does not depend on μ.

Example 3- $X \sim \exp(\lambda)$, $\lambda > 0$. Then $f_X(x) = \begin{cases} \lambda \cdot e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

Compute differential entropy.

$$h(X) = -\int_R f_X(x) \cdot \ln f_X(x)\, dx \quad \text{in nats}$$

$$= -\int_R f_X(x) \cdot [\ln \lambda - \lambda x]\, dx$$

$$= -\ln \lambda + \lambda E[X]$$

$$= -\ln \lambda + \lambda \cdot 1/\lambda$$

$$= 1 - \ln \lambda$$

$$= \ln (e/\lambda) \text{ nats}$$

$$= \log_2 (e/\lambda) \text{ bits}$$

Note -
- $E[X] = 1/\lambda$ $\quad Var[X] = 1/\lambda^2$
- Note as λ increases, $Var[X]$ decreases, concentrating the pmf around mean.

2) Conditional Differential Entropy

X and Y are $\mathbb{R}^k$ and $\mathbb{R}^l$ valued rvs respectively, with joint pdf f_{XY} on $\mathbb{R}^{k+l}$. Let f_X and f_Y be the marginal pdfs of X and Y respectively.

Definition - Conditional Differential Entropy

The conditional differential entropy of X given Y is,

$$h(X|Y) = -\int_{\mathbb{R}^{k+l}} f_{XY}(x,y) \cdot \log \frac{f_{XY}(x,y)}{f_Y(y)} \, dx\, dy$$

assuming that the integral exists.

Remark

- $h(X|Y) = h(X,Y) - h(Y)$

  From ①

  $$h(X|Y) = -\int_{\mathbb{R}^{k+l}} f_{XY}(x,y) \cdot \log f_{XY}(x,y)\, dx\, dy - \left[-\int_{\mathbb{R}^l} f_Y(y) \cdot \log f_Y(y)\, dy\right]$$

  $$= h(X,Y) - h(Y)$$

- Caution if any or boath are infinite.

3) Differential Entropy of a $\mathbb{R}^n$-valued Gaussian RV

Theorem 1- Differential entropy of $X \sim \mathcal{N}(\mu, \Sigma)$, $\mathbb{R}^n$ valued rv. is (Assume Σ^{-1} exists.)

$$h(X) = h(\mathcal{N}(\mu, \Sigma)) = \frac{1}{2} \log \left[(2\pi e)^n |\Sigma| \right] \text{ bits}$$

where $|\Sigma|$ = determinant of Σ.

Note: $h(X)$ does not depend on μ, and depends on Σ through it's determinant.

Proof: From pdf

$$f_X(x) = \frac{1}{(\sqrt{2\pi})^n |\Sigma|^{1/2}} \exp\left[-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu) \right] \quad \forall x$$

$$\Rightarrow h(X) = -E[\ln f_X(X)] \quad \text{in nats}$$

$$= \frac{1}{2} E\left[\underset{1\times n}{(X-\mu)^T} \underset{n\times n}{\Sigma^{-1}} \underset{n\times 1}{(X-\mu)} \right] + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} E\left[\sum_{i,j} (X_i - \mu_i) \underbrace{\Sigma^{-1}_{ij}}_{(\Sigma^{-1})_{ij}} (X_j - \mu_j) \right] + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} E\left[\sum_{i,j} (X_i - \mu_i)(X_j - \mu_j) \Sigma^{-1}_{ij} \right] + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} \left\{ \sum_{i,j} E\left[(X_i - \mu_i)(X_j - \mu_j) \right] \cdot \Sigma^{-1}_{ij} \right\} + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} \left[\sum_{i,j} \left(\sum_{j,i} \Sigma^{-1}_{ij} \right) \right] + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} \left[\sum_{j} \left(\sum_{j,i} \left(\sum_{i,j} \Sigma^{-1}_{ij} \right) \right) \right] + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} \left[\sum_{j} (\Sigma \Sigma^{-1})_{jj} \right] + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} \left(\sum_{j} I_{jj} \right) + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{n}{2} + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{n}{2} \ln e + \frac{1}{2} \ln\left[(2\pi)^n |\Sigma| \right]$$

$$= \frac{1}{2} \ln\left[(2\pi e)^n |\Sigma| \right] \quad \text{nats}$$

$$= \frac{1}{2} \log_2\left[(2\pi e)^n |\Sigma| \right] \quad \text{bits}$$

■

4) Kullback - Liebler Divergence and Mutual Information

Definition - Support of a PDF

Given pdf f on $\mathbb{R}^n$, the support of f is

$$S_f \triangleq \{ x \in \mathbb{R}^n \mid f(x) > 0 \}$$

Definition - Divergence

f, g are pdfs on $\mathbb{R}^n$ with $S_f \subseteq S_g$. KL divergence between f and g, $D(f\|g)$ is,

$$D(f\|g) = \int_{\mathbb{R}^n} f(x) \cdot \log \frac{f(x)}{g(x)} \, dx$$

- Convention is $0 \cdot \log \frac{0}{0} = 0$
- If $S_f \not\subseteq S_g$ we set $D(f\|g) = \infty$.

Theorem 2 - f, g are pdfs on $\mathbb{R}^n$. Then $D(f\|g) \geq 0$ with equality iff $f = g$ allmost everywhere on S_f. (and hence on $\mathbb{R}^n$).

Proof - 1. Using Logsum Inequality

Let f, g be pdfs on $\mathbb{R}^n$. If $g(x) = 0$ on a set where $f(x) > 0$, then

$$D(f\|g) = +\infty,$$

and there is nothing to prove. Hence assume $g(x) > 0$ whenever $f(x) > 0$.

By the integral form of the log-sum inequality,

$$\int f(x) \log \frac{f(x)}{g(x)} \, dx \geq \left(\int f(x) \, dx \right) \log \frac{\int f(x) \, dx}{\int g(x) \, dx}.$$

Since f and g are pdfs,

$$\int f(x) \, dx = 1, \qquad \int g(x) \, dx = 1.$$

Therefore,

$$D(f\|g) = \int f(x) \log \frac{f(x)}{g(x)}\, dx \geq 1 \log \frac{1}{1} = 0.$$

Hence

$$\boxed{D(f\|g) \geq 0.}$$

Equality in the log-sum inequality holds iff

$$\frac{f(x)}{g(x)}$$

is constant a.e. on the support of f. Since both densities integrate to 1, that constant must be 1. Hence

$$f(x) = g(x)$$

a.e. on $S_f = \{x : f(x) > 0\}$. Also, because $\int f = \int g = 1$, this implies $g = 0$ a.e. outside S_f. Thus

$$\boxed{D(f\|g) = 0 \iff f = g \text{ a.e. on } \mathbb{R}^n.}$$

2. Using Jensen's Inequality

Let $S_f = \{x : f(x) > 0\}$. If $g = 0$ on a set where $f > 0$, then $D(f\|g) = +\infty$, and the result is trivial. Otherwise,

$$D(f\|g) = \int_{S_f} f \log \frac{f}{g} = -\int_{S_f} f \log \frac{g}{f}.$$

Since log is concave, Jensen's inequality gives

$$\int_{S_f} f \log \frac{g}{f} \leq \log \int_{S_f} f \frac{g}{f} = \log \int_{S_f} g \leq \log 1 = 0.$$

Hence

$$D(f\|g) \geq 0.$$

Equality holds iff equality holds in Jensen and $\int_{S_f} g = 1$. Thus

$$\frac{g}{f} = c \quad f\text{-a.e. on } S_f.$$

Since $\int f = \int g = 1$, we get $c = 1$, and also $g = 0$ a.e. on S_f^c. Hence

$$D(f\|g) = 0 \iff f = g \text{ a.e. on } \mathbb{R}^n.$$

Definition- Mutual Information

X, Y are $\mathbb{R}^n$ and $\mathbb{R}^1$ valued rvs with joint pdf f_{XY} and marginal pdfs f_X and f_Y. Mutual Information of X and Y is,

$$I(X \wedge Y) = D(f_{XY} \,\|\, f_X f_Y)$$

$$= \int_{\mathbb{R}^{k+1}} f_{XY}(x,y) \log \frac{f_{XY}(x,y)}{f_X(x) \cdot f_Y(y)}\, dx\, dy$$

Remarks -

1) $I(X \wedge Y) = D(f_{xy} \,\|\, f_x f_y) \geq 0$
with equality iff $f_{xy}(x,y) = f_x(x) f_y(y) \quad \forall x \in \mathbb{R}^k, y \in \mathbb{R}^l$
i.e. iff $X \perp\!\!\!\perp Y$

2) $I(X \wedge Y) = I(Y \wedge X)$

3) $I(X \wedge Y) = h(X) - h(X|Y) = h(Y) - h(Y|X)$
Cautionary note is when $X = Y$ we run into few difficulties.
$\Rightarrow h(X|Y) = h(X|X)$ is not defined.
$\Rightarrow I(X \wedge Y) = I(X \wedge X)$ is not defined.
Specifically F_{XY} does not admit a density w.r.t the Lebesgue measure if $X = Y$

$$\begin{aligned} F_{XY}(x,y) &= P(X \leq x, Y \leq y) \\ &= P(X \leq x, X \leq y) \\ &= F_X(\min(x,y)) \end{aligned}$$

this is not differentiable as a function of (x,y).

4) $0 \leq I(X \wedge Y) = h(X) - h(X|Y) \Rightarrow h(X|Y) \leq h(X)$
with equality if $X \perp\!\!\!\perp Y$

5) Other Results

Theorem 3 - Chain Rule for Differential Entropy

$$h(X_1, \ldots, X_n) = \sum_{t=1}^{n} h(X_t \mid X_1, \ldots, X_{t-1})$$

provided all the differential entropies exist.

Corollary - $h(X_1, \ldots, X_n) \leq \sum_{t=1}^{n} h(X_t)$ with equality iff $X_1, \ldots, X_n$ are mutually $\perp\!\!\!\perp$.

Theorem 4 - X is $\mathbb{R}^n$ valued rv and $C \in \mathbb{R}^n$ is a constant. Then $h(X+C) = h(X)$

Theorem 5 - X is a $\mathbb{R}^n$ valued rv. Let $0 \neq a \in \mathbb{R}$ is a constant.

$$h(aX) = h(X) + \log |a|$$

Proof: $Y \triangleq aX$

$$\Rightarrow f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y}{|a|}\right) \qquad y \in \mathbb{R}$$

$$\Rightarrow h(aX) = h(Y) = -\int_{\mathbb{R}} \frac{1}{|a|} f_X\left(\frac{y}{|a|}\right) \cdot \log\left(\frac{1}{|a|} \cdot f_X\left(\frac{y}{|a|}\right)\right) dy$$

Let $z = y/a$

$$\Rightarrow h(aX) = h(Y) = -\int_{\mathbb{R}} f_X(z) \cdot \log\left(\frac{1}{|a|} f_X(z)\right) dz$$

$$= h(X) + \log |a|$$

■

Corollary - Let X be a $\mathbb{R}^n$-valued rv, and A is a $n \times n$ matrix with $\mathbb{R}$-valued entries. Assume A^{-1} exists. Then,

$$h(AX) = h(X) + \log |A|$$

where $|A|$ = absolute value of $\det(A)$.

Theorem 6. Let X is a $\mathbb{R}^n$ valued rv with $\mathbb{E}[X] = 0$ and $\text{Cov}[X] = \mathbb{E}[XX^T] = \Sigma$, where Σ^{-1} is assumed to exist. Then,

$$h(X) \leq \frac{1}{2} \log \left[(2\pi e)^n |\Sigma| \right]$$

with equality iff $X \sim N(0, \Sigma)$

Proof: Let f be the pdf of X Note,

$$\Sigma = \mathbb{E}\left[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^T \right] \quad \cdots \quad (1)$$

Let $g \sim N(0, \Sigma)$ be the gaussian distribution. Then from positivity of divergence,

$$\begin{aligned} 0 \leq D(f \| g) &= \int_{\mathbb{R}^n} f(x) \cdot \log \frac{f(x)}{g(x)} \, dx \\ &= -h(f) - \int_{\mathbb{R}^n} f(x) \cdot \log g(x) \, dx \\ &= -h(f) - \mathbb{E}_f [\log g(x)] \\ &= -h(f) - \mathbb{E}_g [\log g(x)] \quad \cdots \text{ since } \log g(x) \text{ is a quadratic form and } X \text{ has the same second order moments under } f \text{ and } g. \\ &= -h(f) - (-h(g)) \end{aligned}$$

$\Rightarrow h(f) \leq h(g)$ with equality iff $f = g$. ∎

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Chapter 9

Gaussian Channel

- Let $X \sim N(0, \rho)$. Let this signal X encounters a gaussian noise $N \sim N(0, \sigma^2)$ in the channel which is independant of signal X.

$$Y \triangleq X + N \qquad N \perp\!\!\!\perp X$$

- $Y \sim N(0, \rho + \sigma^2)$.

- $E[X] = 0 \qquad E[X^2] = \rho$

- $$\begin{aligned} I(X \wedge Y) &= h(Y) - h(Y|X) \\ &= \frac{1}{2} \log\left[(2\pi e)(\rho+\sigma^2)\right] - h(X+N|X) \\ &= \frac{1}{2} \log\left[(2\pi e)(\rho+\sigma^2)\right] - h(N|X) \\ &= \frac{1}{2} \log\left[(2\pi e)(\rho+\sigma^2)\right] - h(N) \quad \cdots N \perp\!\!\!\perp X \\ &= \frac{1}{2} \log\left[(2\pi e)(\rho+\sigma^2)\right] - \frac{1}{2}\log\left[2\pi e \sigma^2\right] \\ &= \frac{1}{2} \log_2 \left(1 + \rho/\sigma^2\right) \text{ bits} \end{aligned}$$

Interpret SNR

Theorem - X is a $\mathbb{R}$ valued rv with given pdf f, satisfying $E[X^2] < \infty$. Then for every $x \in \mathbb{R}$

$$E[(X-x)^2] \geq \frac{1}{2\pi e} \exp[2h(X)]$$

with equality iff $X \sim N$ and $x = E[X]$.

Proof : ▶ For any $x \in \mathbb{R}$,

$$\mathbb{E}[(X-x)^2] \overset{i}{\geq} \min_{x' \in \mathbb{R}} \mathbb{E}[(X-x')^2]$$

$$= \mathbb{E}\left[(X-\mathbb{E}[X])^2\right]$$

$$= \mathrm{Var}[X]$$

$$\overset{ii}{\geq} \frac{1}{2\pi e} \exp[2h(X)] \quad \because h(X) \leq \frac{1}{2}\log(2\pi e \,\mathrm{Var}(X))$$

▶ For the equality condition, equality should hold in (i) and (ii)

(i) $\iff$ $x = \arg\min_{x' \in \mathbb{R}} \mathbb{E}[(X-x')^2] = \mathbb{E}[X]$

(ii) $\iff$ $X \sim N$

$\therefore$ for the equality $X \sim N$ and $x = \mathbb{E}[X]$ ■

Corollary - Suppose $g(Y)$ is an estimator X on the basis of Y, where (X,Y) have a given joint CDF and X has a pdf. Then, estimation error satisfies,

$$\mathbb{E}\left[(X-g(Y))^2\right] \geq \frac{1}{2\pi e} \exp[2h(X|Y)]$$

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Chapter 10

Rate Distortion Theory

i.e

Source Coding with a Fidelity Criterion

i.e

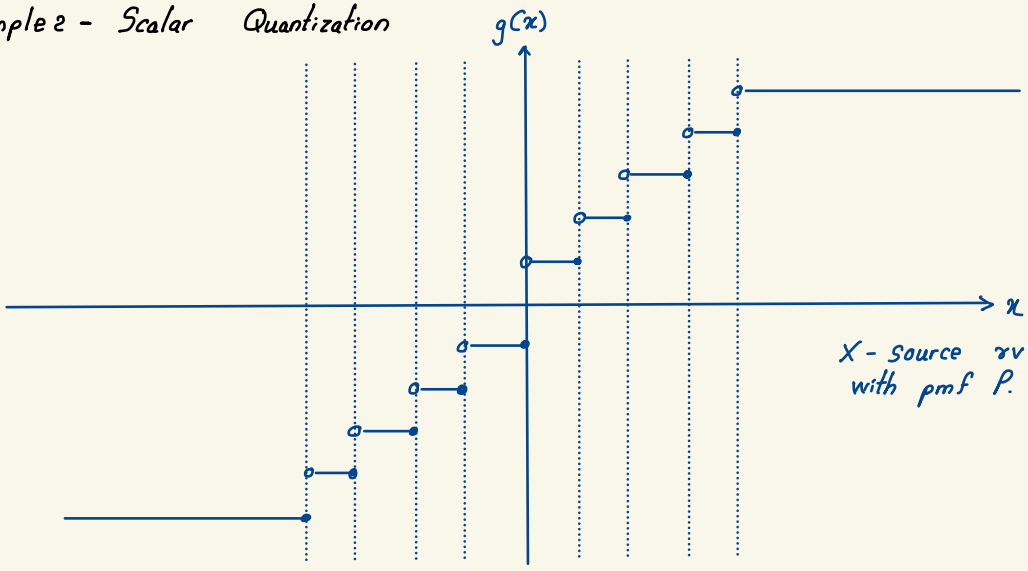
Lossy Source Coding

1) Introduction

Example 1 - Take an arbitary $\mathbb{R}$-valued rv X. Represent the value of X by a finite no. of bits.

More bits $\iff$ Higher rate in bits / rv value.
$\implies$ Less distortion.

Example 2 - Scalar Quantization



If this is a N-level quantizer

R = rate (bits / source rv value) = $\log_2 N \implies N = 2^R$

$g(x)$ = Quantizer output

$\|g\|$ = Cardinality of the range of g $\implies N = \|g\|$
$\implies \log_2 \|g\| = R$

Distortion - Rate

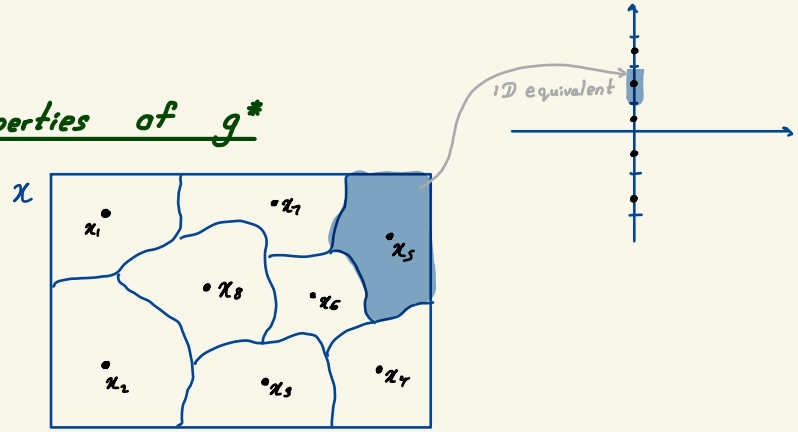
- Find $\min_g \underbrace{\mathbb{E}[(X - g(X))^2]}_{\text{Distortion}}$ subject to $\|g\| \leq 2^R$
- What is the minimum distortion and optimal g^*.

Rate Distortion

A dual formulation.

- $\min_g \underbrace{\|g\|}_{2^{\text{"rate"}}}$ such that $\mathbb{E}_P[(X - g(X))^2] \leq D$
- What is min rate as a function of D and optimal g^*.

Two Usual Properties of g^*



- Given a set $\{\hat{X}(\omega)\}$ of reconstruction points, the distortion is minimized if all points are mapped to the representation $\hat{x}(\omega)$ closest to it. The set of regions of $\mathcal{X}$ defined by this mapping is called Voronoi / Dirichlet partition.

- Given a partition of $\mathcal{X}$, each representation point x_i for region i should minimize,

$$\min_{x_i} \mathbb{E}\left[\text{distortion}(X, x_i) \mid X \in \text{region } i\right]$$

Loyd Algorithm

1) Start with a set of reconstruction points.

2) Find the optimal set of reconstruction regions, which are nearest neighbours to above selected points.

3) Take the optimal reconstruction points (i.e centroids) for above regions

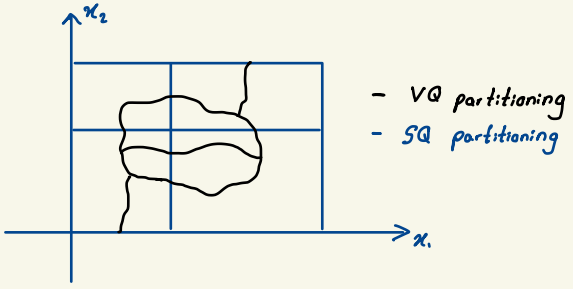
4) Repeat

Scalar vs Vector Quantization

Suppose source has X^n i.i.d source components. It is allowed nR bits to represent $X_1, \ldots, X_n$.

- 2^R Level Scalar Quantizer - Each source rv is quantized using R bits.
- Vector Quantizer - Quantize X^n using nR bits.

VQ outperformce SQ.



2) Definitions

Problem Formulation

- Let $\{X_t\}_{t=1}^{\infty}$ is an i.i.d DMS with finite alphabet $\mathcal{X}$ and pmf P.
- A n-block source $x^n \in \mathcal{X}^n$ will be represented / reproduced as $y^n \in \mathcal{Y}^n$, where $\mathcal{Y}$ is the reproduction alphabet.
- In general y^n is a possibly distorted version of x^n.

Distortion Measure / Distortion Function

Definition - Distortion function / Distortion Measure $\{d_n\}_{n=1}^{\infty}$

The distortion measure $d_n(x^n, y^n)$ is a measure of the cost of representing the source $x^n \in \mathcal{X}^n$ as the reproduction $y^n \in \mathcal{Y}^n$.

$$d_n : \mathcal{X}^n \times \mathcal{Y}^n \longrightarrow \mathbb{R}^+ \cup \{0\}$$

- The distortion of the sequences x^n, y^n is is considered as the average of the per symbol distortion of the elements of the sequence.

Single Letter Distortion : $d : \mathcal{X} \times \mathcal{Y} \longrightarrow \mathbb{R}^+ \cup \{0\}$

Average distortion measure $= d_n(x^n, y^n) = d(x^n, y^n) \triangleq \frac{1}{n} \sum_{t=1}^{n} d(x_t, y_t)$

Assumptions

I) A distortion measure is said to be bounded if the maximum distortion is bounded.

$$d_{max} \triangleq \max_{\substack{x \in \mathcal{X} \\ y \in \mathcal{Y}}} d(x,y) < \infty$$

II) For each $x \in \mathcal{X}$, $\exists$ at least one $y \in \mathcal{Y}$ s.t. $d(x,y) = 0$

Examples

I) Hamming Distortion - need boath assumptions

Assume source and reproduction alphabets $\mathcal{X}, \mathcal{Y}$ are finite and equal. i.e. $\mathcal{X} = \mathcal{Y}$, $|\mathcal{X}| < \infty$

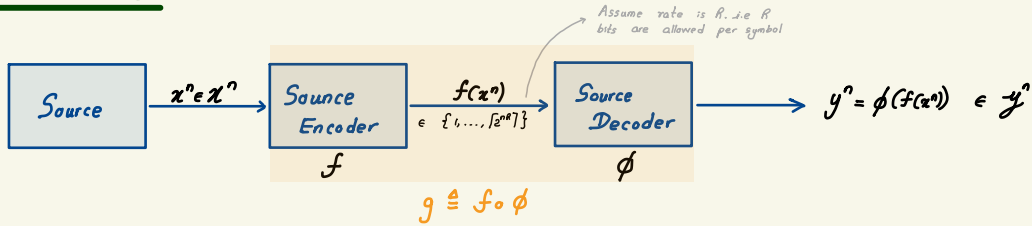
$$d(x,y) = \mathbb{1}(x \neq y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

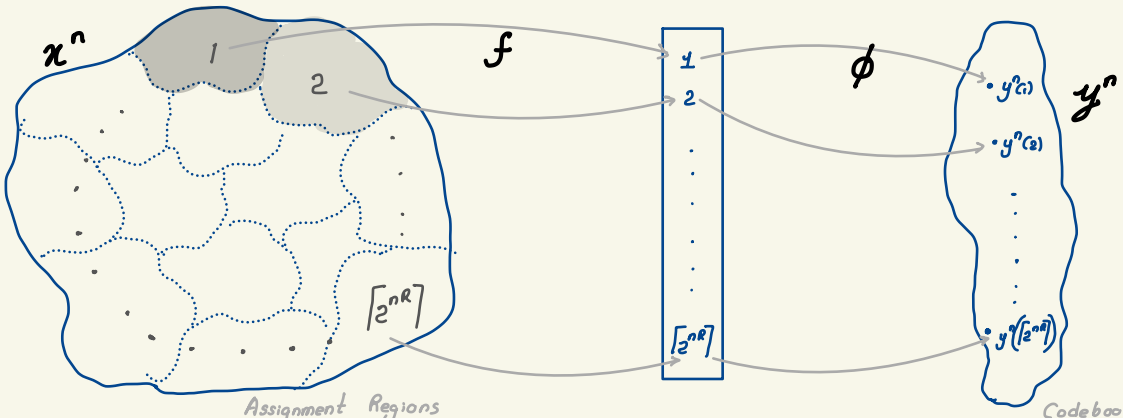
$$\text{Expected distortion} = \mathbb{E}_p[d(X,Y)] = \mathbb{E}_p[\mathbb{1}(X=Y)] = \underbrace{P(X=Y)}_{\text{probability of error}}$$

II) Square Error Distortion - need only the second assumption.

$$d(x,y) = (x-y)^2 \qquad \mathcal{X} = \mathbb{R}, \quad \mathcal{Y} = \mathbb{R}$$

Block Codes





- This is a classic "covering problem" in combinatorics
- Assignment Regions = $\{ f^{-1}(1), \ldots, f^{-1}(\lceil 2^{nR} \rceil) \}$
- Codebook = $\{ y^n(1) = \phi(1), \ldots, y^n(\lceil 2^{nR} \rceil) = \phi(\lceil 2^{nR} \rceil) \}$

Definition - $(2^{nR}, n)$ Block Code

$(2^{nR}, n)$ Block code consists of a encoder and a decoder $(f, \phi) = (f^{(n)}, \phi^{(n)})$, where f maps $\mathcal{X}^n$ into some finite set

$$f: \mathcal{X}^n \longrightarrow \{1, \ldots, \lceil 2^{nR} \rceil\}$$

and ϕ maps the range of f into $\mathcal{Y}^n$

$$\phi: \{1, \ldots, \lceil 2^{nR} \rceil\} \longrightarrow \mathcal{Y}^n$$

- g which is the combination of $f \circ \phi$ maps $\mathcal{X}^n$ to $\mathcal{Y}^n$

Hence

$$g: \mathcal{X}^n \longrightarrow g(x^n) \triangleq \phi(f(x^n)) = y^n \in \mathcal{Y}^n$$

Rate

Definition - R - Rate of Code

The code rate for a $(2^{nR}, n)$ block code (f, ϕ) is

$$R \triangleq \frac{1}{n} \log_2 \|f\|$$

$$= \frac{1}{n} \log_2 (\text{no. of elements in the range of } f)$$

- When R reduces, the number of assignment regions reduce, hence increasing the uncertainty in reproduction, i.e. distortion.

Distortion

Definition - Distortion - D

If $\{X_n\}_{n=1}^{\infty}$ is a i.i.d DMS with pmf P and (f, ϕ) be a $(2^{nR}, n)$ block code. The distortion of it with respect to distortion measure $\{d_n\}_{n=1}^{\infty}$ is

$$D = E_P\left[d_n(X^n, \phi(f(X^n))) \right]$$

Definition - Average Fidelity Criterion - $(\{d_n\}, D)$

$(2^{nR}, n)$ block code meets average fidelity criterion $(\{d_n\}, D)$ is the expected distortion is not greater than D,

$$E_P\left[d_n(X^n, \phi(f(X^n))) \right] = E_P\left[d_n(X^n, g(X^n)) \right] \leq D$$

Definition - ε - fidelity Criterion

$$P\left(d_n(X^n, \phi(f(X^n))) \leq D \right) \geq 1 - \varepsilon$$

- Boath formulations for a DMS yields the same assymptotic rate of compression, R.
- **Problem** - Given the $(\{d_n\}, D)$ - Average Fidelity Criterion, construct codes $(f, \phi) = (f^{(n)}, \phi^{(n)})$ that achieve maximum data compression. (i.e minimal rate R).

Achievability

Definition - Achievable Rate Distortion Pairs (R, D)

For a DMS $\{X_t\}_{t=1}^{\infty}$ and for a given distortion measure $\{d_n\}_{n=1}^{\infty}$, a rate $R \geq 0$ is achievable at distortion level D if,

$\forall\ \varepsilon > 0$ $\exists\ n = n(\varepsilon)$ - sufficiently large such that there exist a n - length block code (f_n, ϕ_n) of

$$\text{rate} = \frac{1}{n} \log \|f\| = \frac{1}{n} \log \|f^{(n)}\| < R + \varepsilon$$

which satisfies the fidelity criterion

$$(\{d_n\}, D + \varepsilon)$$

- Alternate Definition -

  A rate distortion pair (R, D) is said to be achievable if there exists a sequence of $(2^{nR}, n)$ - block codes (f, ϕ) with $\lim_{n\to\infty} \mathbb{E}_p\left[d(x^n, \phi(f(x^n)))\right] \leq D$

- **Claim** - If $R > 0$ is achievable $\Longrightarrow$ $R' > R$ is achievable.

Distortion Functions

Definition - **Rate Distortion Function - R(D)**

Infimum of all achievable rates at distortion level D is called $R(D)$ - Rate Distortion Function for $\{X_t\}_{t=1}^{\infty}$

Definition - **Distortion Rate Function - D(R)**

Infimum of all achievable distortions D, at rate R is called $D(R)$ - Distortion Rate Function for $\{X_t\}_{t=1}^{\infty}$

3) Rate Distortion Theorem (Preview)

Theorem 1- Rate Distortion Theorem

Achievability - Let $\{X_t\}_{t=1}^{\infty}$ be a DMS with source pmf P on $\mathcal{X}$. Let $\{d_n\}_{n=1}^{\infty}$ be the distortion measure. Then $\forall\ D \geq 0$

$$R(D) = R(P,D) = \tilde{R}(P,D) = \min_{\substack{P_X = P \\ \mathbb{E}[d(X,Y)] \leq D}} I(X \wedge Y)$$

Defined in terms of operational significance

$$= \min_{W:\ \sum_{x,y} P(x)\cdot W(y|x)\cdot d(x,y) \leq D} I(P,W)$$

P_{XY}

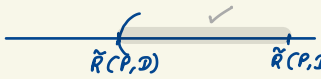
When P_X fixed to P and $d(x,y)$ is fixed over P_{XY}. effectively $P(y|x) = W(y|x)$ is what's changing.

where $0 \leq D \leq D_{max} = \min_{y \in \mathcal{Y}} \sum_x P(x)\cdot d(x,y)$

If distortion is higher than D_{max}, rate becomes zero. Why?

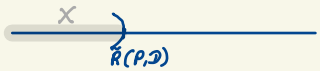
Converse - Any sequence of codes $\{f^{(n)}, \phi^{(n)}\}_{n=1}^{\infty}$ with $\mathbb{E}[d(X^n, g_n(X^n))] \leq D+\delta$ for $(g_n \triangleq \phi_n \circ f_n)$ any n, where $\delta > 0$ is arbitary must have rate $\geq R(P,D)$

- LHS = $R(D) = R(P,D)$ - defined operationally.
- RHS = $\min_{\mathbb{E}[d(X,Y)] \leq D} I(X \wedge Y) \triangleq \tilde{R}(P,D)$ - definition for the sake of precision.
- Rate distortion theorem is about proving LHS $\cong$ RHS.
- Achievability of RDT -



Any rate in this range is achievable

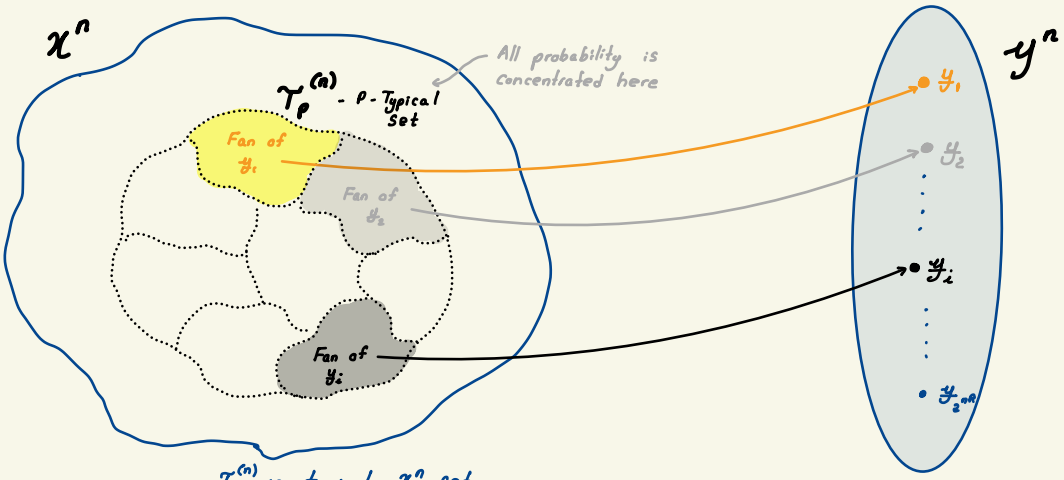
- Converse of RDT -



Any rate in this range is not achievable

Heuristics of RDT

- Similar to the packing argument we had in channel capacity.
- In Lossless compression each fan had only one x^n.
  $\therefore 2^{nR} = 2^{nH(x)} \Rightarrow R \cong H(x)$



$T_\rho^{(n)}$ is typical x^n set
$\therefore$ Cardinality $\simeq 2^{nH(x)}$

- P_{XY} is designers choice, and $P_X \simeq P$.
- From condition 1 and 3 in Joint-Typical set definition, if y_i is reproduced, only certain x^n's are likely and they should be in the fan of y_i and have approximately equal probability of

$$\Rightarrow P(x^n/y_i) = \frac{P(y_i, x^n)}{P(y_i)} \simeq \frac{2^{nH(X,Y)}}{2^{nH(Y)}} \simeq 2^{-nH(X|Y)}$$

$$\Rightarrow \text{Size of fan of } y_i \simeq \frac{1}{P(x^n/y_i)} \simeq 2^{nH(X|Y)}$$

- All the 2^{nR} fans each of size $2^{nH(X|Y)}$ should fit inside the x^n's ρ-typical set of size $2^{nH(x)}$.

$$\Rightarrow 2^{nH(x)} \cong 2^{nR} \cdot 2^{nH(X|Y)}$$
$$\Rightarrow H(x) \cong R + H(X|Y)$$
$$\Rightarrow R \cong I(X;Y)$$
$$\Rightarrow \text{When } I(X;Y) \overset{\text{minimum}}{\simeq} R \overset{\text{minimum}}{\Longrightarrow} 2^{nR} \text{ no. of codewords are minimum}$$
$$\Rightarrow \text{To achieve the minimum rate } R, \text{ we should achieve minimum, mutual information.}$$

- We only consider the x^n that are in ρ-Typical set. From the distortion assumption,

$$\Rightarrow d_n(x^n, y^n) \leq D$$
$$\Rightarrow \frac{1}{n}\sum_{t=1}^{n} d(x_t, y_t) \leq D$$
$$\Rightarrow E[d(x,y)] \leq D$$

Properties of $\tilde{R}(P,D)$

Theorem 2- Lemma 10.4.1 in the book

For a fixed pmf P on finite $\mathcal{X}$, define

$$\tilde{R}(P,D) = \min_{W:\, \mathbb{E}[d(X,Y)] \le D} I(X;Y)$$

where $P_{XY}(x,y) = P_X(x) \cdot W(y|x)$. Then $\tilde{R}(P,D)$ is bounded, non-increasing, convex in D, and continuous in (P,D).

Proof.
Let

$$\rho(P,W) = \sum_{x,y} P(x)W(y|x)d(x,y).$$

Then

$$\tilde{R}(P,D) = \min_{W:\rho(P,W)\le D} I(P,W).$$

First, for every feasible W,

$$0 \le I(P,W) \le H(P) \le \log|\mathcal{X}|.$$

- Bounded above by the entropy of P.
- Corollary: $\tilde{R}(P,D) \le H(P)$
- Above equality holds for $D=0$ when P is Hamming Distortion.

Hence $\tilde{R}(P,D)$ is bounded.
Next, if $D_1 \le D_2$, then

$$\{W : \rho(P,W) \le D_1\} \subseteq \{W : \rho(P,W) \le D_2\}.$$

Thus the minimization for D_2 is over a larger set, and so

$$\tilde{R}(P,D_2) \le \tilde{R}(P,D_1).$$

Therefore $\tilde{R}(P,D)$ is non-increasing in D.
Now let $0 \le \lambda \le 1$ and set

$$D_\lambda = \lambda D_1 + (1-\lambda)D_2.$$

Let W_1, W_2 be optimal for D_1, D_2, respectively. Define

$$W_\lambda = \lambda W_1 + (1-\lambda)W_2.$$

Then

$$\rho(P,W_\lambda) = \lambda\rho(P,W_1) + (1-\lambda)\rho(P,W_2) \le D_\lambda.$$

So W_λ is feasible for D_λ. Also, for fixed P, $I(P,W)$ is convex in W. Hence

$$\tilde{R}(P,D_\lambda) \le I(P,W_\lambda) \le \lambda I(P,W_1) + (1-\lambda)I(P,W_2).$$

Therefore

$$\tilde{R}(P,D_\lambda) \le \lambda\tilde{R}(P,D_1) + (1-\lambda)\tilde{R}(P,D_2),$$

so $\tilde{R}(P,D)$ is convex in D.
Finally, since the alphabets are finite, the set of stochastic matrices W is compact. Also, both

$$(P,W) \mapsto I(P,W) \qquad \text{and} \qquad (P,W) \mapsto \rho(P,W)$$

are continuous. Therefore, by compactness and continuity of the minimization problem, the value

$$(P,D) \mapsto \tilde{R}(P,D)$$

is continuous.

Hence $\tilde{R}(P,D)$ has all the claimed properties.

Lemma 10.4.1 *(Convexity of $R(D)$) The rate distortion function $R(D)$ given in (10.53) is a nonincreasing convex function of D.*

Proof from the book.

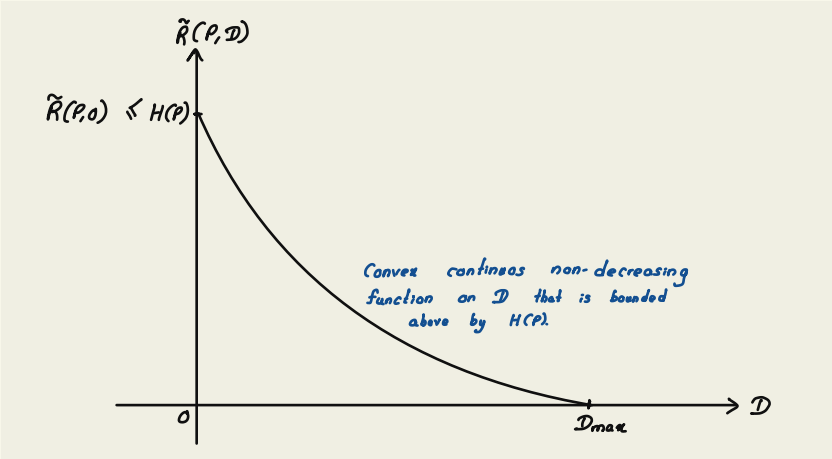
Proof: $R(D)$ is the minimum of the mutual information over increasingly larger sets as D increases. Thus, $R(D)$ is nonincreasing in D. To prove that $R(D)$ is convex, consider two rate distortion pairs, (R_1, D_1) and (R_2, D_2), which lie on the rate distortion curve. Let the joint distributions that achieve these pairs be $p_1(x,\hat{x}) = p(x)p_1(\hat{x}|x)$ and $p_2(x,\hat{x}) = p(x)p_2(\hat{x}|x)$. Consider the distribution $p_\lambda = \lambda p_1 + (1-\lambda)p_2$. Since the distortion is a linear function of the distribution, we have $D(p_\lambda) = \lambda D_1 + (1-\lambda)D_2$. Mutual information, on the other hand, is a convex function of the conditional distribution (Theorem 2.7.4), and hence

$$I_{p_\lambda}(X;\hat{X}) \le \lambda I_{p_1}(X;\hat{X}) + (1-\lambda)I_{p_2}(X;\hat{X}). \tag{10.54}$$

Hence, by the definition of the rate distortion function,

$$\begin{aligned} R(D_\lambda) &\le I_{p_\lambda}(X;\hat{X}) && (10.55)\\ &\le \lambda I_{p_1}(X;\hat{X}) + (1-\lambda)I_{p_2}(X;\hat{X}) && (10.56)\\ &= \lambda R(D_1) + (1-\lambda)R(D_2), && (10.57) \end{aligned}$$

which proves that $R(D)$ is a convex function of D. □



Corollary : $\tilde{R}(P,0) \leq H(P)$

Proof : Straightforward from the bounded above proof in Theorem 2.

Exercise - Show that $\tilde{R}(P,D)$ is the $H(P)$ for hamming distortion at distortion level $D = 0$.

Proof : For hamming distortion, $x = y$

$$\Rightarrow d(x,y) = d_H(x,y) = \mathbb{1}(x \neq y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

Since distortion level is '0'

$$\Rightarrow \mathbb{E}[d(x,y)] \leq D = 0$$

$$\Rightarrow \mathbb{E}_p[d_H(x,y)] = 0$$

$$\Rightarrow \mathbb{E}_p[\mathbb{1}(x \neq y)] = 0 \quad \Rightarrow 1 \cdot P(x \neq y) + 0 \cdot P(x = y) = 0$$

$$\Rightarrow P(x \neq y) = 0 \quad \cdots\cdots (1)$$

Hence from Rate Distortion Theorem,

$$\Rightarrow \tilde{R}(P,0) = \min_{\substack{p_x = p \\ \mathbb{E}[d(x,y)] \leq D}} I(x \wedge y)$$

$$= \min_{P(x \neq y) = 0} \Big[H(x) - \underbrace{H(x|y)}_{0 \text{ when } P(x \neq y) = 0} \Big] \quad \cdots\cdots \text{ from (1)}$$

$$= H(x)$$

$$= H(P).$$

∎

Exercise - For Hamming Distortion

Expected block distortion = Average bit error probability

Proof: For hamming distortion, $x = y$

$$\Rightarrow d(x,y) = d_H(x,y) = \mathbb{1}(x \neq y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

$$\text{Expected block distortion} = \mathbb{E}\left[d_H(x^n, g(x^n)) \right]$$

$$= \mathbb{E}\left[\frac{1}{n} \sum_{t=1}^{n} d_H(x_t, g(x^n)_t) \right]$$

$$= \frac{1}{n} \sum_{t=1}^{n} \mathbb{E}\left[\mathbb{1}(x_t \neq g(x^n)_t) \right]$$

$$= \frac{1}{n} \sum_{t=1}^{n} P(x_t \neq g(x^n)_t)$$

$$= \text{Average bit error probability.}$$

■

$g = f \circ \phi$

Lemma - Average bit error probability $\frac{1}{n} \sum_{t=1}^{n} P(x_t \neq (g(x^n))_t)$ $\leq$ Block error probability $P(g(x^n) \neq x^n)$

Proof: Let $g = \varphi \circ f$, and define

$$Y^n = g(X^n) = \varphi(f(X^n)).$$

For each $t = 1, \ldots, n$,

$$\{X_t \neq Y_t\} \subseteq \{X^n \neq Y^n\}.$$

Therefore

$$P(X_t \neq Y_t) \leq P(X^n \neq Y^n), \qquad t = 1, \ldots, n.$$

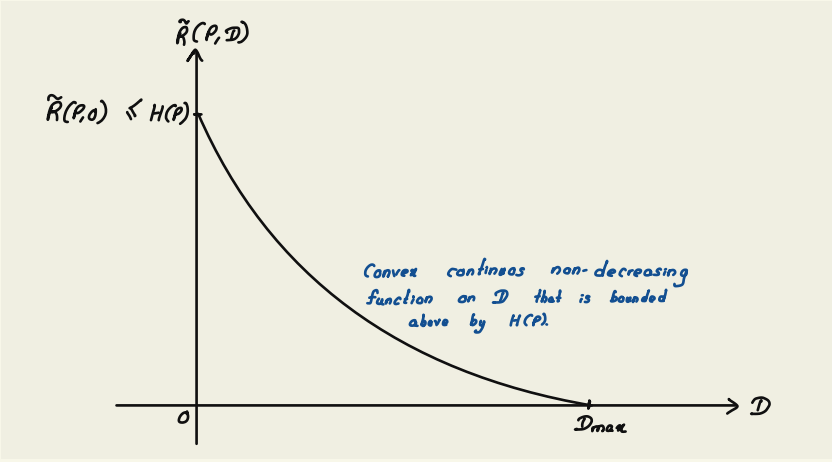
Averaging over t, we get

$$\frac{1}{n} \sum_{t=1}^{n} P(X_t \neq Y_t) \leq \frac{1}{n} \sum_{t=1}^{n} P(X^n \neq Y^n) = P(X^n \neq Y^n).$$

Since $Y^n = g(X^n)$, this gives

$$\frac{1}{n} \sum_{t=1}^{n} P\big(X_t \neq g(X^n)_t\big) \leq P\big(g(X^n) \neq X^n\big).$$

Properties of D_{max}



Definition - $D_{max} \triangleq \min \{ D : \tilde{R}(P,D) = 0 \}$

$=$ Smallest D for which $\exists P_{xy}$ with $P_{xy} = P_x P_y$, $P_x = P$, $\mathbb{E}[d(x,y)] \leq D$

$=$ Smallest value of $\mathbb{E}[d(x,y)]$ (smallst w.r.t P_{xy}) such that $P_{xy} = P_x P_y$

Claim - $\boxed{D_{max} = \min_{y} \sum_{x'} P(x')\, d(x',y)}$

Proof :- According to the above definition of D_{max}

$$D_{max} \triangleq \min_{\substack{P_{xy} = P_x P_y \\ P_x = P}} \mathbb{E}[d(x,y)]$$

$$= \min_{\substack{P_x, P_y \\ P_x = P}} \sum_{x,y} \underbrace{P_x(x) \cdot P_y(y)}_{P_{xy}(x,y)} \cdot d(x,y)$$

$$= \min_{P_y} \sum_{y} \left[P_y(y) \cdot \left(\underbrace{\sum_{x} P_x(x) \cdot d(x,y)}_{\text{A function of } y.} \right) \right]$$

$$= \min_{y} \sum_{x} P_x(x) \cdot d(x,y)$$

■

Interpretation :- At $D = D_{max}$, the minimum mutual information ($=0$) is achieved by mapping all xs to that y for which $\sum_{x'} p(x')\, d(x',y)$ is minimal.

4) Calculation of Rate Distortion Function R(D)

Theorem 3- The rate distortion function for a Bernoulli(p) source with Hamming distortion is given by,

$$R(D) = \begin{cases} h(p) - h(D) & 0 \leq D \leq D_{max} = \min\{p, 1-p\} \\ 0 & D > D_{max} = \min\{p, 1-p\} \end{cases}$$

Proof:- ▸ Let $\{X_t\}_{t=1}^{\infty}$ is i.i.d and has the pmf $P = (1-p, p)$, where $p = P_X(x)$, $0 < p < 1$. $\mathcal{Y} = \{0, 1\}$. Since it has Hamming distortion,

$$d(x,y) = d_H(x,y) = \mathbb{1}(x \neq y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

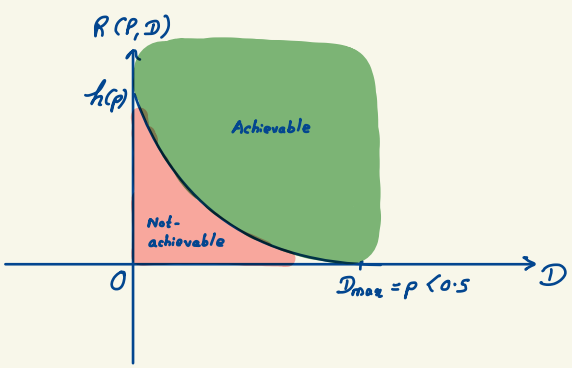
Finding D_{max}

▸ $$D_{max} = \min_{y} \sum_{x} p(x) \cdot d(x,y)$$

$$= \min \left\{ \sum_{x} p(x) \cdot d(x,0), \sum_{x} p(x) \cdot d(x,1) \right\} \quad \cdots \cdots \; y \in \{0,1\}$$

$$= \min\{ p(x=1) \cdot 1, p(x=0) \cdot 1 \} = \min(p, 1-p)$$

$$= \underline{\underline{p}} \quad \cdots \cdots \text{ w.l.o.g we assume } p \leq 0.5$$



▸ The minimizing constraints in RDT becomes,

$$E[d(X,Y)] = P(X \neq Y) \leq D \quad \cdots \cdots (1)$$

$$0 \leq D \leq D_{max} = p \leq 0.5$$

- Method : Find a Lower Bound for $R(P,D)$, (called Shanon Lower Bound) then a feasible P_{XY} that attains it.

Finding Shanon's Lower Bound

- Let $\oplus$ = addition modulo 2. Then

$$X \oplus Y = 1 \iff X \neq Y \quad \cdots\cdots (2)$$

- Method 1

$$\begin{aligned} I(X \wedge Y) &= H(X) - H(X|Y) \\ &= h(p) - H(X \oplus Y | Y) \quad \cdots\cdots \because \text{Given } Y, \text{ knowing } Y \text{ is exactly same as knowing what } X \oplus Y \text{ is.} \\ &\geq h(p) - H(X \oplus Y) \\ &= h(p) - h(P(X \neq Y)) \quad \cdots\cdots \text{from (1) and Binary Entropy} \\ &\geq h(p) - h(D) \quad \cdots\cdots \text{from (1), } D \geq P(X \neq Y) \text{ and } D \leq p \leq 0.5 \text{ and } h(D) \text{ is increasing in range } 0 \leq D \leq 0.5. \end{aligned}$$

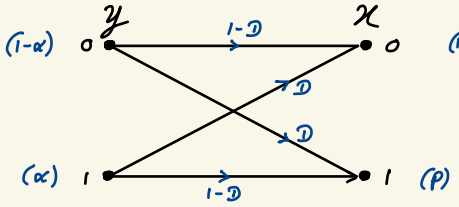
$$\begin{aligned} R(P,D) &= \min_{\substack{P_X = P \\ E[d(x,y)] \leq D}} I(X \wedge Y) \\ &= h(p) - h(D) \quad \cdots\cdots 0 \leq D \leq D_{max} = p = 0.5 \end{aligned}$$

- Method 2 - From Fano's Inequality,

$$\begin{aligned} H(X|Y) &\leq P(X \neq Y) \cdot \log(|\mathcal{X}|-1) + h(P(X \neq Y)) \\ &\leq D \cdot \log(2-1) + h(D) \\ &= h(D) \end{aligned}$$

Finding P_{XY} that Attains Feasible LB

- Consider the following "backward" channel, which is a BSC.



$$P(X \neq Y) = \alpha \cdot D + (1-\alpha) \cdot D = D$$

- To ensure that the input pmf of this backward channel is indeed a pmf, from ①,

$$\Rightarrow 0 \leq D \leq p \leq \tfrac{1}{2}$$
$$\Rightarrow D \leq p \leq 1-D$$
$$\Rightarrow 0 \leq p-D \leq 1-2D$$
$$\Rightarrow 0 \leq \alpha = \frac{p-D}{1-2D} \leq 1 \quad \cdots\cdots \left(\because \begin{matrix} p = (1-\alpha)D + \alpha(1-D) \\ \alpha = \frac{p-D}{1-2D} \end{matrix}\right)$$

- The rate distortion function for the channel designed here is

$$\begin{aligned} H(X|Y) &= P(Y=0)\cdot H(X|Y=0) + P(Y=1)\cdot H(X|Y=1) \\ &= (1-p)\cdot h(D) + p\cdot h(D) \\ &= h(D) \end{aligned}$$

$$\begin{aligned} I(X;Y) &= H(X) - H(X|Y) \\ &= h(p) - h(D) \\ &= \text{Shannon's lower bound} \end{aligned}$$

Hence, $R(p,D) = \begin{cases} h(p) - h(D), & \cdots 0 \leq D \leq p \\ 0 & \cdots \text{o.w} \end{cases}$

■

5) Converse of Rate Distortion Theorem

Theorem 1- Rate Distortion Theorem

Achievability - Let $\{X_t\}_{t=1}^{\infty}$ be a DMS with source pmf P on $\mathcal{X}$. Let $\{d_n\}_{n=1}^{\infty}$ be the distortion measure. Then $\forall\ D \geq 0$

$$R(D) = R(P,D) = \tilde{R}(P,D) = \min_{\substack{P_X = P \\ \mathbb{E}[d(X,Y)] \leq D}} I(X \wedge Y)$$

$$= \min_{W:\ \sum_{x,y} P(x)\cdot W(y|x)\cdot d(x,y) \leq D} I(P,W)$$

where $0 \leq D \leq D_{max} = \min_{y \in \mathcal{Y}} \sum_x P(x)\cdot d(x,y)$

Converse - Any sequence of codes $\{f^{(n)}, \phi^{(n)}\}_{n=1}^{\infty}$ with $\mathbb{E}[d(X^n, g_n(X^n))] \leq D+\delta$ for $(g_n \triangleq \phi_n \circ f_n)$ any n, where $\delta > 0$ is arbitary must have rate $\geq R(P,D)$

Proof : Consider $(f^{(n)}, \phi^{(n)})$ of rate R, with expected distortion being $\mathbb{E}[d(X^n, g_n(X^n))] \leq D+\delta$. Then for that n,

$$\Rightarrow R = \frac{1}{n} \log_2 \|f^{(n)}\|$$

$$\Rightarrow nR = \log_2 \|f^{(n)}\|$$

$$\geq H\left(f^{(n)}(X^n)\right) \quad \cdots\cdots \text{ upper bound for entropy.}$$

$$\geq H\left(\phi^{(n)}(f^{(n)}(X^n))\right) \quad \cdots\cdots \text{ Since } f^{(n)} \text{ determines } \phi^{(n)}(f^{(n)}(\cdot)) \text{ uncertainity in } f^{(n)} \text{ is lesser than that of the other.}$$

$$= H\left(g_n(X^n)\right)$$

$$= H\left(g_n(X^n)\right) - \underbrace{H\left(g_n(X^n) \mid X^n\right)}_{=0}$$

$$= I\left(g_n(x^n) \wedge x^n\right)$$

$$= H(x^n) - H(x^n \mid g_n(x^n))$$

Since X's are i.i.d; from chain rule

$$= \sum_{t=1}^{n} H(x_t) - \sum_{t=1}^{n} H\left(x_t \mid x^{t-1}, g_n(x^n)\right)$$

$$\geq \sum_{t=1}^{n} H(x_t) - \sum_{t=1}^{n} H\left(x_t \mid (g_n(x^n))_t\right) \quad \cdots \text{reducing conditioning terms}$$

$$= \sum_{t=1}^{n} I\left(x_t \wedge (g_n(x^n))_t\right) \quad \cdots\cdots\cdots (1)$$

From the definition $R(P, \mathcal{D}) = \min_{P_{x=P},\ \mathbb{E}[d(x,y)] \leq \mathcal{D}} I(x,y)$, taking $\mathcal{D}$ to be $\mathbb{E}[d(x_t, y_t)]$

$$\Rightarrow R(P, \mathcal{D}) \leq I(x_t, y_t)$$

$$\Rightarrow R(P, \mathbb{E}[d(x_t, y_t)]) \leq I(x_t, y_t)$$

$$\Rightarrow R(P, \mathbb{E}[d(x_t, (g_n(x^n))_t)]) \leq I(x_t, (g_n(x^n))_t)$$

Hence (1) becomes,

$$\Rightarrow nR \geq \sum_{t=1}^{n} R\left(P, \mathbb{E}[d(x_t, (g_n(x^n))_t)]\right)$$

$$= n \sum_{t=1}^{n} \frac{1}{n} \cdot R\left(P, \mathbb{E}[d(x_t, (g_n(x^n))_t)]\right)$$

$$\geq n \cdot R\left(P, \frac{1}{n} \sum_{t=1}^{n} \mathbb{E}[d(x_t, (g_n(x^n))_t)]\right) \quad \cdots \because \text{Convexity of Rate Distortion}$$

$$= n \cdot R\left(P, \underbrace{\mathbb{E}[d(x^n, g_n(x^n))]}_{\leq \mathcal{D}+\delta}\right) \quad \cdots \text{per symbol vs average distortion measure}$$

$$= n \cdot R(P, \mathcal{D}+\delta) \quad \cdots \quad R(P) \text{ is non increasing in } \mathcal{D} \text{ and } \mathbb{E}[d(\cdot)] \leq \mathcal{D}+\delta$$

$$\Rightarrow R \geq R(P, \mathcal{D}+\delta)$$

Hence $\delta > 0$ can be made arbitarily small. $\therefore$ From the continuity of $R(P, \cdot)$ we get that $R \geq R(P, \mathcal{D})$.

■

6) Distortion Typical Sequences

Definition - Distortion Typical Sequences - $A_{d,\varepsilon}^{(n)}$

Given $\varepsilon > 0$, the Distortion ε-Typical Set (Distortion Typical Set, Set of Distortion Typical Sequences) $A_{d,\varepsilon}^{(n)} \subseteq \mathcal{X}^n \times \mathcal{Y}^n$ with respect to P_{XY} on $\mathcal{X} \times \mathcal{Y}$ and distortion measure $d: \mathcal{X} \times \mathcal{Y} \longrightarrow \mathbb{R}^+ \cup \{0\}$ is defined as,

$$A_{d,\varepsilon}^{(n)} = \left\{ (x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n \;\middle|\; \begin{array}{l} \left| -\frac{1}{n} \cdot \log P_{X^n}(x^n) - H(X) \right| \leq \varepsilon, \\ \left| -\frac{1}{n} \cdot \log P_{Y^n}(y^n) - H(Y) \right| \leq \varepsilon, \\ \left| -\frac{1}{n} \cdot \log P_{X^nY^n}(x^n, y^n) - H(X,Y) \right| \leq \varepsilon, \\ \left| d(x^n, y^n) - \mathbb{E}[d(X,Y)] \right| \leq \varepsilon \end{array} \right\}$$

($H(X) \to H(P_X)$, $H(Y) \to H(P_Y)$, $H(X,Y) \to H(P_{XY})$)

where $\forall\ x^n \in \mathcal{X}^n \quad y^n \in \mathcal{Y}^n$

$$P_{X^nY^n}(x^n, y^n) = \prod_{t=1}^{n} P_{XY}(x_t, y_t)$$

- Clearly $A_{d,\varepsilon}^{(n)} \subseteq A_{\varepsilon}^{(n)}$

Theorem 4 - Distortion Assymptotic Equipartition Property (AEP)

$\{X_t, Y_t\}_{t=1}^{\infty}$ is a $\mathcal{X} \times \mathcal{Y}$ valued i.i.d process with p.m.f. P_{XY} on $\mathcal{X} \times \mathcal{Y}$. Let $d: \mathcal{X} \times \mathcal{Y} \longrightarrow \mathbb{R}^+ \cup \{0\}$ be a distortion measure. Then, given $\varepsilon > 0$

I) $\lim_{n \to \infty} P\left((X^n, Y^n) \in A_{d,\varepsilon}^{(n)} \right) = 1$

II) $\forall\ (x^n, y^n) \in A_{d,\varepsilon}^{(n)}$ it holds $\quad P_{Y^n}(y^n) \geq P_{Y^n|X^n}(y^n|x^n) \cdot 2^{-n(I(X \wedge Y) + 3\varepsilon)}$

Proofs

I) Since $\{X_n, Y_n\}_{n=1}^{\infty}$ is i.i.d with joint distribution P_{XY} by WLLN,

$$\Rightarrow d(x^n, y^n) = \frac{1}{n}\sum_{t=1}^{n} d(x_t, y_t) \xrightarrow[\text{probability}]{\text{in}} \mathbb{E}[d(X,Y)]$$

$$\Rightarrow \forall n \geq N_1(\varepsilon) \quad P\left((x^n, y^n) \;\middle|\; d(x^n, y^n) - \mathbb{E}[d(X,Y)] > \varepsilon\right) \leq \varepsilon/4$$

Combine the above fact with a modified version of the proof of part I of Joint AEP by replacing $\varepsilon/3$ with $\varepsilon/4$, we get

$$\lim_{n} P\left((x^n, y^n) \in A_{d,\varepsilon}^{(n)}\right) = 1$$

■

II) Since $(x^n, y^n) \in A_{d,\varepsilon}^{(n)} \in A_{\varepsilon}^{(n)}$ we can bound the probabilities of $p(x^n)$, $p(y^n)$ and $p(x^n, y^n)$.

$$\Rightarrow P_{Y^n|X^n}(y^n|x^n) = \frac{P_{X^nY^n}(x^n, y^n)}{P_{X^n}(x^n)}$$

$$= \frac{P_{Y^n}(y^n) \cdot P_{X^nY^n}(x^n, y^n)}{P_{X^n}(x^n) \cdot P_{Y^n}(y^n)}$$

$$\leq P_{Y^n}(y^n) \cdot \frac{2^{-n(H(X,Y)-\varepsilon)}}{2^{-n(H(X)-\varepsilon)} \cdot 2^{-n(H(Y)-\varepsilon)}}$$

$$= P_{Y^n}(y^n) \cdot 2^{n(I(X,Y)+3\varepsilon)}$$

$$\Rightarrow P_{Y^n}(y^n) \geq P_{Y^n|X^n}(y^n|x^n) \cdot 2^{-n(I(X,Y)+3\varepsilon)}$$

■

7) Achievability of Rate Distortion Theorem

Theorem 1- Rate Distortion Theorem

Achievability - Let $\{X_t\}_{t=1}^{\infty}$ be a DMS with source pmf P on $\mathcal{X}$. Let $\{d_n\}_{n=1}^{\infty}$ be the distortion measure. Then $\forall\ D \geq 0$

$$R(D) = R(P,D) = \tilde{R}(P,D) = \min_{\substack{P_X = P \\ E[d(X,Y)] \leq D}} I(X \wedge Y)$$

$$= \min_{W:\ \sum_{x,y} P(x)\cdot W(y|x)\cdot d(x,y) \leq D} I(P,W)$$

where $0 \leq D \leq D_{max} = \min_{y \in \mathcal{Y}} \sum_x P(x)\cdot d(x,y)$

Converse - Any sequence of codes $\{f^{(n)}, \phi^{(n)}\}_{n=1}^{\infty}$ with $E[d(x^n, g_n(x^n))] \leq D+\delta$ for $(g_n \triangleq \phi_n \circ f_n)$ any n, where $\delta > 0$ is arbitary must have rate $\geq R(P,D)$

Lemma - For $0 \leq u, v \leq 1$, $n > 0$ $u, v \in \mathbb{R}$, $n \in \mathbb{Z}$ it holds that $(1-uv)^n \leq 1 - u + e^{-nv}$

Proof :

Lemma 10.5.3 *For $0 \leq x, y \leq 1$, $n > 0$,*

$$(1-xy)^n \leq 1 - x + e^{-yn}. \tag{10.82}$$

Proof: Let $f(y) = e^{-y} - 1 + y$. Then $f(0) = 0$ and $f'(y) = -e^{-y} + 1 > 0$ for $y > 0$, and hence $f(y) > 0$ for $y > 0$. Hence for $0 \leq y \leq 1$, we have $1 - y \leq e^{-y}$, and raising this to the nth power, we obtain

$$(1-y)^n \leq e^{-yn}. \tag{10.83}$$

Thus, the lemma is satisfied for $x = 1$. By examination, it is clear that the inequality is also satisfied for $x = 0$. By differentiation, it is easy to see that $g_y(x) = (1-xy)^n$ is a convex function of x, and hence for $0 \leq x \leq 1$, we have

$$(1-xy)^n = g_y(x) \tag{10.84}$$

$$\leq (1-x)g_y(0) + xg_y(1) \tag{10.85}$$

$$= (1-x)1 + x(1-y)^n \tag{10.86}$$

$$\leq 1 - x + xe^{-yn} \tag{10.87}$$

$$\leq 1 - x + e^{-yn}. \quad \square \tag{10.88}$$

Proof of Achievability

Setting

- Assume W^* is the optimal W required. i.e.

$$\Rightarrow W^* = \underset{W:\ \sum_{x,y} P(x)\cdot W(y|x)\cdot d(x,y) \le D}{\arg\min}\ I(P,W)$$

$$\Rightarrow R(P,D) = I(P,W^*)$$

- Let Q be the p.m.f on $\mathcal{Y}$ obtained by,

$$Q(y) = \sum_x P(x)\ W^*(y|x) \qquad y \in \mathcal{Y}$$

Codebook Generation

- Pick $R > R(P,D) = I(P,W^*)$

- Let $Y(1), \ldots, Y(\lceil 2^{nR} \rceil)$ be i.i.d $\mathcal{Y}^n$-valued rvs each with i.i.d components $\sim$ pmf Q on $\mathcal{Y}$. Then,

$$P(Y(m) = y^n) = \prod_{t=1}^{n} Q(y_t) \qquad \cdots \quad m = 1, \ldots, \lceil 2^{nR} \rceil$$

$$P\left(Y(1) = y^n(1), \ldots, Y(\lfloor 2^{nR} \rfloor) = y^n(\lfloor 2^{nR} \rfloor)\right) = \prod_{m=1}^{\lfloor 2^{nR} \rfloor} P(Y(m) = y^n(m)) \quad \cdots (1)$$

- A realization of $(Y^n(1), \ldots, Y^n(\lceil 2^{nR} \rceil))$ constitutes a codebook $\mathcal{C} = (y^n(1), \ldots, y^n(\lceil 2^{nR} \rceil))$ and the $P(\mathcal{C})$ is given by (1).

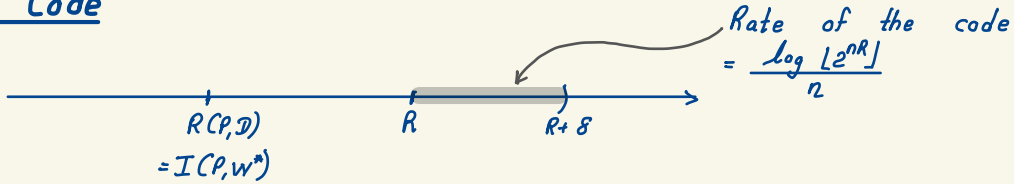
- $\mathcal{C}$ defines a code $(f^{(n)}, \phi^{(n)})$ as follows,

$$f^n(x^n) = \begin{cases} \text{Smallest } m \in \{1, \ldots, \lfloor 2^{nR} \rfloor\} \text{ such that} \\ (x^n, y^n(m)) \in A_{d,\varepsilon}^{(n)} \\ \\ 1 \text{ if } \nexists\ y^n(m) \in \mathcal{C} \text{ such that} \\ (x^n, y^n(m)) \in A_{d,\varepsilon}^{(n)} \end{cases}$$

$$\phi^n(m) = y^n(m)$$

- The constitution of decoding and encoding is mapping g. $\phi_o^{(n)} f_o^{(n)} \triangleq g_o^{\mathcal{C}}$

Rate of Code



Claim - $R < \frac{\log \lfloor 2^{nR} \rfloor}{n} < R+\delta$ $\forall$ n large enough $\forall$ $\delta > 0$

Proof:

$$M_n = \lceil 2^{nR} \rceil .$$

Then the rate of the code is

$$\frac{1}{n} \log M_n = \frac{1}{n} \log \lceil 2^{nR} \rceil .$$

Since

$$2^{nR} \leq \lceil 2^{nR} \rceil ,$$

$$R = \frac{1}{n} \log 2^{nR} \leq \frac{1}{n} \log \lceil 2^{nR} \rceil .$$

Also,

$$\lceil 2^{nR} \rceil \leq 2^{nR} + 1 \leq 2 \cdot 2^{nR} = 2^{nR+1}.$$

Therefore,

$$\frac{1}{n} \log \lceil 2^{nR} \rceil \leq \frac{1}{n} \log 2^{nR+1} = R + \frac{1}{n}.$$

Given any $\delta > 0$, choose n large enough so that

$$\frac{1}{n} < \delta.$$

Hence, for all sufficiently large n,

$$\boxed{R \leq \frac{1}{n} \log \lceil 2^{nR} \rceil < R + \delta.}$$

Calculation of Expected Distortion

- To Prove : $\exists$ a deterministic code $\mathcal{C}$ if rate $\geq R$ and $< R+\delta$ with $\mathbb{E}[d(x^n, g_n^{\mathcal{C}}(x^n))] \leq D+\varepsilon$ $\forall$ n sufficiently large.

- Let's compute the expected distortion over the ensemble of codebooks.

$$\therefore \bar{d} \triangleq \sum_{C \in \mathcal{C}} P(C) \cdot \mathbb{E}[d(x^n, g_n^C(x^n))]$$

$$= \sum_{C \in \mathcal{C}} P(C) \cdot \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot d(x^n, g_n^C(x^n)) \quad \cdots \textcircled{2}$$

Definition - J(C)

For each codebook $C \in \mathcal{C}$, define

$$J(C) = \left\{ x^n \in \mathcal{X}^n \;\middle|\; \exists\, y^n \in C \text{ such that } (x^n, y^n) \in A_{d,\varepsilon}^{(n)} \right\}$$

- This definition converts $\tilde{d}$ to,

$$\tilde{d} = \sum_{C \in \mathcal{C}} P(C) \cdot \left[\sum_{x^n \in J(C)} P(x^n) \cdot d\left(x^n, g_n^C(x^n)\right) + \sum_{x^n \notin J(C)} P(x^n) \cdot d\left(x^n, g_n^C(x^n)\right) \right]$$

$\leq \mathbb{E}[d(X,Y)] + \varepsilon$ ... by definition of Distortion Typical sets $A_{d,\varepsilon}^{(n)}$

$\leq d_{max}$

$\leq \mathcal{D} + \varepsilon$ ... since $\mathbb{E}[d(X,Y)] = \sum_{x,y} P(x) \cdot W^*(y,x) \cdot d(x,y) \leq \mathcal{D}$

$$\leq \mathcal{D} + \varepsilon + d_{max} \left(\sum_{C \in \mathcal{C}} P(C) \sum_{x^n \notin J(C)} P(x^n) \right) \quad \cdots\cdots \textcircled{3}$$

Definition - $P_e \triangleq \sum_{C \in \mathcal{C}} P(C) \sum_{x^n \notin J(C)} P(x^n)$

$$P_e \triangleq \sum_{C \in \mathcal{C}} P(C) \sum_{x^n \notin J(C)} P(x^n)$$

$$= \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \sum_{C \,:\, x^n \notin J(C)} P(C) \quad \cdots\cdots \text{change of order of Summation.}$$

$$= \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \left[P\left[(x^n, Y^n(m)) \in A_{d,\varepsilon}^{(n)} \right] \;\forall\; m = 1, \ldots, \lceil 2^{nR} \rceil \right]$$

$$= \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \left(\prod_{m=1}^{\lceil 2^{nR} \rceil} P\left[(x^n, Y^n(m)) \notin A_{d,\varepsilon}^{(n)} \right] \right) \quad \cdots\cdots \because Y^n(1), \ldots, Y^n(\lceil 2^{nR} \rceil) \text{ are mutually } \perp\!\!\!\perp.$$

$$= \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \left(P\left[(x^n, Y^n) \notin A_{d,\varepsilon}^{(n)} \right] \right)^{\lceil 2^{nR} \rceil} \quad \cdots\cdots \text{where } Y^n \sim Y^n(1) \text{ Since codewords are identically distributed.}$$

$$\leq \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \left(P\left[(x^n, Y^n) \notin A_{d,\varepsilon}^{(n)} \right] \right)^{2^{nR}} \quad \cdots\cdots \lceil 2^{nR} \rceil \geq 2^{nR}$$

$$\leq \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \left(1 - P\left[(x^n, Y^n) \in A_{d,\varepsilon}^{(n)} \right] \right)^{2^{nR}} \quad \cdots\cdots \textcircled{4}$$

- From part II of Distortion AEP,

$$P\left[(x^n, y^n) \in \mathcal{A}_{d,\varepsilon}^{(n)}\right] = \sum_{\{y^n \mid (x^n, y^n) \in \mathcal{A}_{d,\varepsilon}^{(n)}\}} Q(y^n)$$

$$\geq \sum_{\{y^n \mid (x^n, y^n) \in \mathcal{A}_{d,\varepsilon}^{(n)}\}} W^*(y^n \mid x^n) \cdot 2^{-n[I(P,W^*)+3\varepsilon]}$$

- By substituting above to ⊛

$$P_e \leq \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \left(1 - 2^{-n[I(P,W^*)+3\varepsilon]} \sum_{\{y^n \mid (x^n, y^n) \in \mathcal{A}_{d,\varepsilon}^{(n)}\}} W^*(y^n \mid x^n)\right)^{2^{nR}}$$

from Lemma

$$\leq \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \left(1 - \sum_{\{y^n \mid (x^n, y^n) \in \mathcal{A}_{d,\varepsilon}^{(n)}\}} W^*(y^n \mid x^n) + e^{\left(-2^{nR}\, 2^{-n[I(P,W^*)+3\varepsilon]}\right)}\right)^{2^{nR}}$$

$$= 1 - \sum_{x^n \in \mathcal{X}^n} P(x^n) \cdot \sum_{\{y^n \mid (x^n, y^n) \in \mathcal{A}_{d,\varepsilon}^{(n)}\}} W^*(y^n \mid x^n) + e^{-2^{n\left[R - (I(P,W^*)+3\varepsilon)\right]}}$$

$\leq P\left((X^n, Y^n) \in \mathcal{A}_{d,\varepsilon}^{(n)}\right)$ with $P_X = P$, $P_{Y|X} = W^*$

This goes to '0' super exponentially in n if $R > I(P,W^*) + 3\varepsilon$

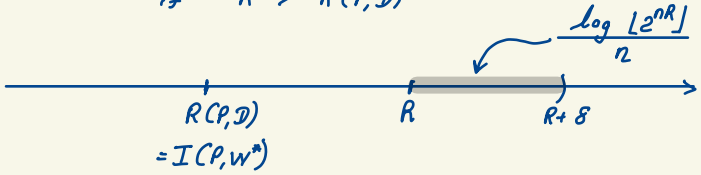
$\leq \varepsilon$ ... if n is large enough provided that .... ④

$R > I(P,W^*)$

- Since $\varepsilon > 0$ can be choosen arbitarily small, we see if it is given $R > I(P,W^*) = R(P,D)$. i.e P_e can be arbitarily small if n is sufficiently large.

$\therefore$ ④ $\Longrightarrow$ $\tilde{d} \leq D + 2\varepsilon$ $\forall n$ sufficiently large

② $\Longrightarrow$ $\exists$ at least one codebook C^* with $\mathbb{E}[d(X^n, g_n^C(X^n))] \leq D + 3\varepsilon$ $\forall$ n large if $R > R(P,D)$



Q.E.D

Corollary - Strong Achievability Condition for RDT

For $R > R(P,D)$, can find a sequence of codes $\{f^{(n)}, \phi^{(n)}\}$ of rate $\geq R$ such that,

$$P\left(d\left(x^n, \phi^{(n)}(f^{(n)}(x^n))\right) > D\right) \downarrow 0 \text{ as } n \uparrow \infty$$

Setting

Definition A sequence $x^n \in \mathcal{X}^n$ is said to be ϵ-*strongly typical* with respect to a distribution $p(x)$ on $\mathcal{X}$ if:

1. For all $a \in \mathcal{X}$ with $p(a) > 0$, we have
$$\left|\frac{1}{n}N(a|x^n) - p(a)\right| < \frac{\epsilon}{|\mathcal{X}|}. \tag{10.106}$$
2. For all $a \in \mathcal{X}$ with $p(a) = 0$, $N(a|x^n) = 0$.

$N(a|x^n)$ is the number of occurrences of the symbol a in the sequence x^n.

The set of sequences $x^n \in \mathcal{X}^n$ such that x^n is strongly typical is called the *strongly typical* set and is denoted $A_\epsilon^{*(n)}(X)$ or $A_\epsilon^{*(n)}$ when the random variable is understood from the context.

Definition A pair of sequences $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ is said to be ϵ-*strongly typical* with respect to a distribution $p(x, y)$ on $\mathcal{X} \times \mathcal{Y}$ if:

1. For all $(a, b) \in \mathcal{X} \times \mathcal{Y}$ with $p(a, b) > 0$, we have
$$\left|\frac{1}{n}N(a, b|x^n, y^n) - p(a, b)\right| < \frac{\epsilon}{|\mathcal{X}||\mathcal{Y}|}. \tag{10.107}$$
2. For all $(a, b) \in \mathcal{X} \times \mathcal{Y}$ with $p(a, b) = 0$, $N(a, b|x^n, y^n) = 0$.

$N(a, b|x^n, y^n)$ is the number of occurrences of the pair (a, b) in the pair of sequences (x^n, y^n).

Lemma 10.6.1 *Let (X_i, Y_i) be drawn i.i.d. $\sim p(x, y)$. Then* $\Pr(A_\epsilon^{*(n)}) \to 1$ *as* $n \to \infty$.

The set of sequences $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ such that (x^n, y^n) is strongly typical is called the *strongly typical* set and is denoted $A_\epsilon^{*(n)}(X, Y)$ or $A_\epsilon^{*(n)}$. From the definition, it follows that if $(x^n, y^n) \in A_\epsilon^{*(n)}(X, Y)$, then $x^n \in A_\epsilon^{*(n)}(X)$. From the strong law of large numbers, the following lemma is immediate.

Lemma 10.6.2 *Let $Y_1, Y_2, \ldots, Y_n$ be drawn i.i.d. $\sim p(y)$. For $x^n \in A_\epsilon^{*(n)}(X)$, the probability that $(x^n, Y^n) \in A_\epsilon^{*(n)}$ is bounded by*

$$2^{-n(I(X;Y)+\epsilon_1)} \leq \Pr((x^n, Y^n) \in A_\epsilon^{*(n)}) \leq 2^{-n(I(X;Y)-\epsilon_1)}, \tag{10.108}$$

where ϵ_1 goes to 0 as $\epsilon \to 0$ and $n \to \infty$.

Proof Using Strongly Typical Sequences

Proof: Fix $p(\hat{x}|x)$. Calculate $p(\hat{x}) = \sum_x p(x)p(\hat{x}|x)$. Fix $\epsilon > 0$. Later we will choose ϵ appropriately to achieve an expected distortion less than $D + \delta$.

Generation of codebook: Generate a rate distortion codebook $\mathcal{C}$ consisting of 2^{nR} sequences $\hat{X}^n$ drawn i.i.d. $\sim \prod_i p(\hat{x}_i)$. Denote the sequences $\hat{X}^n(1), \ldots, \hat{X}^n(2^{nR})$.

Encoding: Given a sequence X^n, index it by w if there exists a w such that $(X^n, \hat{X}^n(w)) \in A_\epsilon^{*(n)}$, the strongly jointly typical set. If there is more than one such w, send the first in lexicographic order. If there is no such w, let $w = 1$.

Decoding: Let the reproduced sequence be $\hat{X}^n(w)$.

Calculation of distortion: As in the case of the proof in Section 10.5, we calculate the expected distortion over the random choice of codebook as

$$D = E_{X^n,\mathcal{C}}d(X^n, \hat{X}^n) \tag{10.109}$$

$$= E_{\mathcal{C}} \sum_{x^n} p(x^n)d(x^n, \hat{X}^n(x^n)) \tag{10.110}$$

$$= \sum_{x^n} p(x^n)E_{\mathcal{C}}d(x^n, \hat{X}^n), \tag{10.111}$$

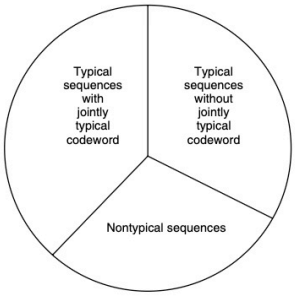


FIGURE 10.8. Classes of source sequences in rate distortion theorem.

where the expectation is over the random choice of codebook. For a fixed codebook $\mathcal{C}$, we divide the sequences $x^n \in \mathcal{X}^n$ into three categories, as shown in Figure 10.8.

- *Nontypical sequences* $x^n \notin A_\epsilon^{*(n)}$. The total probability of these sequences can be made less than ϵ by choosing n large enough. Since the individual distortion between any two sequences is bounded by $d_{\max}$, the nontypical sequences can contribute at most $\epsilon d_{\max}$ to the expected distortion.
- *Typical sequences $x^n \in A_\epsilon^{*(n)}$ such that there exists a codeword $\hat{X}^n(w)$ that is jointly typical with x^n.* In this case, since the source sequence and the codeword are strongly jointly typical, the continuity of the distortion as a function of the joint distribution ensures that they are also distortion typical. Hence, the distortion between these x^n and their codewords is bounded by $D + \epsilon d_{\max}$, and since the total probability of these sequences is at most 1, these sequences contribute at most $D + \epsilon d_{\max}$ to the expected distortion.
- *Typical sequences $x^n \in A_\epsilon^{*(n)}$ such that there does not exist a codeword $\hat{X}^n$ that is jointly typical with x^n.* Let P_e be the total probability of these sequences. Since the distortion for any individual sequence is bounded by $d_{\max}$, these sequences contribute at most $P_e d_{\max}$ to the expected distortion.

The sequences in the first and third categories are the sequences that may not be well represented by this rate distortion code. The probability of the first category of sequences is less than ϵ for sufficiently large n. The probability of the last category is P_e, which we will show can be made small. This will prove the theorem that the total probability of sequences that are not well represented is small. In turn, we use this to show that the average distortion is close to D.

Calculation of P_e: We must bound the probability that there is no codeword that is jointly typical with the given sequence X^n. From the joint AEP, we know that the probability that X^n and any $\hat{X}^n$ are jointly typical is $\doteq 2^{-nI(X;\hat{X})}$. Hence the expected number of jointly typical $\hat{X}^n(w)$ is $2^{nR}2^{-nI(X;\hat{X})}$, which is exponentially large if $R > I(X; \hat{X})$.

But this is not sufficient to show that $P_e \to 0$. We must show that the probability that there is no codeword that is jointly typical with X^n goes to zero. The fact that the expected number of jointly typical codewords is exponentially large does not ensure that there will at least one with high probability. Just as in (10.94), we can expand the probability of error as

$$P_e = \sum_{x^n \in A_\epsilon^{*(n)}} p(x^n)\left[1 - \Pr((x^n, \hat{X}^n) \in A_\epsilon^{*(n)})\right]^{2^{nR}}. \tag{10.112}$$

From Lemma 10.6.2 we have

$$\Pr((x^n, \hat{X}^n) \in A_\epsilon^{*(n)}) \geq 2^{-n(I(X;\hat{X})+\epsilon_1)}. \tag{10.113}$$

Substituting this in (10.112) and using the inequality $(1 - x)^n \leq e^{-nx}$, we have

$$P_e \leq e^{-(2^{nR}2^{-n(I(X;\hat{X})+\epsilon_1)})}, \tag{10.114}$$

which goes to 0 as $n \to \infty$ if $R > I(X; \hat{X}) + \epsilon_1$. Hence for an appropriate choice of ϵ and n, we can get the total probability of all badly represented sequences to be as small as we want. Not only is the expected distortion close to D, but with probability going to 1, we will find a codeword whose distortion with respect to the given sequence is less than $D + \delta$. □

8) Distortion Rate Function

Definition - Given a rate $R \geq 0$, the infimum of all distortions D such that (R, D) constitutes achievable rate - distortion pairs is termed the minimum R- rate distortion for $\{X_t\}_{t=1}^{\infty}$. This function,

$D(R)$ = Distortion rate function of the DMS $\{X_t\}_{t=1}^{\infty}$

Notation - For a pmf $P_X = P$ define a quantity $d(P, W)$ as we did for $R(P, W)$ for RDT where $W : \mathcal{X} \to \mathcal{Y}$ is the stochastic matrix.

$$d(P, W) \triangleq \underset{(P,W)}{\mathbb{E}}[d(X, Y)] = \sum_{x,y} P(x) \cdot W(y|x) \cdot d(x, y)$$

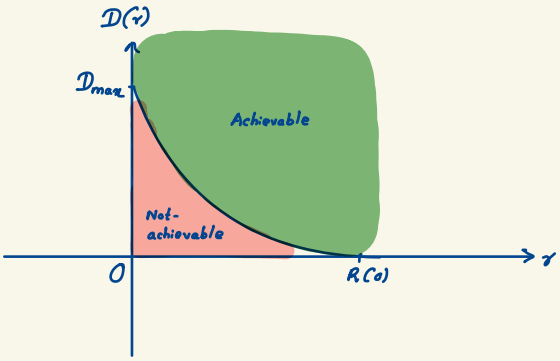
Theorem 5 - Distortion Rate Function Theorem

Let $\{X_t\}_{t=1}^{\infty}$ be a DMS with pmf P on $\mathcal{X}$. Then for $\forall\ R \geq 0$

$$D(R) = \min_{W : I(P,W) \leq R} d(P, W) = \underset{(P,W)}{\mathbb{E}}[d(X, Y)]$$

Proof :- Not done in the course. But similar to RDT.

Lemma 1 - $D(R) = D(P, R)$ is finite value non-increasing in R and convex function of $R \geq 0$.



Lemma 2 - Let $\{X_t\}_{t=1}^{\infty}$ be DMS with pmf P on $\mathcal{X}$. Let $R(D)$ and $D(R)$ be the rate - distortion and distortion - rate functions for this DMS. Then $R(\cdot)$ and $D(\cdot)$ are inverse of each other.

i.e. $R(D(\delta)) = \delta \qquad 0 \leq \gamma \leq R(0)$

$D(R(\Delta)) = \Delta \qquad 0 \leq \Delta \leq D_{max}$

Proof :- Not done in the course.

9) Joint Source - Channel Coding Theorem and Seperation Theorem with Distortion

Theorem 6 - Achievability - If $R(D) < C$, can find a source encoder, channel encoder, channel decoder, source decoder such that,

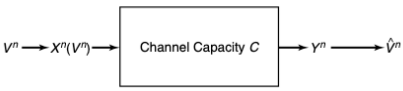
$$\mathbb{E}[\, d(U^n, g_n(U^n))] < D + \varepsilon$$

with source - channel seperation in coding.

Converse - For any n, if code exists with $\mathbb{E}[\, d(U^n, g_n(U^n))] < D + \varepsilon$ then $R(D) < C$

Proof:

10.17 *Source–channel separation theorem with distortion.* Let $V_1, V_2, \ldots, V_n$ be a finite alphabet i.i.d. source which is encoded as a sequence of n input symbols X^n of a discrete memoryless channel. The output of the channel Y^n is mapped onto the reconstruction alphabet $\hat{V}^n = g(Y^n)$. Let $D = Ed(V^n, \hat{V}^n) = \frac{1}{n}\sum_{i=1}^{n} Ed(V_i, \hat{V}_i)$ be the average distortion achieved by this combined source and channel coding scheme.



(a) Show that if $C > R(D)$, where $R(D)$ is the rate distortion function for V, it is possible to find encoders and decoders that achieve a average distortion arbitrarily close to D.

(b) (Converse) Show that if the average distortion is equal to D, the capacity of the channel C must be greater than $R(D)$.

Proof

Achievability

Assume

$$R(P, D) < C.$$

Choose a number R such that

$$R(P, D) < R < C.$$

We shall construct the source-channel code using separation.

Step 1: Source Coding

Since

$$R > R(P, D),$$

by the achievability part of the rate-distortion theorem, for every $\epsilon > 0$ and for all sufficiently large n, there exists a source encoder-decoder pair

$$f_s^{(n)} : \mathcal{U}^n \to \{1, \ldots, M_n\},$$
$$\phi_s^{(n)} : \{1, \ldots, M_n\} \to \hat{\mathcal{U}}^n$$

such that

$$\frac{1}{n} \log M_n < R$$

and

$$\mathbb{E}\left[d\left(U^n, \phi_s^{(n)}(f_s^{(n)}(U^n))\right)\right] < D + \frac{\epsilon}{2}.$$

Let

$$J = f_s^{(n)}(U^n)$$

be the source-code index. Thus the source encoder maps U^n to one of M_n possible indices.

Step 2: Channel Coding

Since

$$R < C,$$

by Shannon's channel coding theorem, for all sufficiently large n, there exists a channel encoder-decoder pair

$$f_c^{(n)} : \{1, \ldots, M_n\} \to \mathcal{X}^n,$$
$$\phi_c^{(n)} : \mathcal{Y}^n \to \{1, \ldots, M_n\}$$

such that, if

$$X^n = f_c^{(n)}(J)$$

is transmitted through the channel and

$$\hat{J} = \phi_c^{(n)}(Y^n)$$

is the decoded index, then

$$P(\hat{J} \neq J) < \eta,$$

where $\eta > 0$ will be chosen later.

Step 3: Composition

The complete source-channel encoder is the composition

$$U^n \xmapsto{f_s^{(n)}} J \xmapsto{f_c^{(n)}} X^n.$$

The channel produces Y^n, and the complete source-channel decoder is the composition

$$Y^n \xmapsto{\phi_c^{(n)}} \hat{J} \xmapsto{\phi_s^{(n)}} \hat{U}^n.$$

Thus

$$\hat{U}^n = \phi_s^{(n)}(\phi_c^{(n)}(Y^n)).$$

Let

$$\tilde{U}^n = \phi_s^{(n)}(J)$$

be the reproduction sequence that would be produced if the channel index were decoded correctly.

If

$$\hat{J} = J,$$

then

$$\hat{U}^n = \tilde{U}^n.$$

Therefore,

$$d(U^n, \hat{U}^n) \le d(U^n, \tilde{U}^n) + d_{\max}\mathbf{1}\{\hat{J} \neq J\},$$

where

$$d_{\max} = \max_{u,\hat{u}} d(u, \hat{u}) < \infty.$$

Taking expectation,

$$\mathbb{E}[d(U^n, \hat{U}^n)] \le \mathbb{E}[d(U^n, \tilde{U}^n)] + d_{\max} P(\hat{J} \neq J).$$

Using the guarantees from source coding and channel coding,

$$\mathbb{E}[d(U^n, \hat{U}^n)] < D + \frac{\epsilon}{2} + d_{\max}\eta.$$

Choose $\eta > 0$ small enough so that

$$d_{\max}\eta < \frac{\epsilon}{2}.$$

Then

$$\mathbb{E}[d(U^n, \hat{U}^n)] < D + \epsilon.$$

Hence, if

$$R(P, D) < C,$$

then distortion D is achievable over the channel.

$$\boxed{R(P, D) < C \implies \exists \text{ source-channel codes with } \mathbb{E}[d(U^n, \hat{U}^n)] < D + \epsilon}$$

This proves achievability.

Converse

Let

$$\{F^{(n)}, G^{(n)}\}_{n=1}^{\infty}$$

be any sequence of source-channel codes, where

$$F^{(n)} : \mathcal{U}^n \to \mathcal{X}^n$$

is the source-channel encoder and

$$G^{(n)} : \mathcal{Y}^n \to \hat{\mathcal{U}}^n$$

is the source-channel decoder.

Set

$$X^n = F^{(n)}(U^n),$$

and

$$\hat{U}^n = G^{(n)}(Y^n).$$

Assume that for some $\epsilon > 0$,

$$\mathbb{E}[d(U^n, \hat{U}^n)] \le D + \epsilon.$$

By the converse part of the rate-distortion theorem,

$$I(U^n; \hat{U}^n) \ge nR(P, D + \epsilon).$$

Now observe that the variables form the Markov chain

$$U^n \longrightarrow X^n \longrightarrow Y^n \longrightarrow \hat{U}^n.$$

Therefore, by the data processing inequality,

$$I(U^n; \hat{U}^n) \le I(X^n; Y^n).$$

Since the channel is a DMC with capacity C,

$$I(X^n; Y^n) \le nC.$$

Combining the previous inequalities gives

$$nR(P, D + \epsilon) \le I(U^n; \hat{U}^n) \le I(X^n; Y^n) \le nC.$$

Hence

$$R(P, D + \epsilon) \le C.$$

Now suppose there exists a sequence of codes such that

$$\limsup_{n\to\infty} \mathbb{E}[d(U^n, \hat{U}^n)] \le D.$$

Then, for every $\epsilon > 0$, for all sufficiently large n,

$$\mathbb{E}[d(U^n, \hat{U}^n)] \le D + \epsilon.$$

Thus, by the argument above,

$$R(P, D + \epsilon) \le C.$$

Letting $\epsilon \downarrow 0$, and using the continuity of $R(P, D)$ in D, we get

$$R(P, D) \le C.$$

This proves the converse.

$$\boxed{\text{If distortion } D \text{ is achievable, then } R(P, D) \le C.}$$

- In the above case we are using channel n-times to transmit n symbols. Now we are generalizing for the case n channel uses to send k source symbols.

Theorem 7 - Joint Source-Channel Coding Theorem

digits/channel use — digits/Source symbol

Achievability $\{U_t\}_{t=1}^{\infty}$ is a DMS with rate-distortion function $R(D)$. W is the matrix of DMC with capacity C. Then, for every (d, D), $\exists$ a sequence of k-to-n_k block codes $(f^{(k)}, \phi^{(k)})$ that satisfy the average fidelity criterion (d, D) such that,

$$\lim_{k\to\infty} \frac{n_k}{k} = \frac{R(D)}{C}$$

Converse If a k-to-n code satisfies the average fidelity criterion (d, D) then,

$$\frac{n}{k} \geq \frac{R(D)}{C}$$

Proof :- Not done in the course. Achievability proof uses "Source Channel Seperation".

Clock Rates

- Suppose that the source has a clock rate of 1 symbol every T_s seconds.

$\iff$ $1/T_s$ Source symbols / Unit time (s)

- Suppose that channel is used once, every T_c seconds.

$\iff$ $1/T_c$ channel uses / Unit time (s)

Minimum Distortion Theoretically Attainable

$$\Delta > D\left(\frac{T_s}{T_c} C\right)$$

$kT_s = nT_c$ Case

- Assume for every k-to-n block code. $\longrightarrow$ $kT_s = nT_c$
- From the converse of the Theorem 7 $\longrightarrow$ $\frac{n}{k} \geq \frac{R(\Delta)}{C}$

$$\Rightarrow nC \geq kR(\Delta)$$

$$\Rightarrow \boxed{\frac{C}{T_c} \geq \frac{R(\Delta)}{T_s}}$$

- From the achievability of Theorem 7 $\longrightarrow$ $\frac{R(\Delta)}{T_s} < \frac{C}{T_c}$

$$\Rightarrow R(\Delta) < C \cdot \frac{T_s}{T_c}$$

Distortion-rate function is decreasing

$$\Rightarrow D(R(\Delta)) > D\left(C \cdot \frac{T_s}{T_c}\right)$$

$$\Rightarrow \boxed{\Delta > D\left(\frac{T_s}{T_c} C\right)}$$

Minimum Distortion Theoretically Attainable

$T_s = T_c$ Case

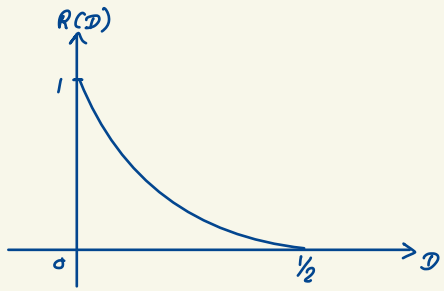
- Then since $kT_s = nT_c \Longrightarrow n = k$
- Then the minimum distortion theoretically attainable is, $\Longrightarrow \boxed{\Delta > D(C)}$

To Code or Not to Code

Example - Transmit a DMS $\{X_t\}_{t=1}^{\infty}$ Bernoulli $(1/2)$ rvs that are $\{0,1\}$ valued with $p = P_X(1) = 1 - P_X(0) = 1/2$. Let d_H = Hamming Distortion measure. This is transmitted over a BSC (ε), $0 < \varepsilon < 1/2$. Show that Optimal Distortion Theoretically Achievable is achieved without any coding.

- From Theorem 3, since $h(p) = h(1/2) = 1$, we have,

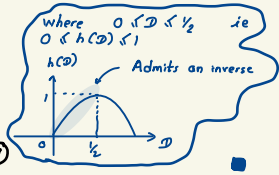
$$R(D) = \begin{cases} h(p) - h(D) = 1 - h(D) & 0 \leq D \leq D_{max} = \min\{p, 1-p\} = 1/2 \\ 0 & D > D_{max} = \min\{p, 1-p\} = 1/2 \end{cases}$$

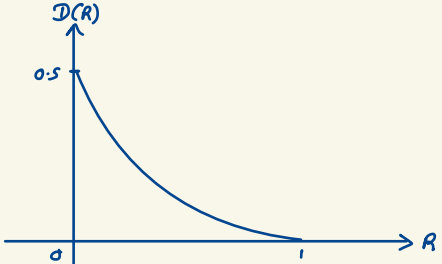


- Claim : $D(R) = \begin{cases} h^{-1}(1-R) & 0 \leq R \leq 1 \\ 0 & R \geq 1 \end{cases}$

Proof: From lemma 2 under distortion rate function, we can get the invers of $R(D)$ to find $D(R)$.

$$\Rightarrow R(D) = 1 - h(D)$$
$$\Rightarrow h(D) = 1 - R(D)$$
$$= 1 - R$$
$$\Rightarrow D = h^{-1}(1-R) \quad \cdots \quad 0 \leq R \leq 1 \quad \cdots (1)$$





- Earlier in the course we saw the channel capacity of a BSC as

$$C = 1 - h(\varepsilon) \quad \text{bits/channel use} \quad \cdots (2)$$

- Consider the special case of $T_s = T_c$. Under this, the minimum distortion theoretically attainable is

$$\begin{aligned} &= D(c) \\ &= D(1 - h(\varepsilon)) \quad \cdots \because (1) \\ &= h^{-1}(1 - (1 - h(\varepsilon))) \quad \cdots \because (2) \\ &= h^{-1}(h(\varepsilon)) \\ &= \varepsilon \quad \cdots\cdots (3) \end{aligned}$$

- Consider the expected average distortion upon connecting DMS directly to DMC for each $k = 1, 2, \ldots$

$$\begin{aligned} \mathbb{E}[d(U^k, V^k)] &= \frac{1}{k} \sum_{i=1}^{k} \mathbb{E}[d_H(u_i, v_i)] \\ &= \frac{1}{k} \sum_{i=1}^{k} P(u_i \neq v_i) \\ &= \frac{1}{k} \cdot k \cdot \varepsilon \\ &= \varepsilon \quad \cdots (4) \end{aligned}$$

- Hence, optimum distortion performance can be attained without any source encoding / decoding or channel encoding / decoding if we have $T_s = T_c$.

■

Remark - Why didn't we need any coding?
DMS and DMC are "matched" to each other.
DMS : Optimal W^* is BSC(ε) for distortion ε.
DMC : Optimal P^* is $(0.5, 0.5)$

Reading Assignment
Gastpar, Rimoldi, Vetterli:
"To code or not to code"